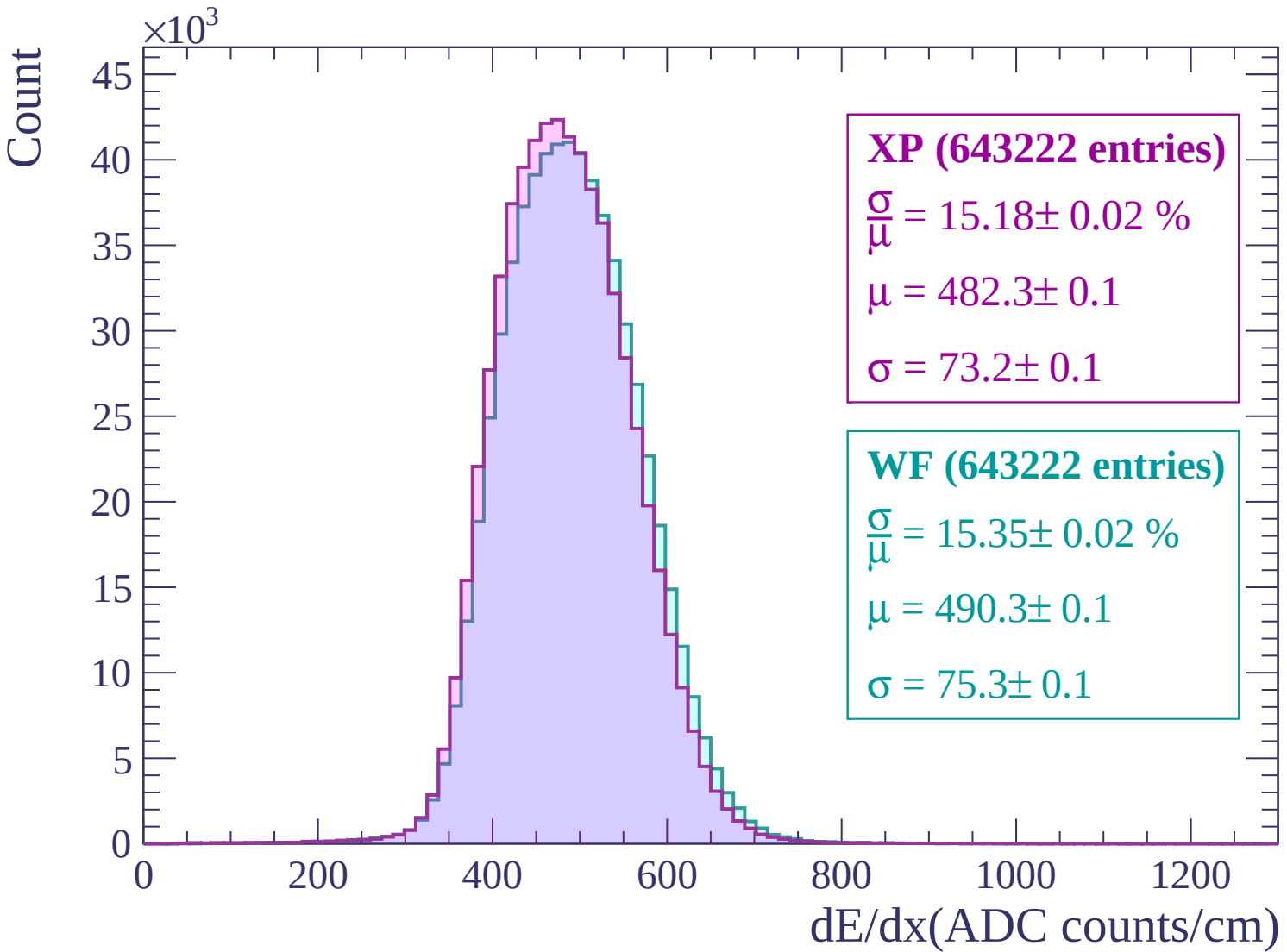
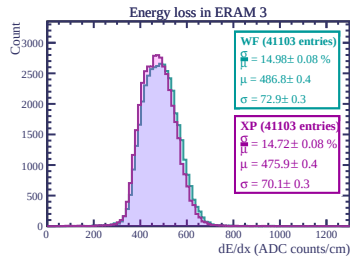
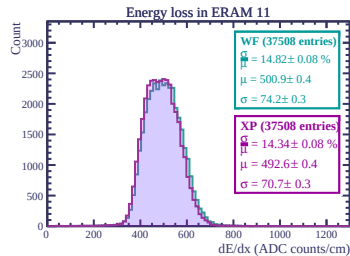
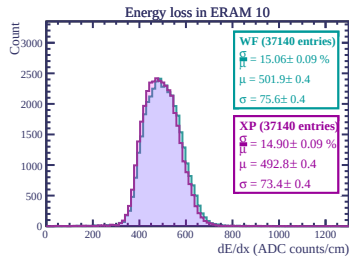
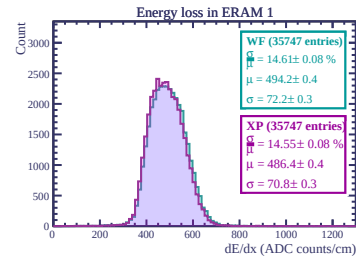
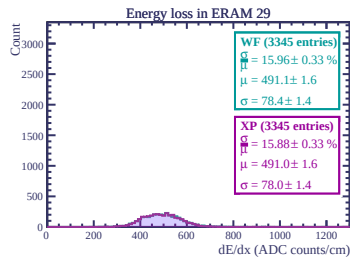
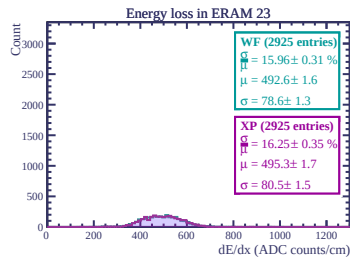
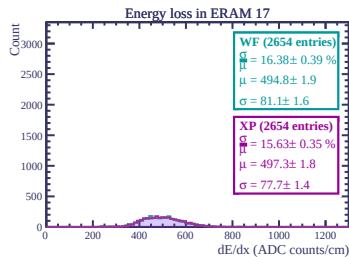
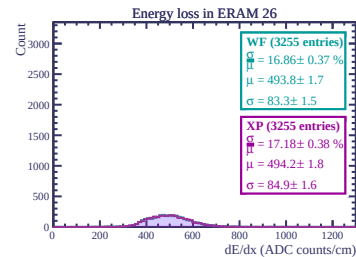
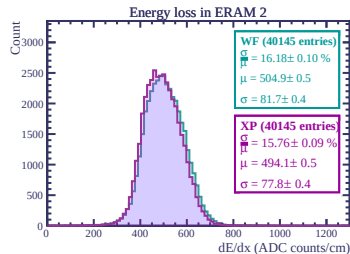
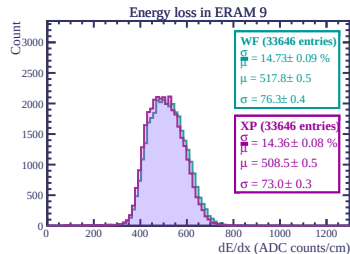
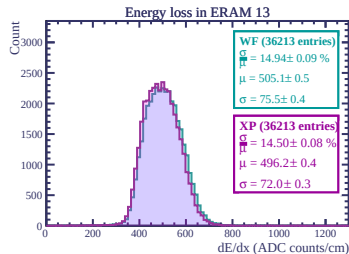
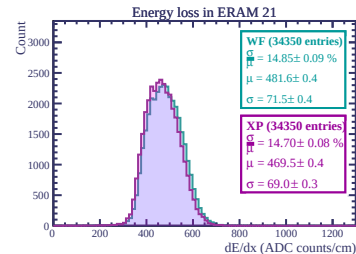
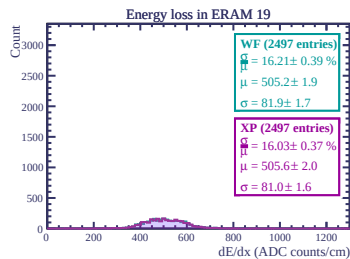
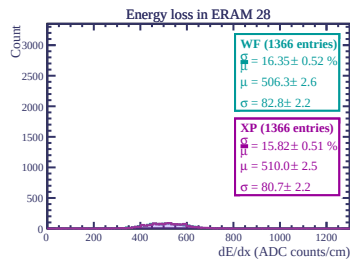
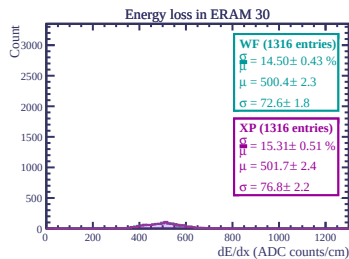
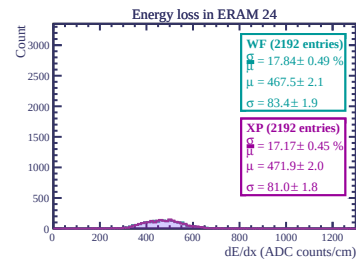
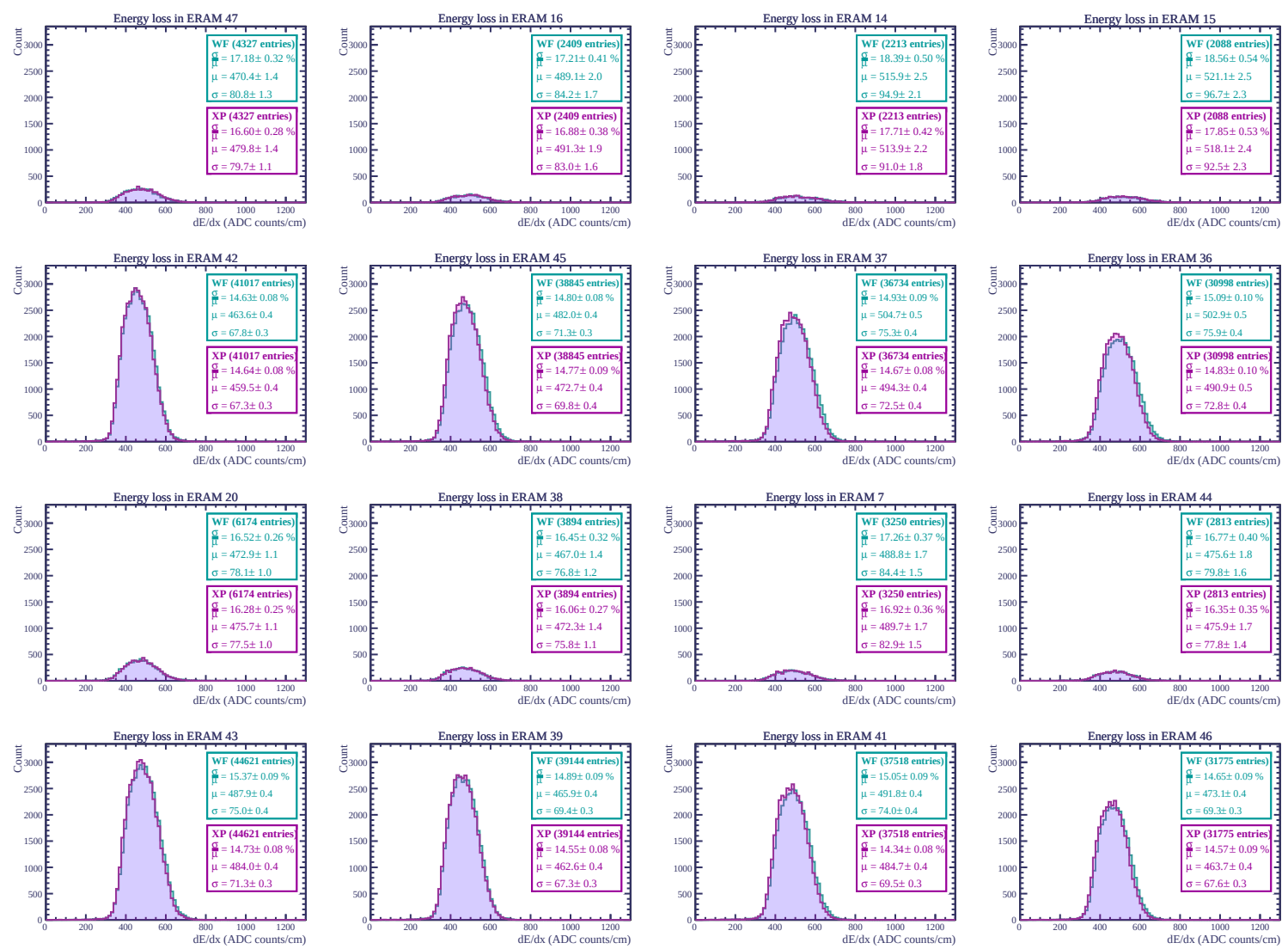
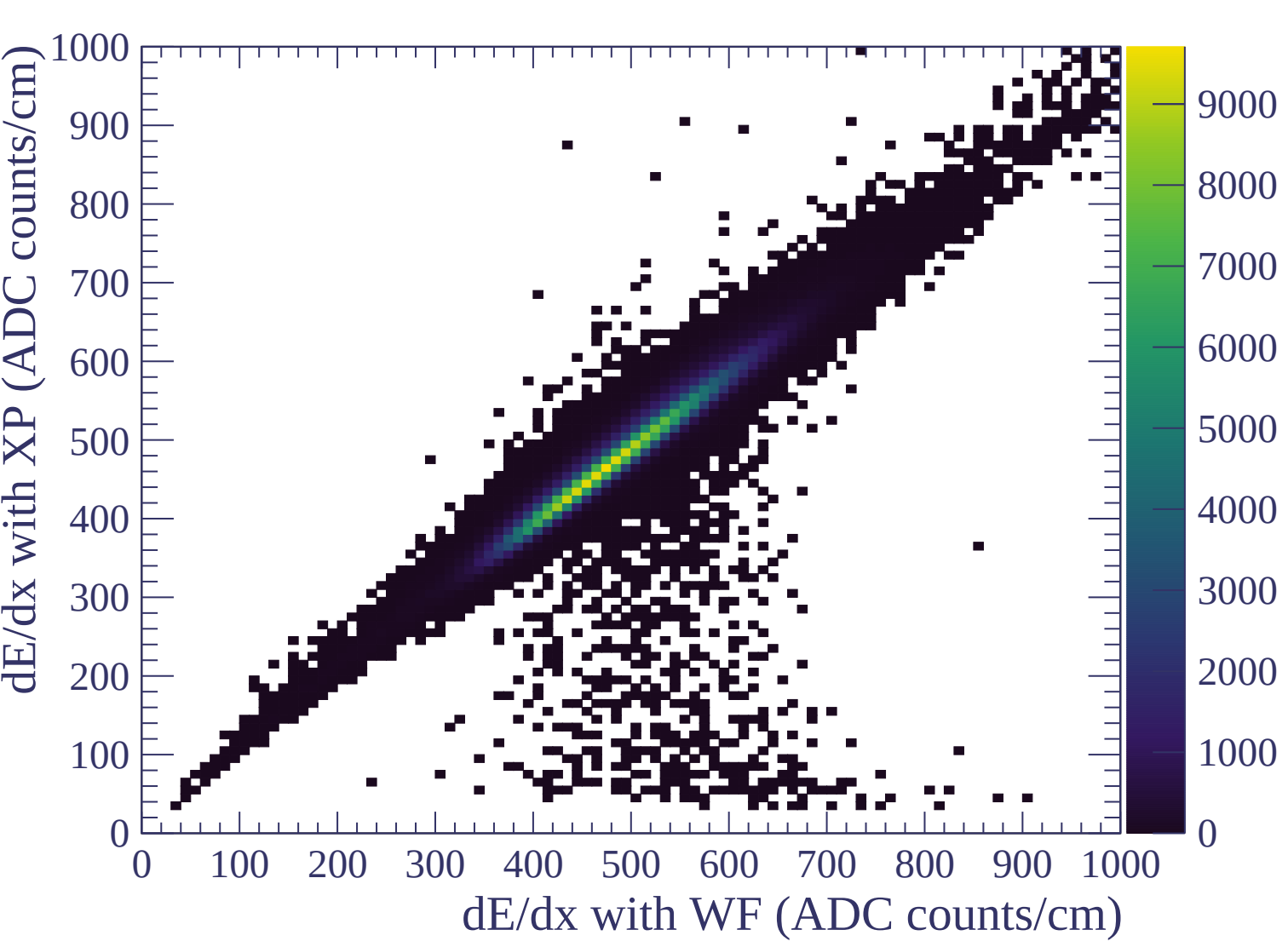
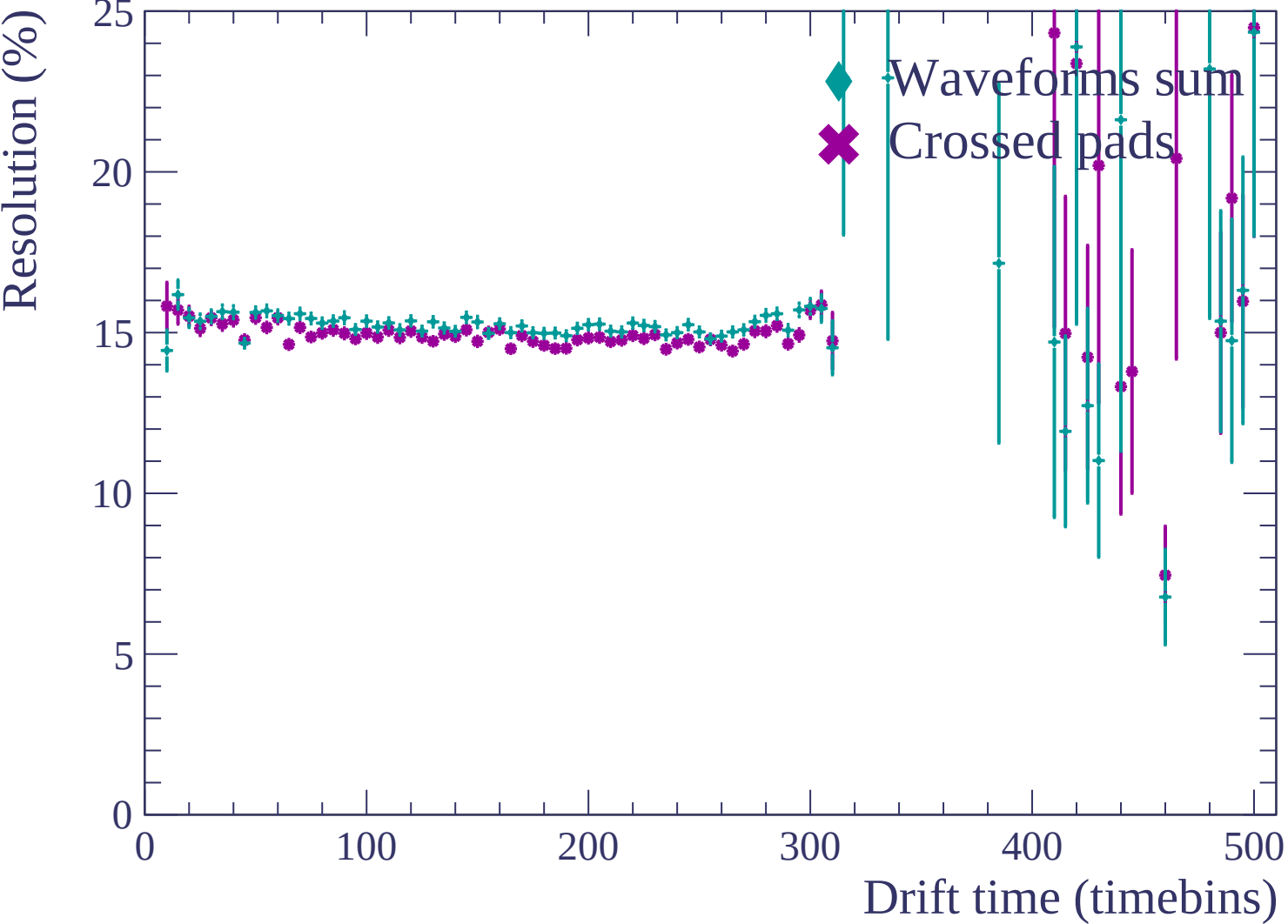


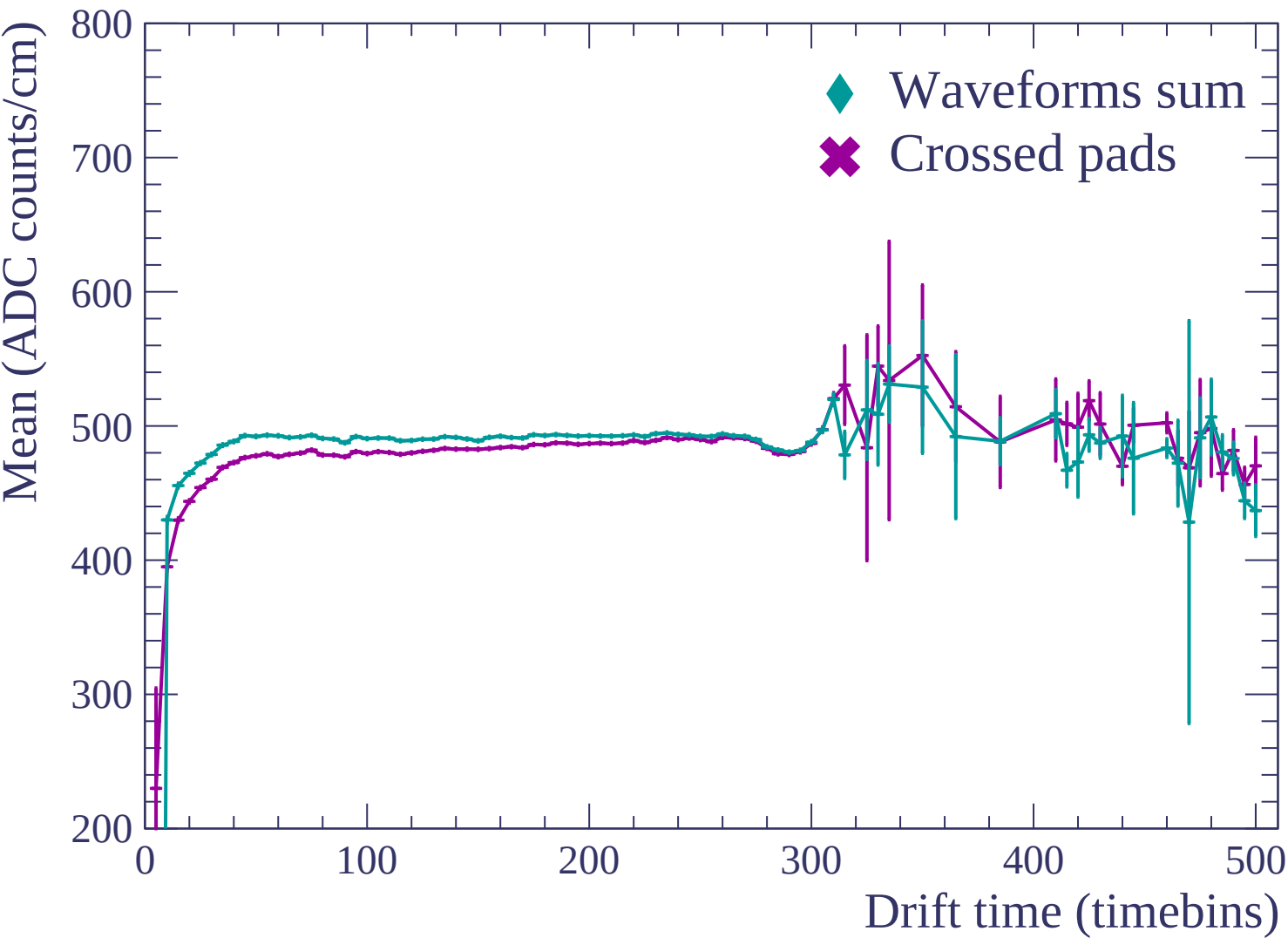


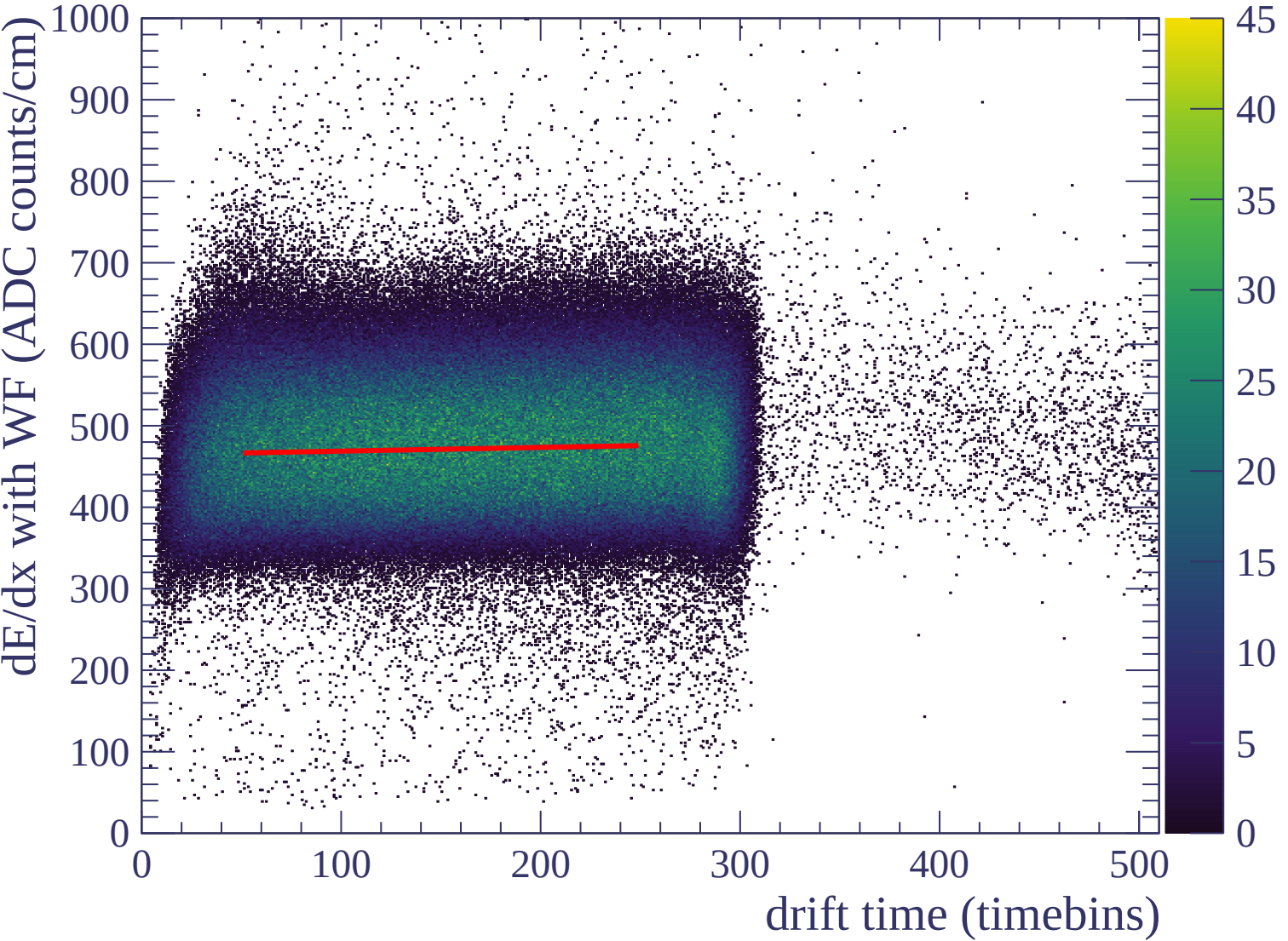


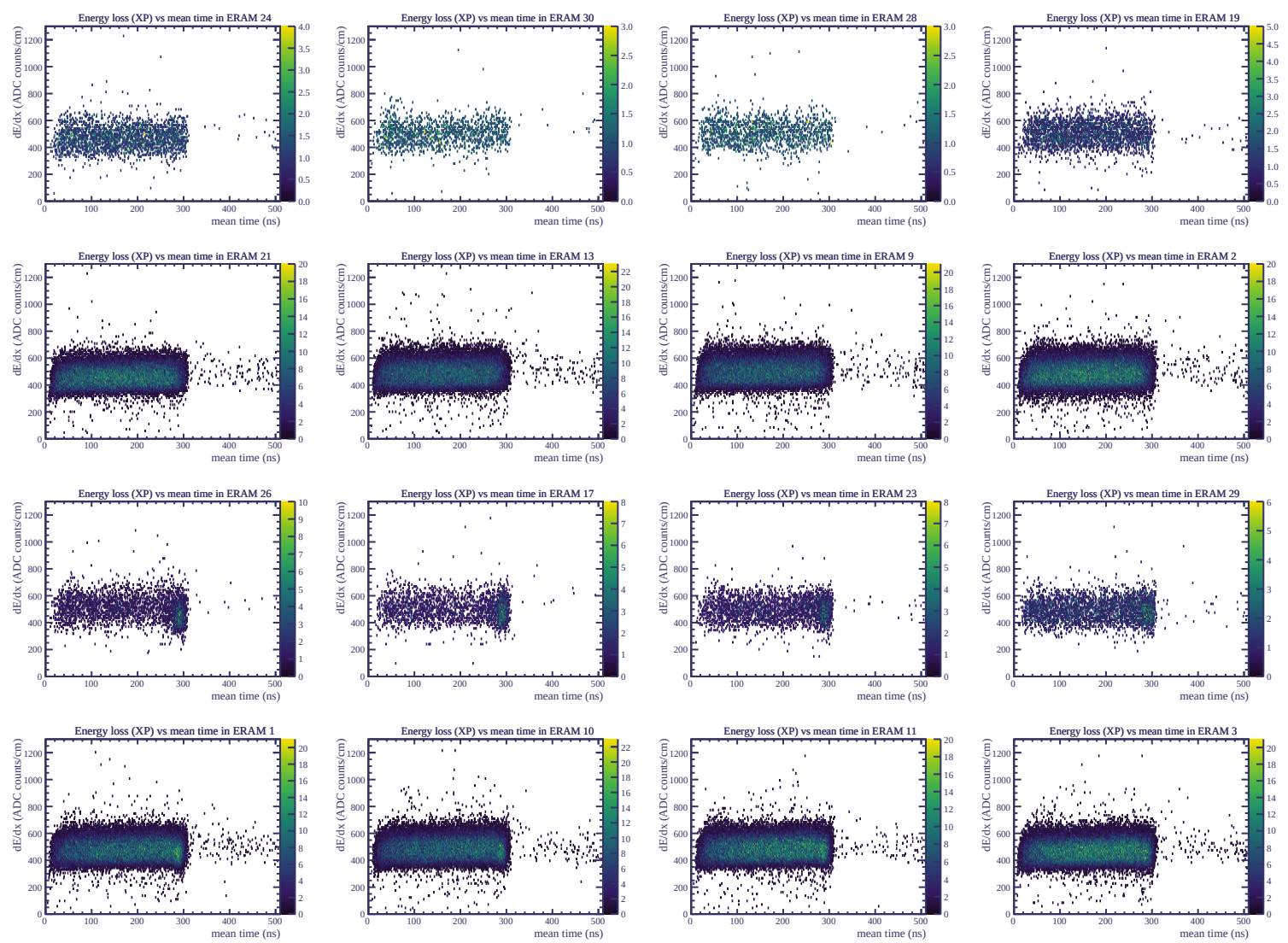




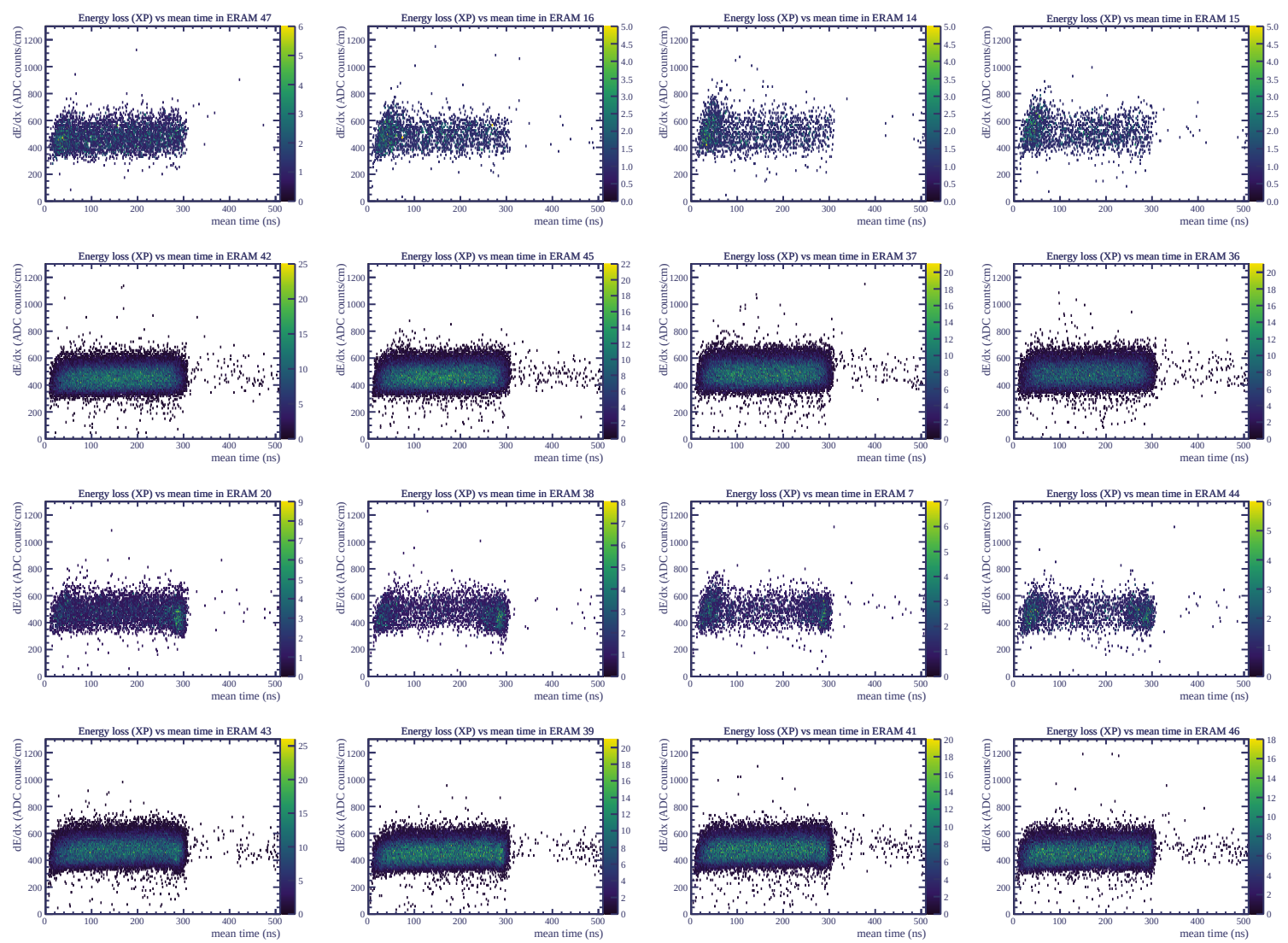


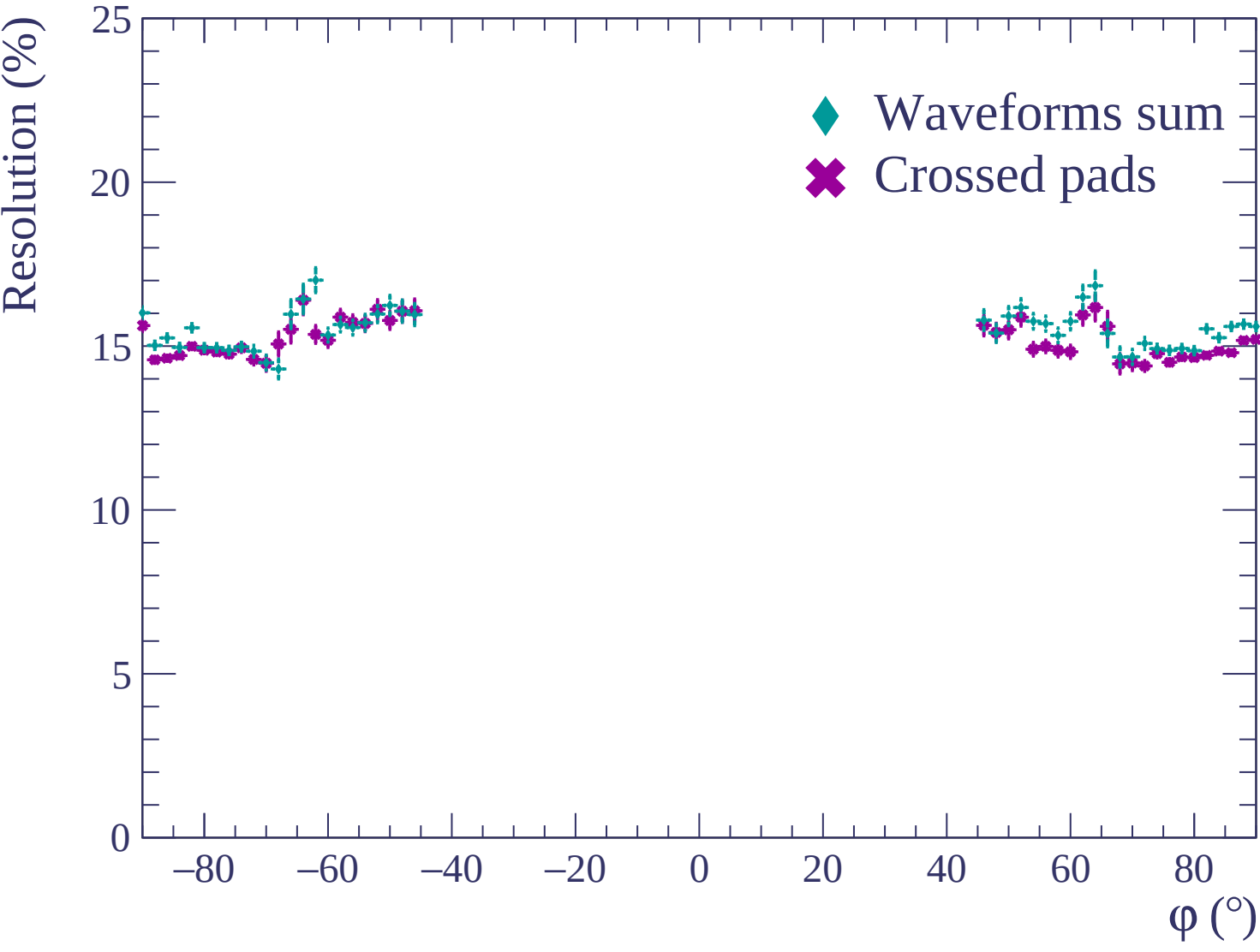


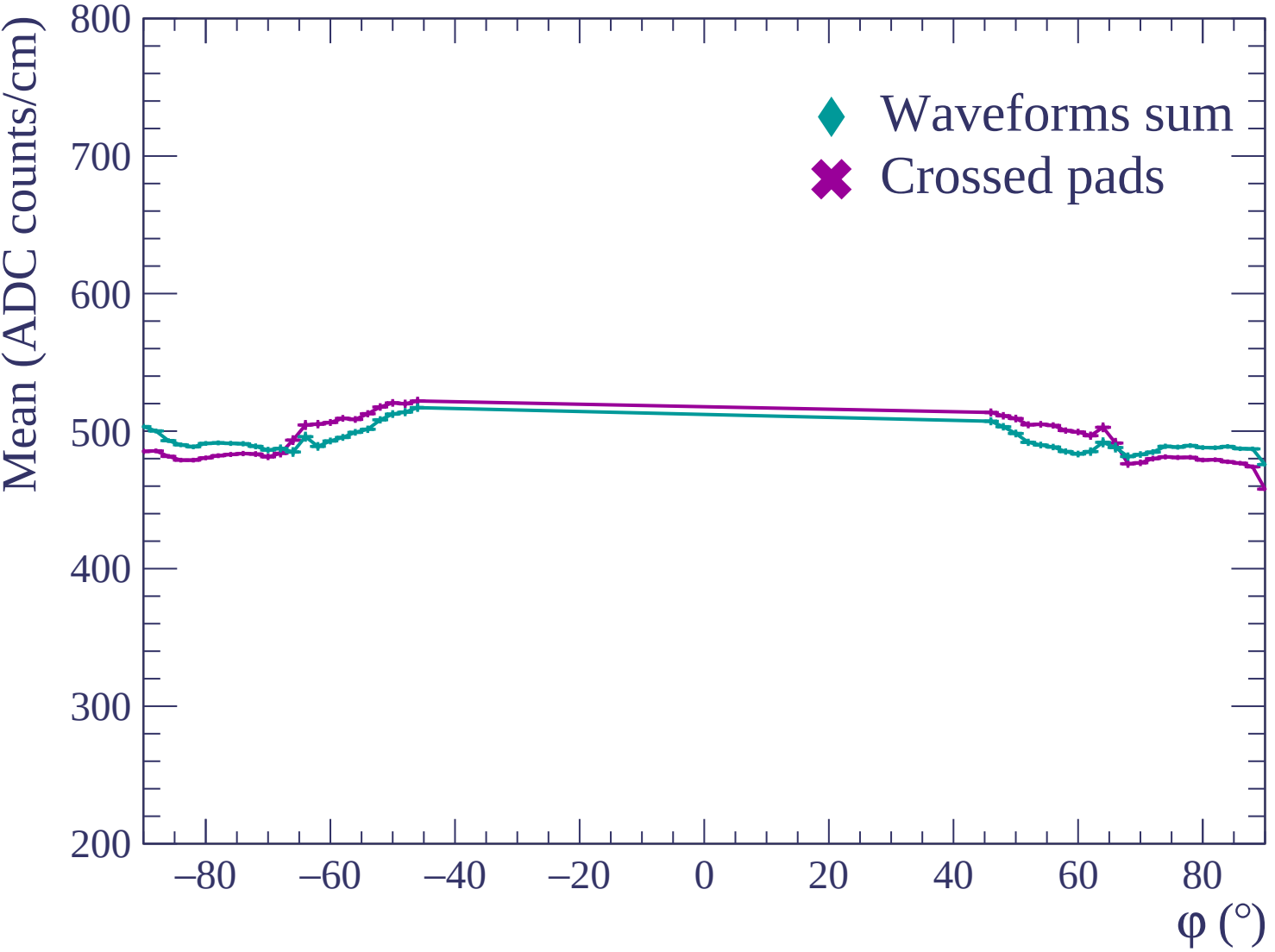


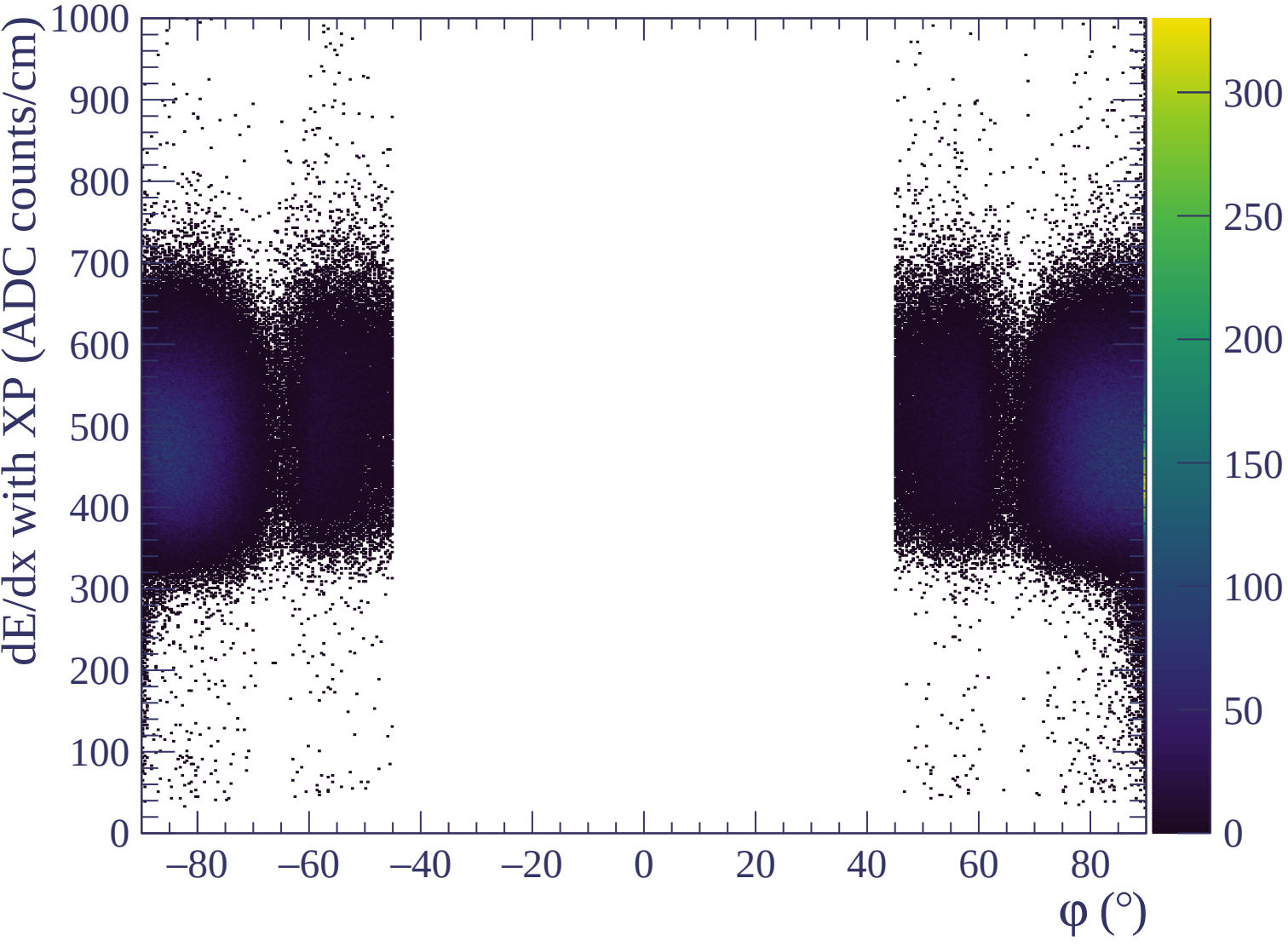


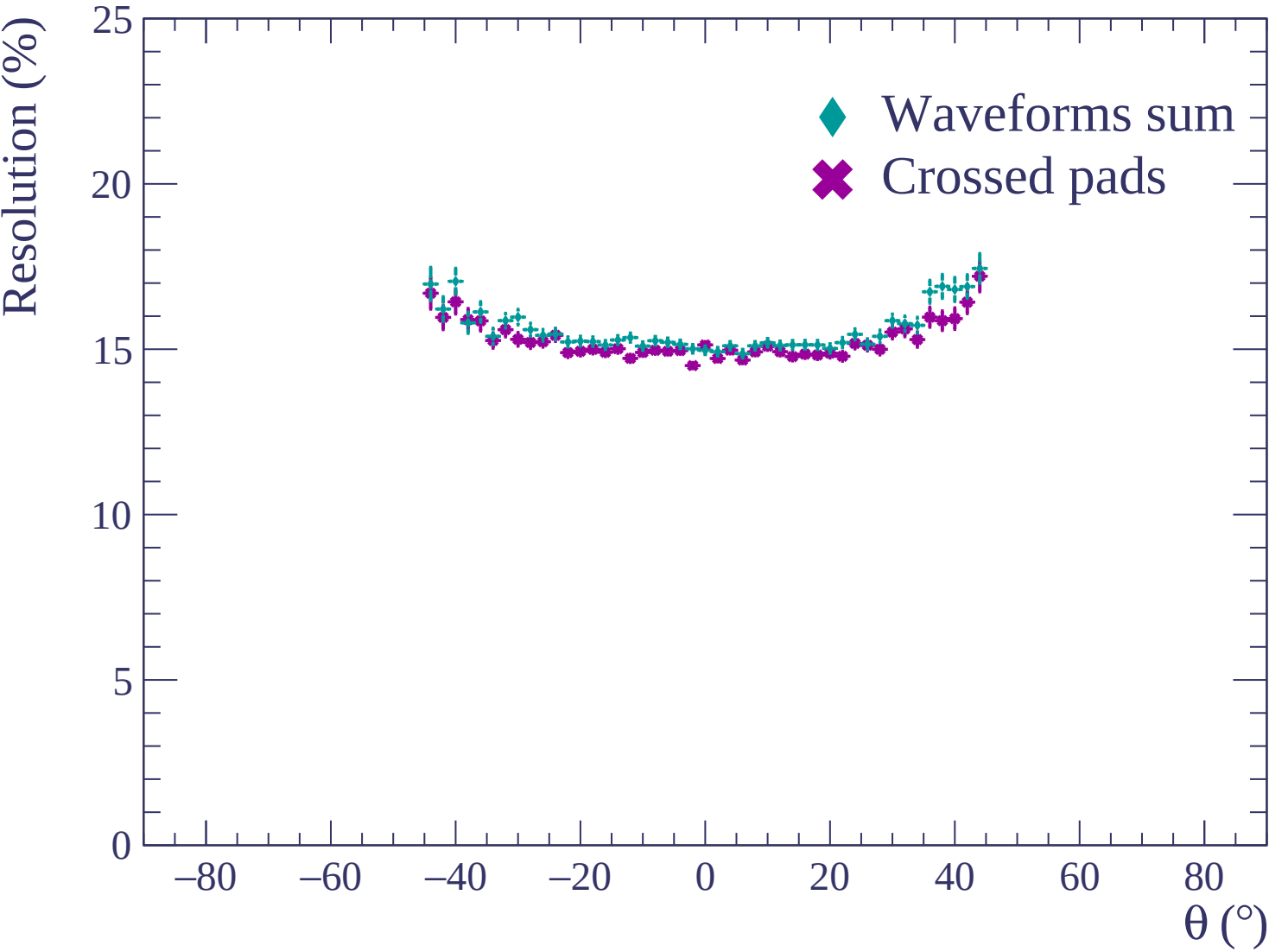


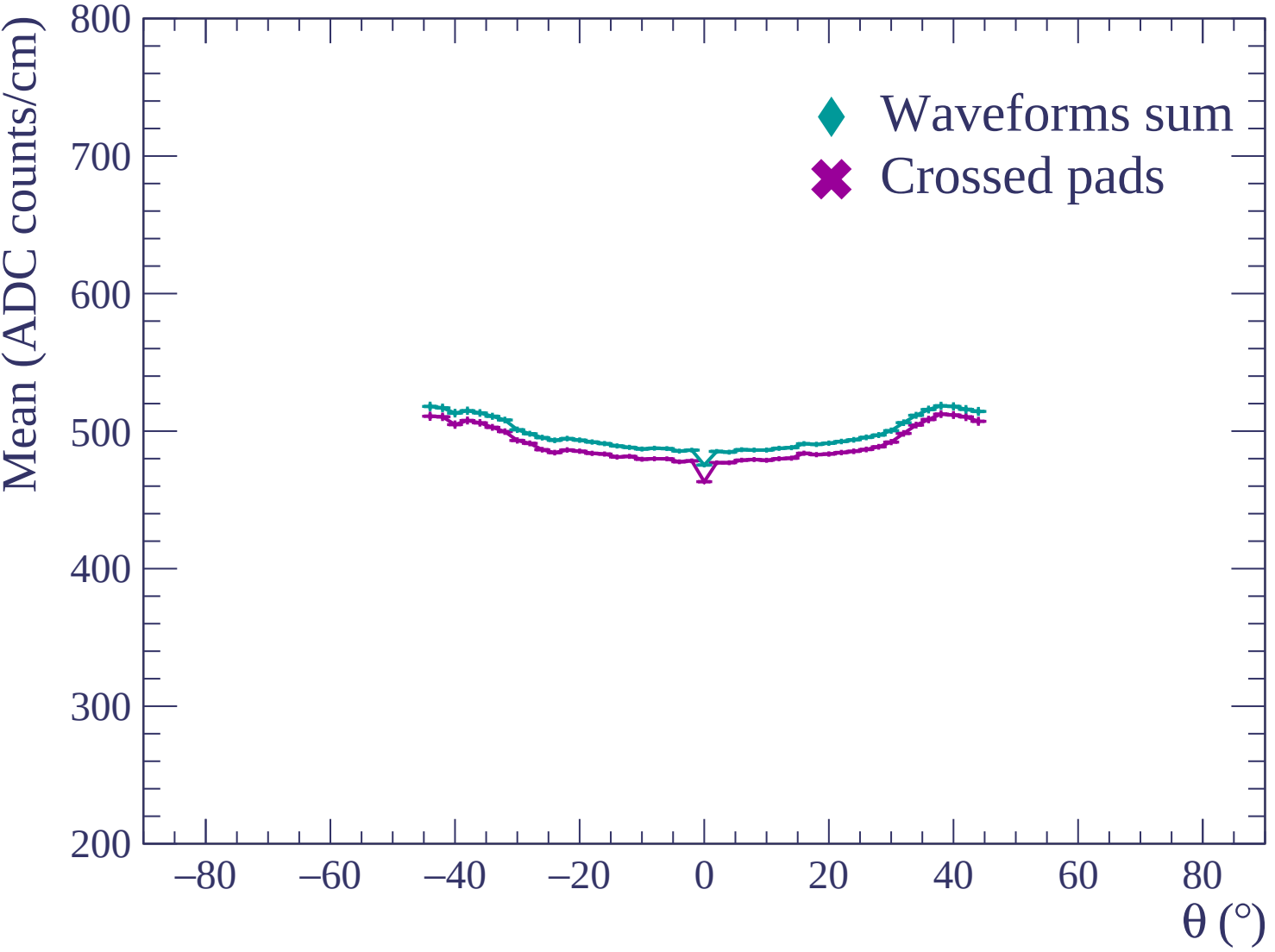


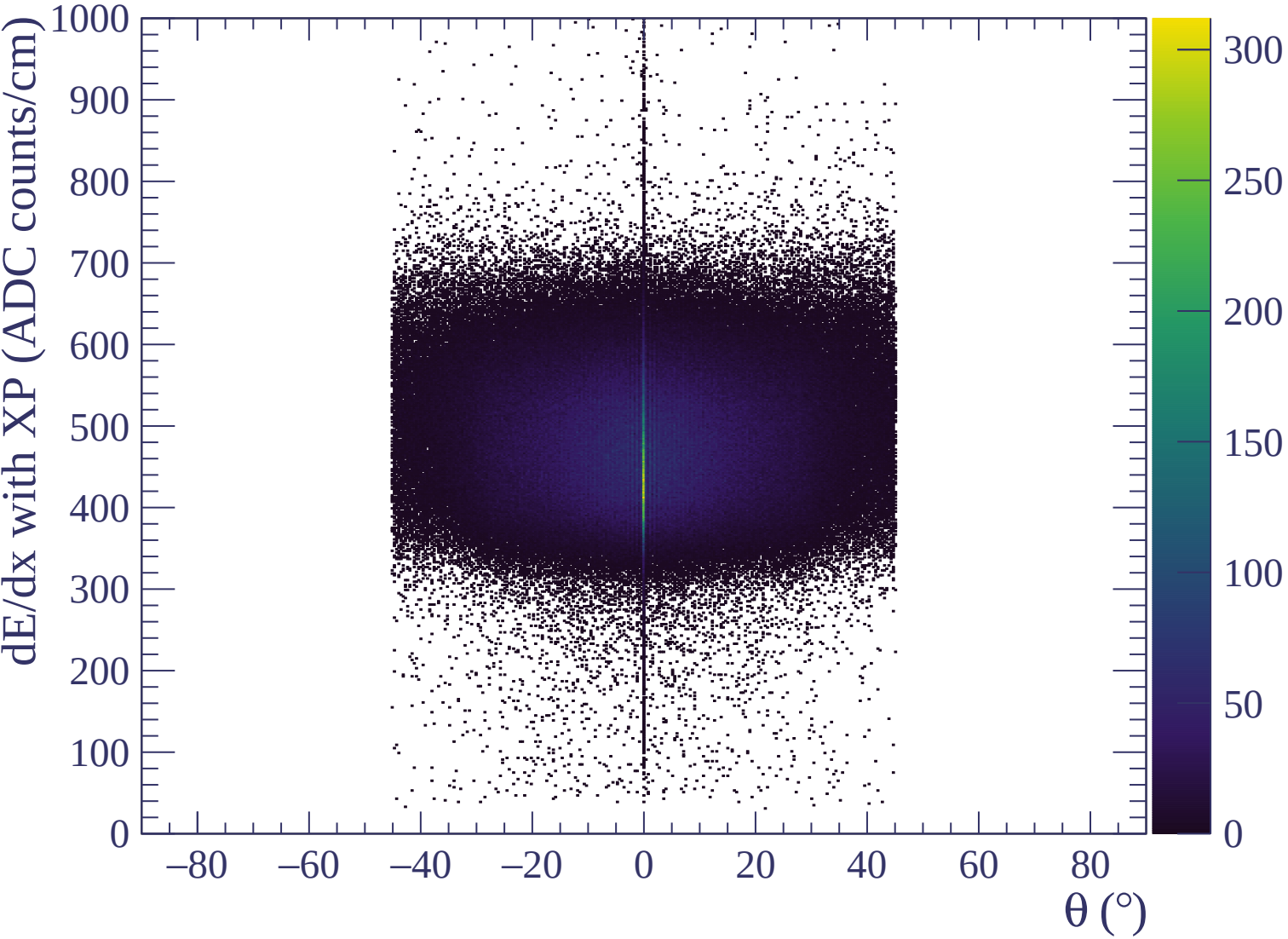


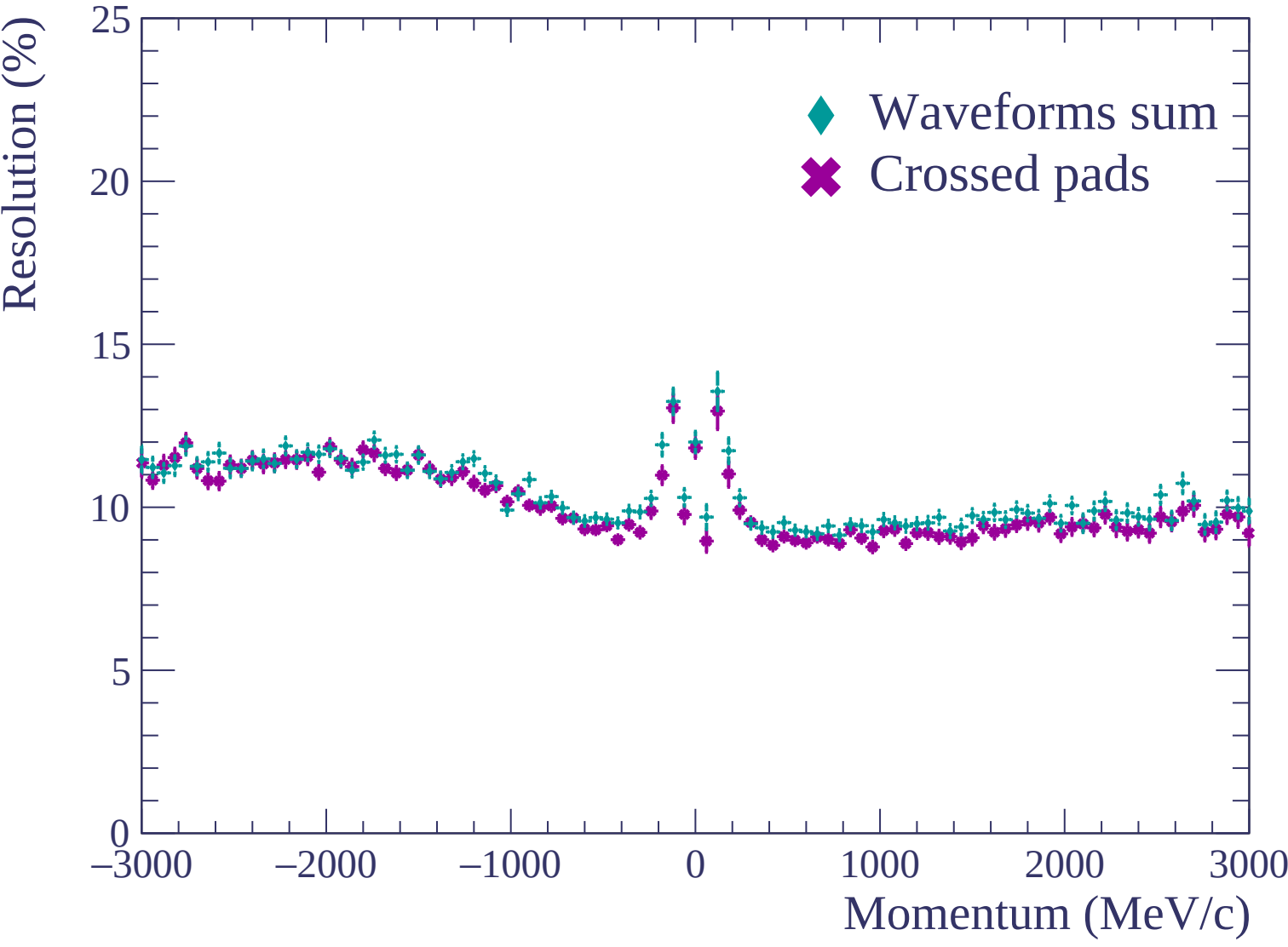




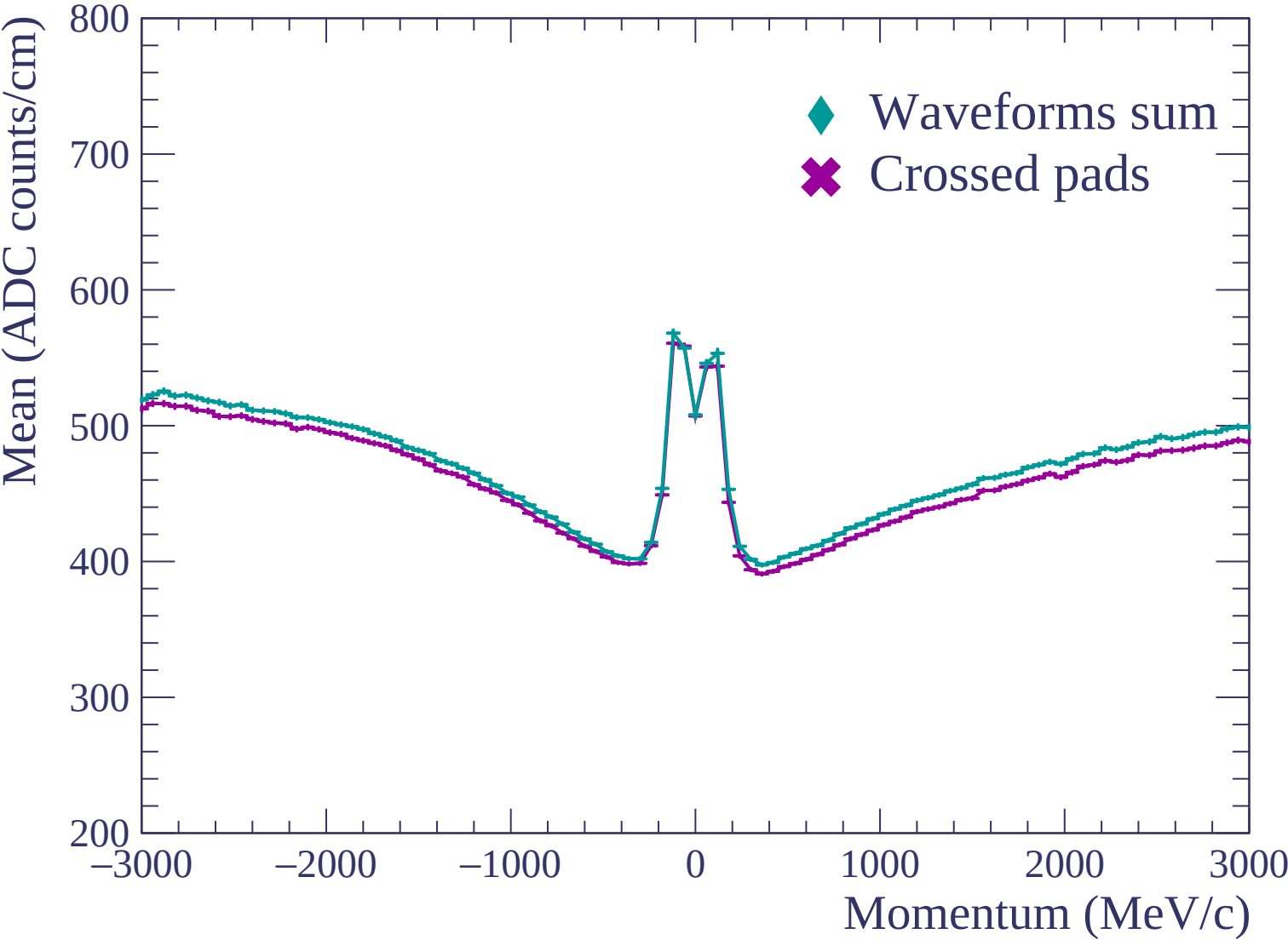


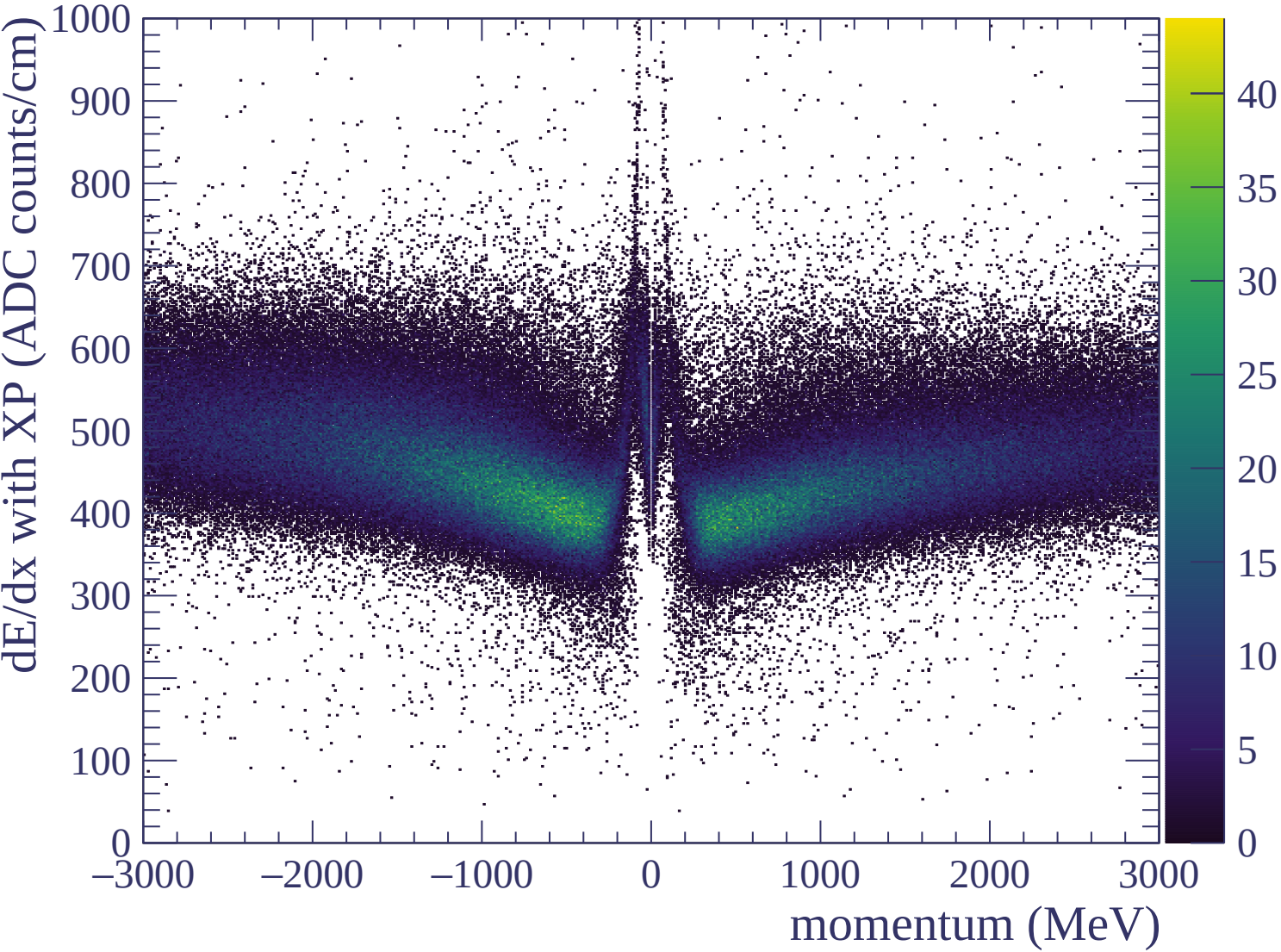


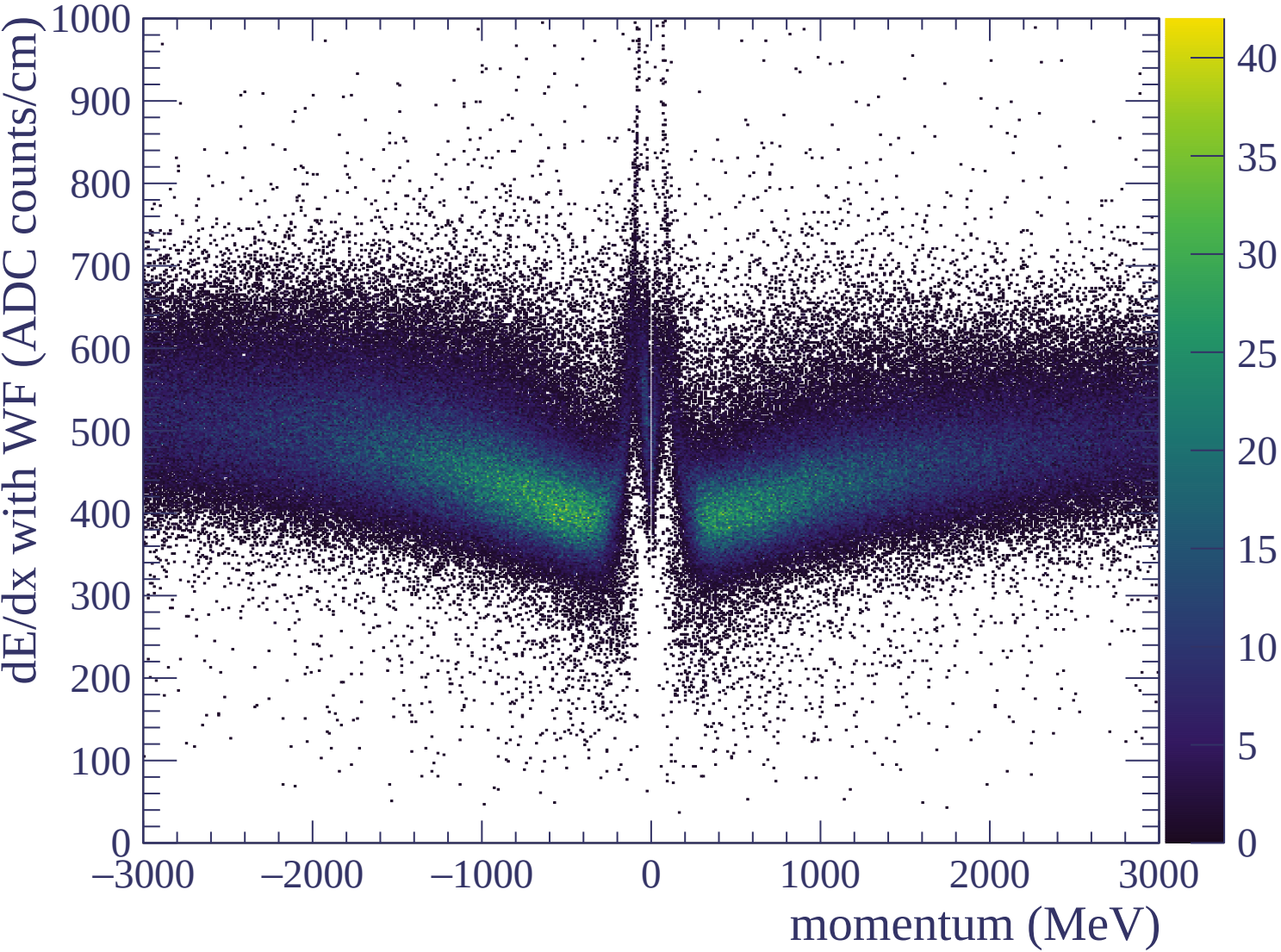


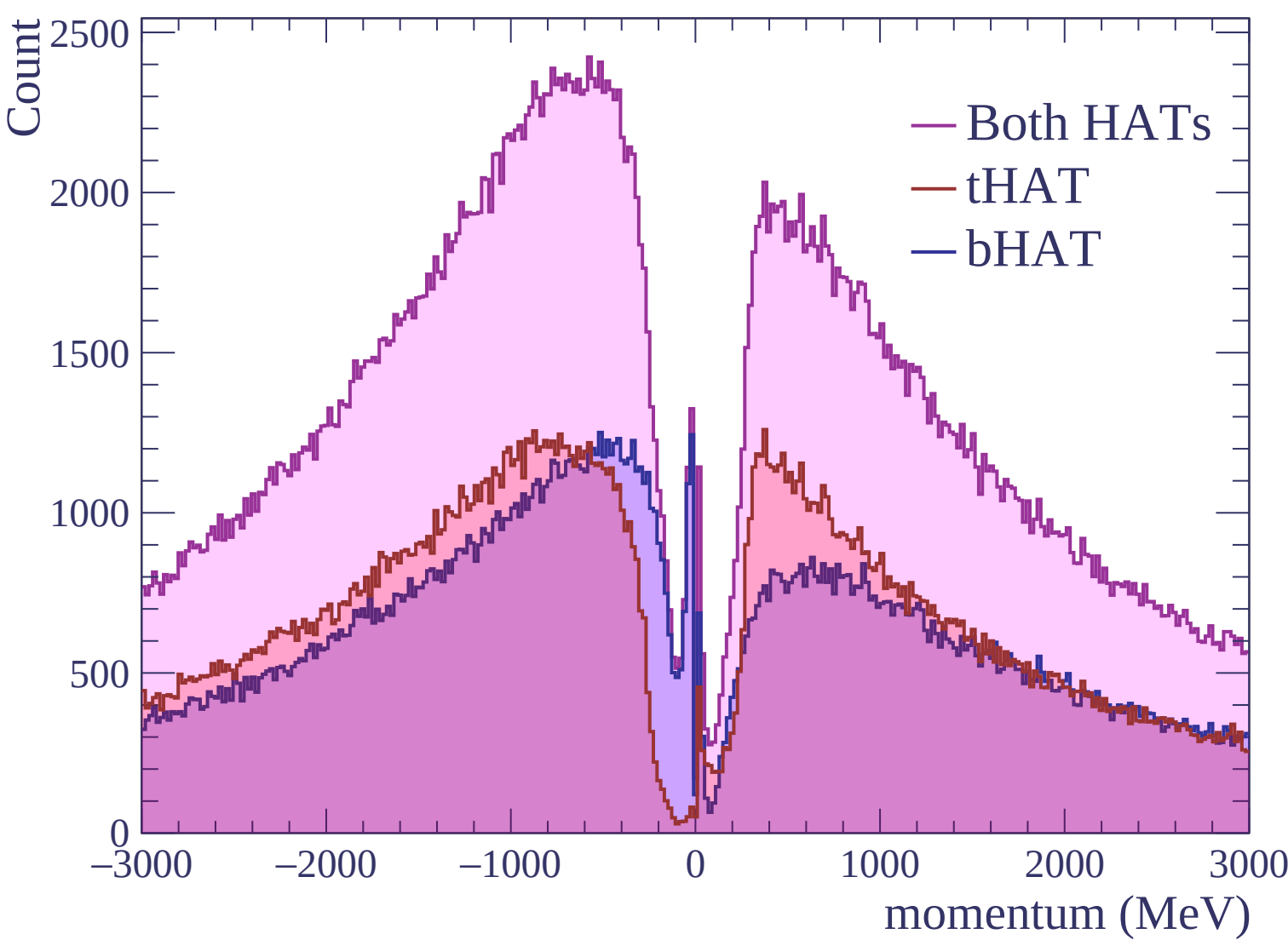


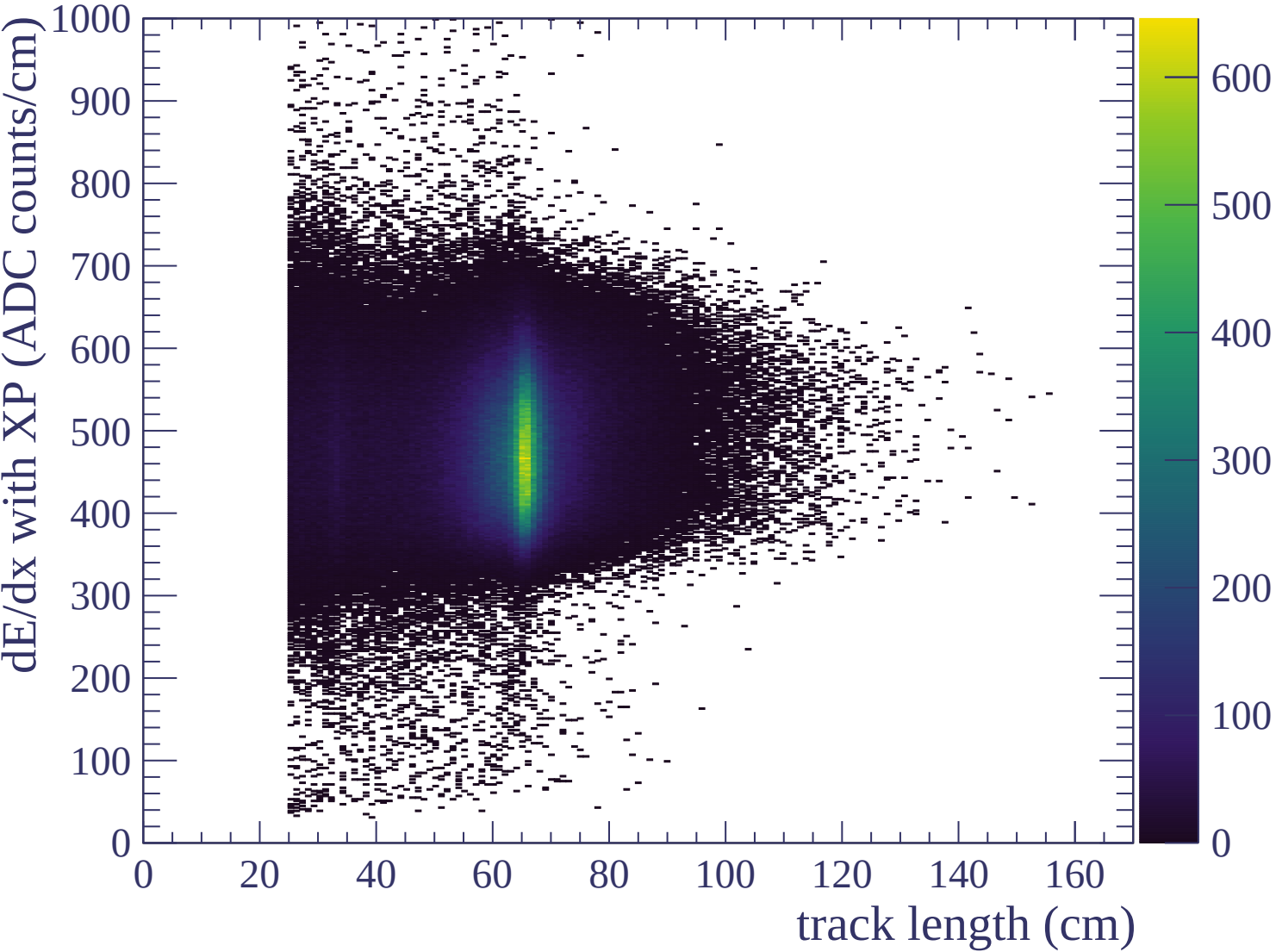


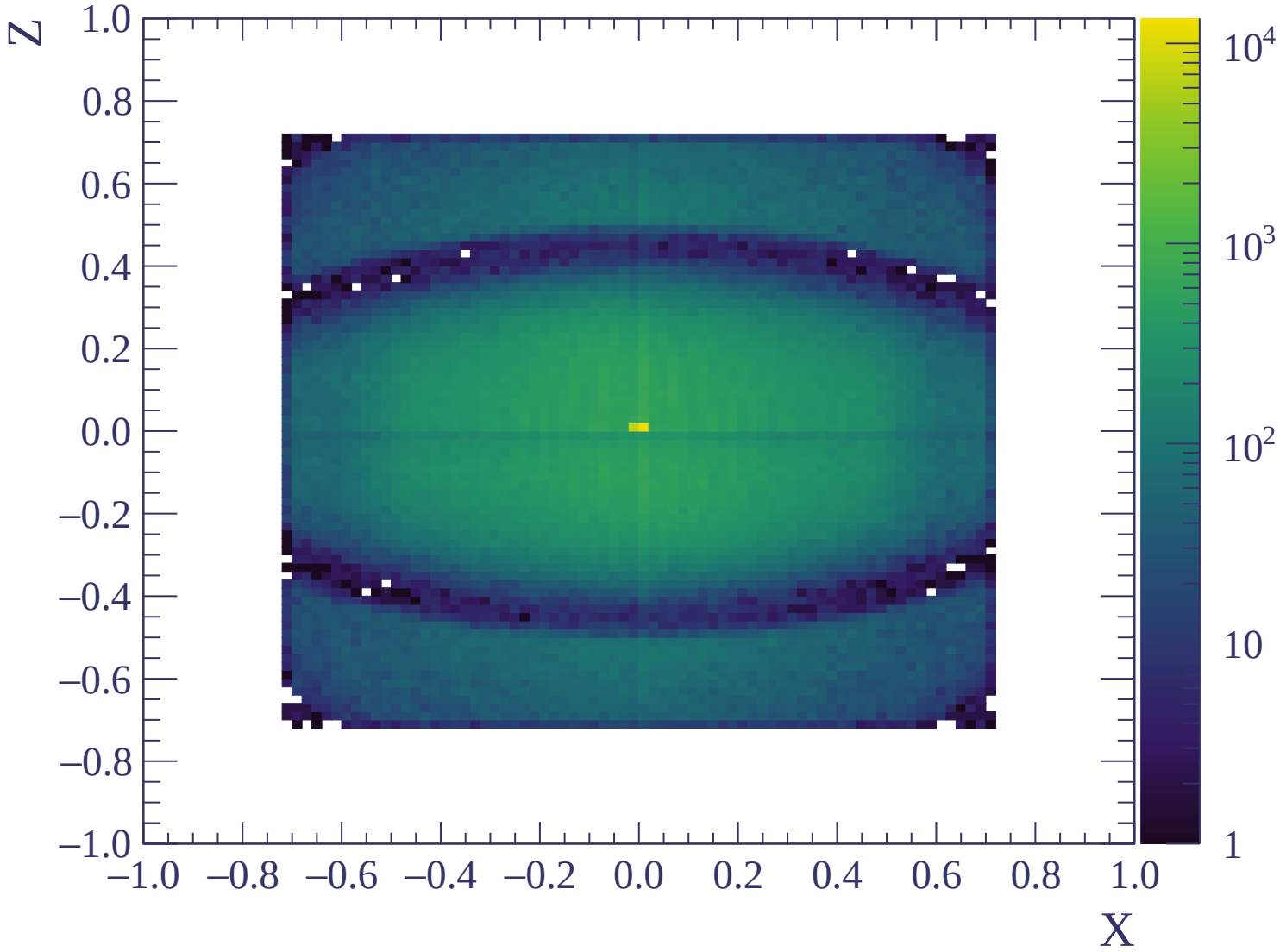


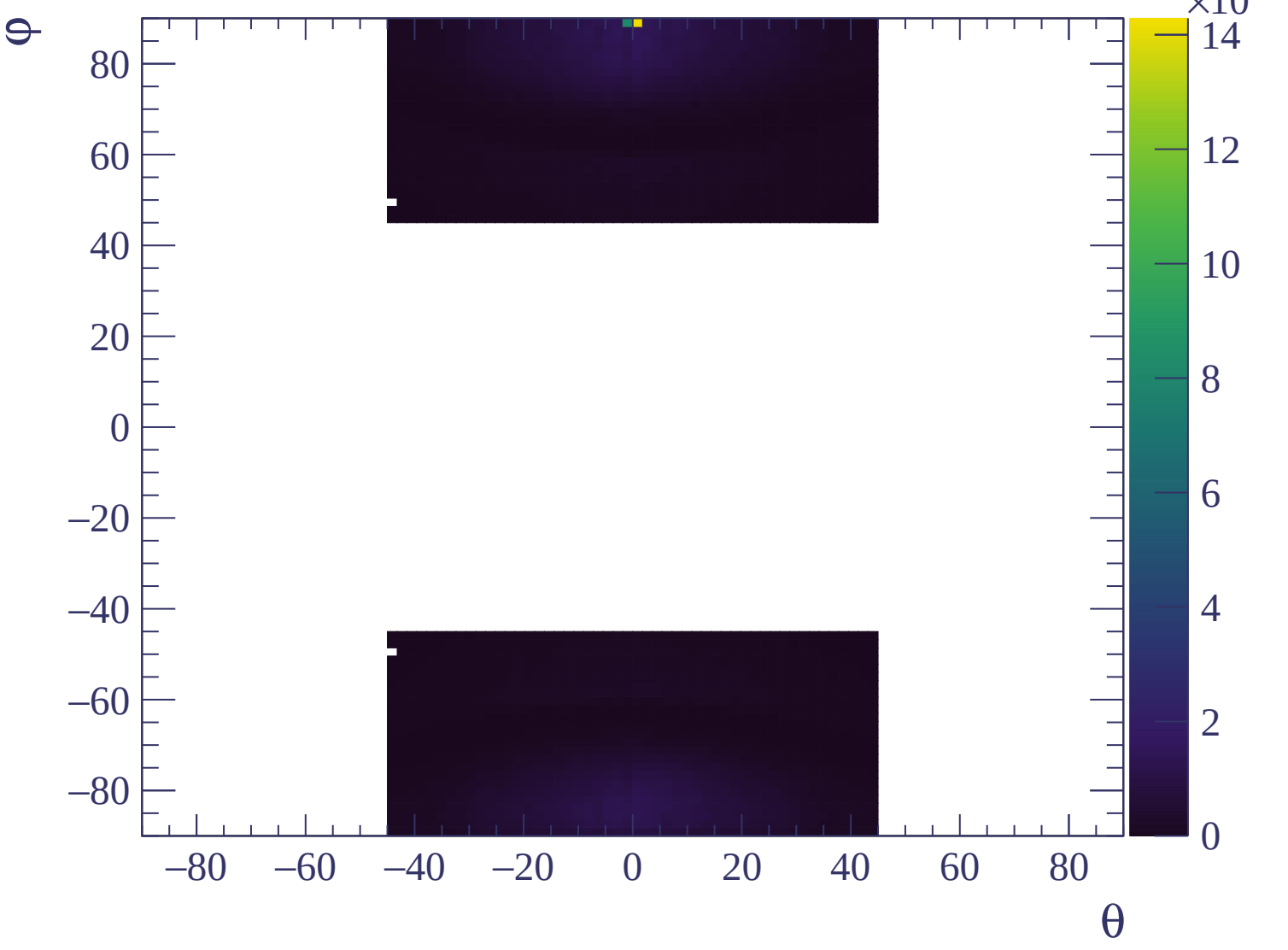


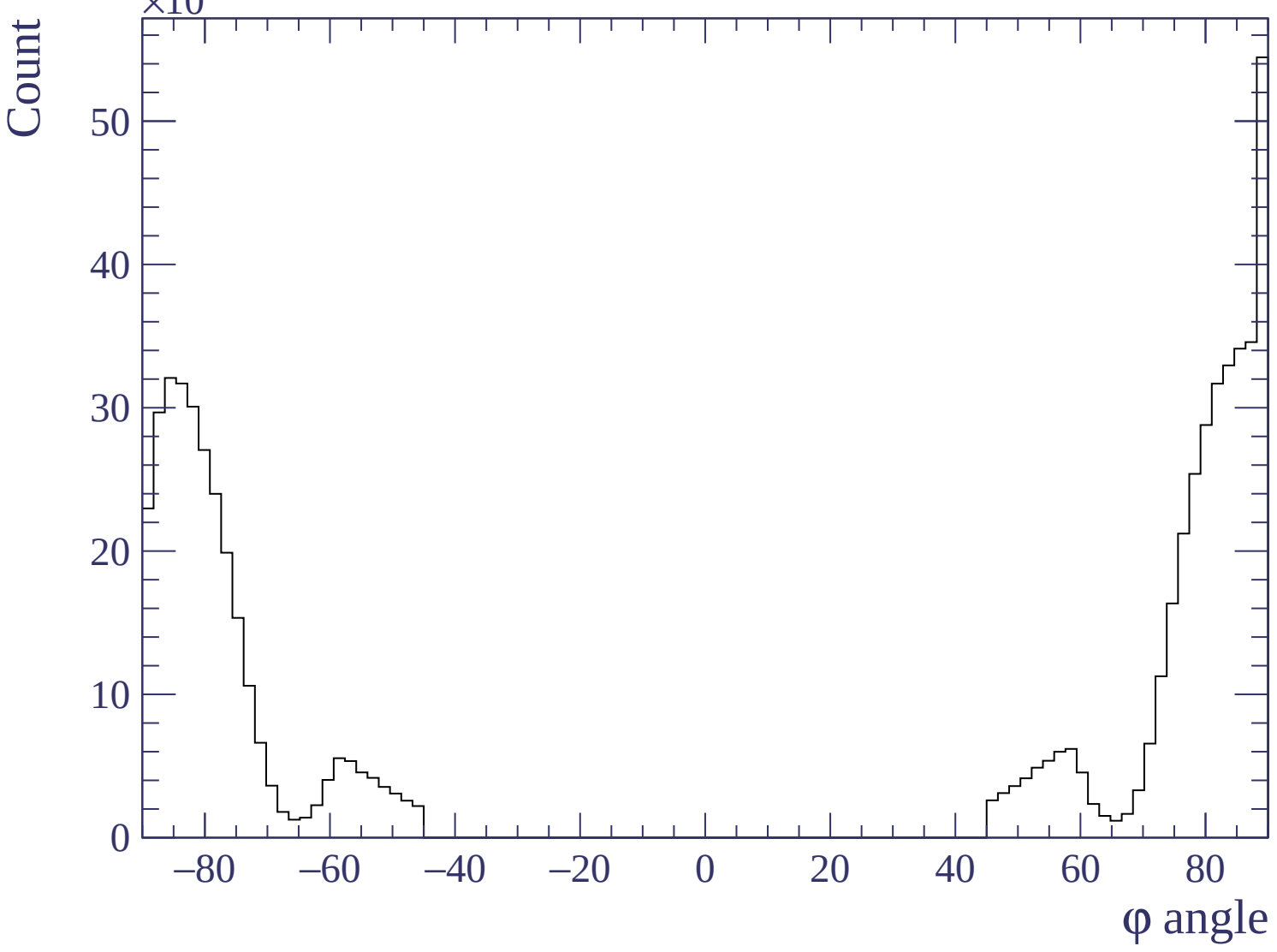




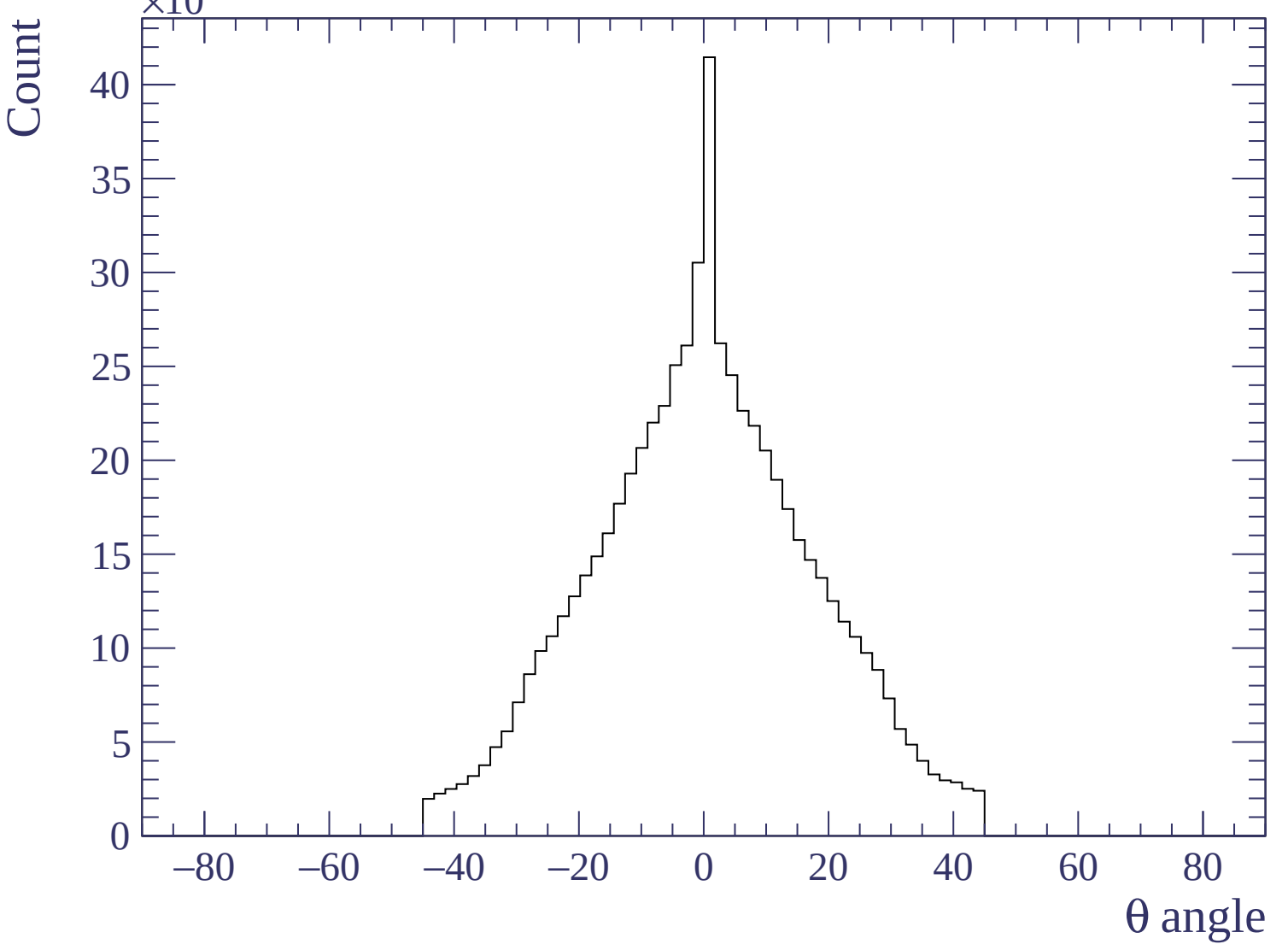


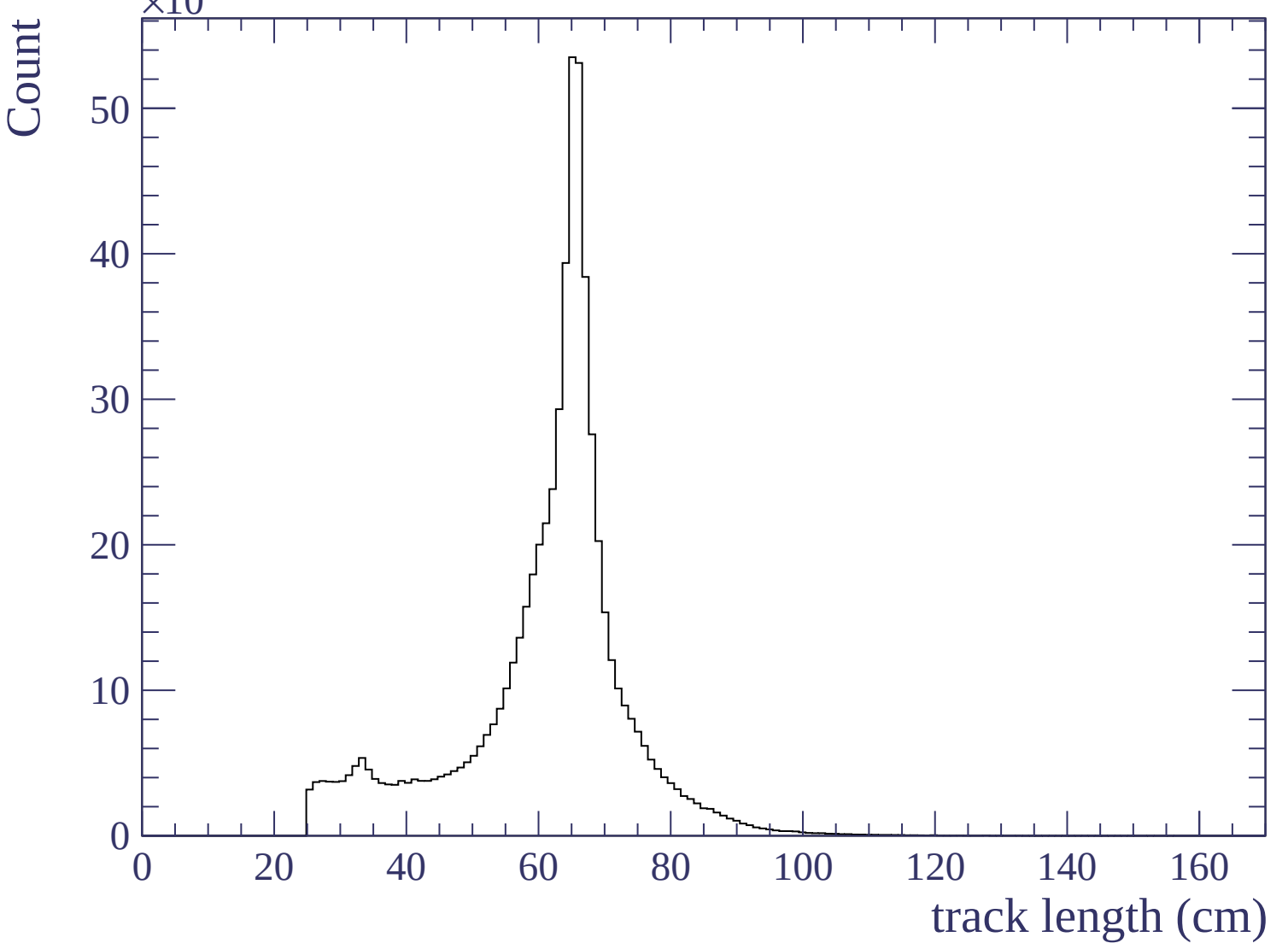


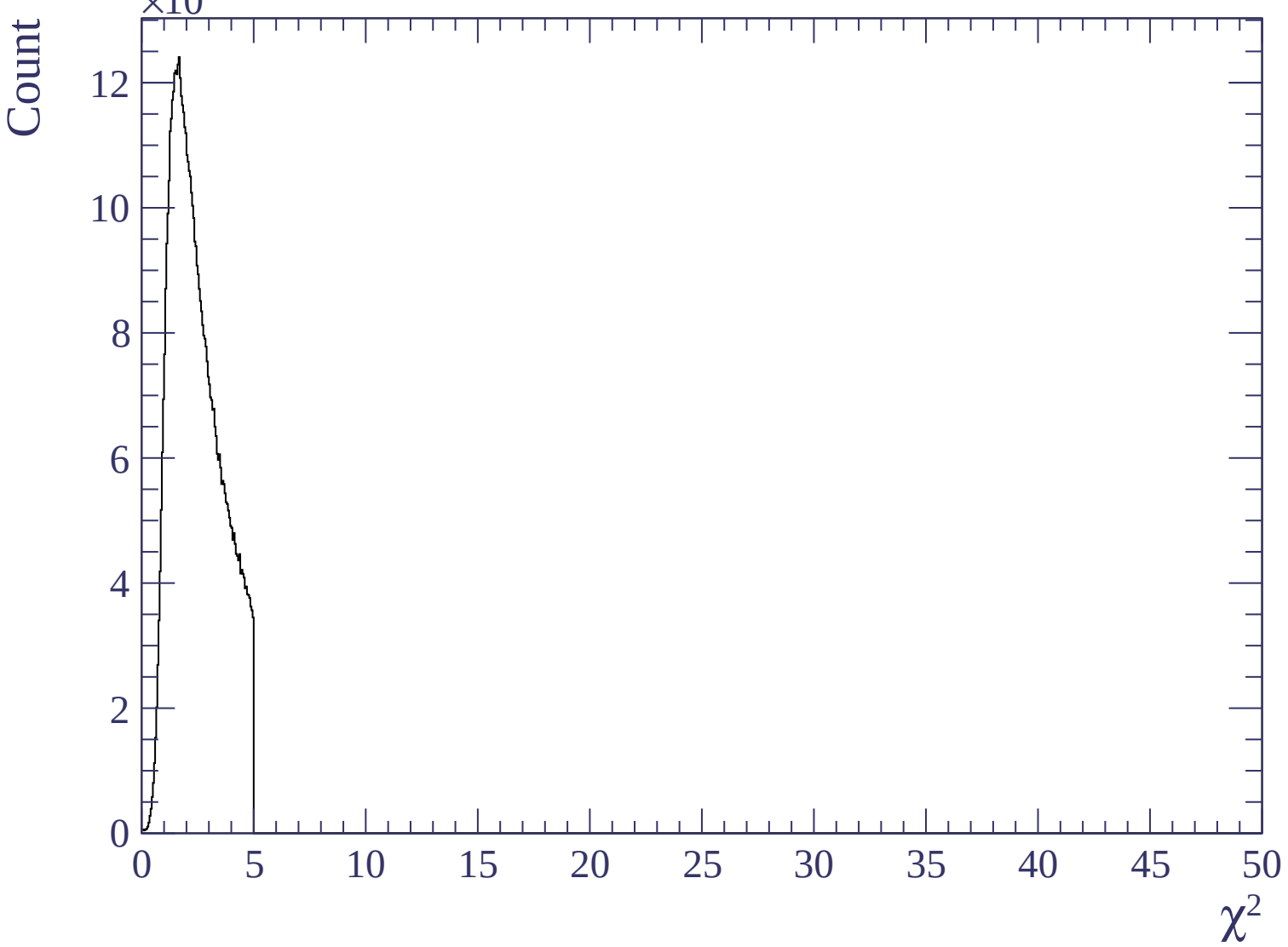


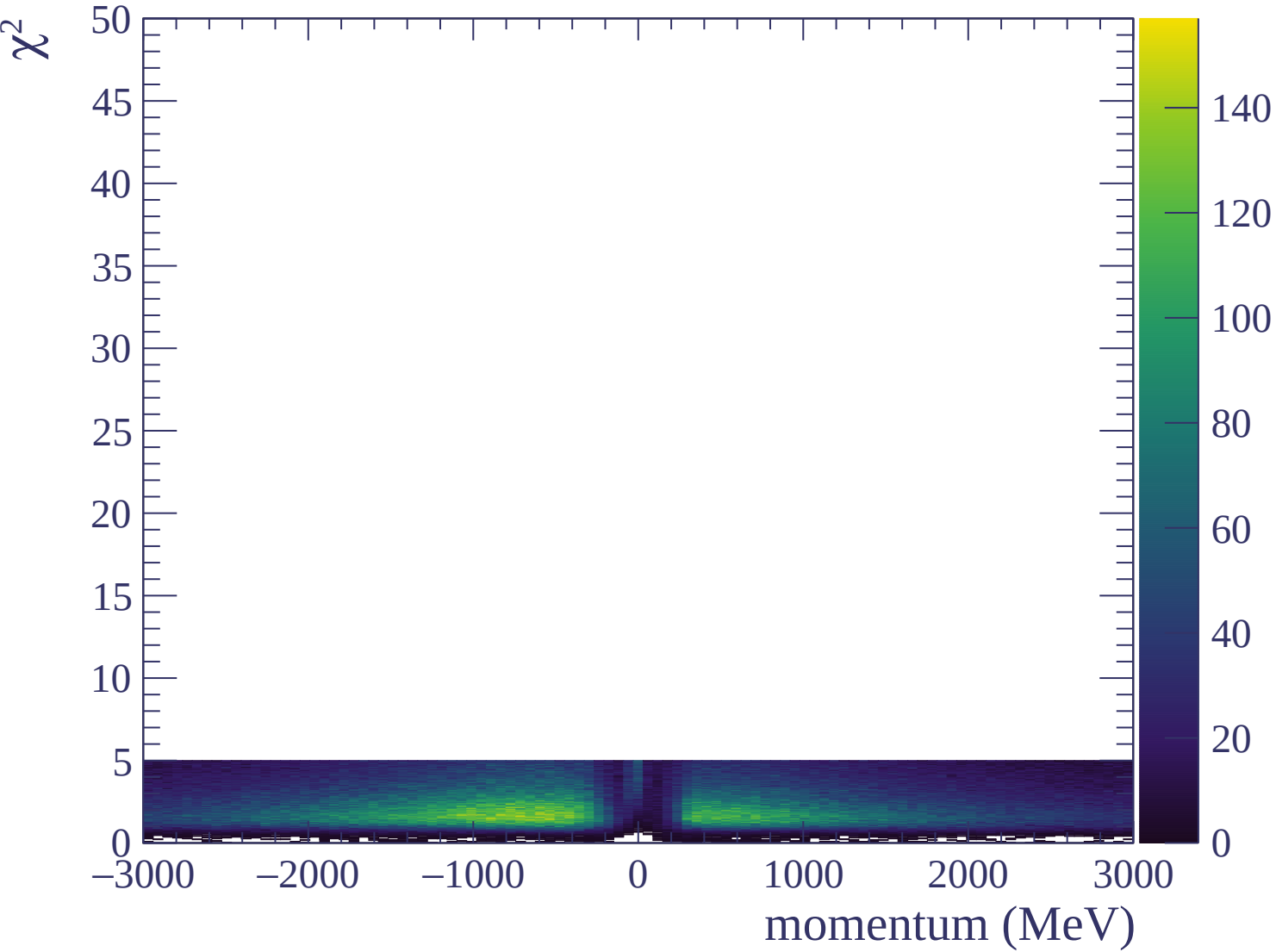


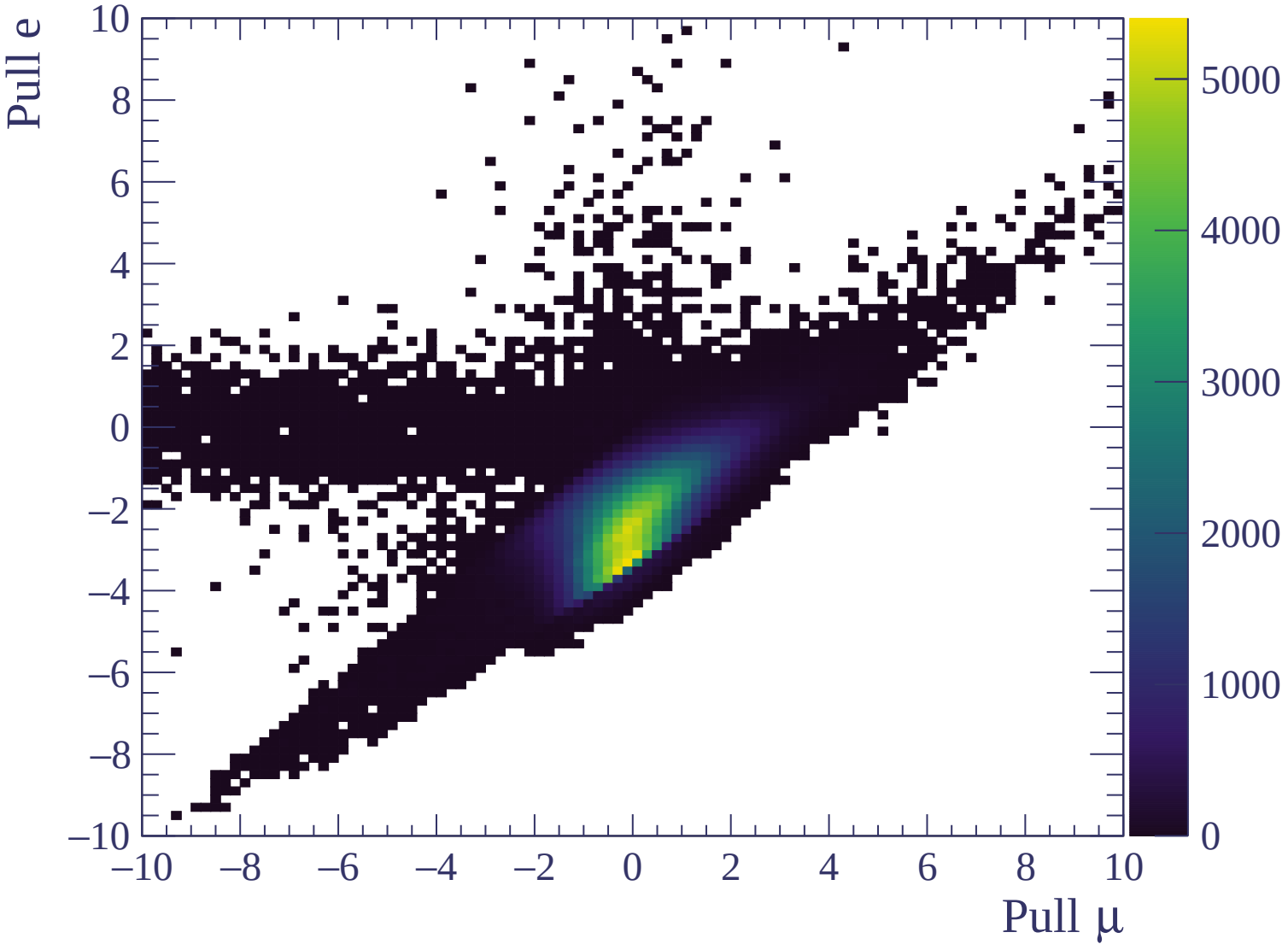


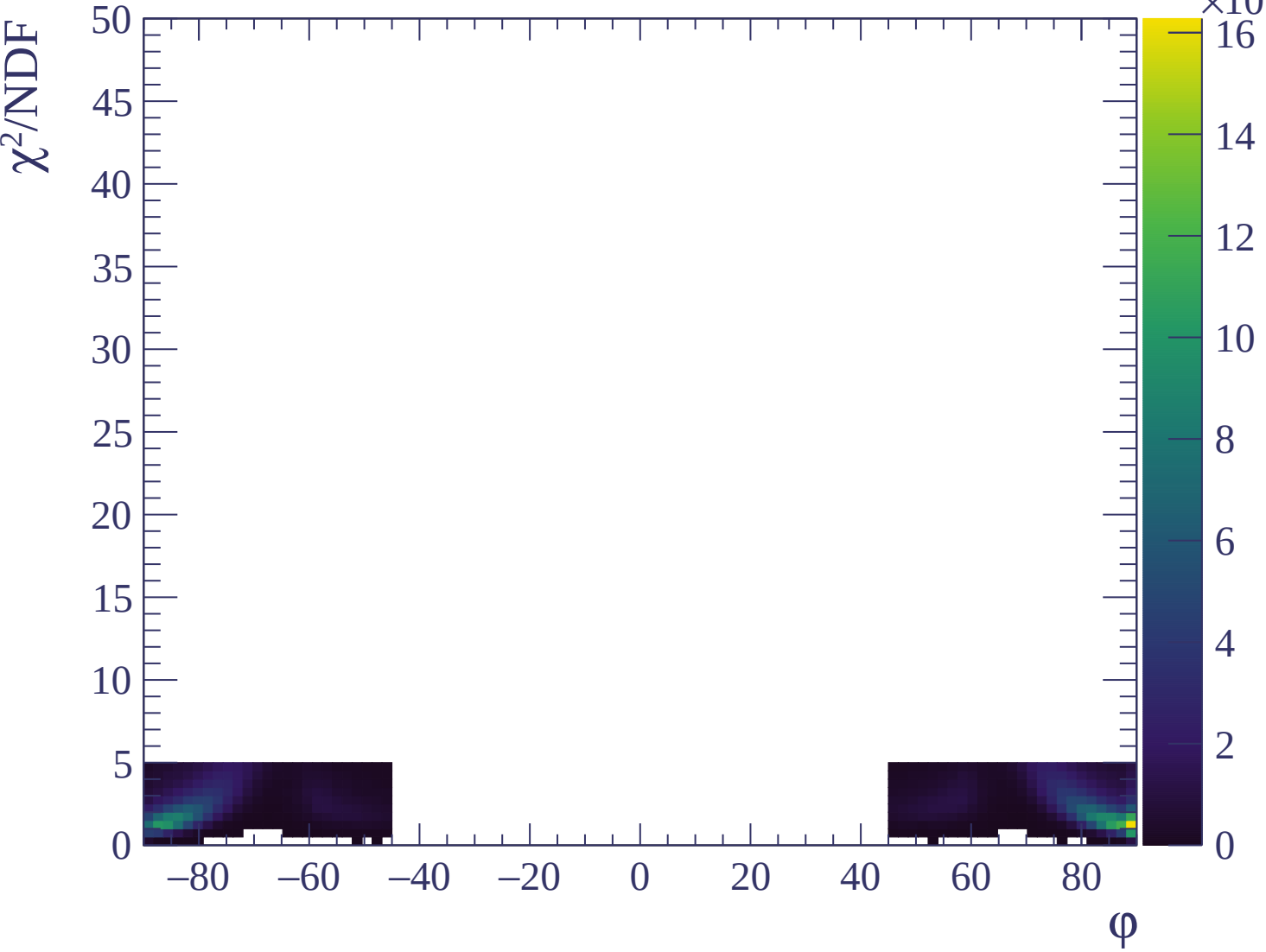


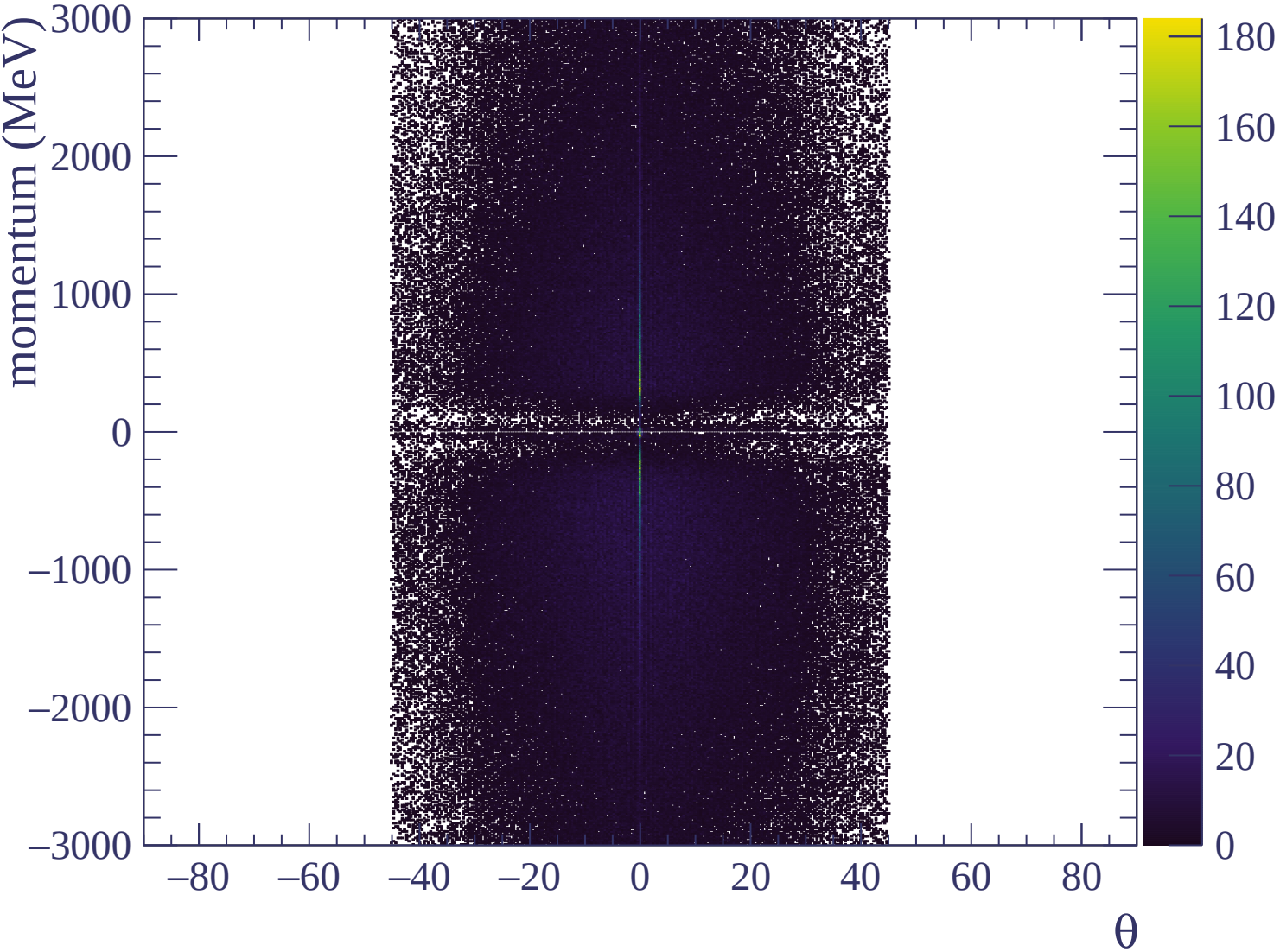


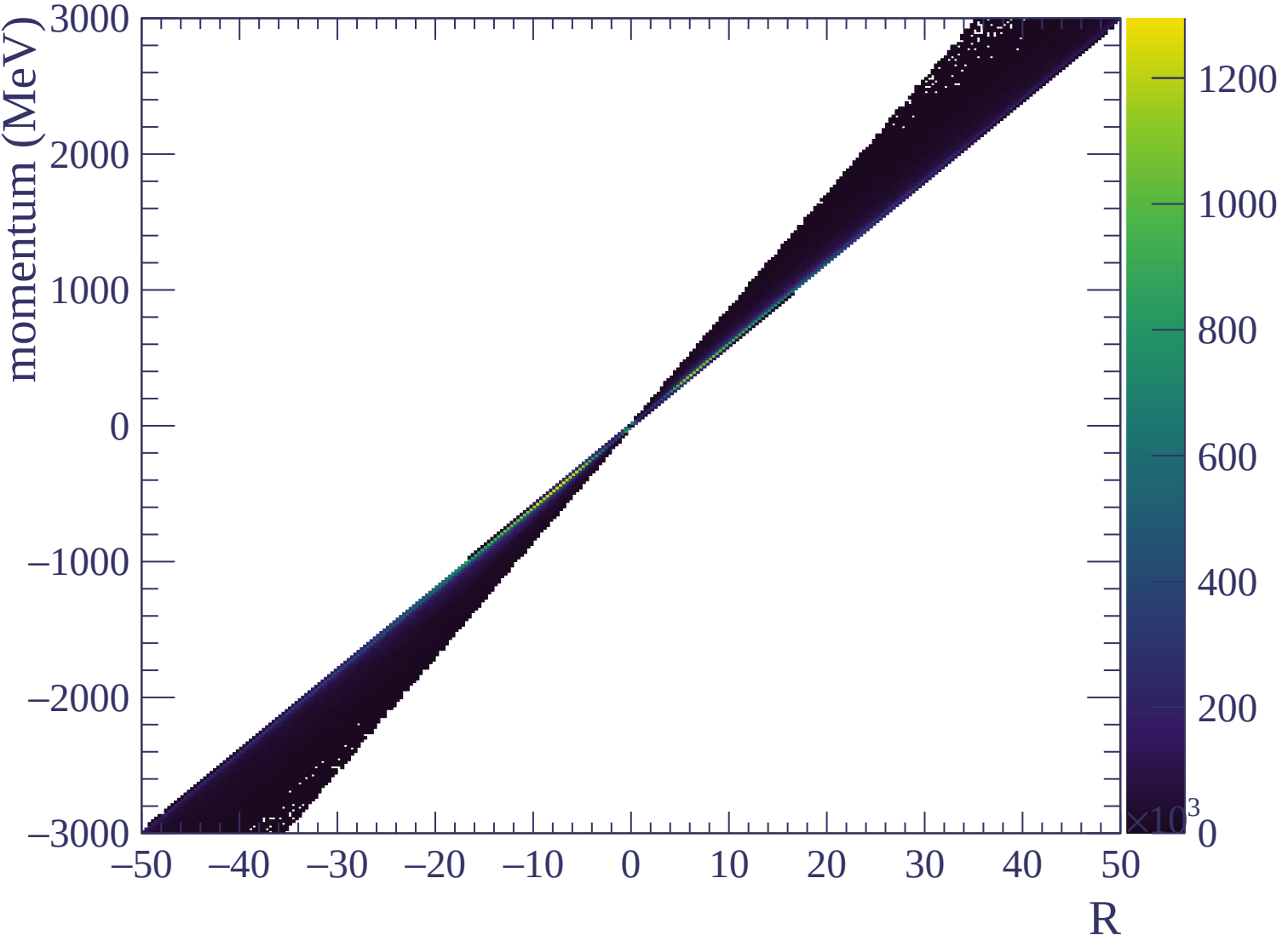




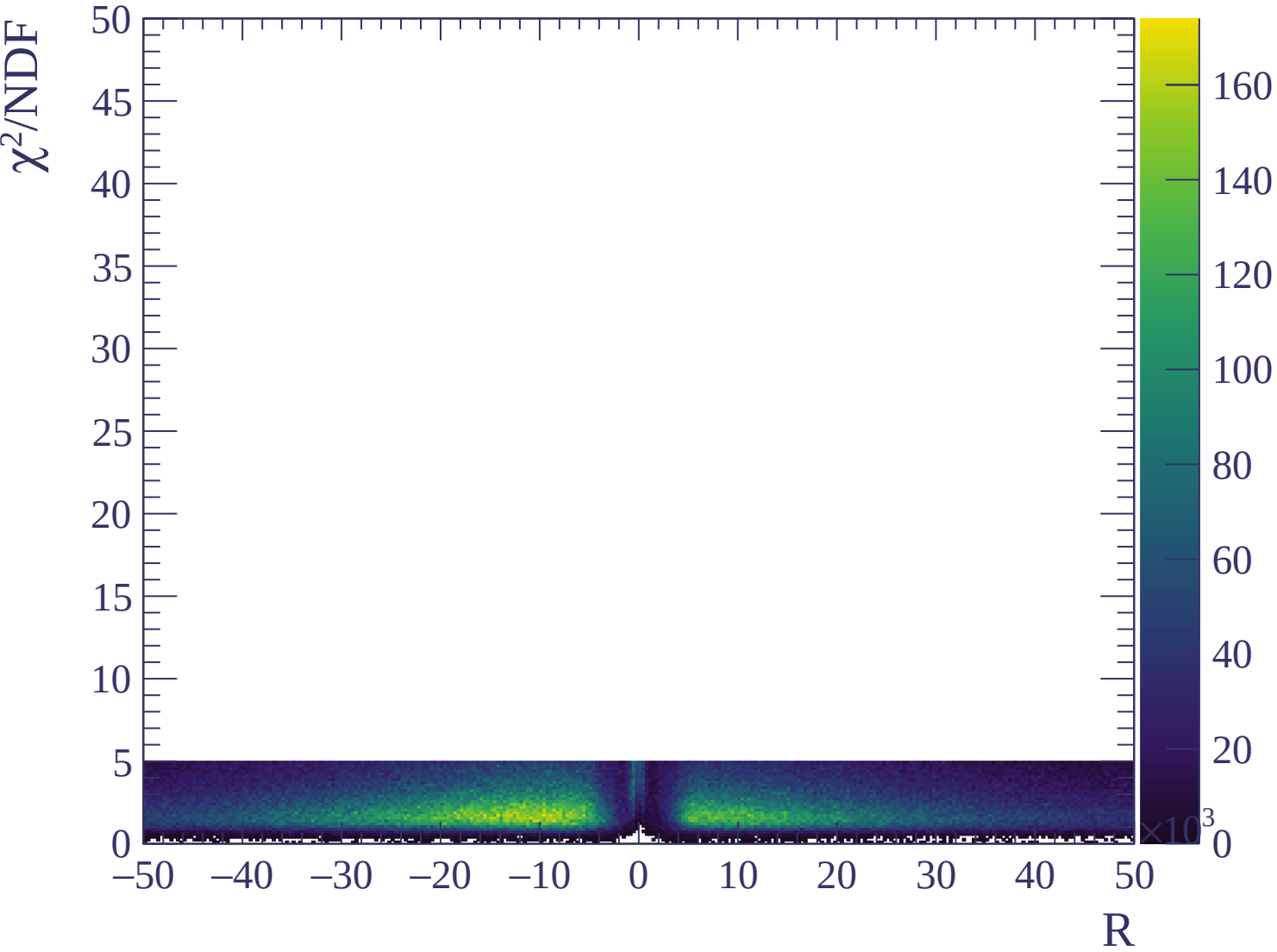


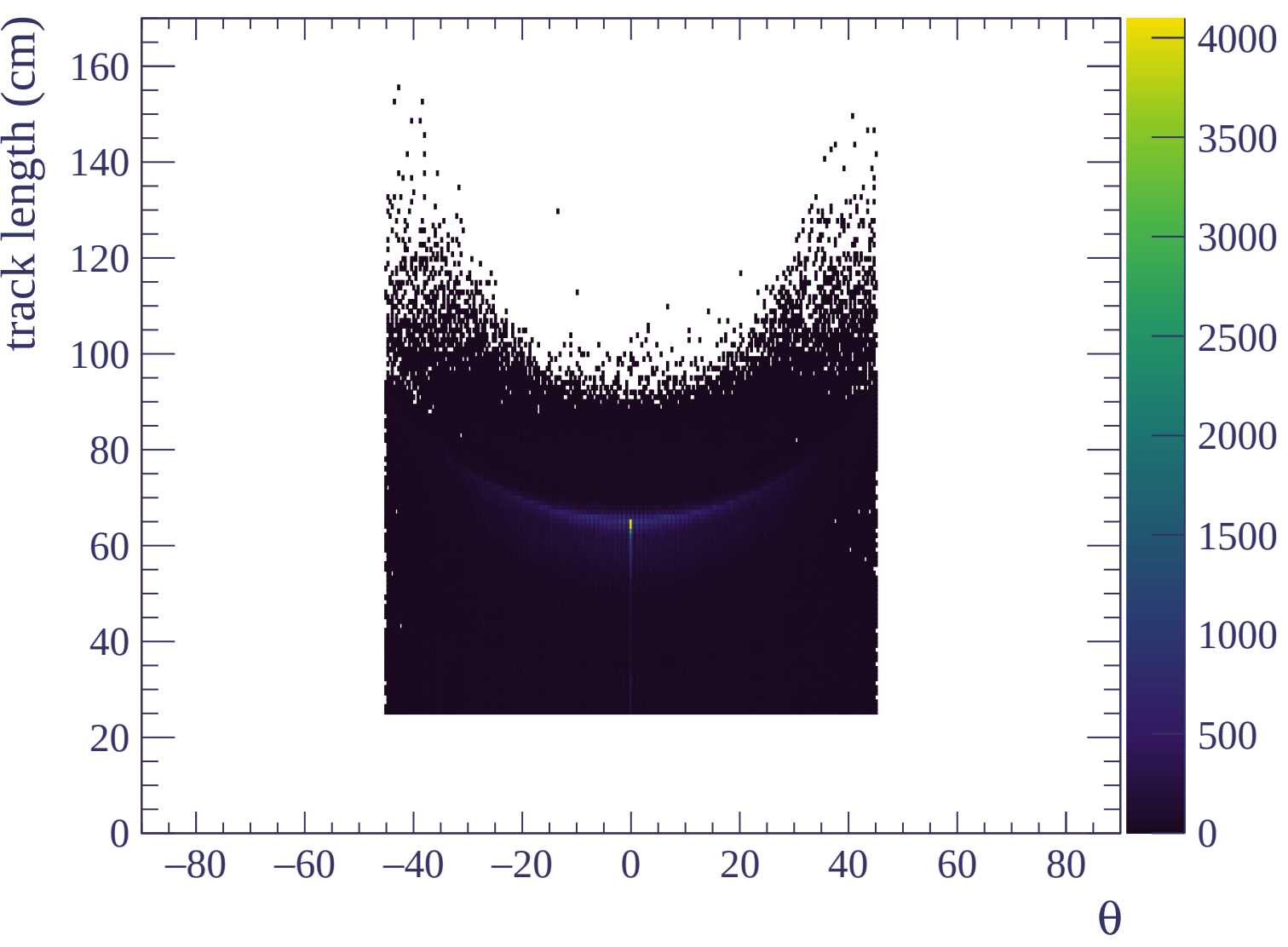


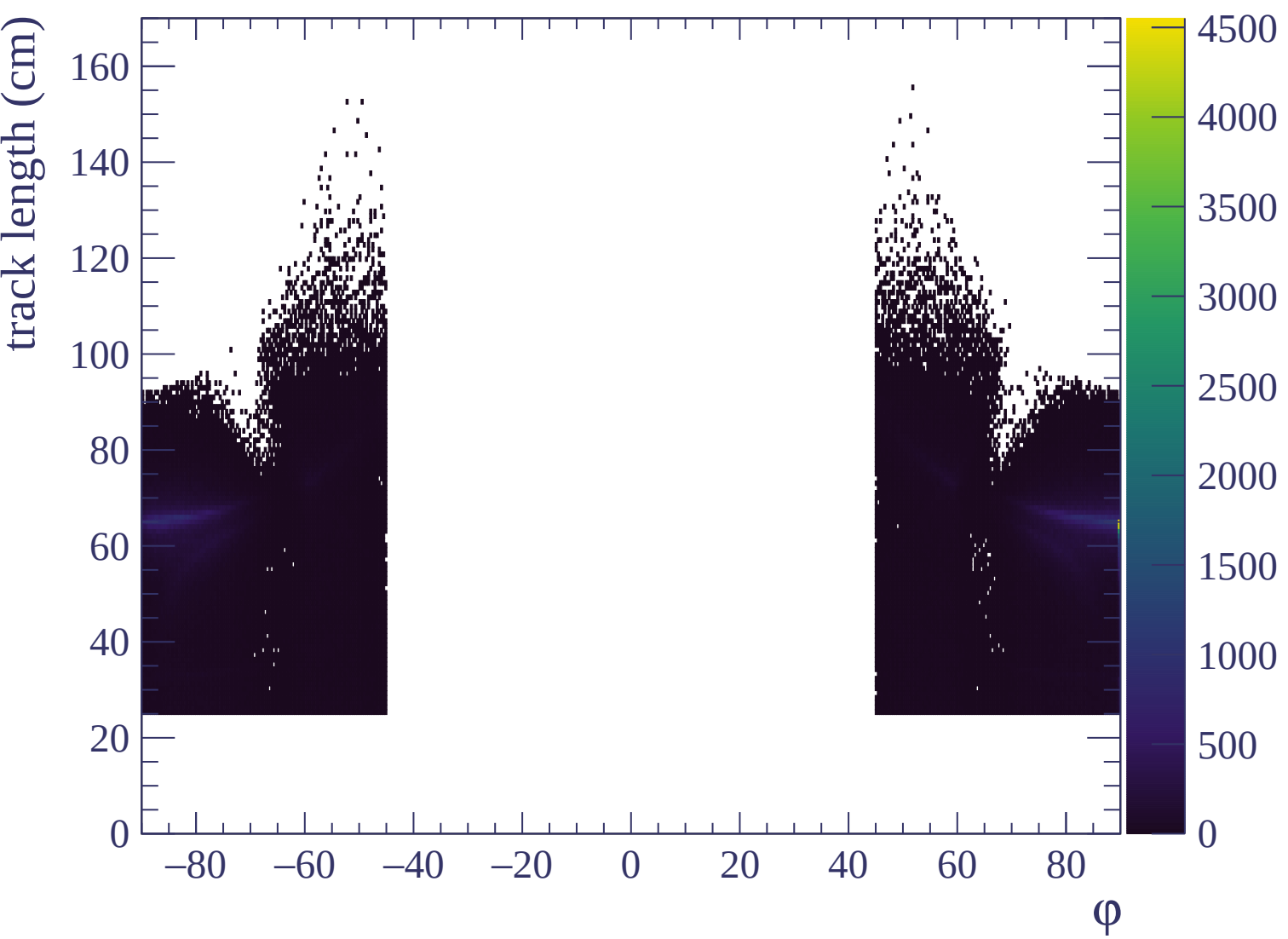






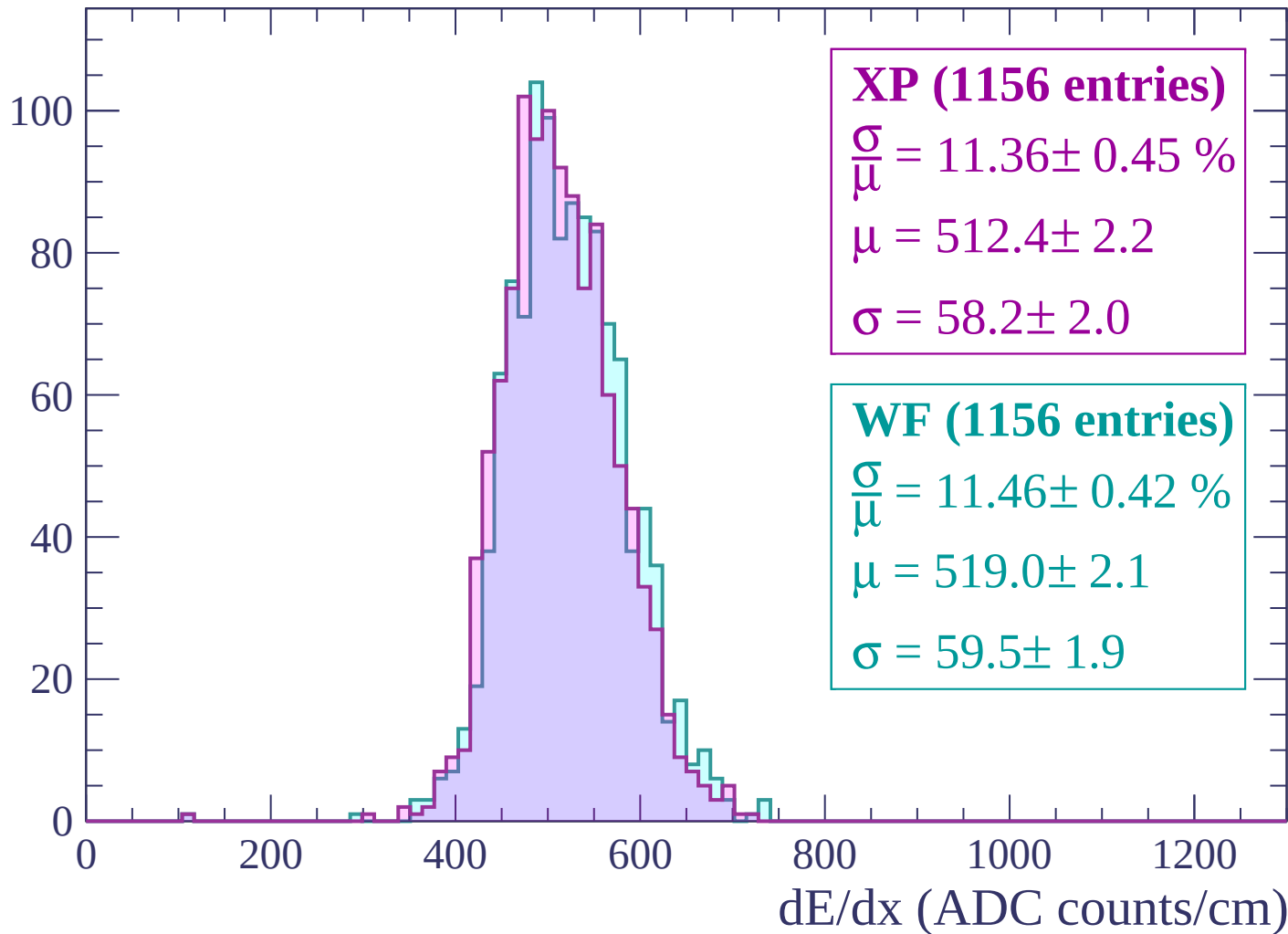






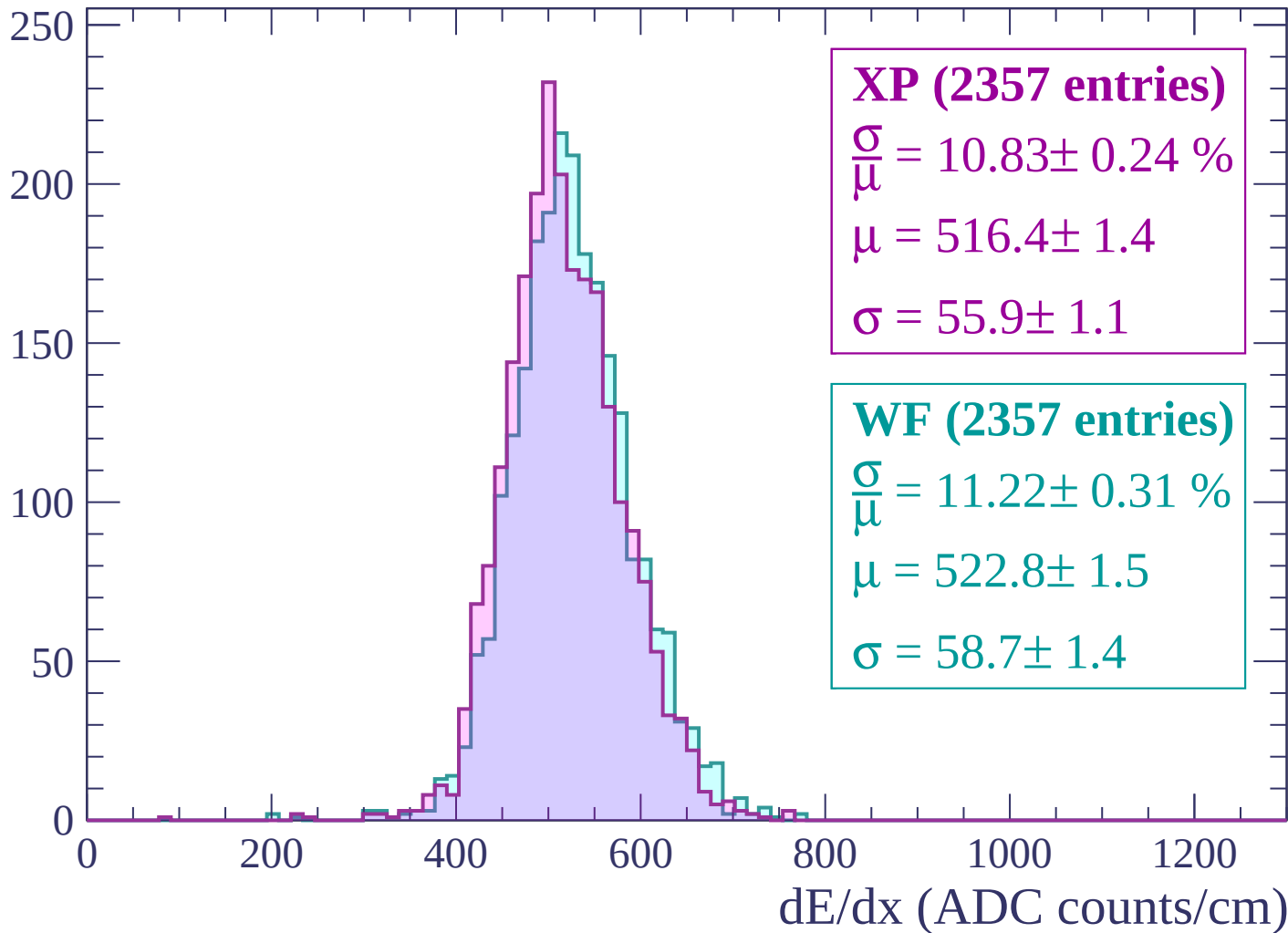
# Energy loss | $-3000 < p < -2940$

Count



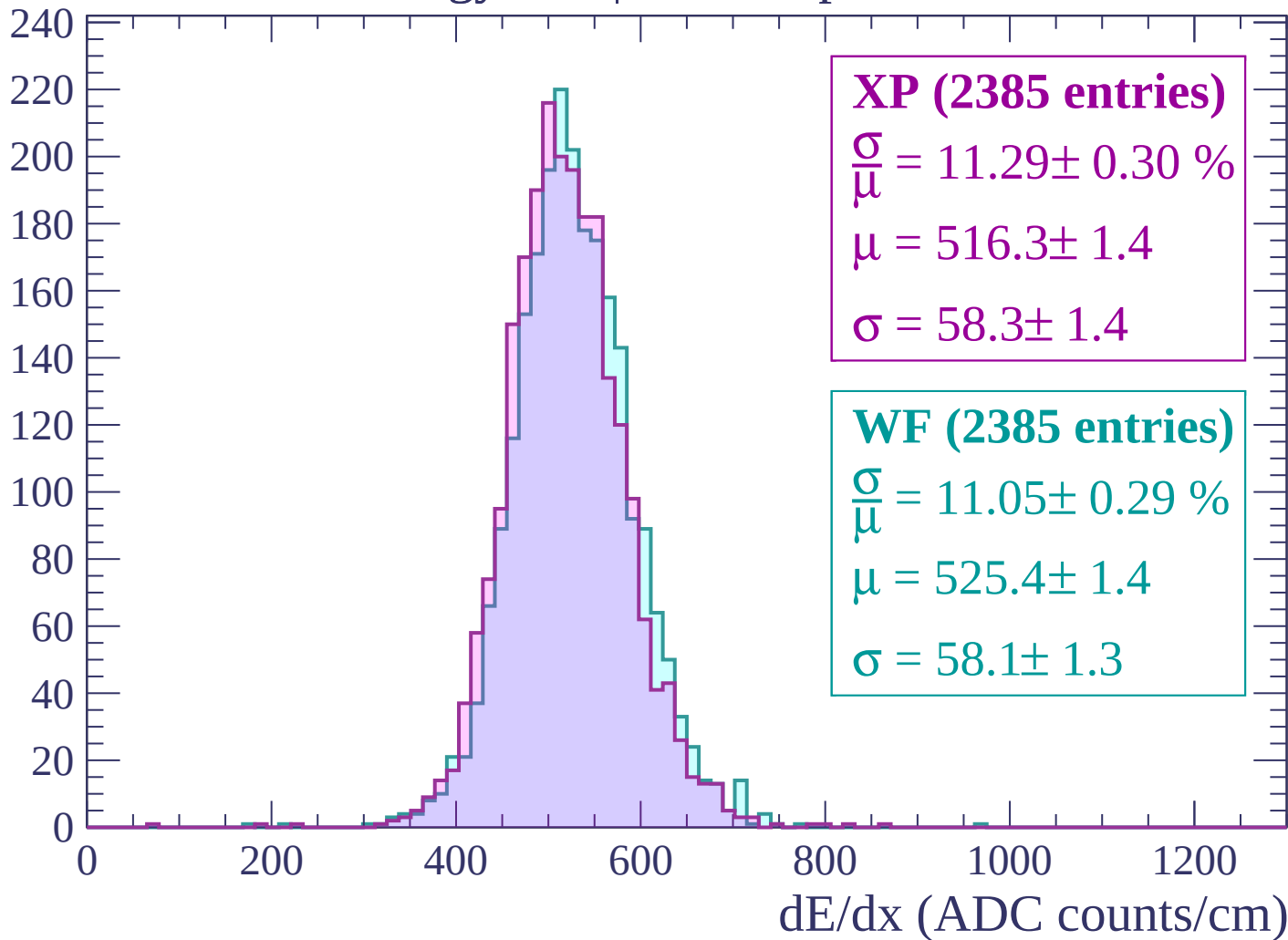
Energy loss |  $-2940 < p < -2880$

Count



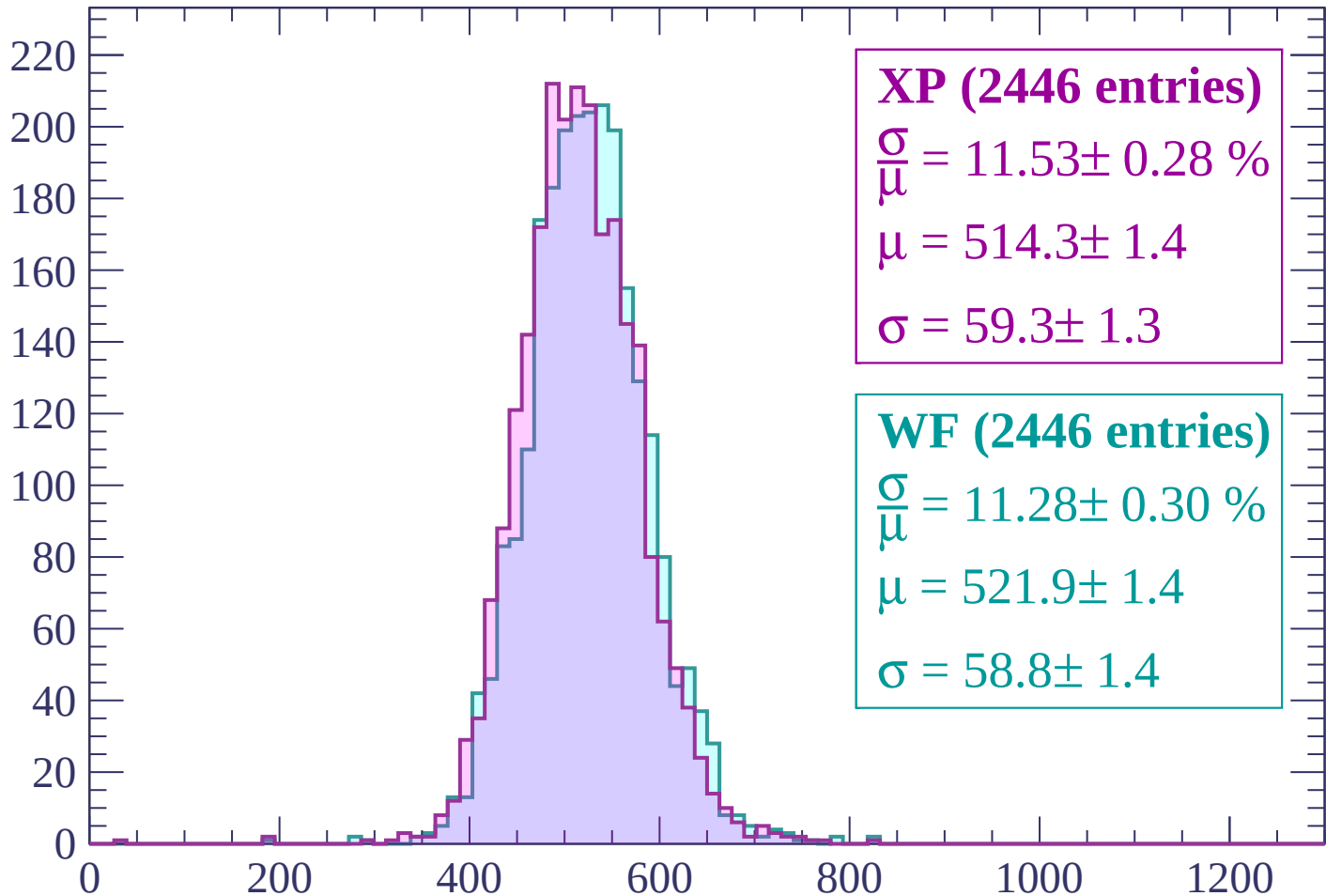
Energy loss |  $-2880 < p < -2820$

Count



Energy loss |  $-2820 < p < -2760$

Count



**XP (2446 entries)**

$$\frac{\sigma}{\mu} = 11.53 \pm 0.28 \%$$

$$\mu = 514.3 \pm 1.4$$

$$\sigma = 59.3 \pm 1.3$$

**WF (2446 entries)**

$$\frac{\sigma}{\mu} = 11.28 \pm 0.30 \%$$

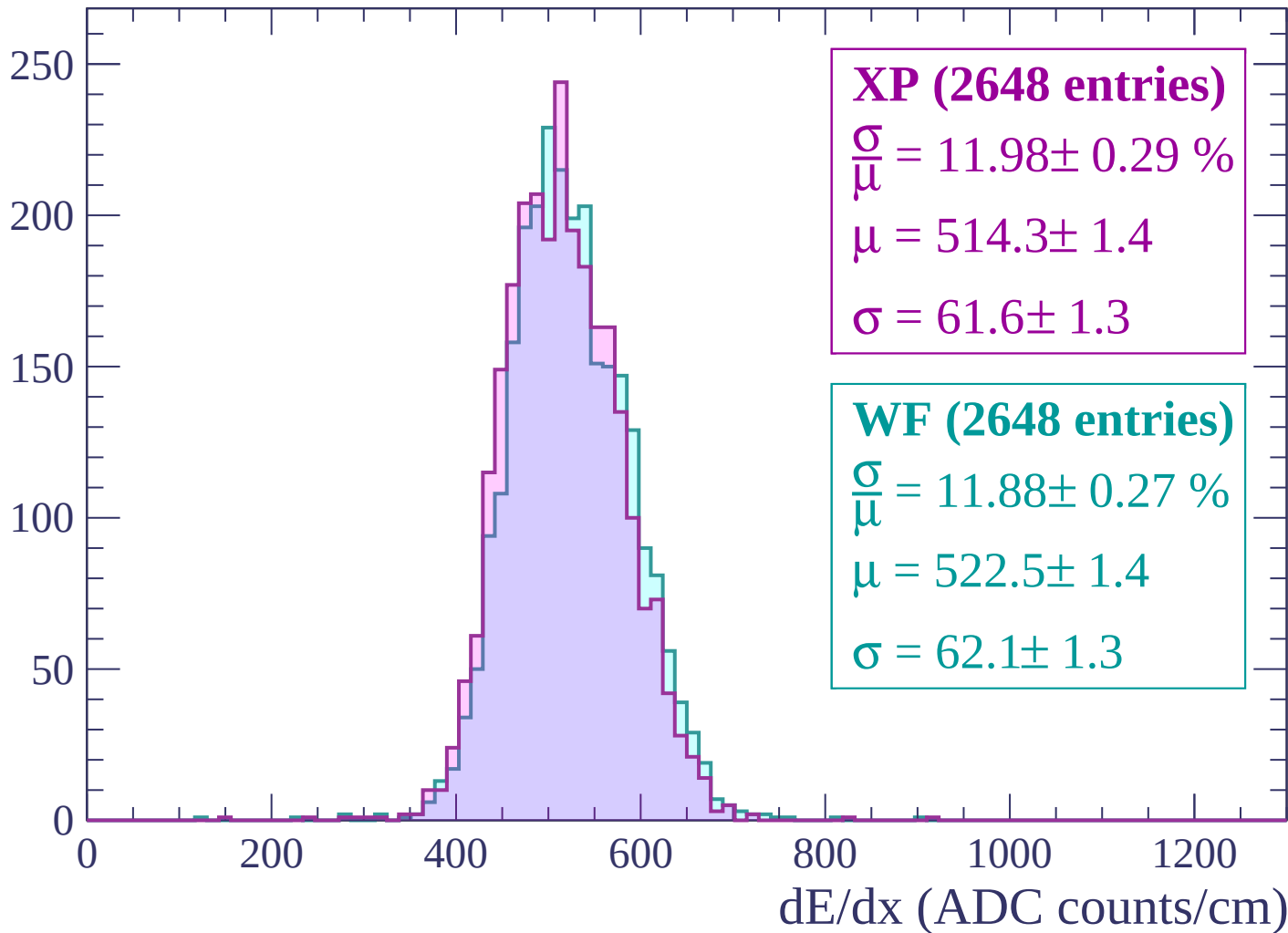
$$\mu = 521.9 \pm 1.4$$

$$\sigma = 58.8 \pm 1.4$$

dE/dx (ADC counts/cm)

# Energy loss | -2760 < p < -2700

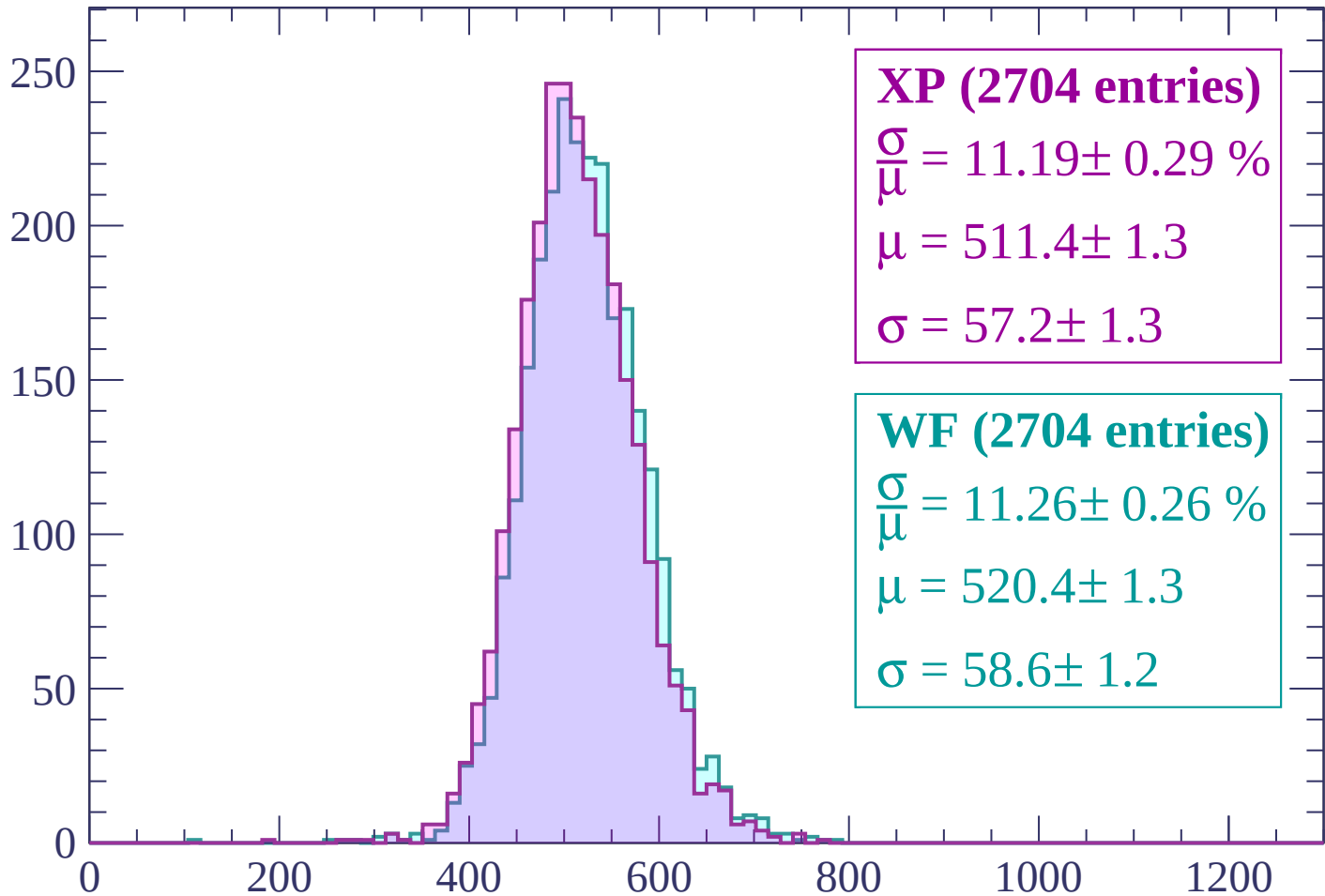
Count





# Energy loss | -2700 < p < -2640

Count



**XP (2704 entries)**

$$\frac{\sigma}{\mu} = 11.19 \pm 0.29 \%$$

$$\mu = 511.4 \pm 1.3$$

$$\sigma = 57.2 \pm 1.3$$

**WF (2704 entries)**

$$\frac{\sigma}{\mu} = 11.26 \pm 0.26 \%$$

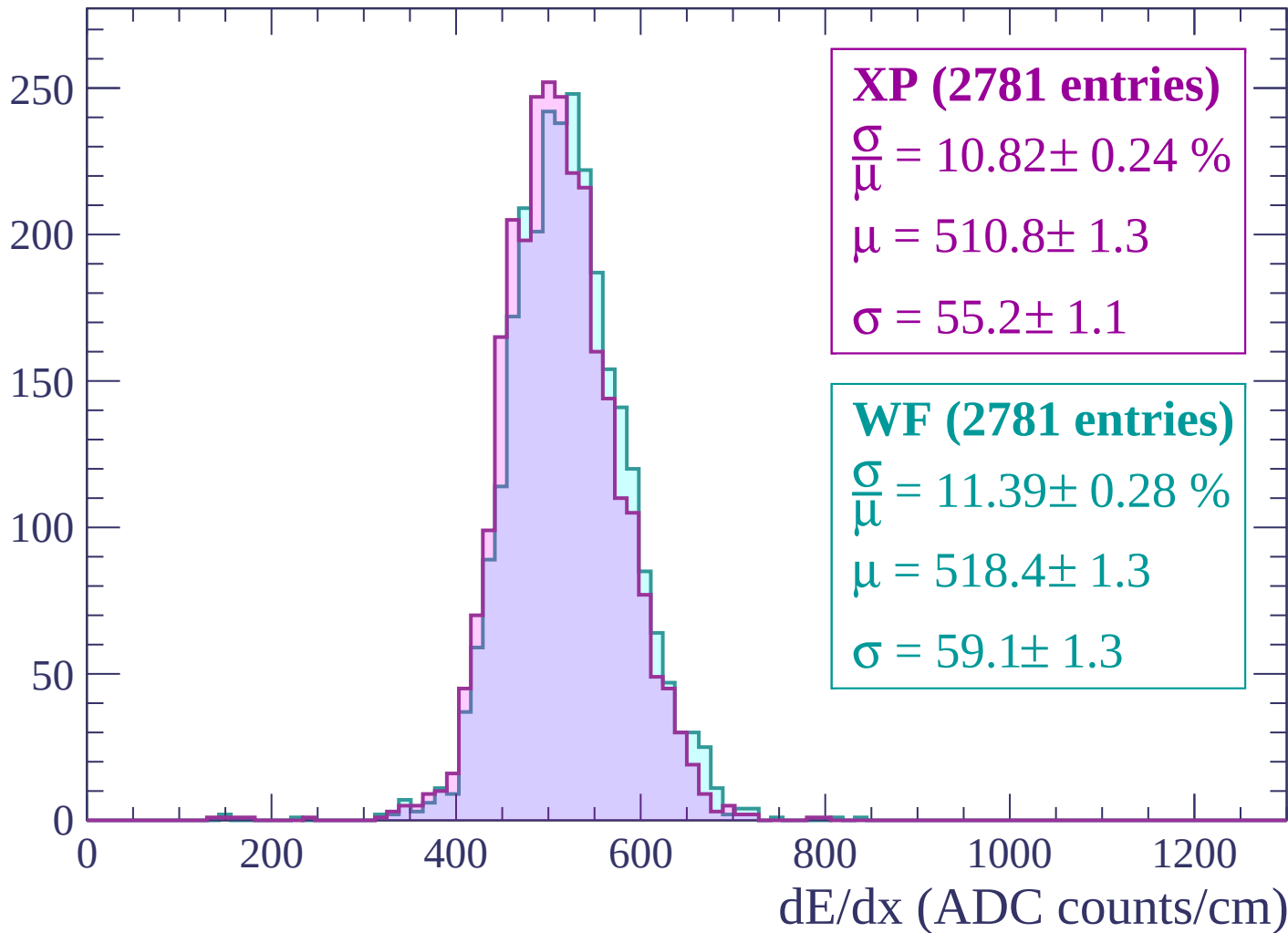
$$\mu = 520.4 \pm 1.3$$

$$\sigma = 58.6 \pm 1.2$$

dE/dx (ADC counts/cm)

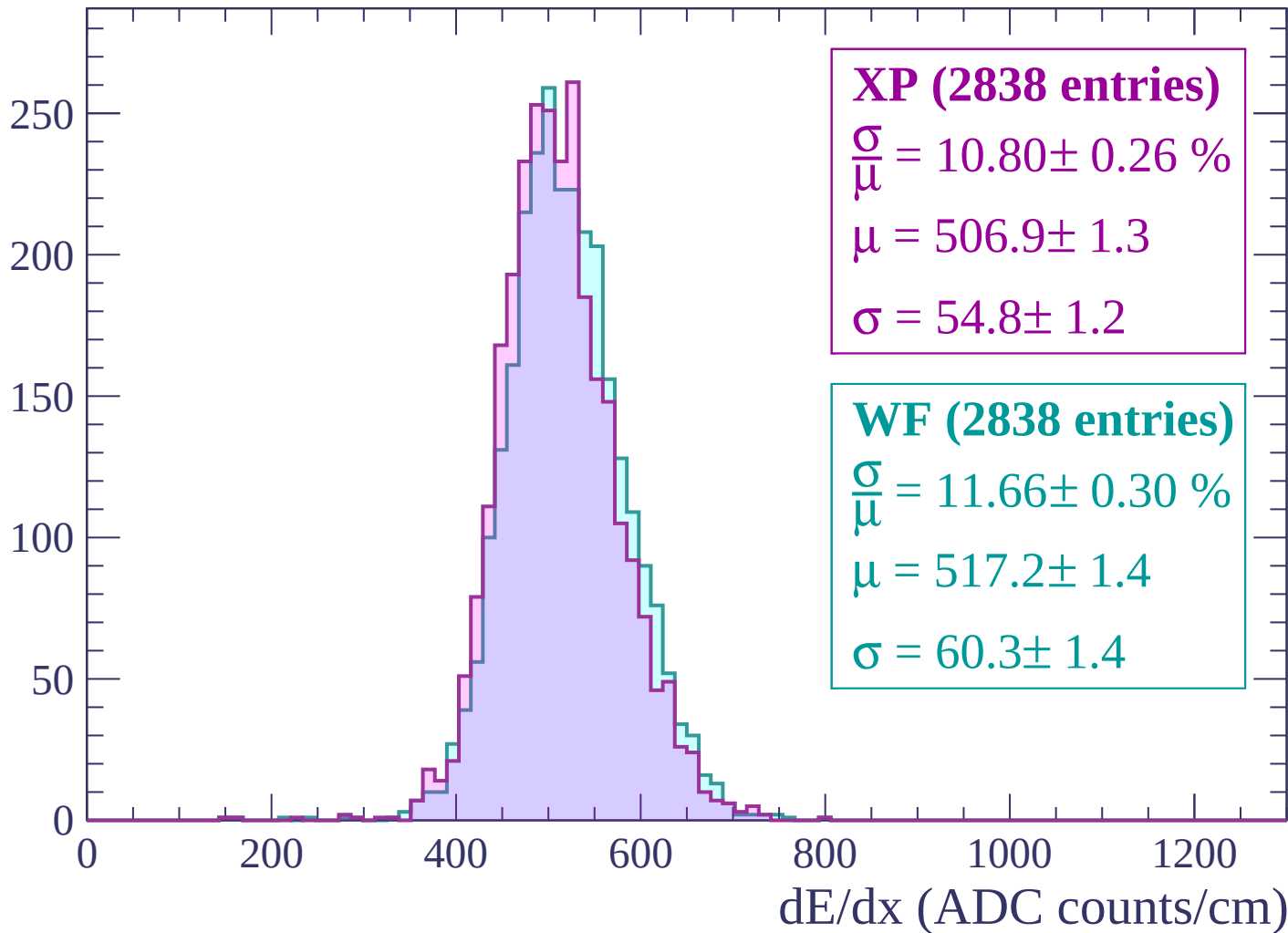
Energy loss |  $-2640 < p < -2580$

Count



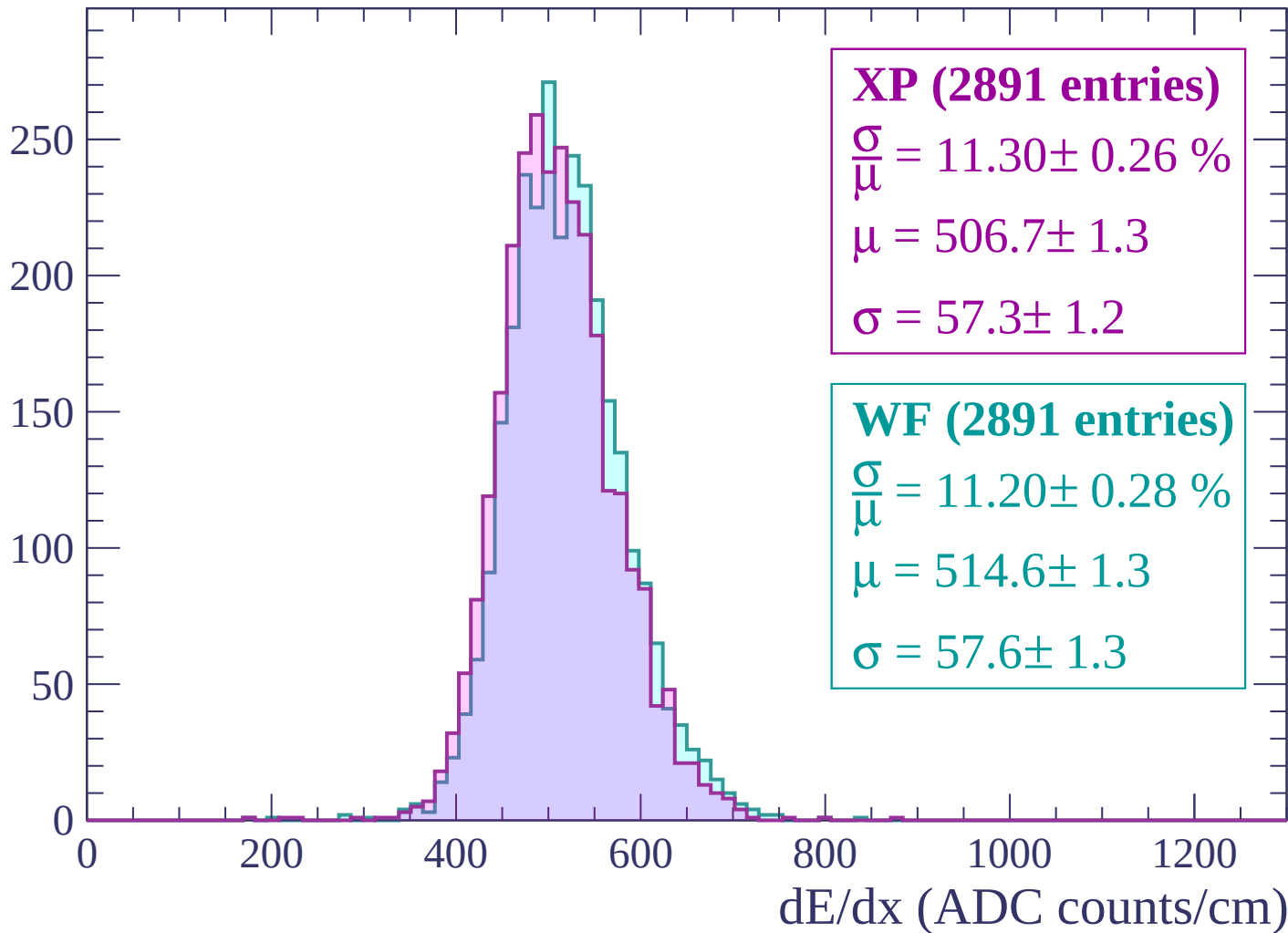
Energy loss |  $-2580 < p < -2520$

Count



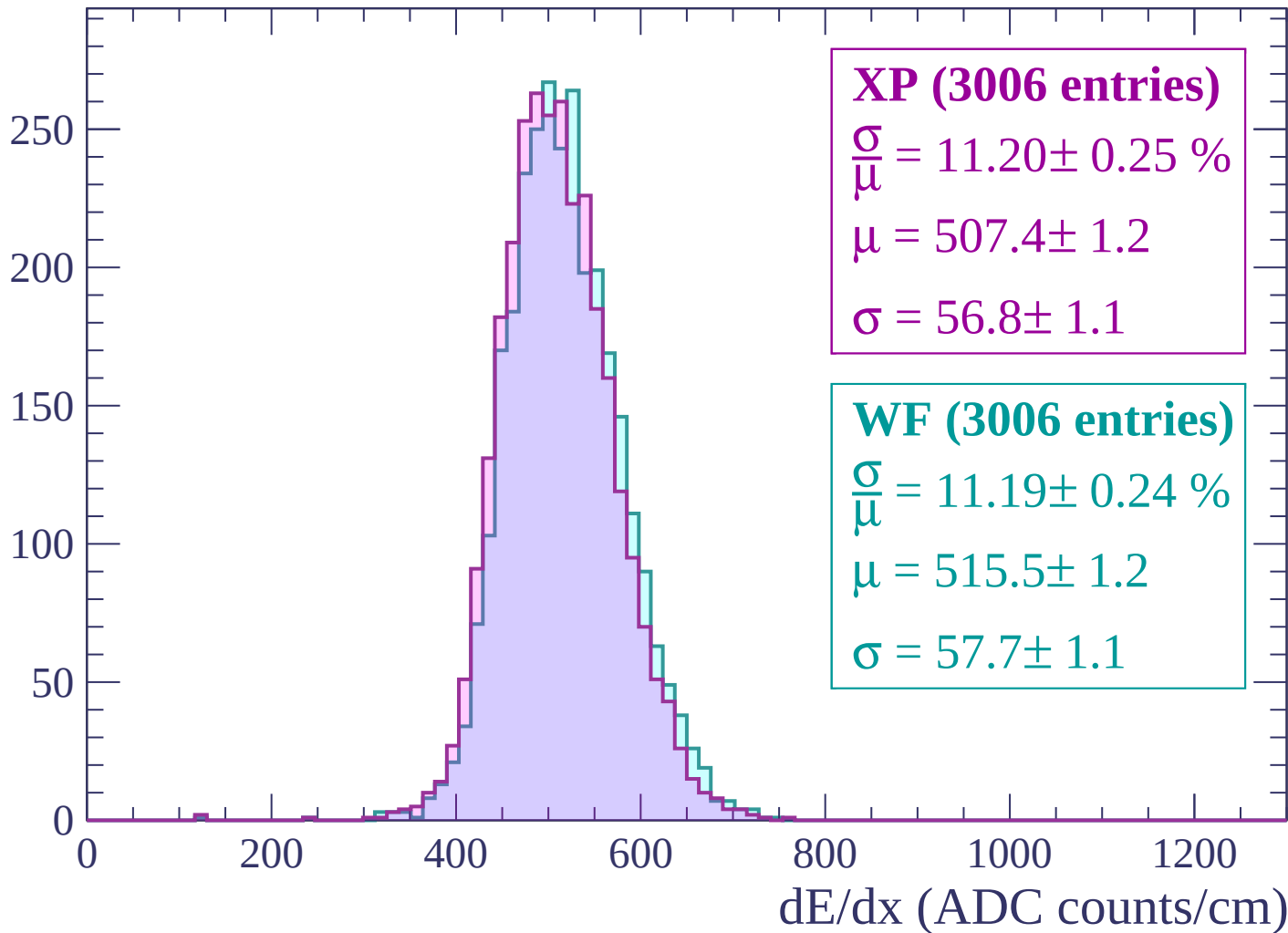
# Energy loss | -2520 < p < -2460

Count



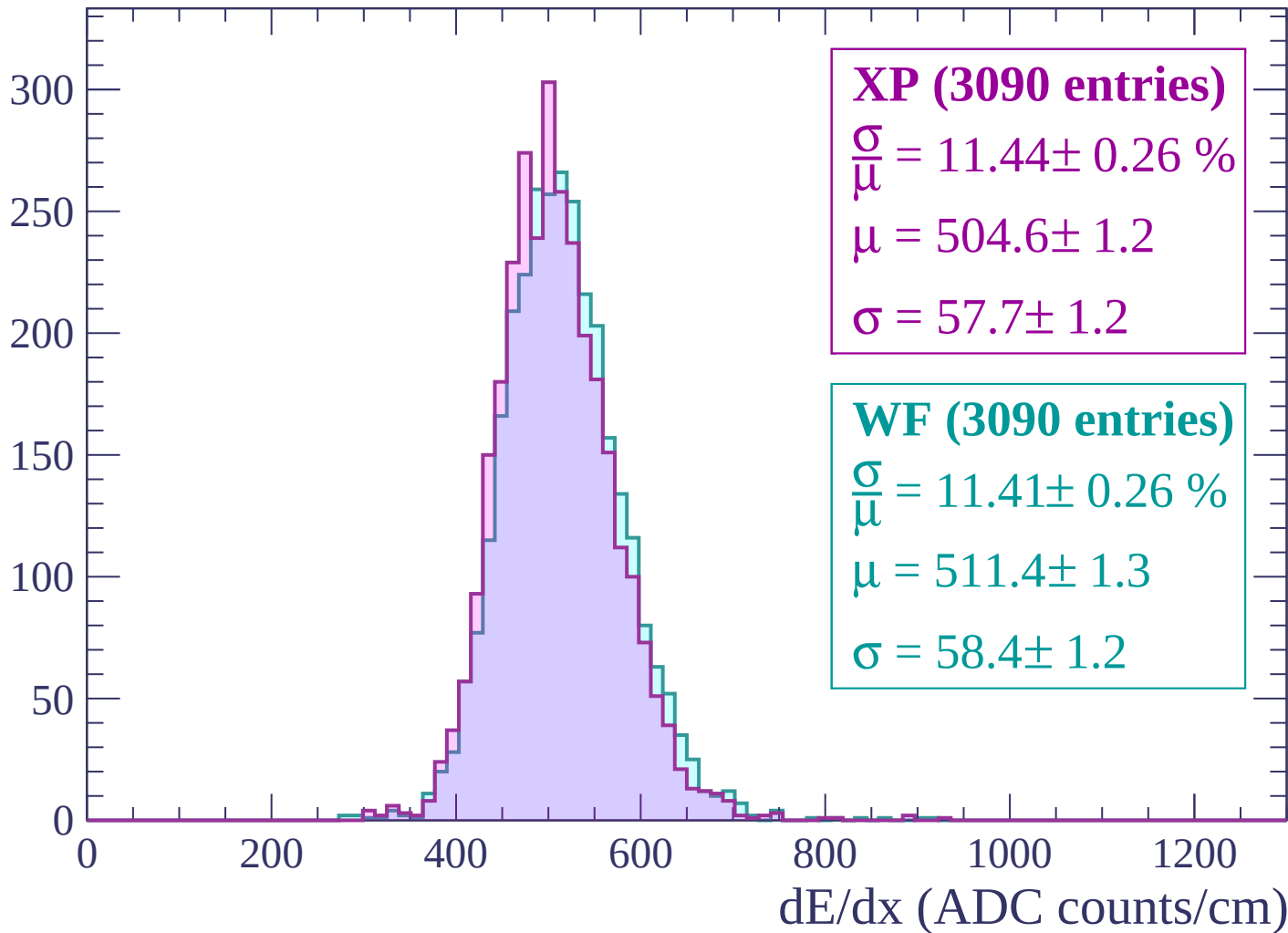
# Energy loss | -2460 < p < -2400

Count



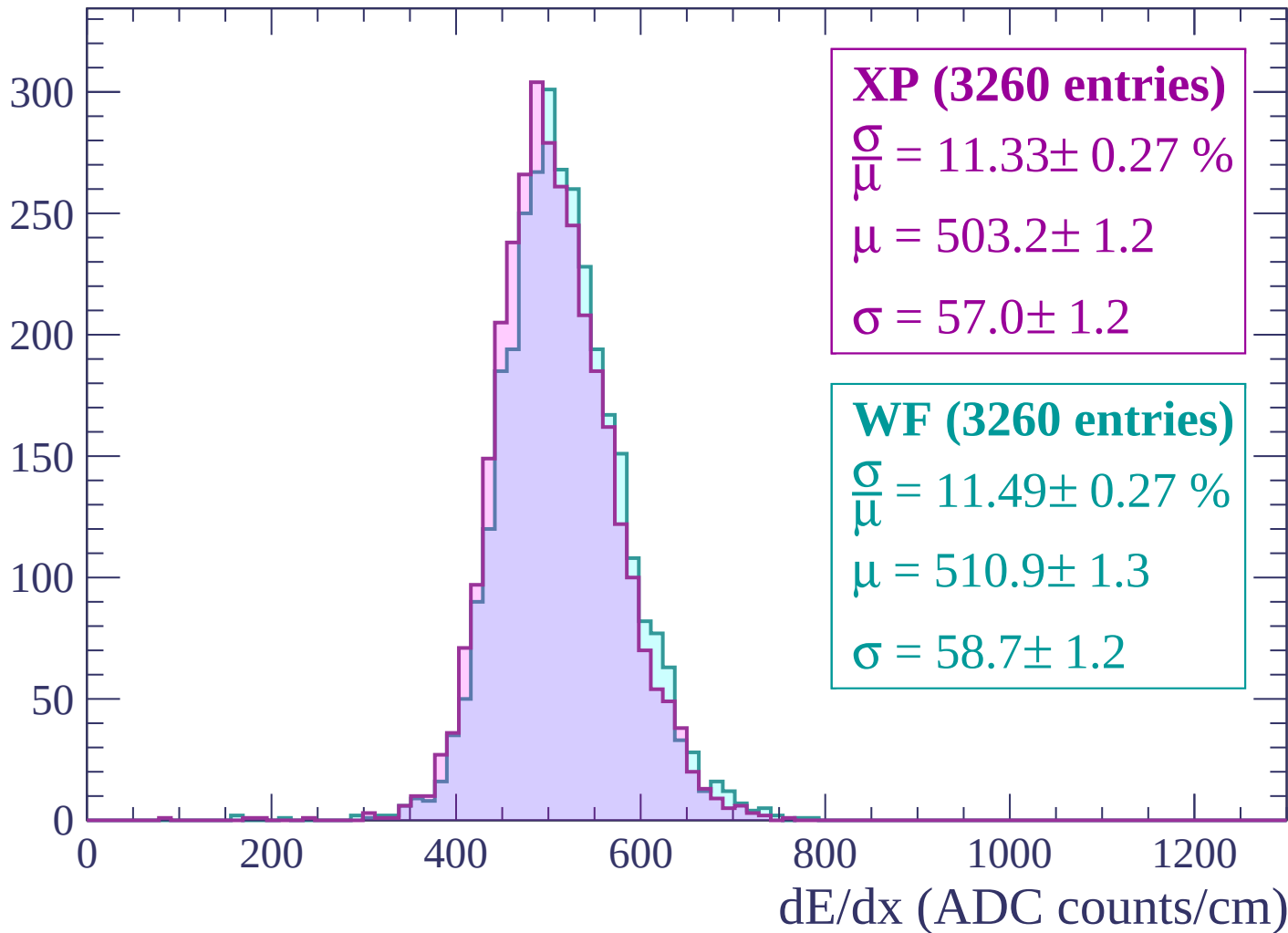
# Energy loss | $-2400 < p < -2340$

Count



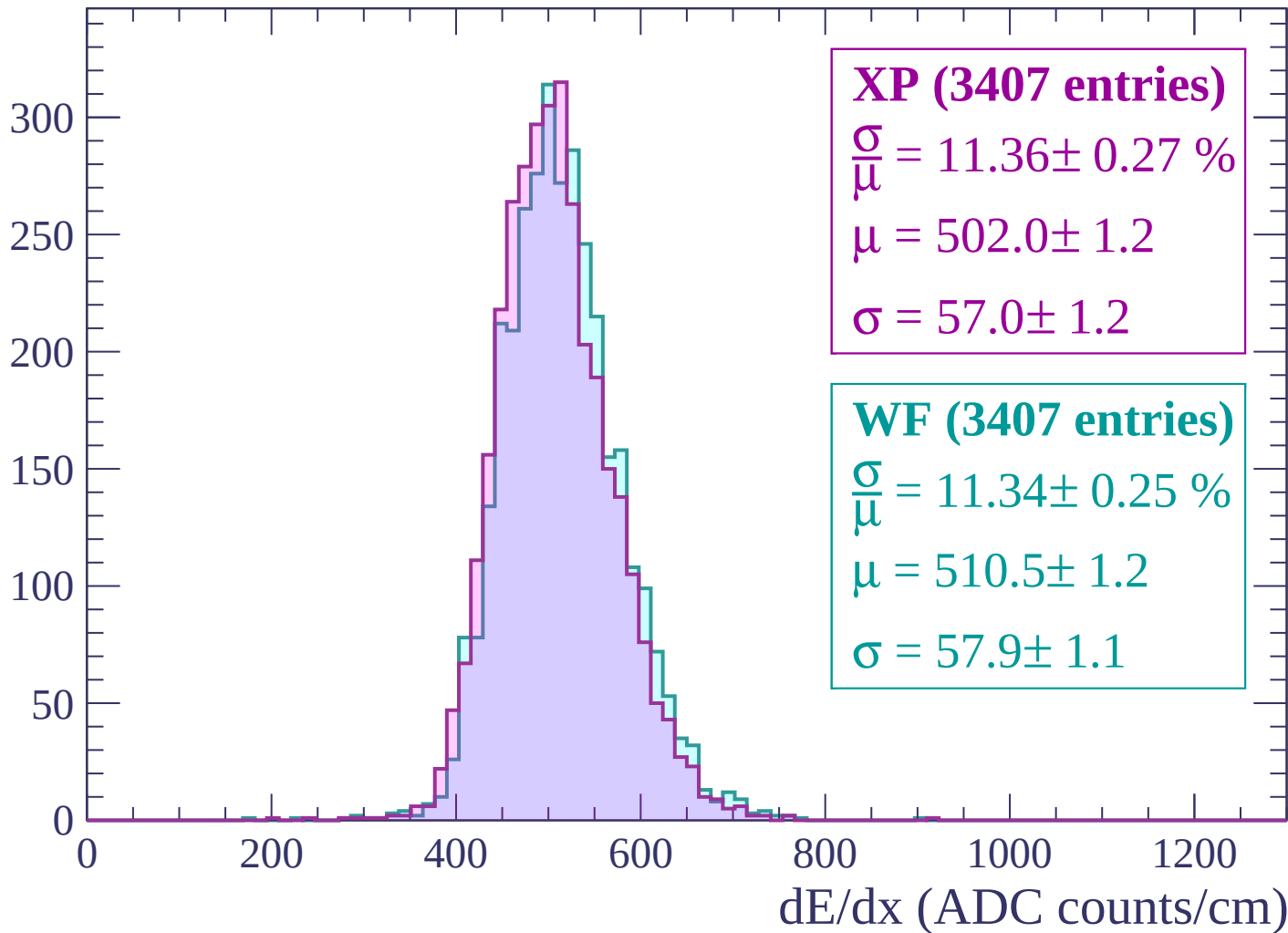
Energy loss |  $-2340 < p < -2280$

Count



# Energy loss | -2280 < p < -2220

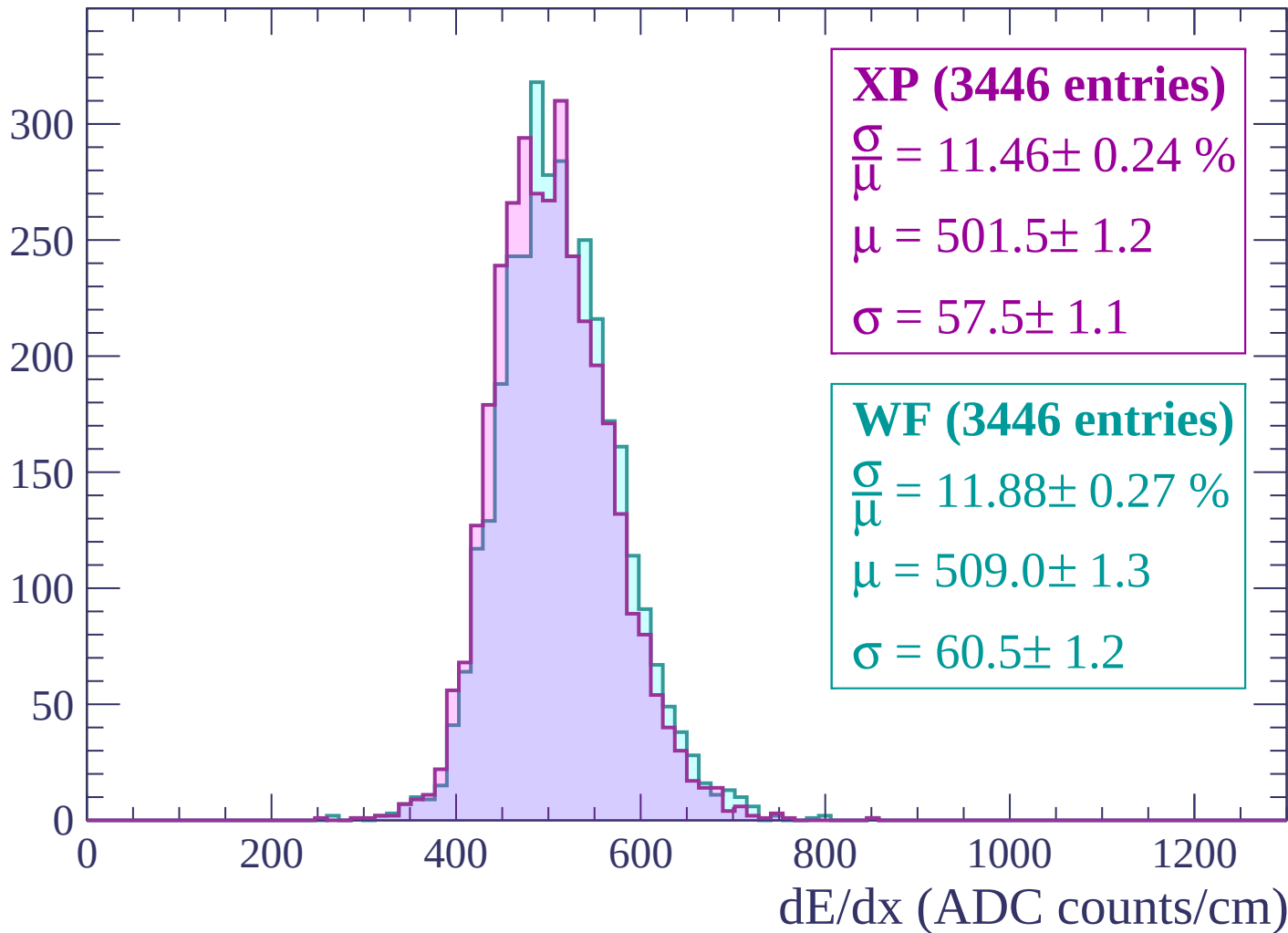
Count





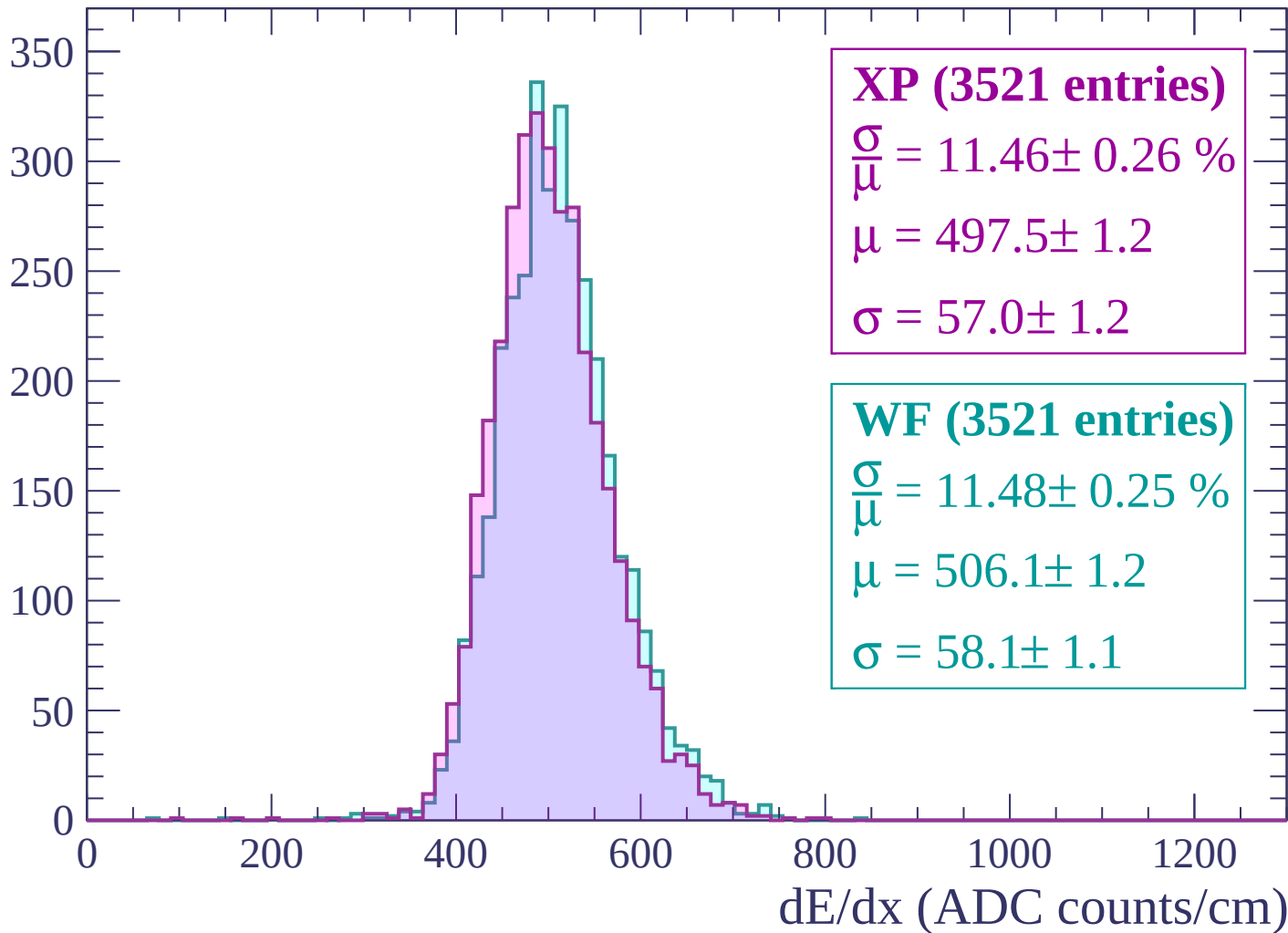
# Energy loss | -2220 < p < -2160

Count



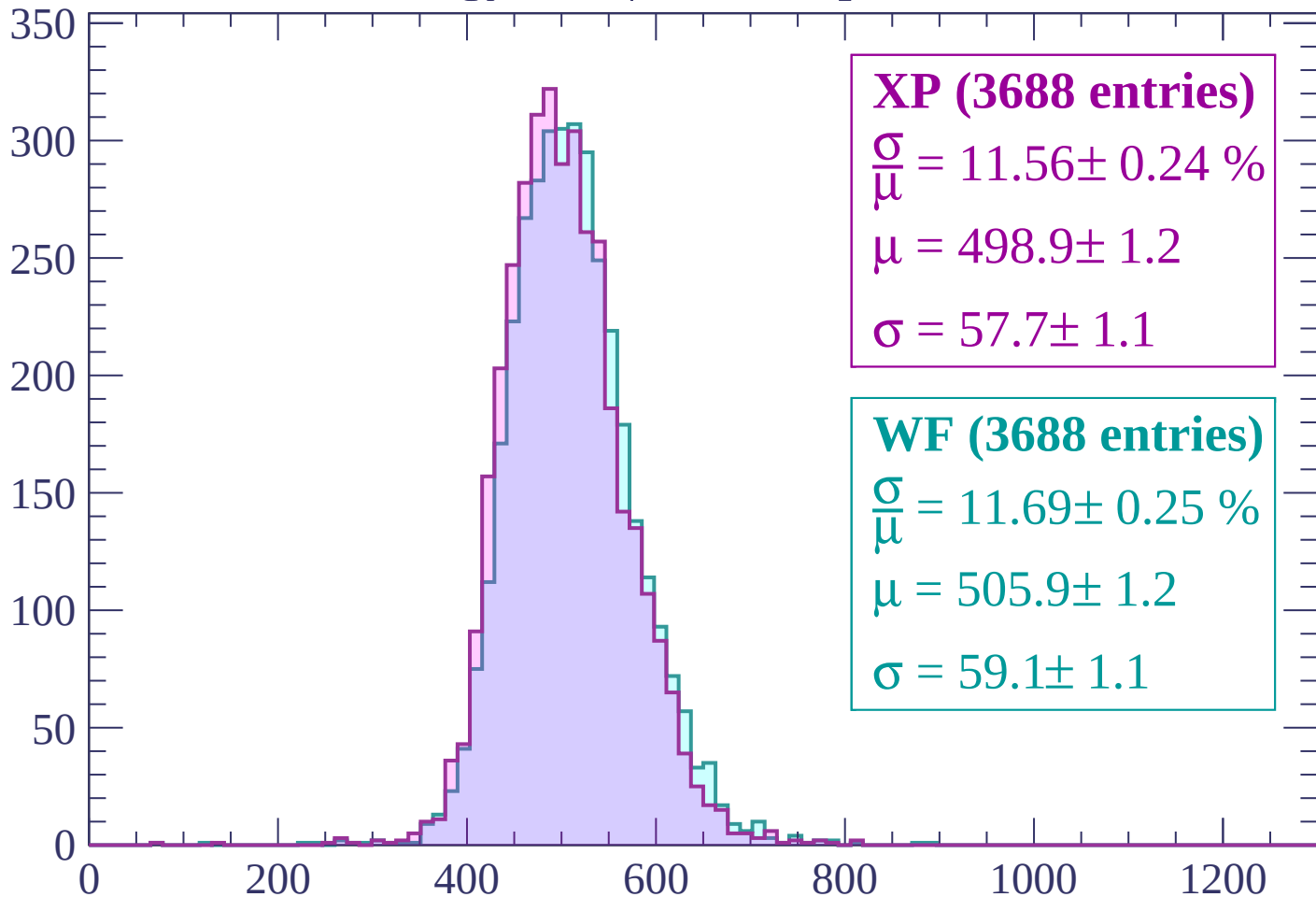
# Energy loss | -2160 < p < -2100

Count



# Energy loss | -2100 < p < -2040

Count



**XP (3688 entries)**

$$\frac{\sigma}{\mu} = 11.56 \pm 0.24 \%$$

$$\mu = 498.9 \pm 1.2$$

$$\sigma = 57.7 \pm 1.1$$

**WF (3688 entries)**

$$\frac{\sigma}{\mu} = 11.69 \pm 0.25 \%$$

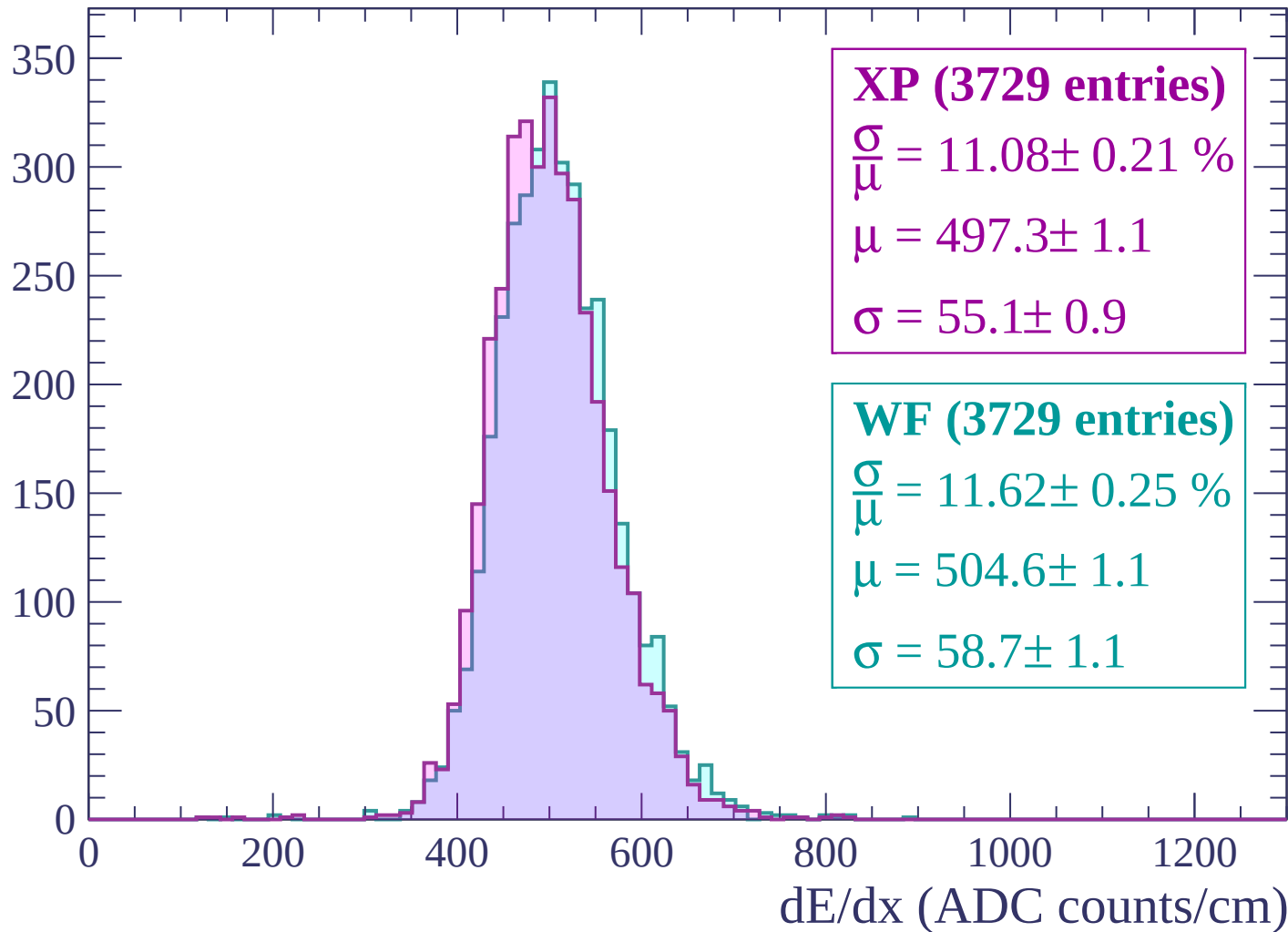
$$\mu = 505.9 \pm 1.2$$

$$\sigma = 59.1 \pm 1.1$$

dE/dx (ADC counts/cm)

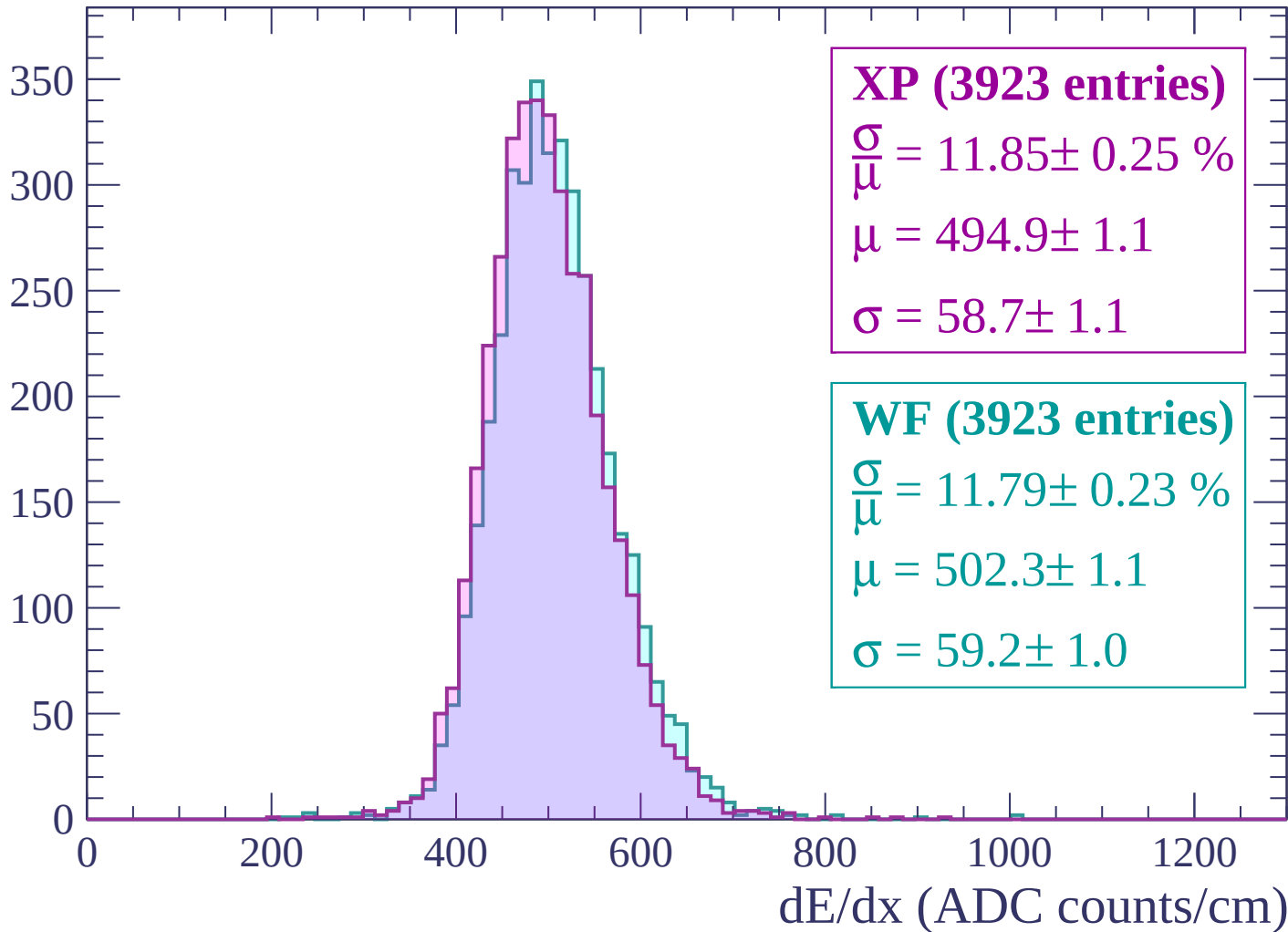
# Energy loss | -2040 < p < -1980

Count



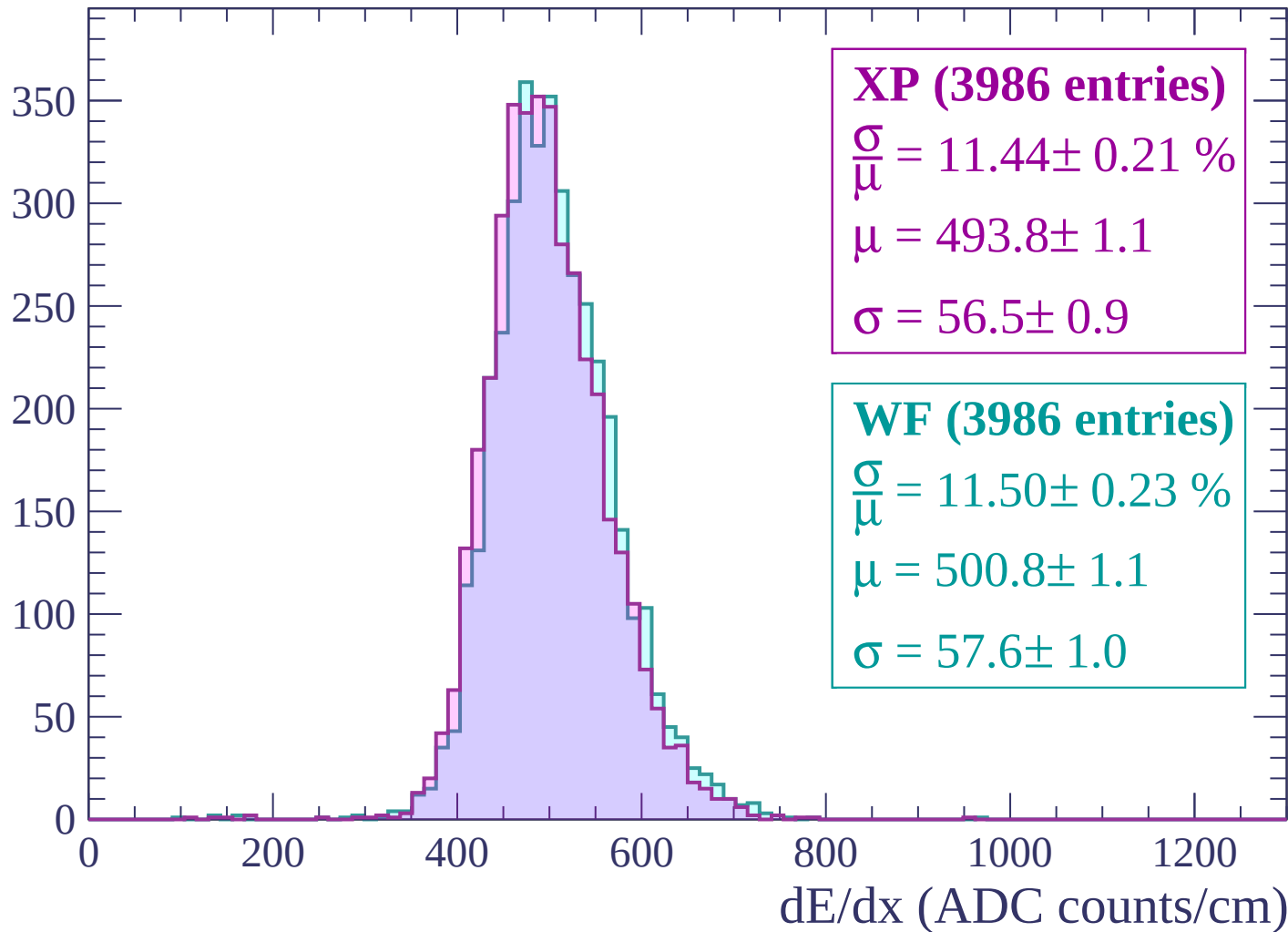
# Energy loss | -1980 < p < -1920

Count



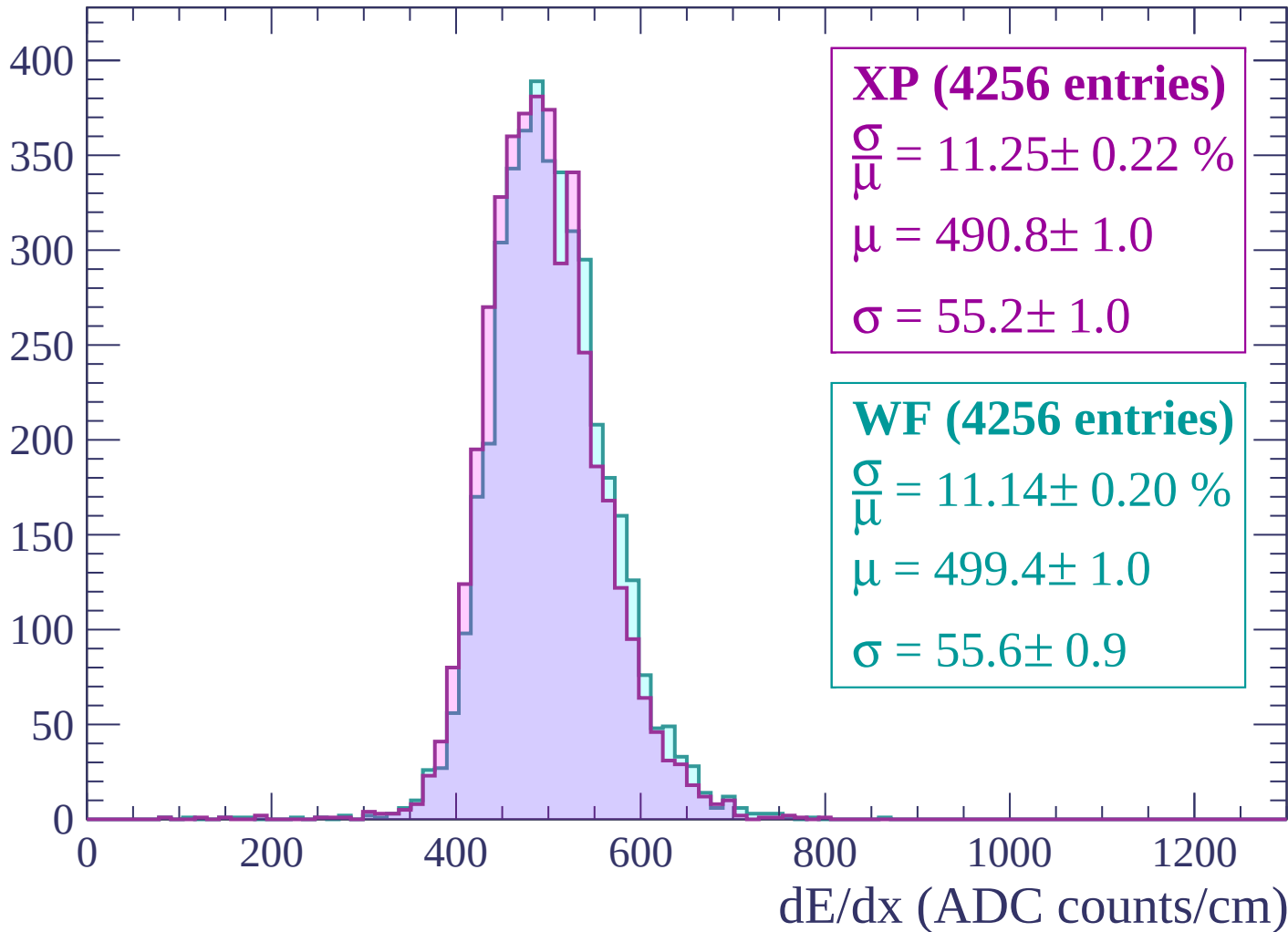
# Energy loss | -1920 < p < -1860

Count



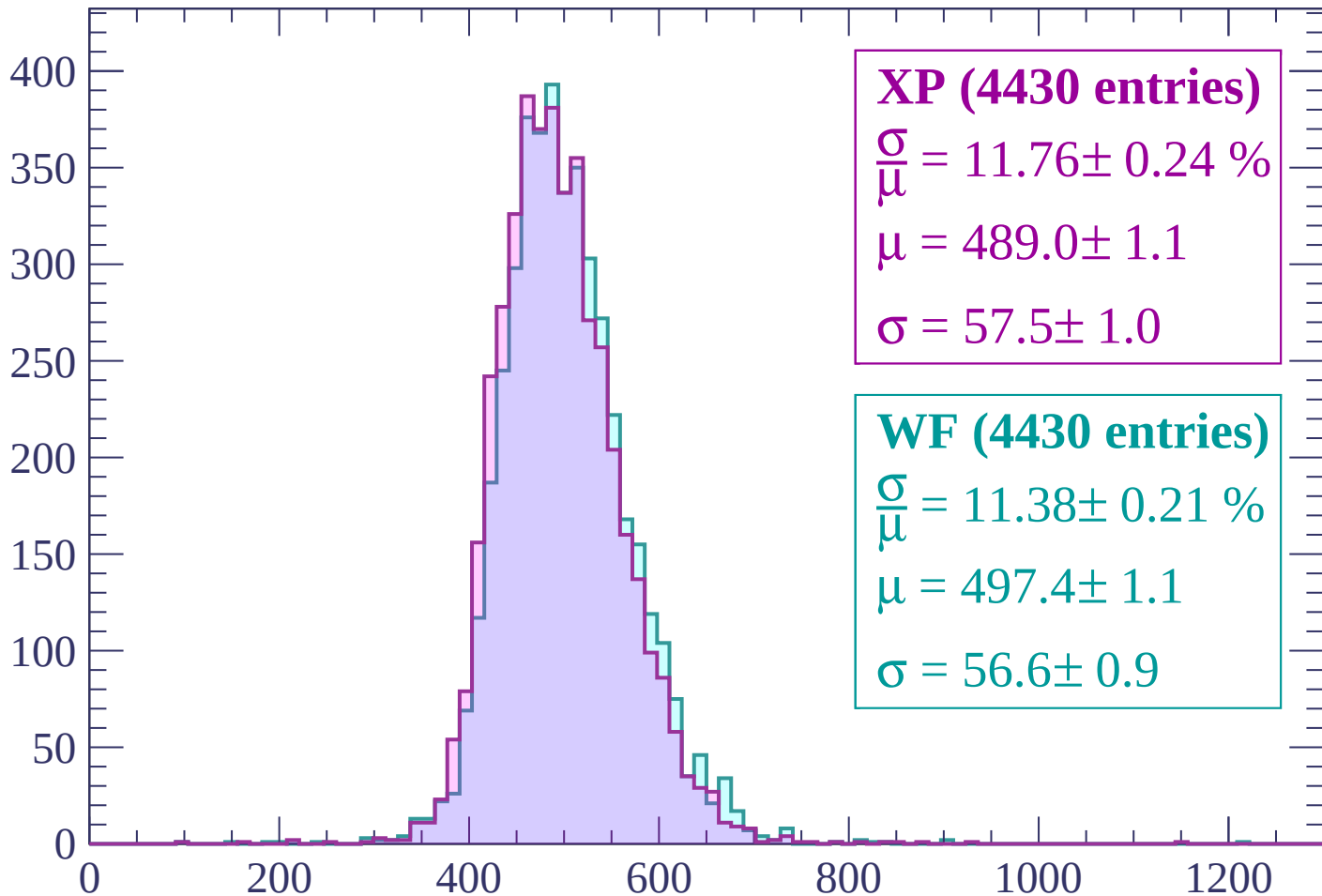
# Energy loss | -1860 < p < -1800

Count



Energy loss |  $-1800 < p < -1740$

Count



**XP (4430 entries)**

$$\frac{\sigma}{\mu} = 11.76 \pm 0.24 \%$$

$$\mu = 489.0 \pm 1.1$$

$$\sigma = 57.5 \pm 1.0$$

**WF (4430 entries)**

$$\frac{\sigma}{\mu} = 11.38 \pm 0.21 \%$$

$$\mu = 497.4 \pm 1.1$$

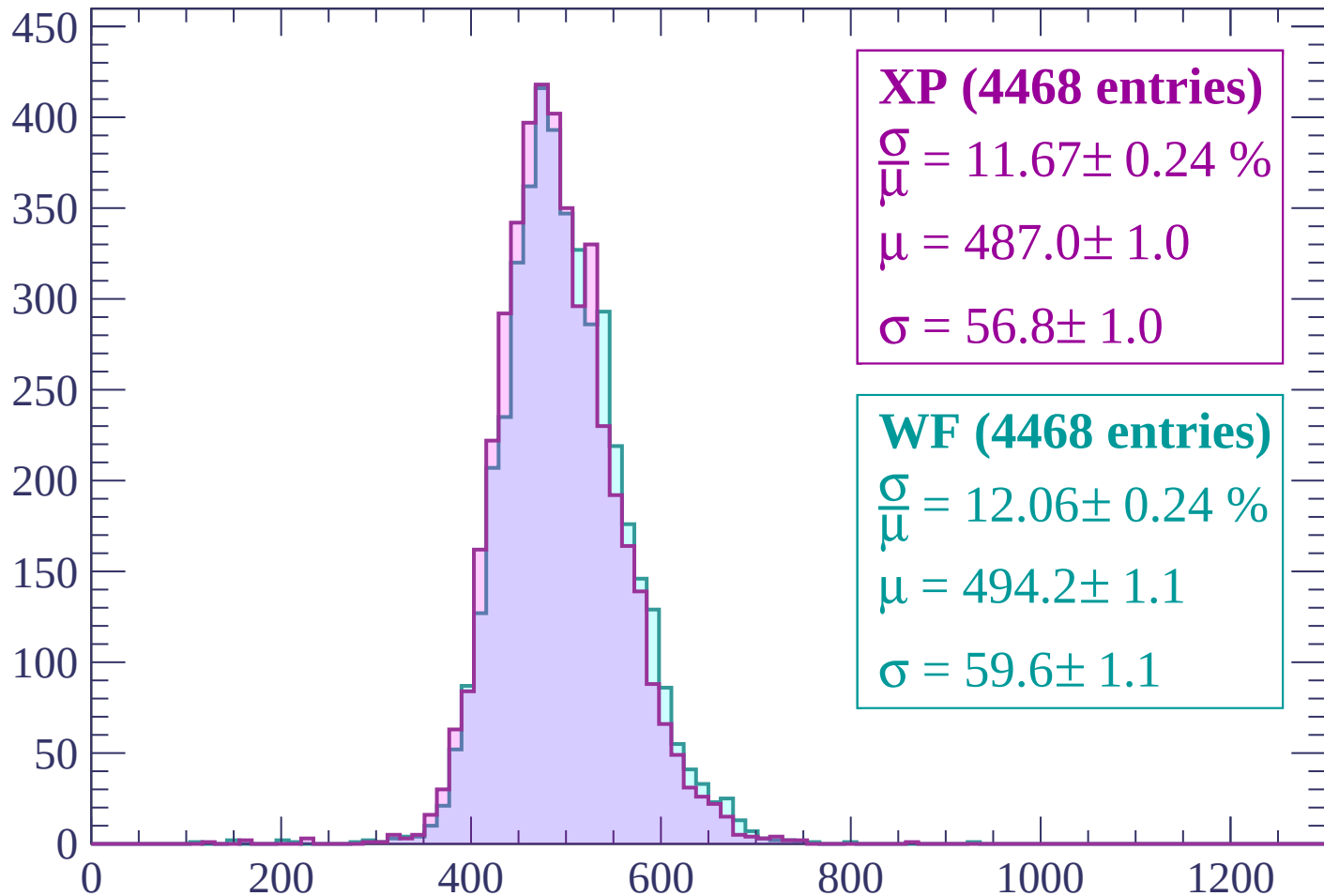
$$\sigma = 56.6 \pm 0.9$$

dE/dx (ADC counts/cm)



Energy loss |  $-1740 < p < -1680$

Count



**XP (4468 entries)**

$$\frac{\sigma}{\mu} = 11.67 \pm 0.24 \%$$

$$\mu = 487.0 \pm 1.0$$

$$\sigma = 56.8 \pm 1.0$$

**WF (4468 entries)**

$$\frac{\sigma}{\mu} = 12.06 \pm 0.24 \%$$

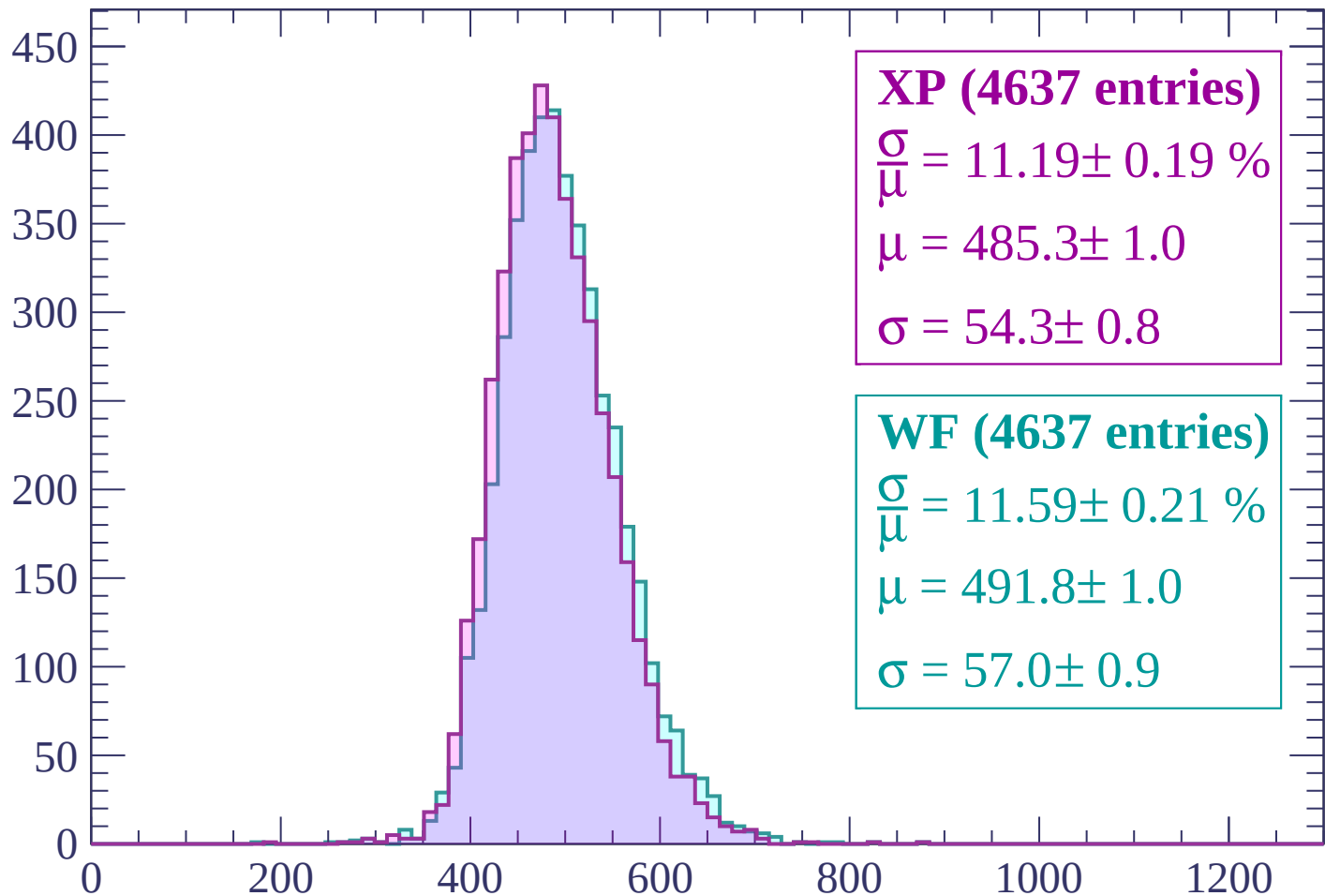
$$\mu = 494.2 \pm 1.1$$

$$\sigma = 59.6 \pm 1.1$$

dE/dx (ADC counts/cm)

# Energy loss | -1680 < p < -1620

Count



**XP (4637 entries)**

$\frac{\sigma}{\mu} = 11.19 \pm 0.19 \%$

$\mu = 485.3 \pm 1.0$

$\sigma = 54.3 \pm 0.8$

**WF (4637 entries)**

$\frac{\sigma}{\mu} = 11.59 \pm 0.21 \%$

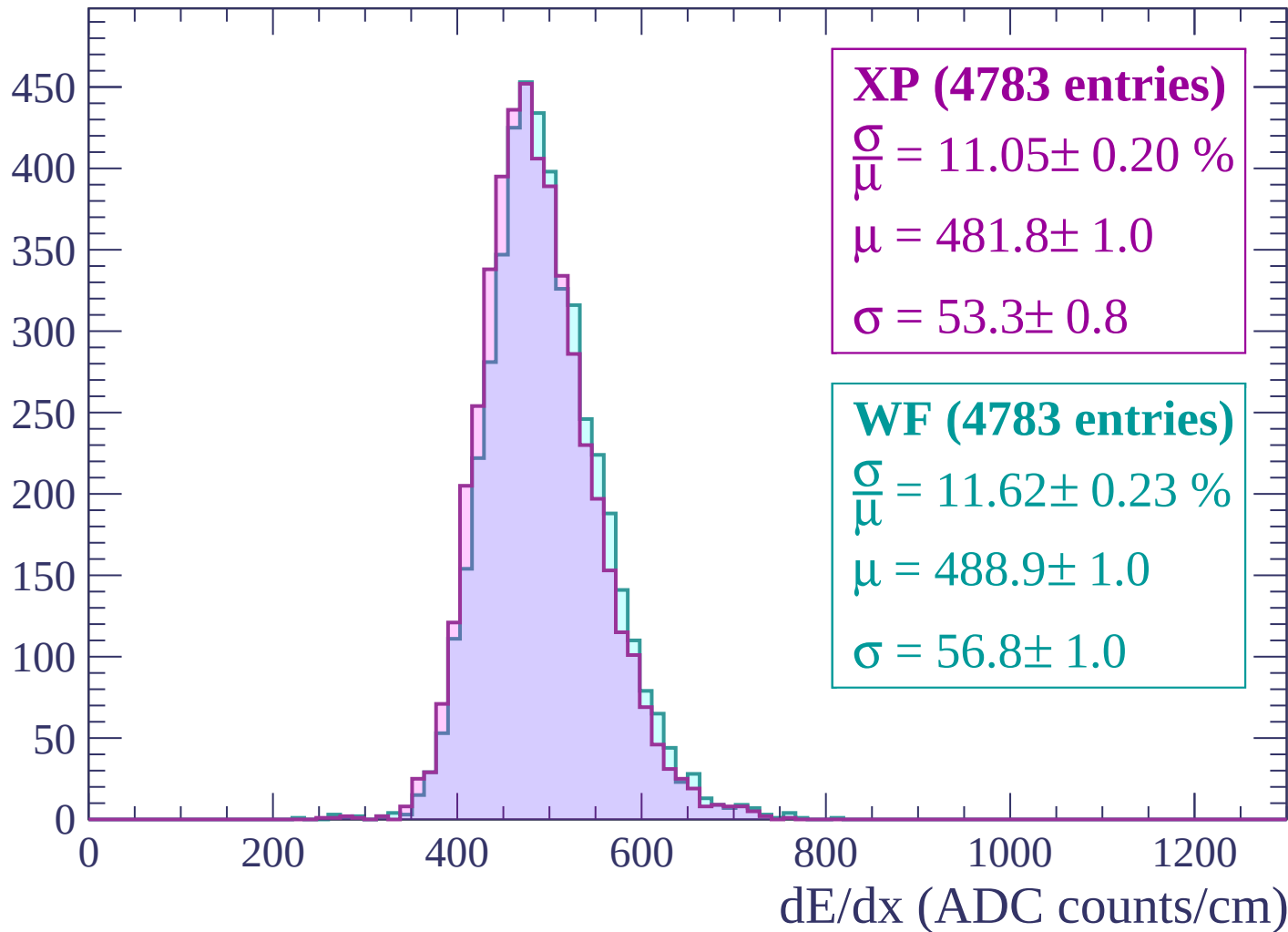
$\mu = 491.8 \pm 1.0$

$\sigma = 57.0 \pm 0.9$

dE/dx (ADC counts/cm)

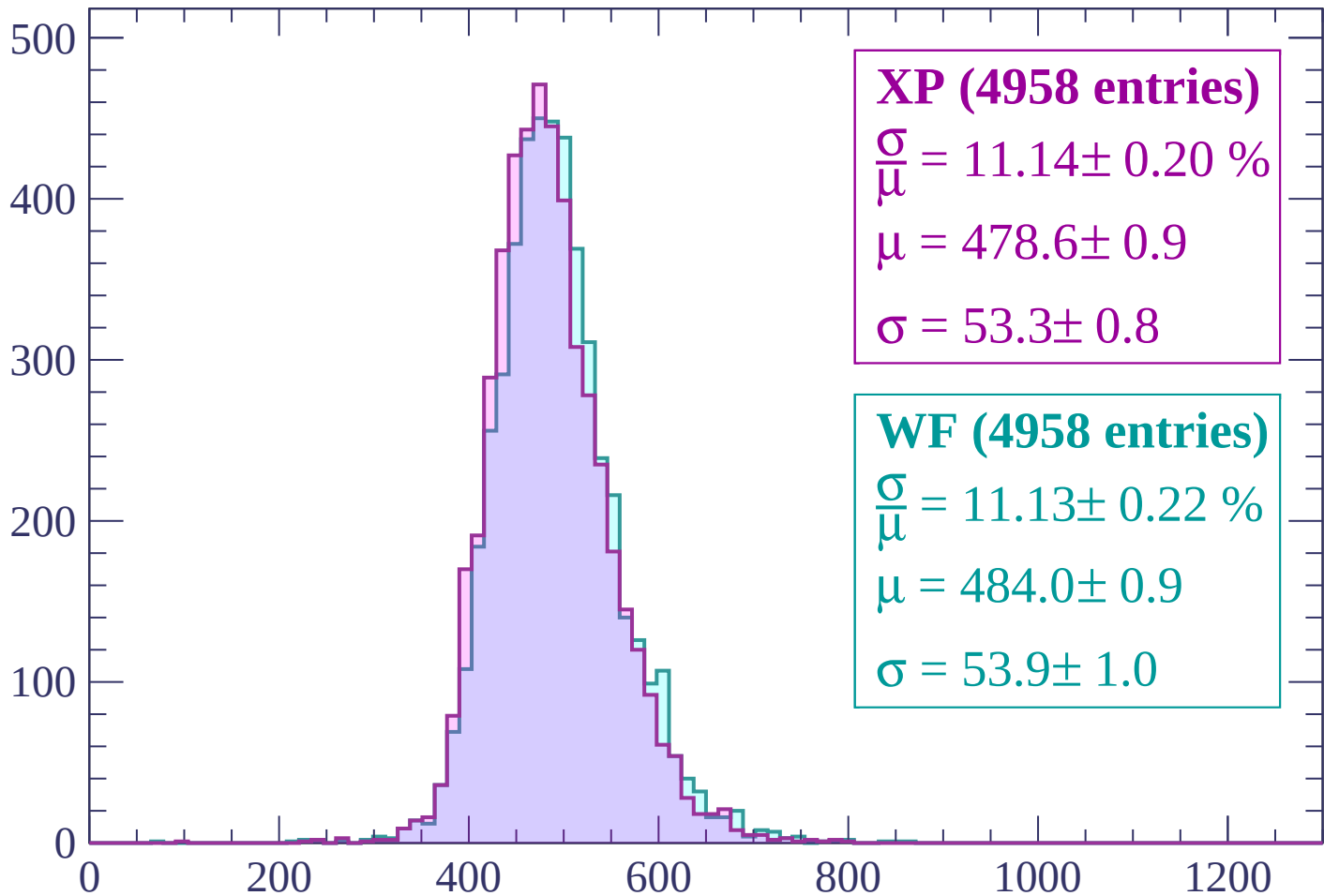
Energy loss |  $-1620 < p < -1560$

Count



# Energy loss | -1560 < p < -1500

Count



**XP (4958 entries)**

$$\frac{\sigma}{\mu} = 11.14 \pm 0.20 \%$$

$$\mu = 478.6 \pm 0.9$$

$$\sigma = 53.3 \pm 0.8$$

**WF (4958 entries)**

$$\frac{\sigma}{\mu} = 11.13 \pm 0.22 \%$$

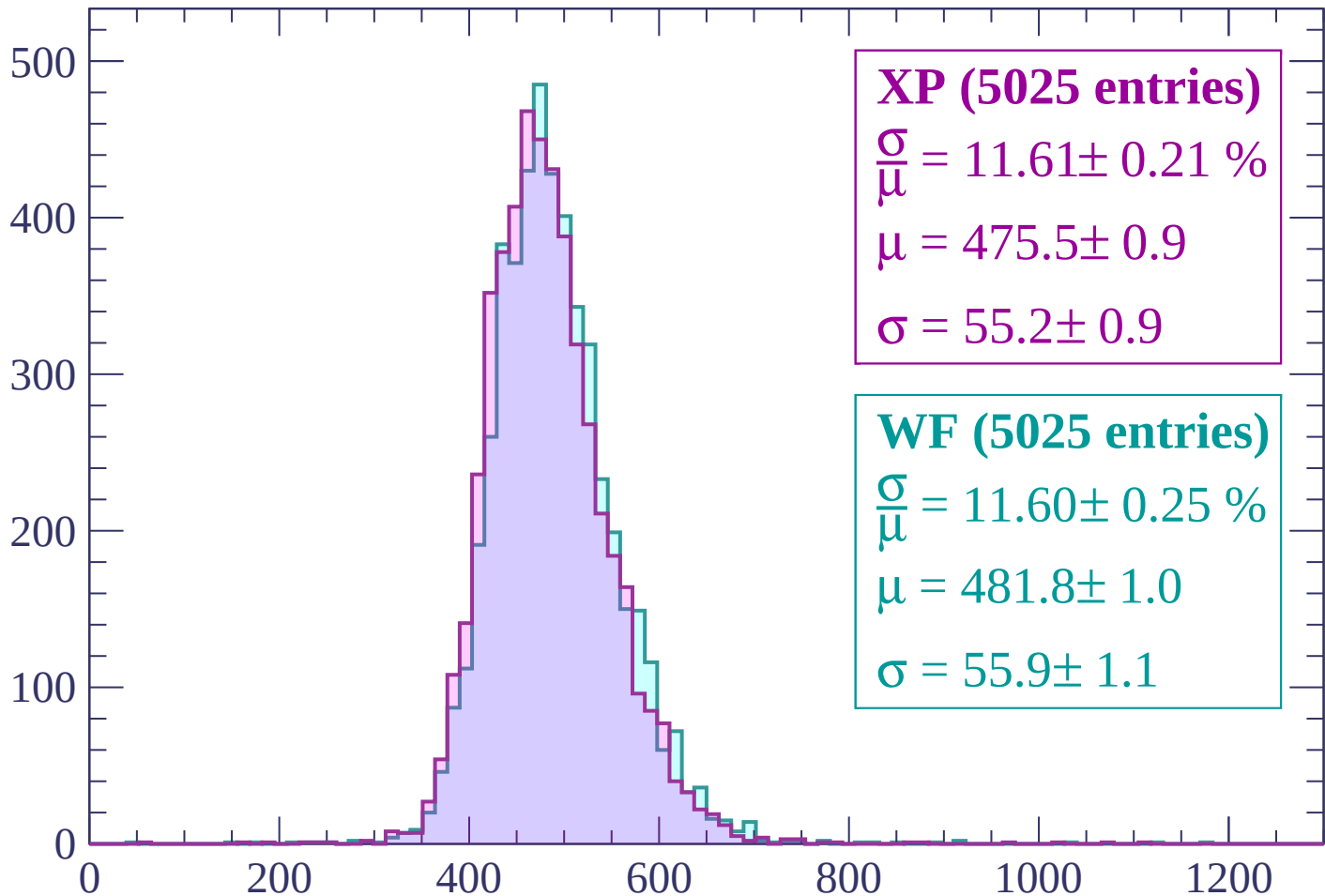
$$\mu = 484.0 \pm 0.9$$

$$\sigma = 53.9 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | $-1500 < p < -1440$

Count



**XP (5025 entries)**

$\frac{\sigma}{\mu} = 11.61 \pm 0.21 \%$

$\mu = 475.5 \pm 0.9$

$\sigma = 55.2 \pm 0.9$

**WF (5025 entries)**

$\frac{\sigma}{\mu} = 11.60 \pm 0.25 \%$

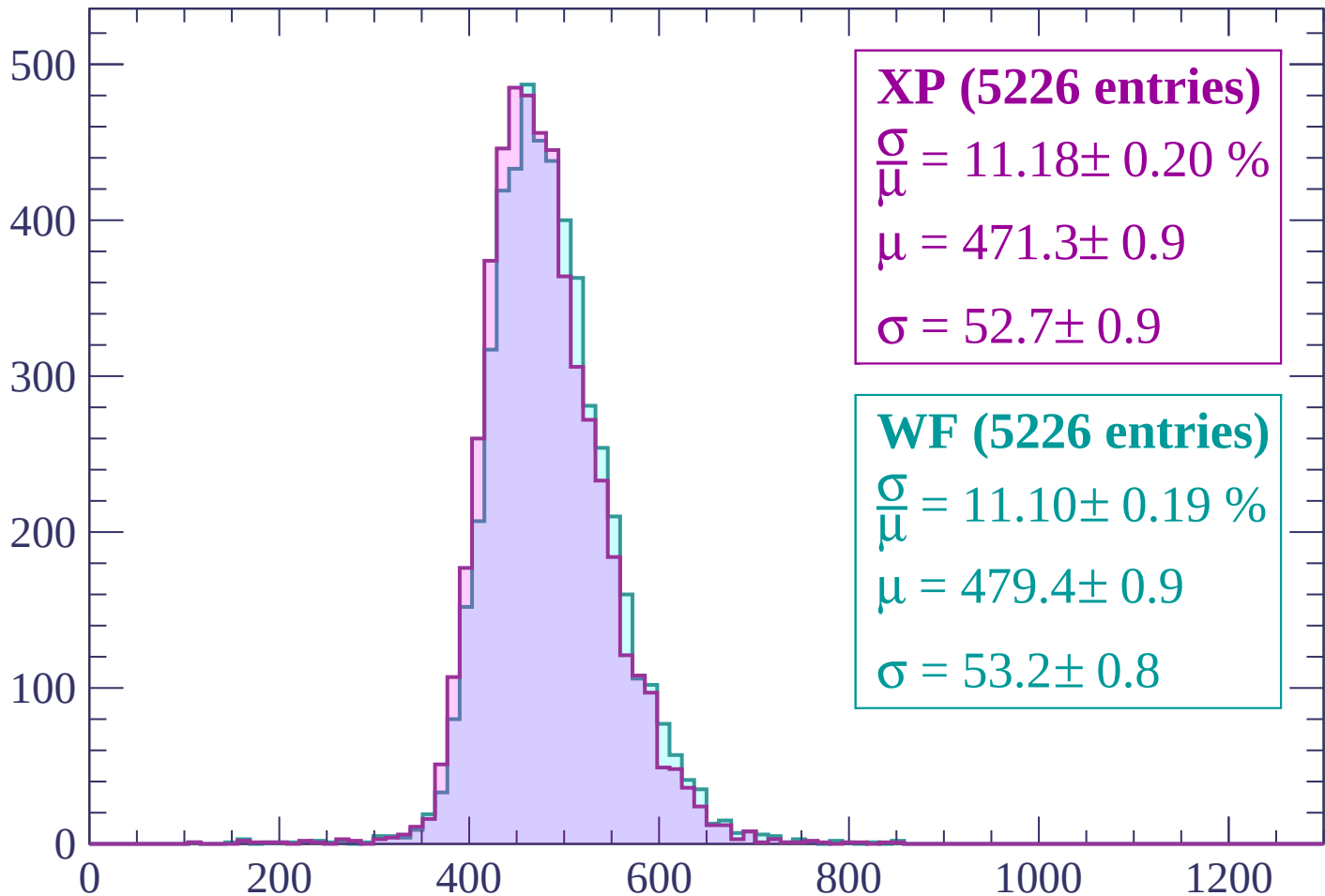
$\mu = 481.8 \pm 1.0$

$\sigma = 55.9 \pm 1.1$

dE/dx (ADC counts/cm)

# Energy loss | -1440 < p < -1380

Count



**XP (5226 entries)**

$$\frac{\sigma}{\mu} = 11.18 \pm 0.20 \%$$

$$\mu = 471.3 \pm 0.9$$

$$\sigma = 52.7 \pm 0.9$$

**WF (5226 entries)**

$$\frac{\sigma}{\mu} = 11.10 \pm 0.19 \%$$

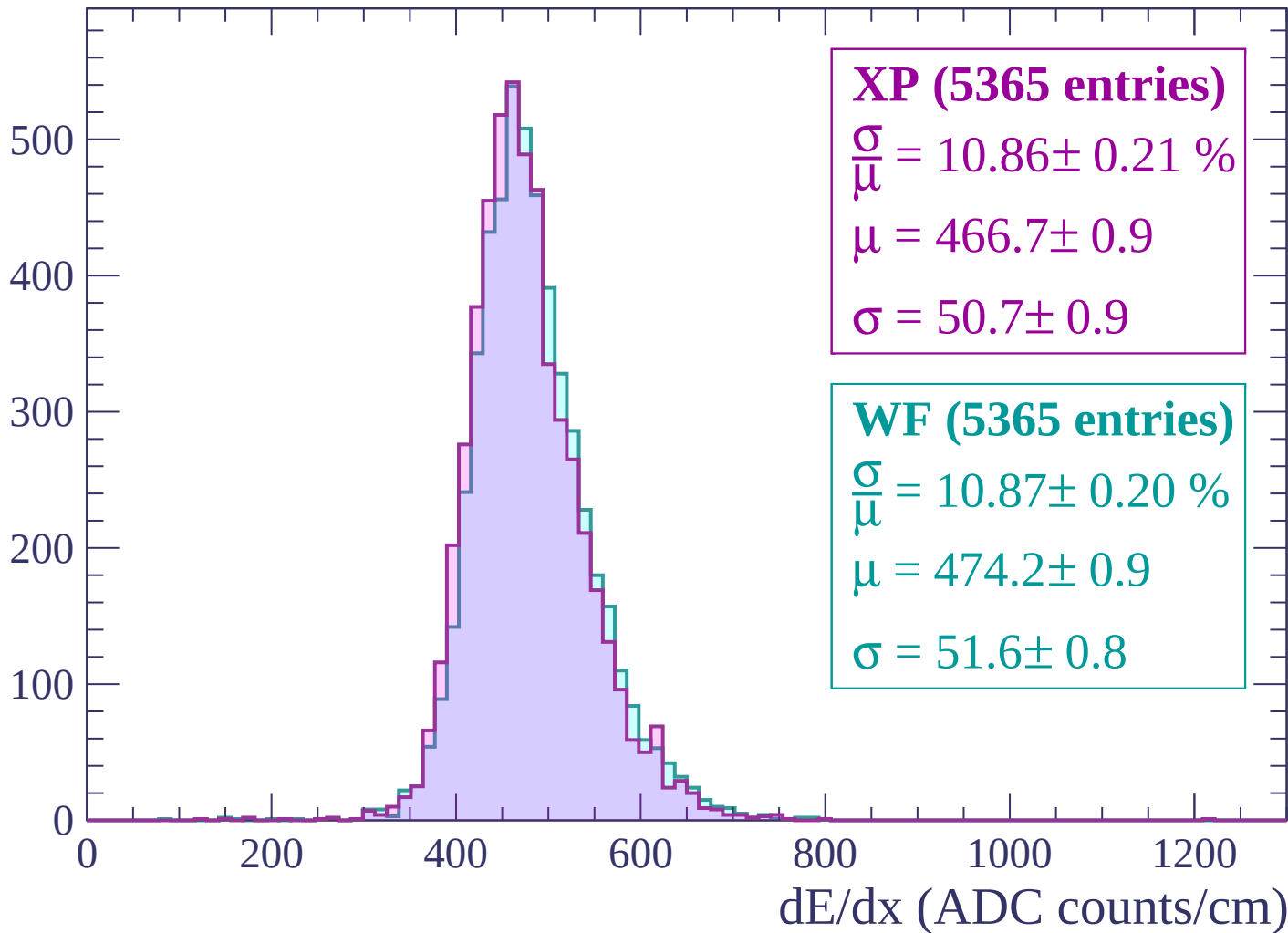
$$\mu = 479.4 \pm 0.9$$

$$\sigma = 53.2 \pm 0.8$$

dE/dx (ADC counts/cm)

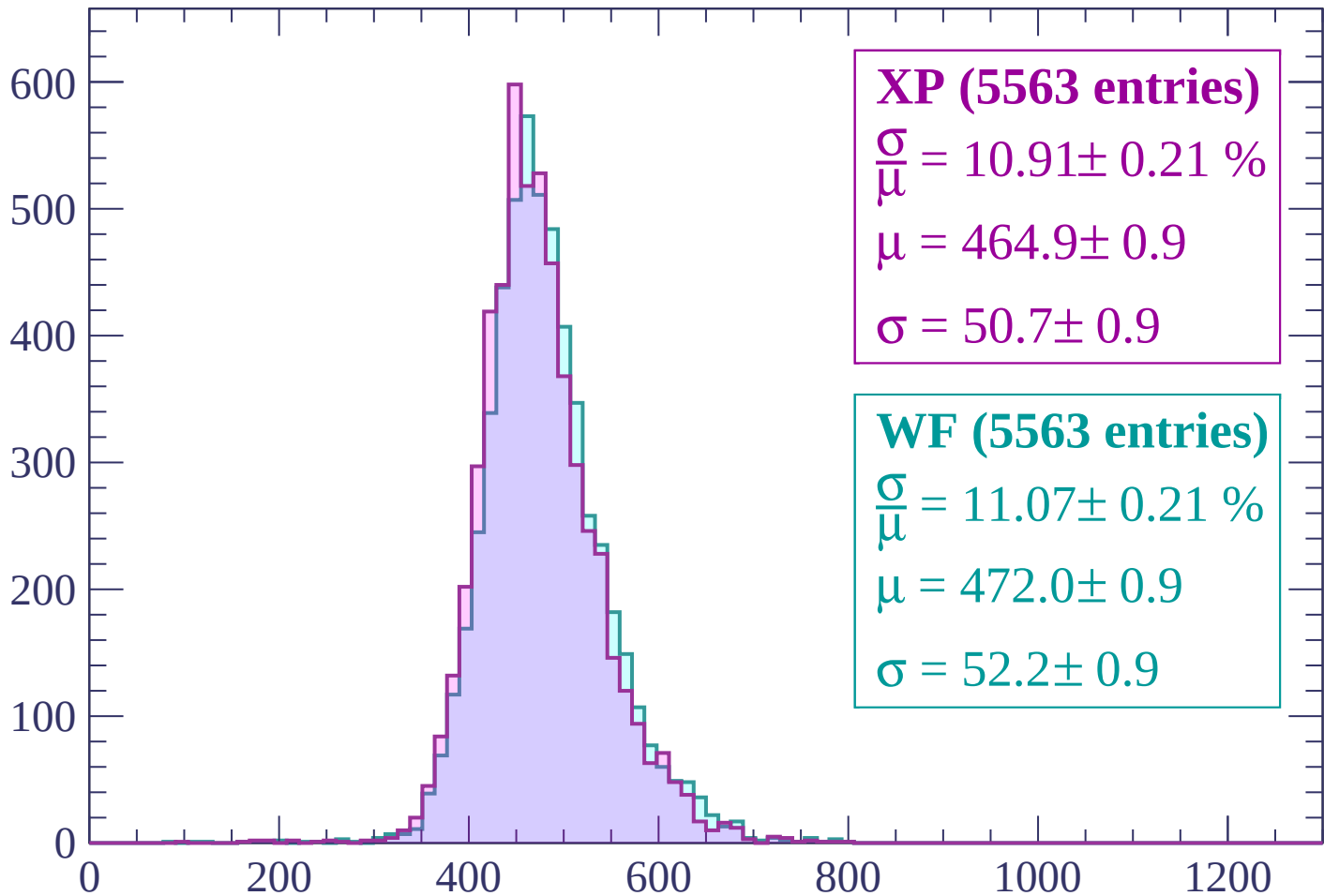
# Energy loss | $-1380 < p < -1320$

Count



Energy loss |  $-1320 < p < -1260$

Count



**XP (5563 entries)**

$\frac{\sigma}{\mu} = 10.91 \pm 0.21 \%$

$\mu = 464.9 \pm 0.9$

$\sigma = 50.7 \pm 0.9$

**WF (5563 entries)**

$\frac{\sigma}{\mu} = 11.07 \pm 0.21 \%$

$\mu = 472.0 \pm 0.9$

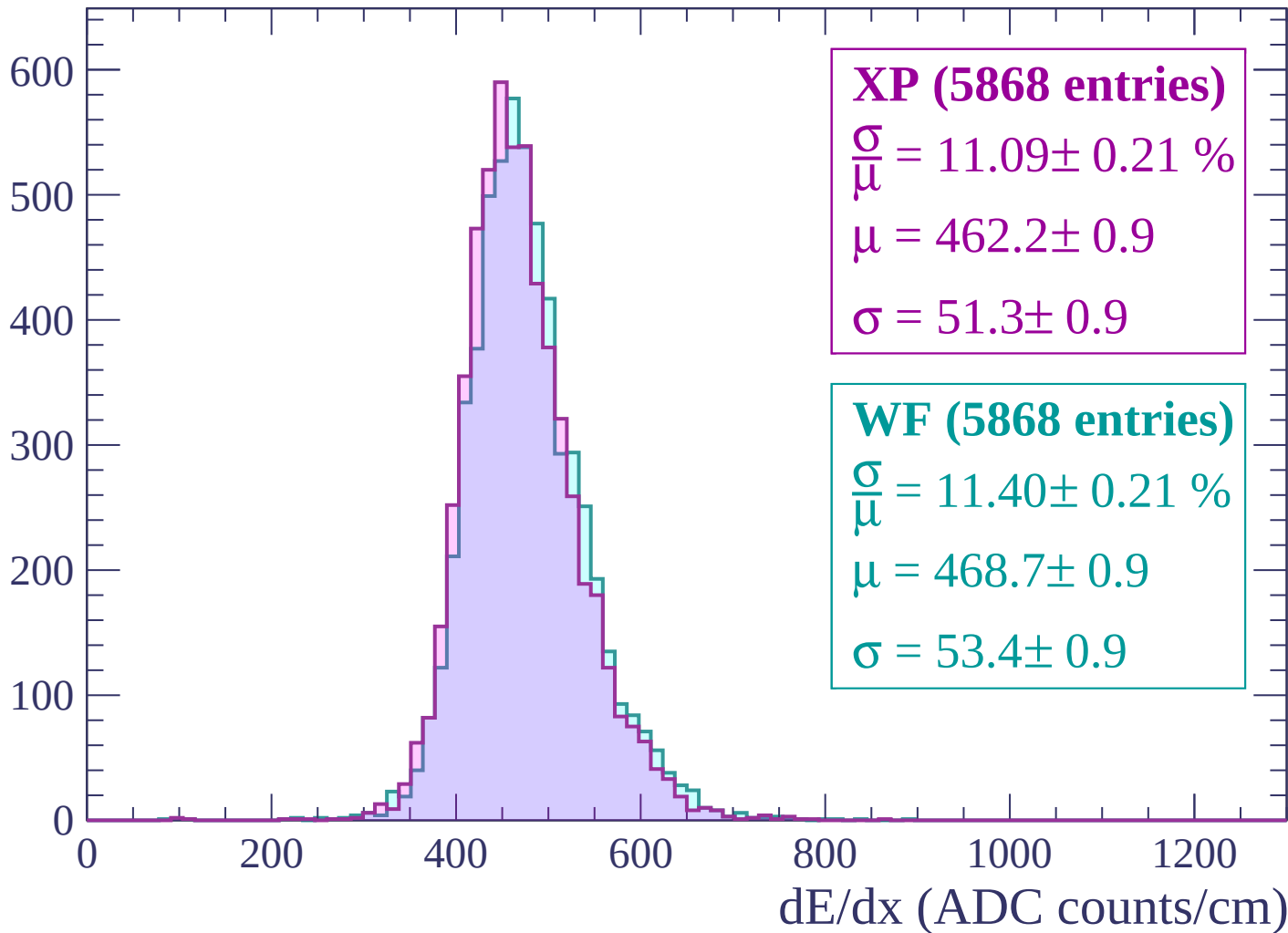
$\sigma = 52.2 \pm 0.9$

dE/dx (ADC counts/cm)



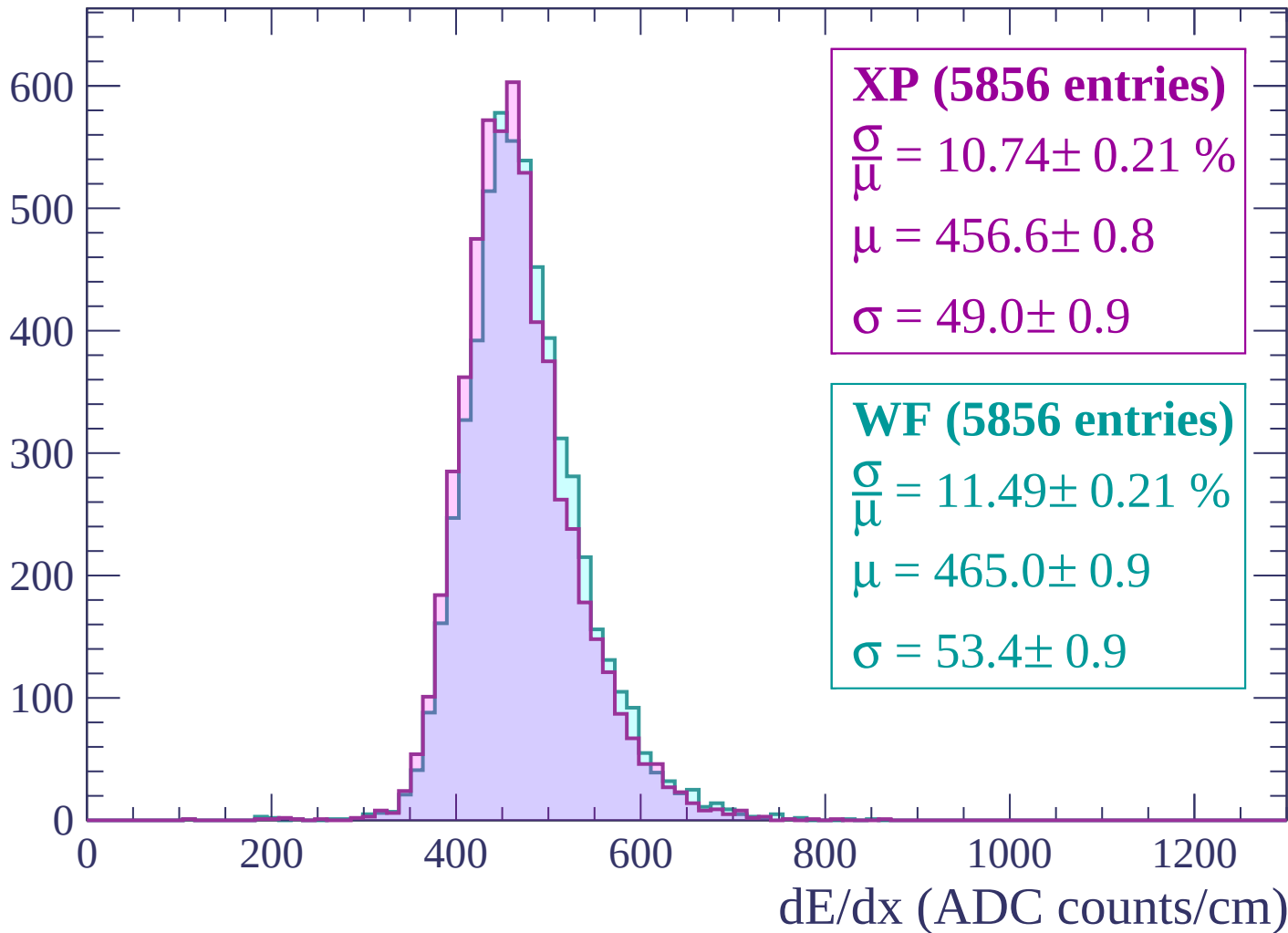
# Energy loss | $-1260 < p < -1200$

Count



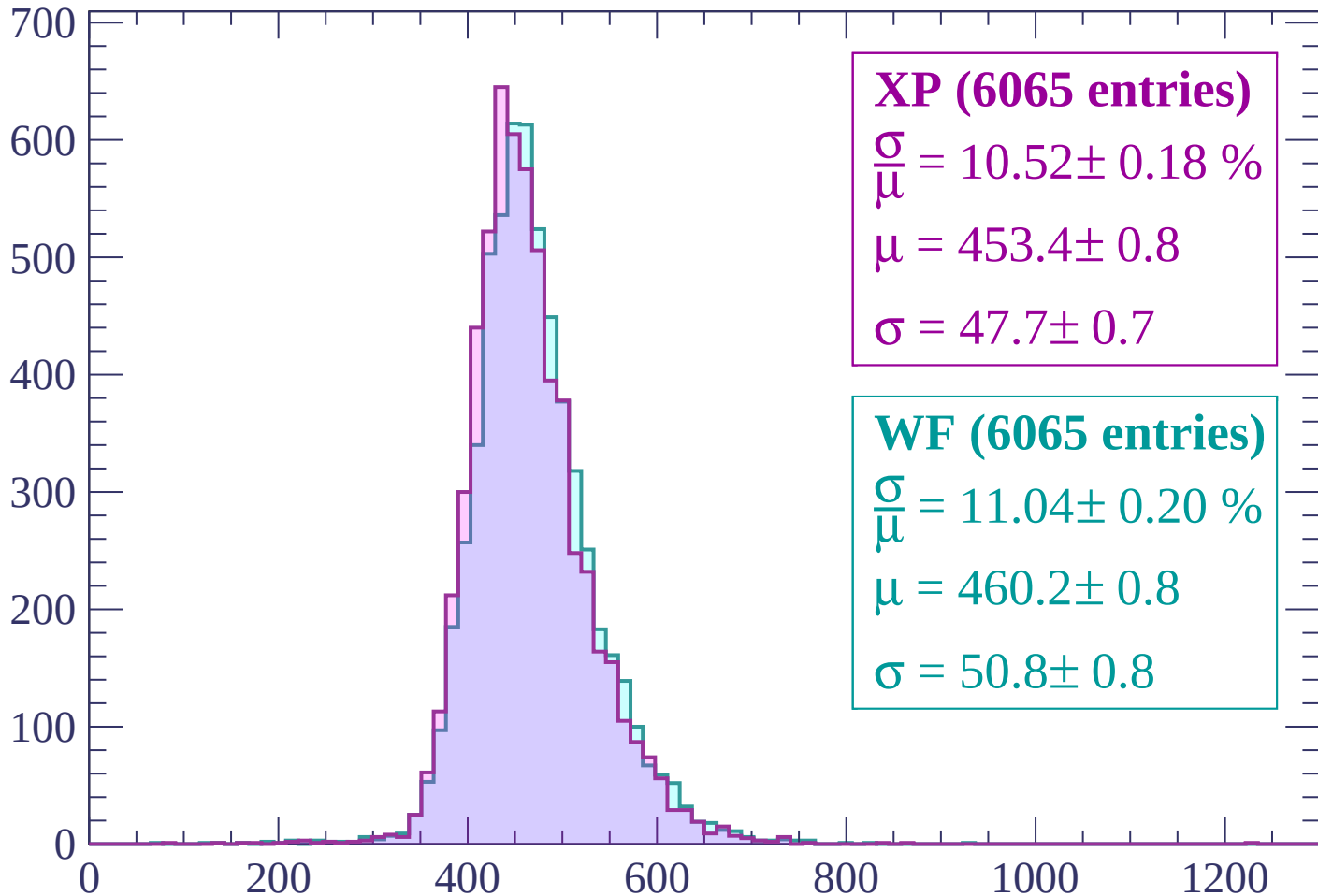
# Energy loss | -1200 < p < -1140

Count



Energy loss |  $-1140 < p < -1080$

Count



**XP (6065 entries)**

$$\frac{\sigma}{\mu} = 10.52 \pm 0.18 \%$$

$$\mu = 453.4 \pm 0.8$$

$$\sigma = 47.7 \pm 0.7$$

**WF (6065 entries)**

$$\frac{\sigma}{\mu} = 11.04 \pm 0.20 \%$$

$$\mu = 460.2 \pm 0.8$$

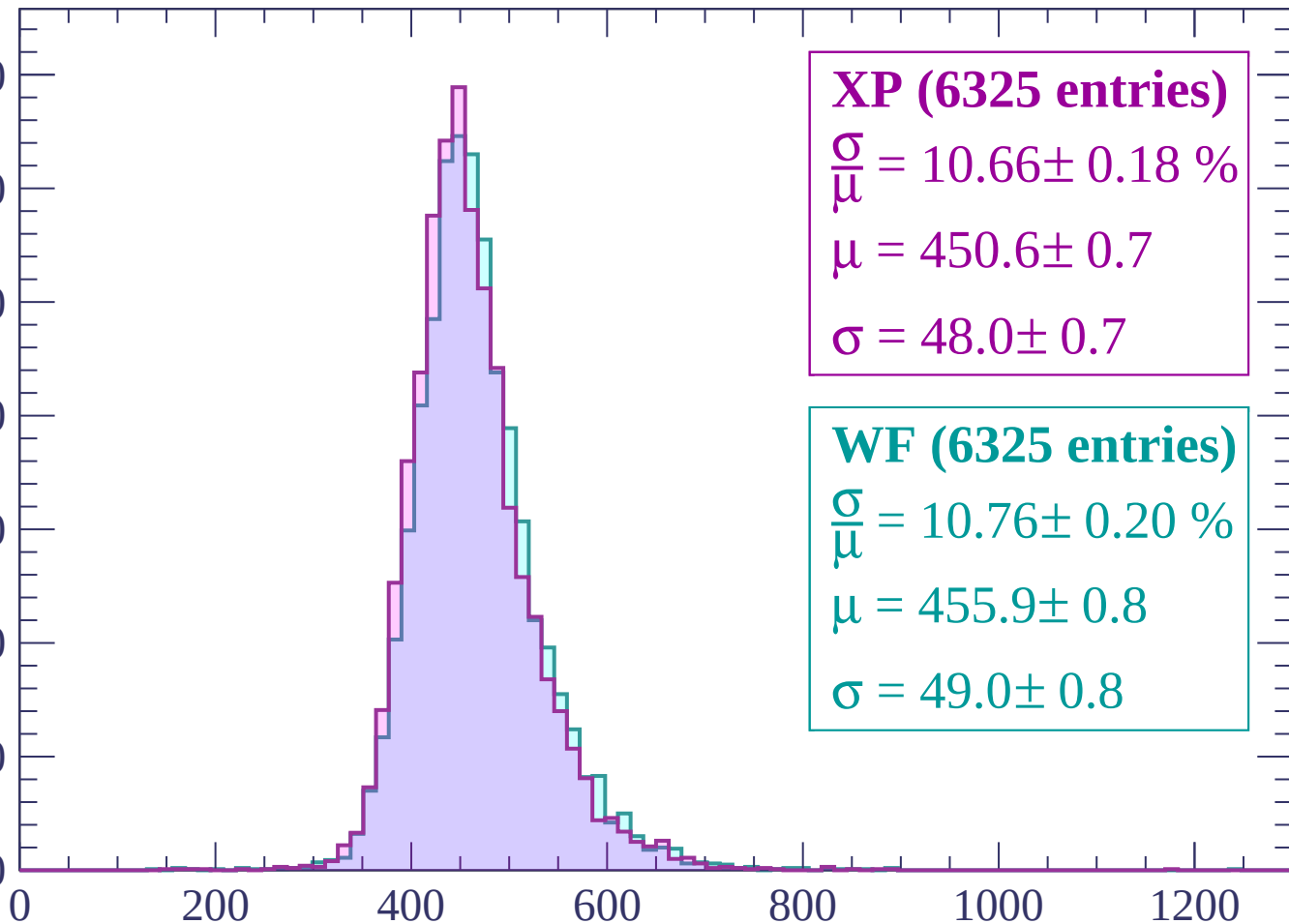
$$\sigma = 50.8 \pm 0.8$$

dE/dx (ADC counts/cm)

# Energy loss | -1080 < p < -1020

Count

700  
600  
500  
400  
300  
200  
100  
0



**XP (6325 entries)**

$\frac{\sigma}{\mu} = 10.66 \pm 0.18 \%$

$\mu = 450.6 \pm 0.7$

$\sigma = 48.0 \pm 0.7$

**WF (6325 entries)**

$\frac{\sigma}{\mu} = 10.76 \pm 0.20 \%$

$\mu = 455.9 \pm 0.8$

$\sigma = 49.0 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss |  $-1020 < p < -960$

Count

700  
600  
500  
400  
300  
200  
100  
0

**XP (6513 entries)**

$$\frac{\sigma}{\mu} = 10.17 \pm 0.17 \%$$

$$\mu = 445.0 \pm 0.7$$

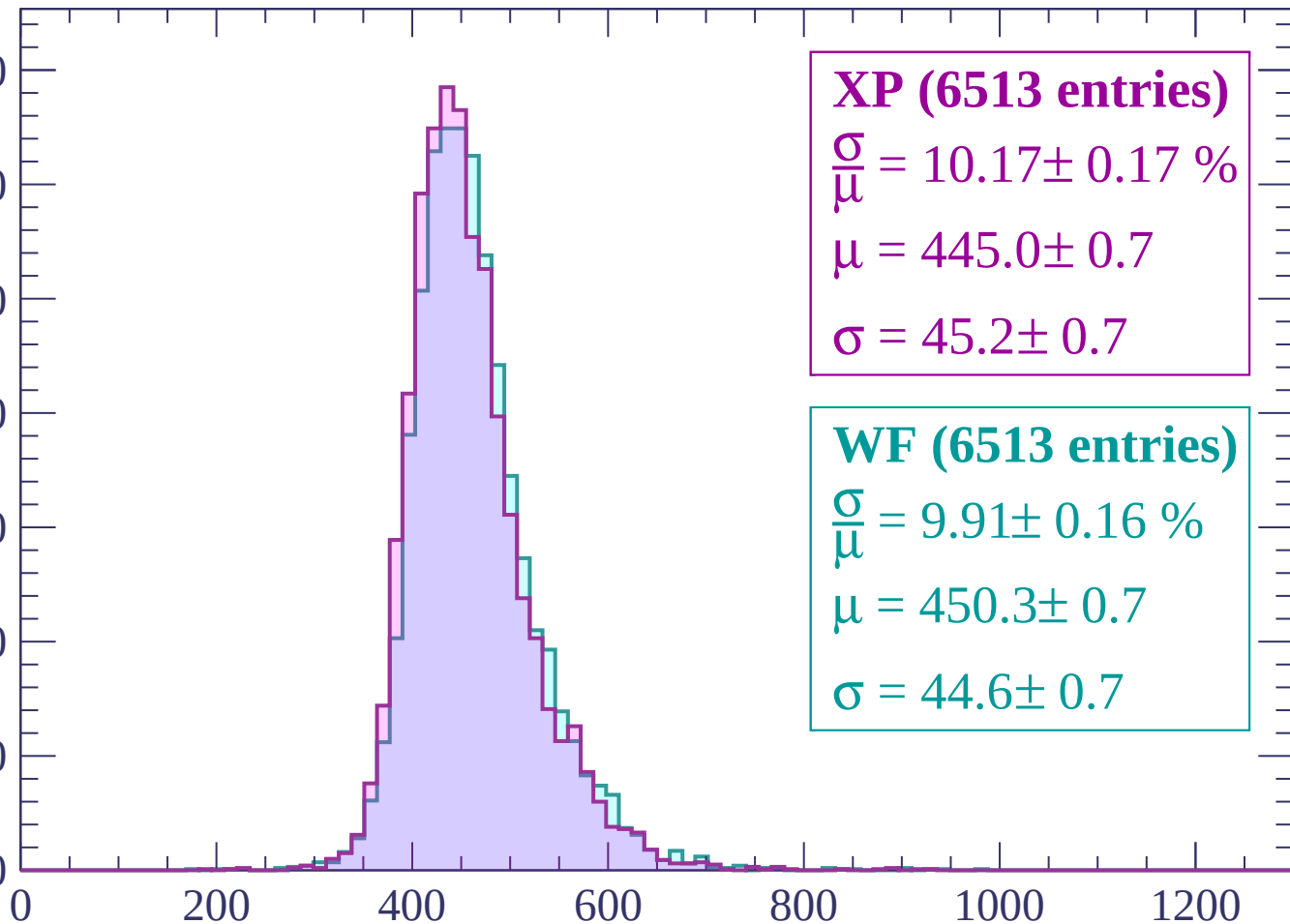
$$\sigma = 45.2 \pm 0.7$$

**WF (6513 entries)**

$$\frac{\sigma}{\mu} = 9.91 \pm 0.16 \%$$

$$\mu = 450.3 \pm 0.7$$

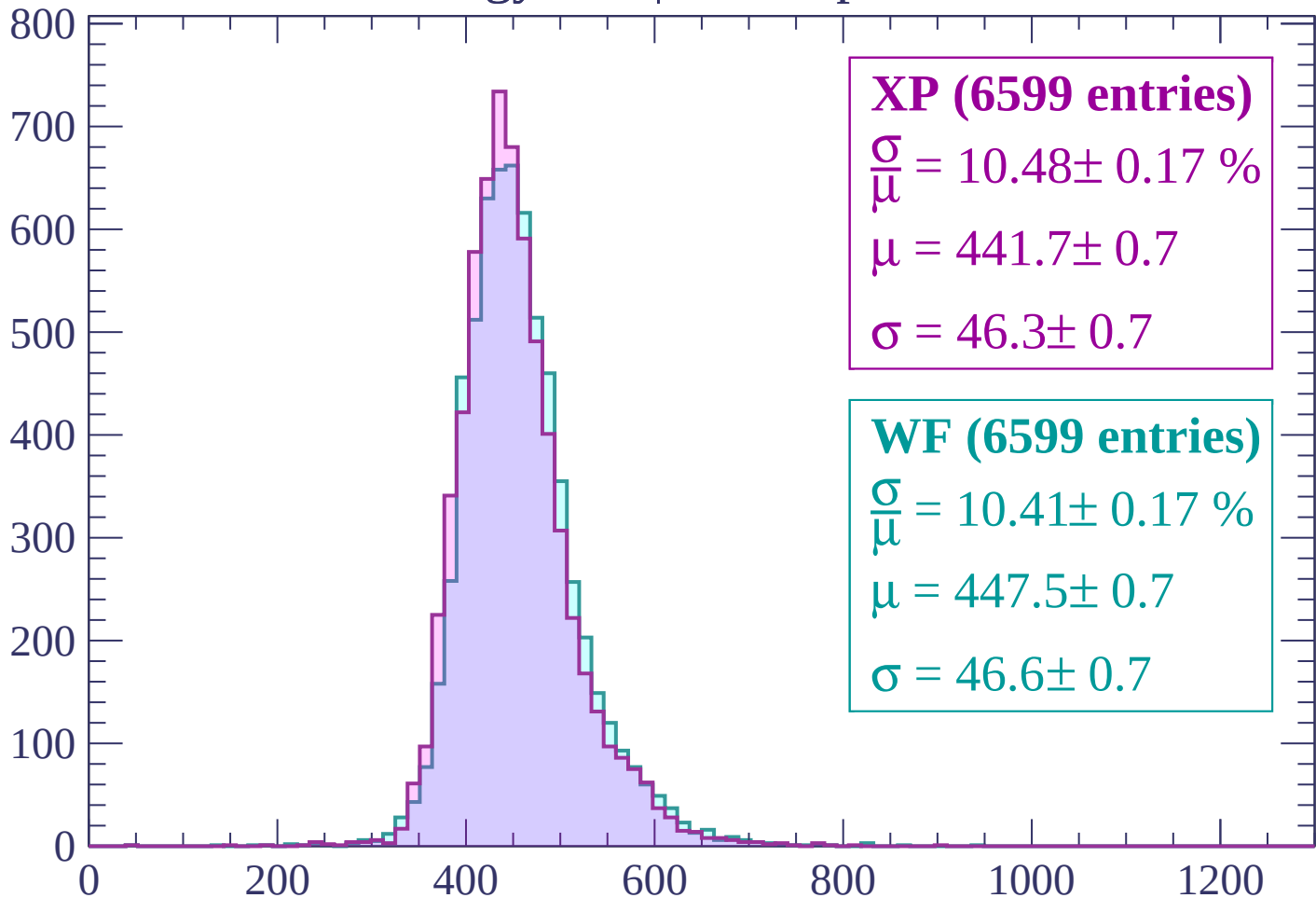
$$\sigma = 44.6 \pm 0.7$$



dE/dx (ADC counts/cm)

# Energy loss | -960 < p < -900

Count



**XP (6599 entries)**

$$\frac{\sigma}{\mu} = 10.48 \pm 0.17 \%$$

$$\mu = 441.7 \pm 0.7$$

$$\sigma = 46.3 \pm 0.7$$

**WF (6599 entries)**

$$\frac{\sigma}{\mu} = 10.41 \pm 0.17 \%$$

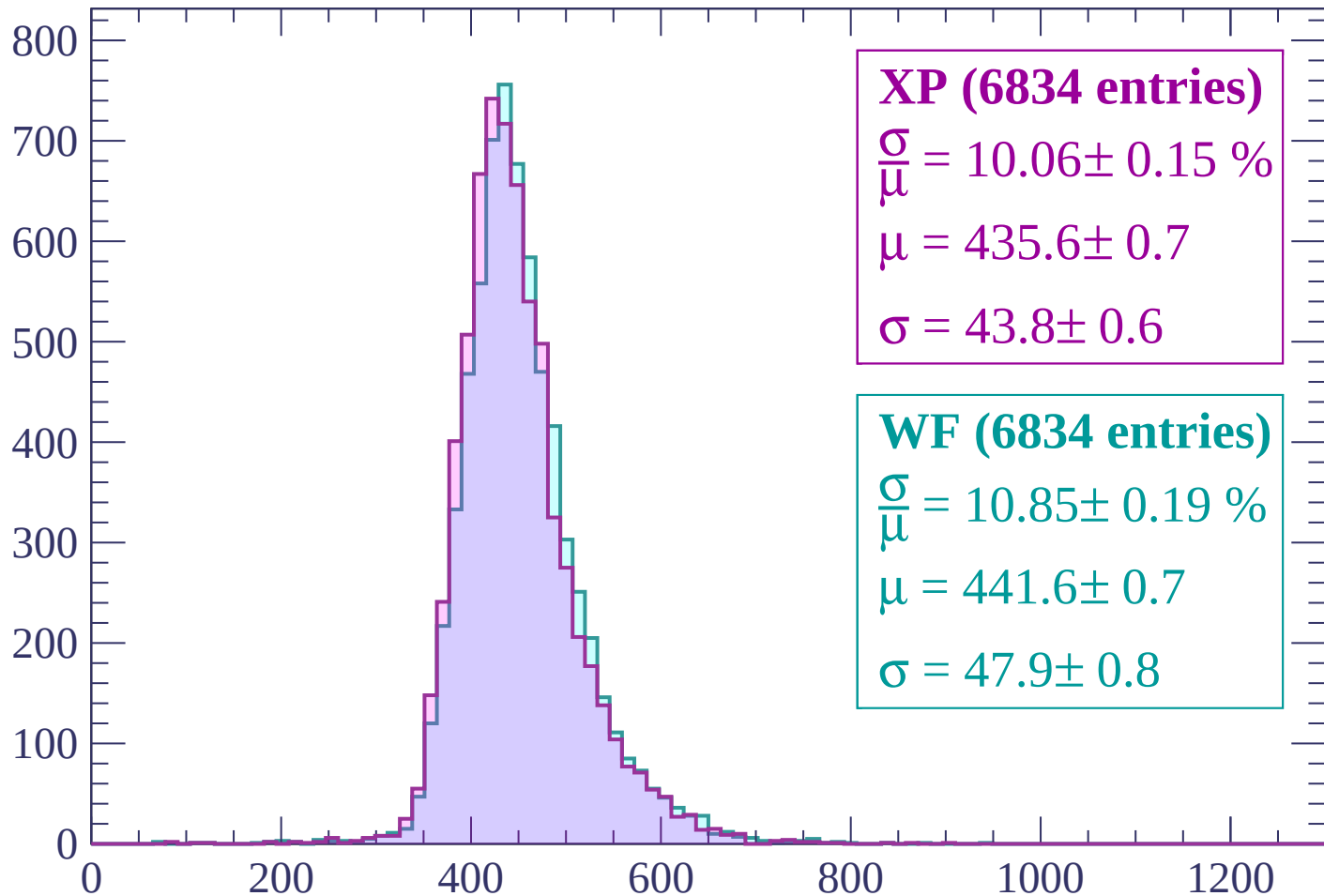
$$\mu = 447.5 \pm 0.7$$

$$\sigma = 46.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-900 < p < -840$

Count



**XP (6834 entries)**

$$\frac{\sigma}{\mu} = 10.06 \pm 0.15 \%$$

$$\mu = 435.6 \pm 0.7$$

$$\sigma = 43.8 \pm 0.6$$

**WF (6834 entries)**

$$\frac{\sigma}{\mu} = 10.85 \pm 0.19 \%$$

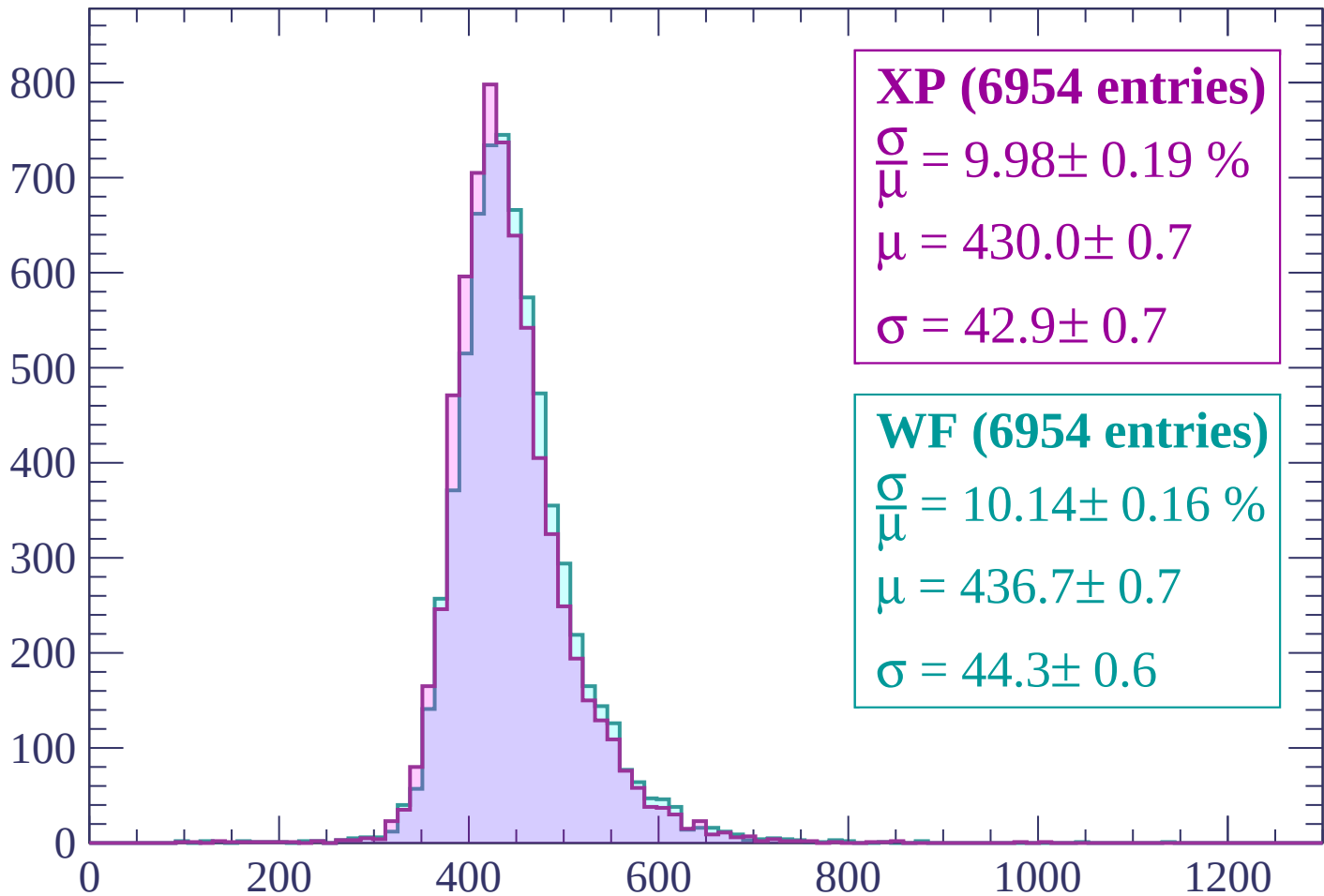
$$\mu = 441.6 \pm 0.7$$

$$\sigma = 47.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-840 < p < -780$

Count



**XP (6954 entries)**

$$\frac{\sigma}{\mu} = 9.98 \pm 0.19 \%$$

$$\mu = 430.0 \pm 0.7$$

$$\sigma = 42.9 \pm 0.7$$

**WF (6954 entries)**

$$\frac{\sigma}{\mu} = 10.14 \pm 0.16 \%$$

$$\mu = 436.7 \pm 0.7$$

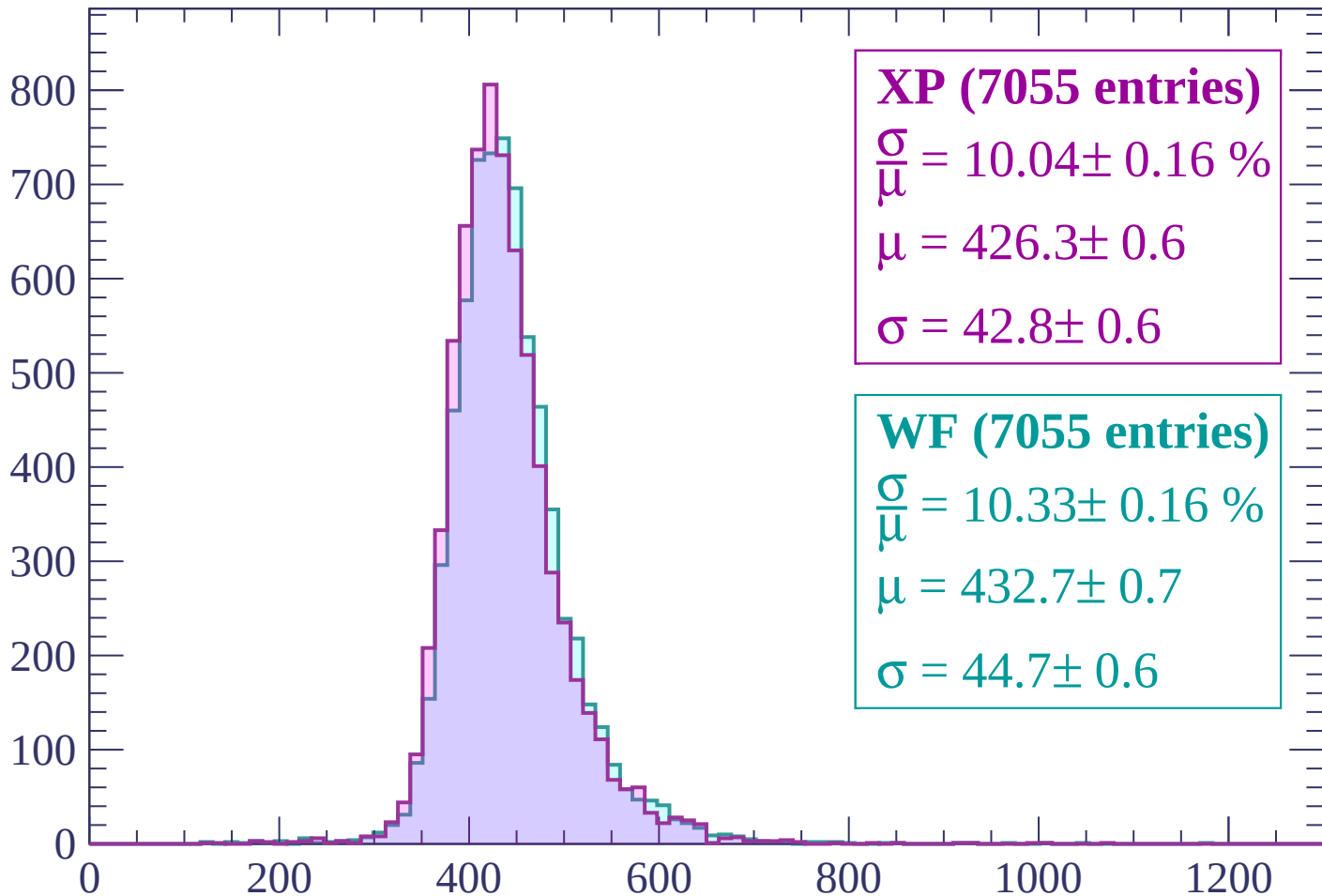
$$\sigma = 44.3 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $-780 < p < -720$

Count



**XP (7055 entries)**

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

$$\mu = 426.3 \pm 0.6$$

$$\sigma = 42.8 \pm 0.6$$

**WF (7055 entries)**

$$\frac{\sigma}{\mu} = 10.33 \pm 0.16 \%$$

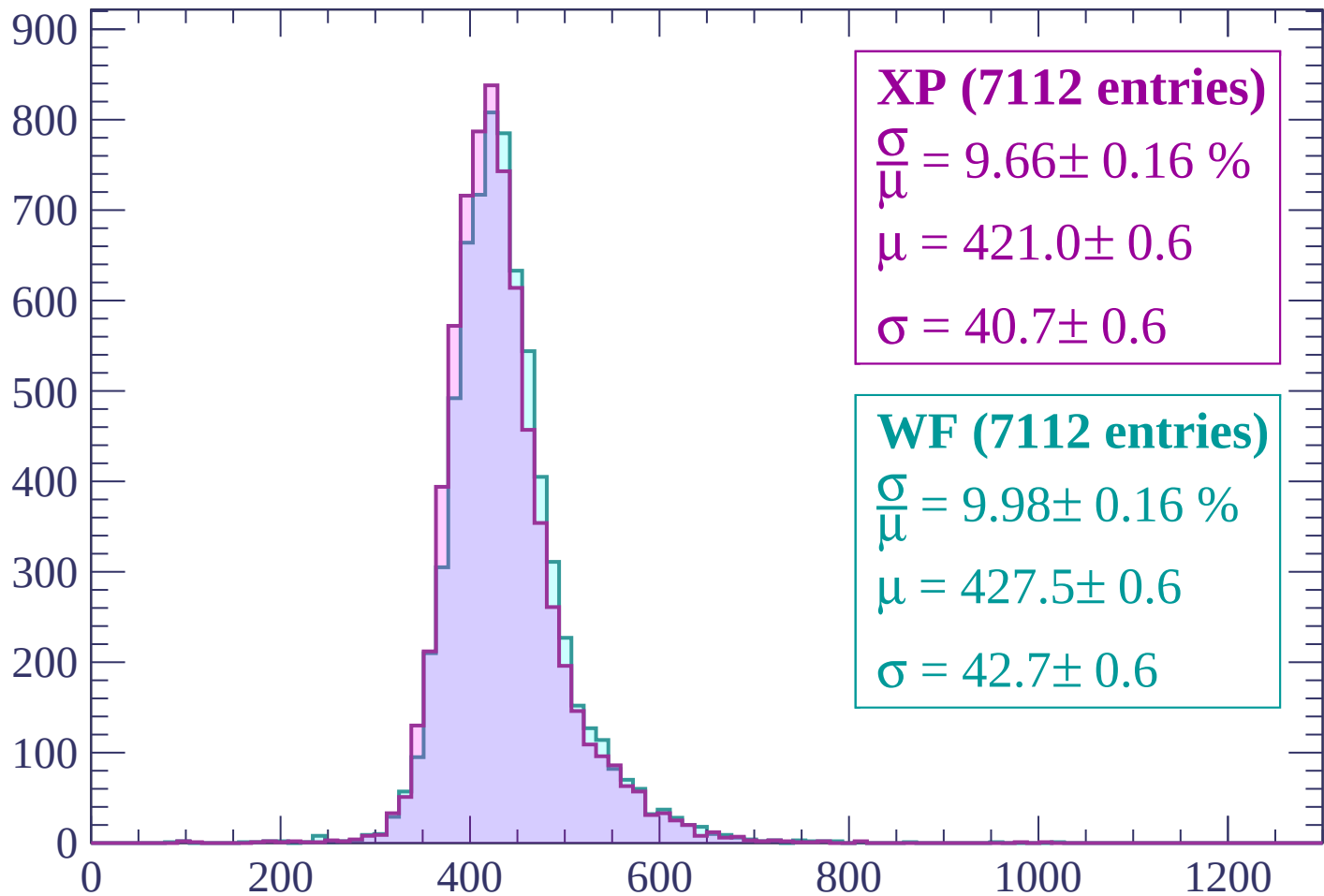
$$\mu = 432.7 \pm 0.7$$

$$\sigma = 44.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-720 < p < -660$

Count



**XP (7112 entries)**

$\frac{\sigma}{\mu} = 9.66 \pm 0.16 \%$

$\mu = 421.0 \pm 0.6$

$\sigma = 40.7 \pm 0.6$

**WF (7112 entries)**

$\frac{\sigma}{\mu} = 9.98 \pm 0.16 \%$

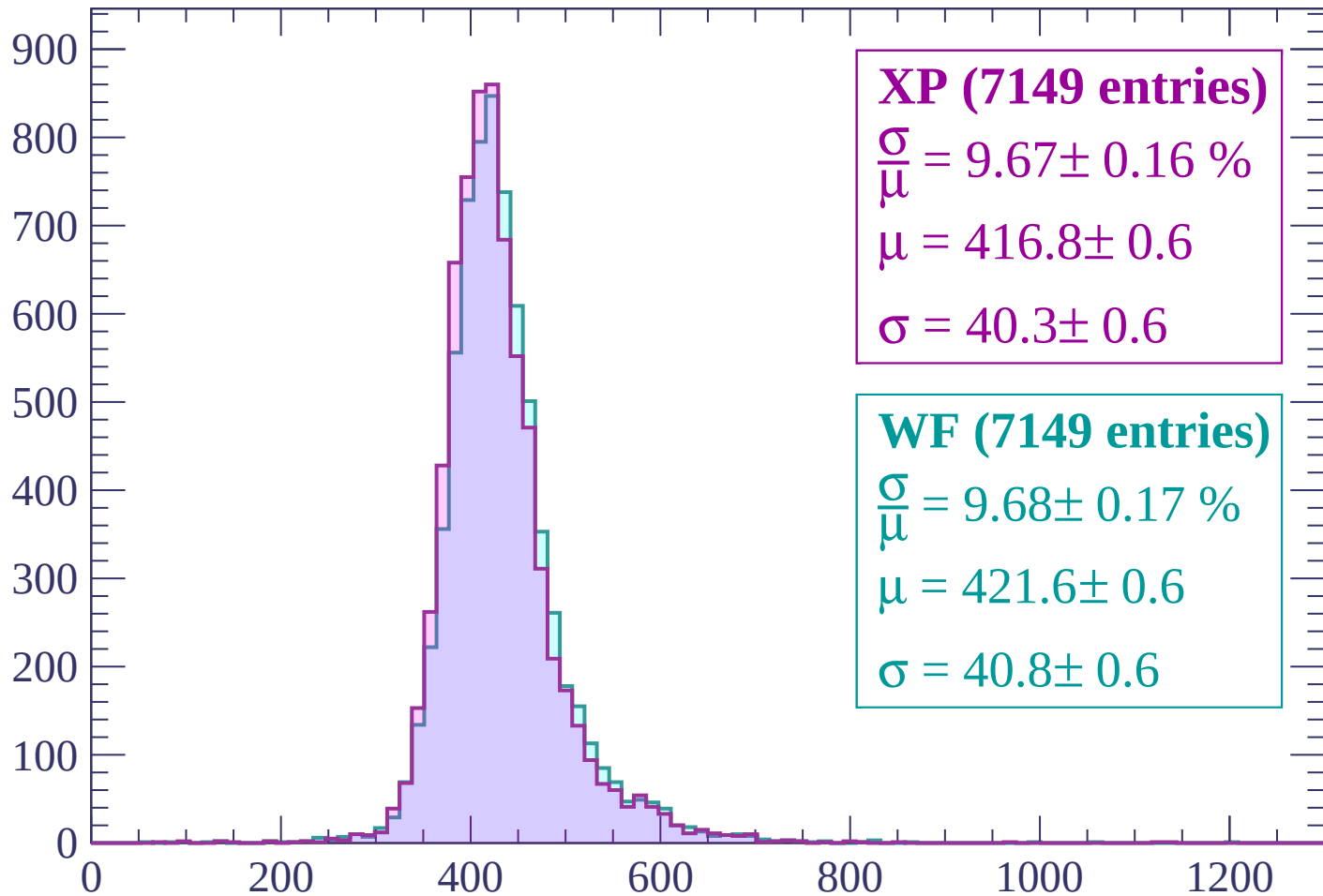
$\mu = 427.5 \pm 0.6$

$\sigma = 42.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $-660 < p < -600$

Count



**XP (7149 entries)**

$$\frac{\sigma}{\mu} = 9.67 \pm 0.16 \%$$

$$\mu = 416.8 \pm 0.6$$

$$\sigma = 40.3 \pm 0.6$$

**WF (7149 entries)**

$$\frac{\sigma}{\mu} = 9.68 \pm 0.17 \%$$

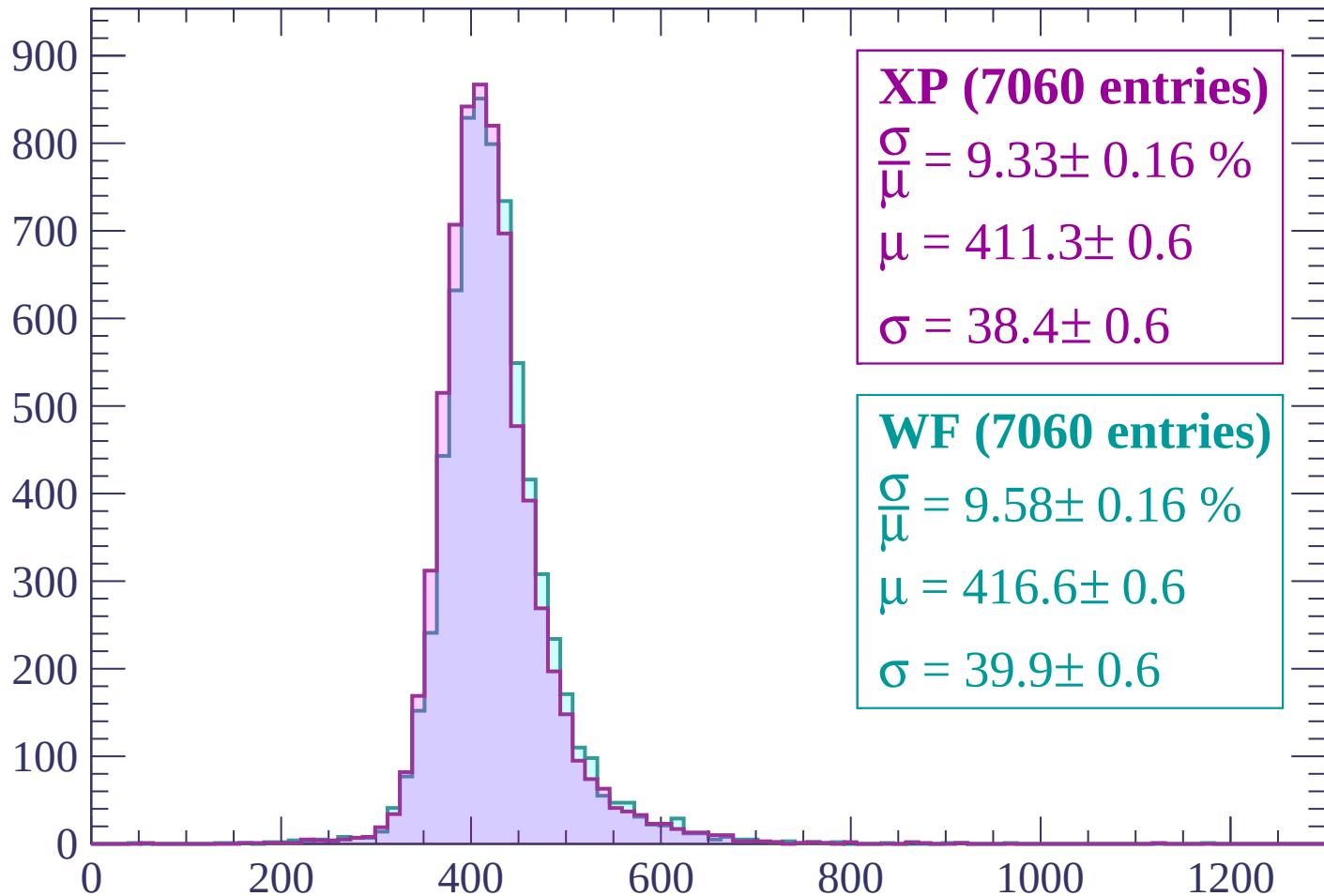
$$\mu = 421.6 \pm 0.6$$

$$\sigma = 40.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-600 < p < -540$

Count



**XP (7060 entries)**

$$\frac{\sigma}{\mu} = 9.33 \pm 0.16 \%$$

$$\mu = 411.3 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

**WF (7060 entries)**

$$\frac{\sigma}{\mu} = 9.58 \pm 0.16 \%$$

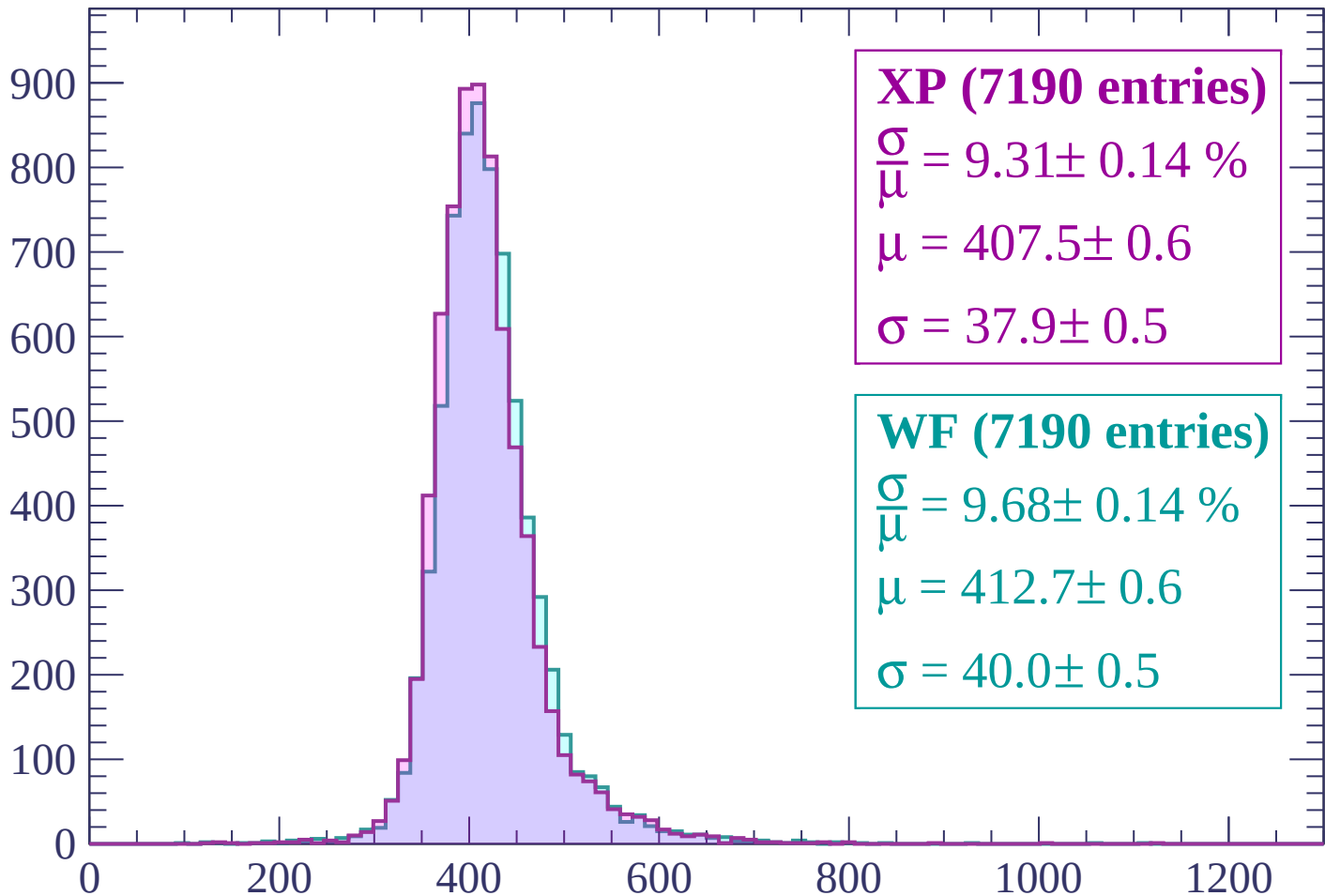
$$\mu = 416.6 \pm 0.6$$

$$\sigma = 39.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-540 < p < -480$

Count



**XP (7190 entries)**

$$\frac{\sigma}{\mu} = 9.31 \pm 0.14 \%$$

$$\mu = 407.5 \pm 0.6$$

$$\sigma = 37.9 \pm 0.5$$

**WF (7190 entries)**

$$\frac{\sigma}{\mu} = 9.68 \pm 0.14 \%$$

$$\mu = 412.7 \pm 0.6$$

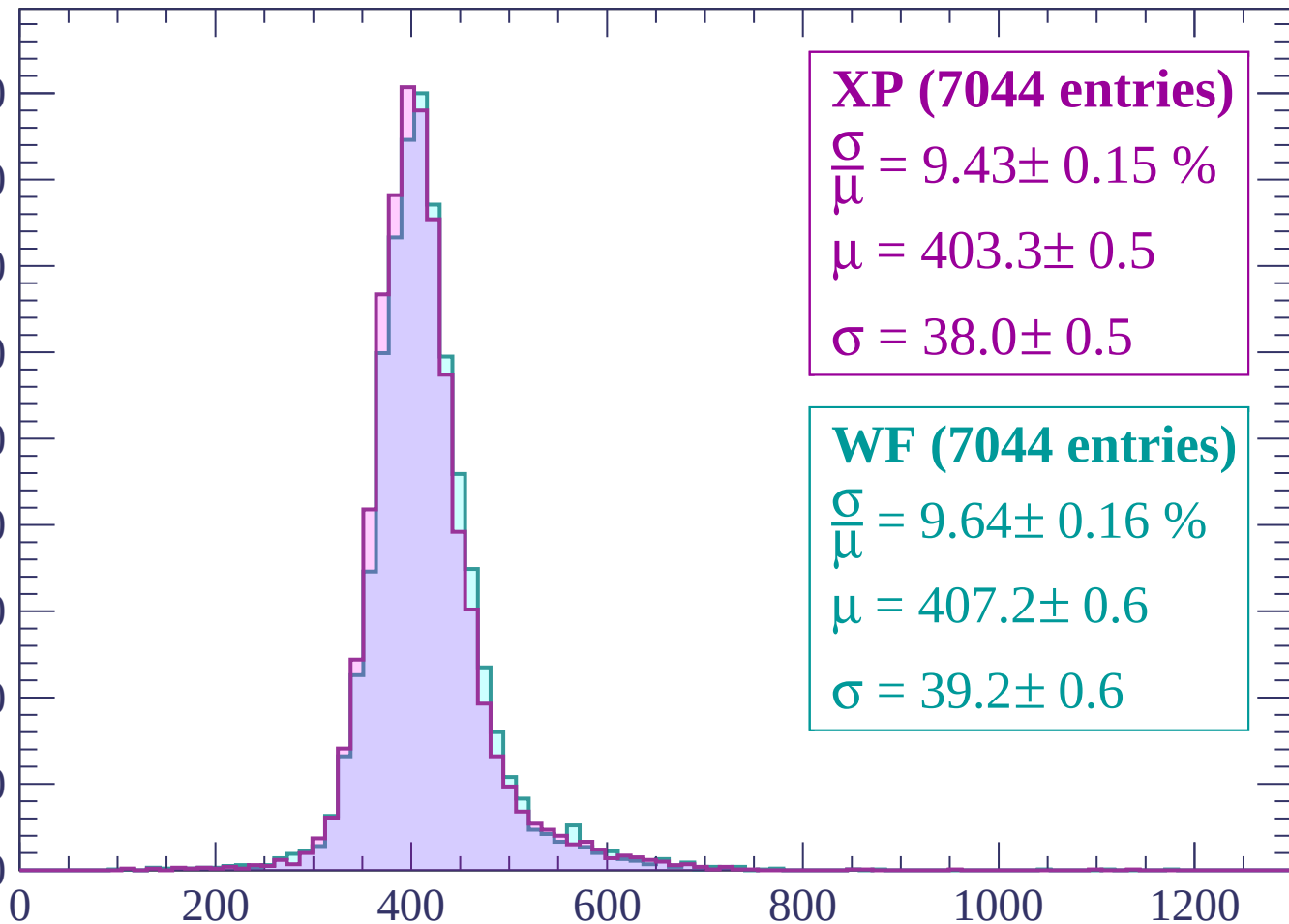
$$\sigma = 40.0 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss |  $-480 < p < -420$

Count

900  
800  
700  
600  
500  
400  
300  
200  
100  
0



**XP (7044 entries)**

$\frac{\sigma}{\mu} = 9.43 \pm 0.15 \%$

$\mu = 403.3 \pm 0.5$

$\sigma = 38.0 \pm 0.5$

**WF (7044 entries)**

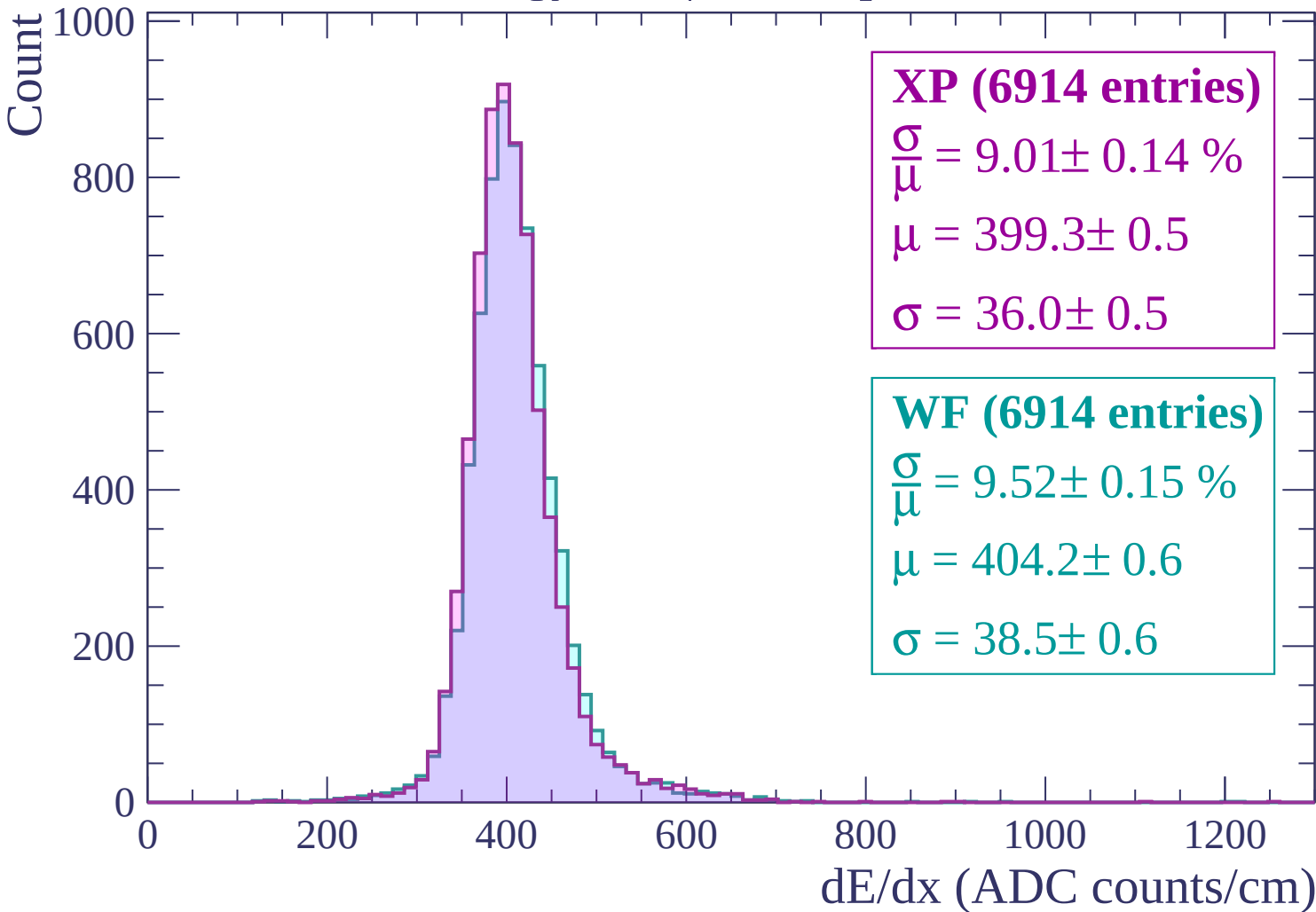
$\frac{\sigma}{\mu} = 9.64 \pm 0.16 \%$

$\mu = 407.2 \pm 0.6$

$\sigma = 39.2 \pm 0.6$

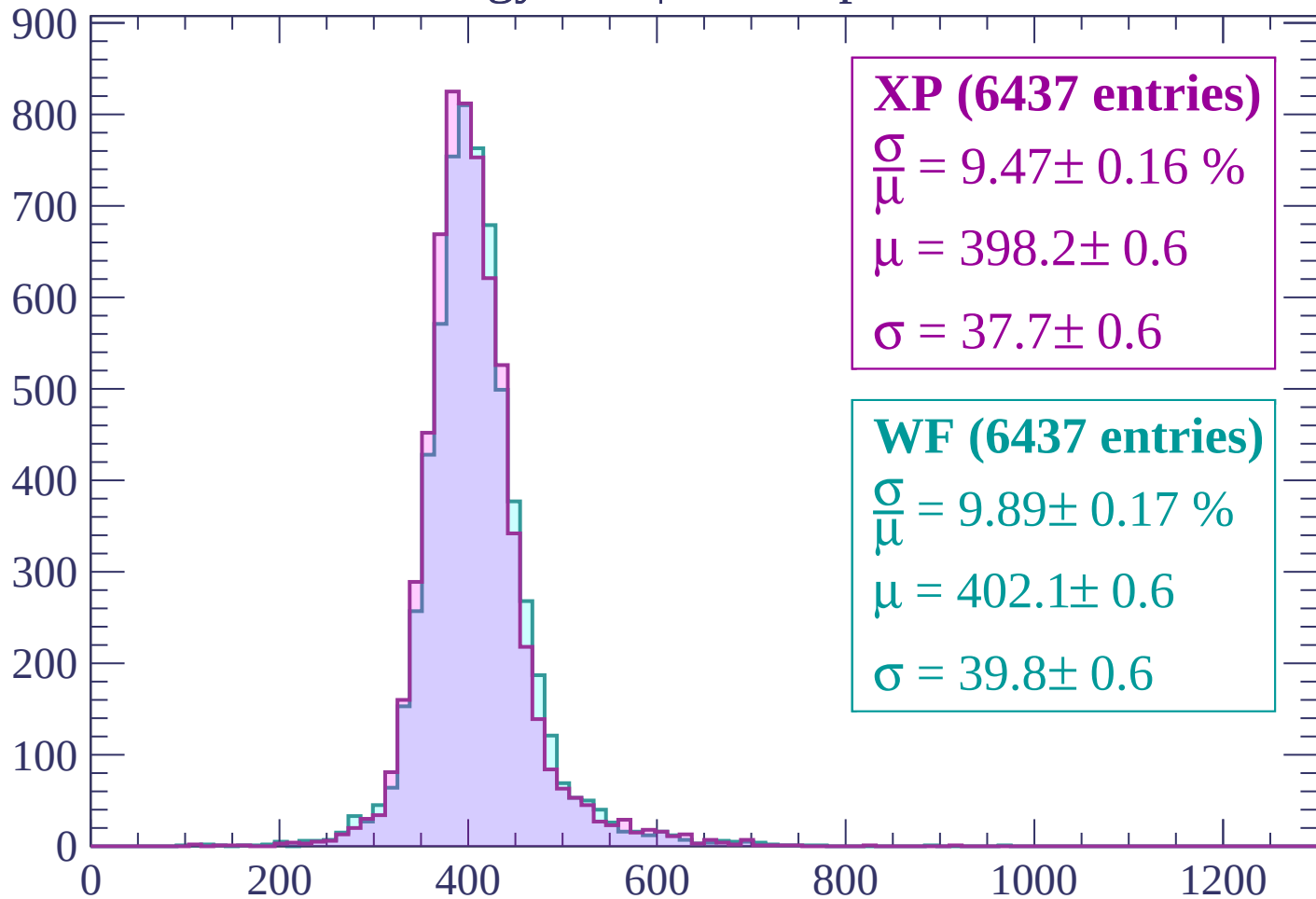
dE/dx (ADC counts/cm)

Energy loss |  $-420 < p < -360$



Energy loss |  $-360 < p < -300$

Count



**XP (6437 entries)**

$$\frac{\sigma}{\mu} = 9.47 \pm 0.16 \%$$

$$\mu = 398.2 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

**WF (6437 entries)**

$$\frac{\sigma}{\mu} = 9.89 \pm 0.17 \%$$

$$\mu = 402.1 \pm 0.6$$

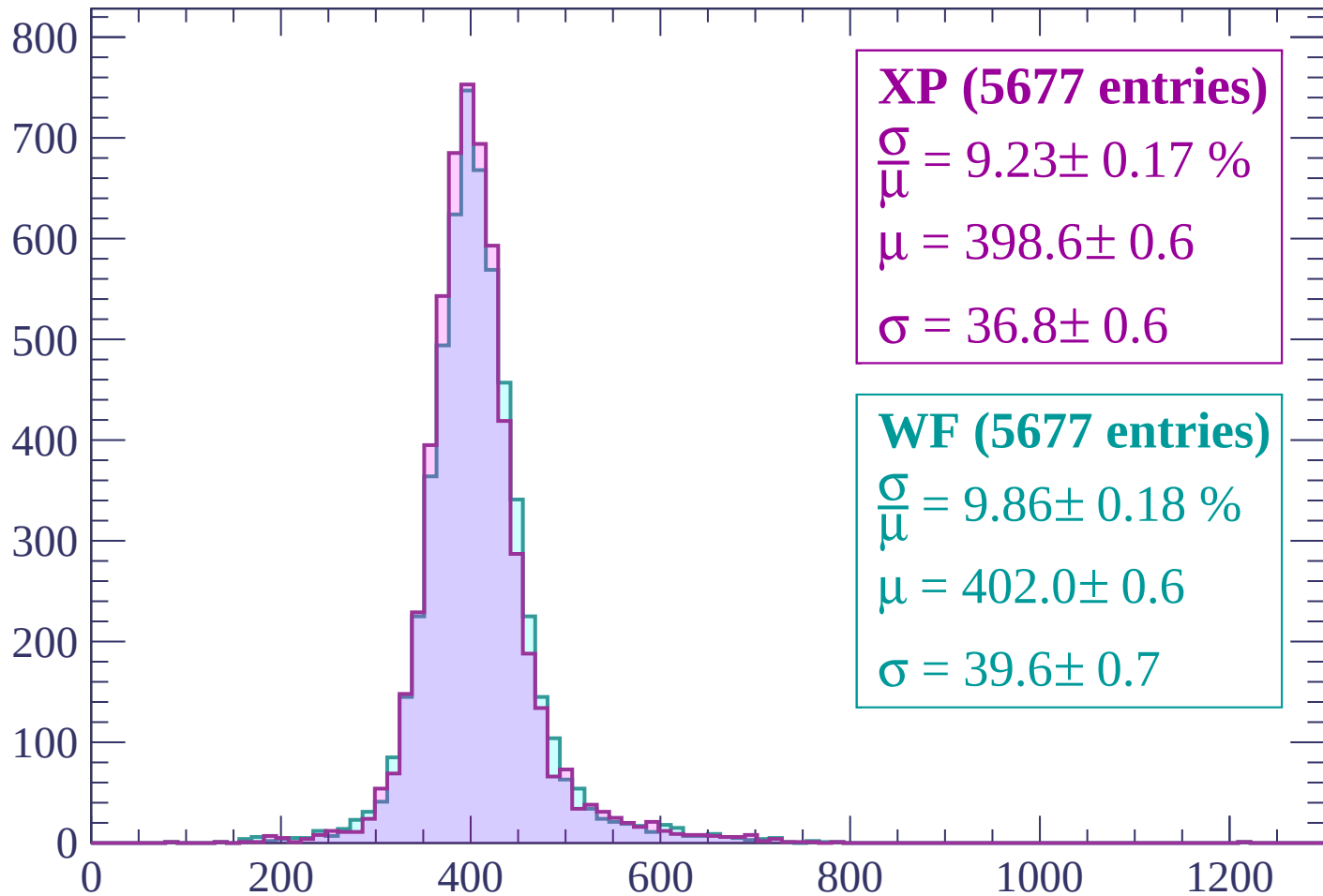
$$\sigma = 39.8 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $-300 < p < -240$

Count



**XP (5677 entries)**

$$\frac{\sigma}{\mu} = 9.23 \pm 0.17 \%$$

$$\mu = 398.6 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

**WF (5677 entries)**

$$\frac{\sigma}{\mu} = 9.86 \pm 0.18 \%$$

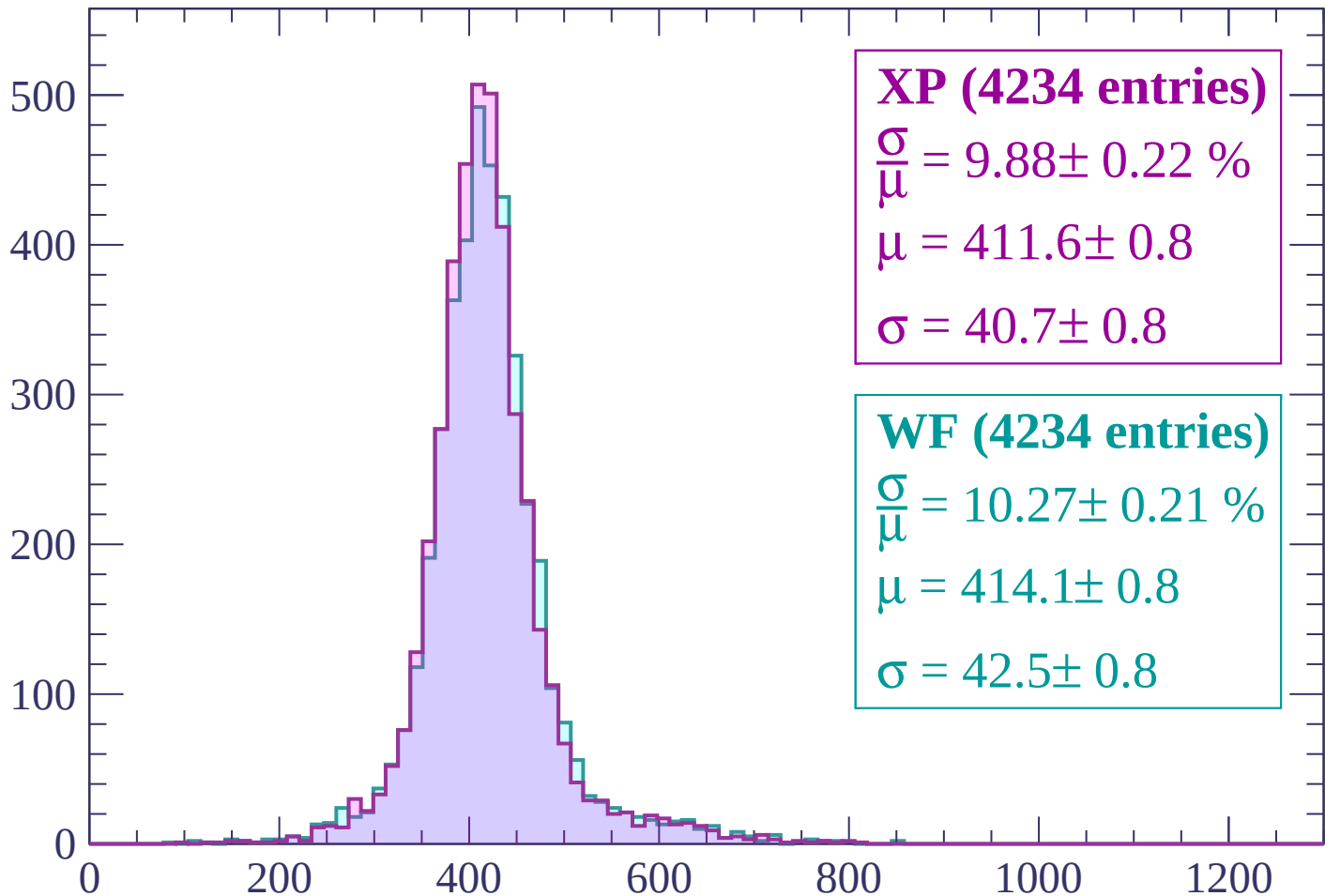
$$\mu = 402.0 \pm 0.6$$

$$\sigma = 39.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-240 < p < -180$

Count



**XP (4234 entries)**

$$\frac{\sigma}{\mu} = 9.88 \pm 0.22 \%$$

$$\mu = 411.6 \pm 0.8$$

$$\sigma = 40.7 \pm 0.8$$

**WF (4234 entries)**

$$\frac{\sigma}{\mu} = 10.27 \pm 0.21 \%$$

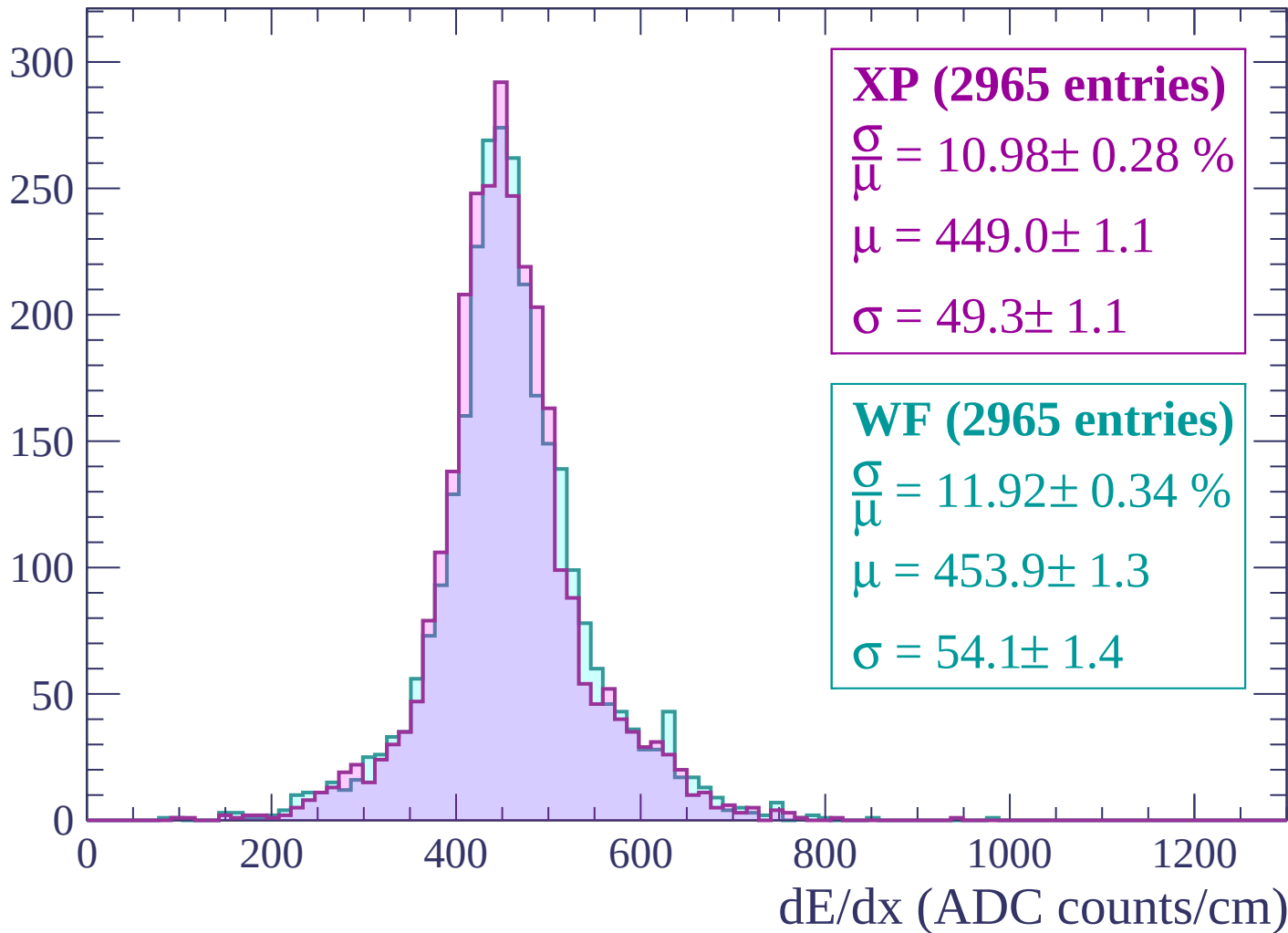
$$\mu = 414.1 \pm 0.8$$

$$\sigma = 42.5 \pm 0.8$$

dE/dx (ADC counts/cm)

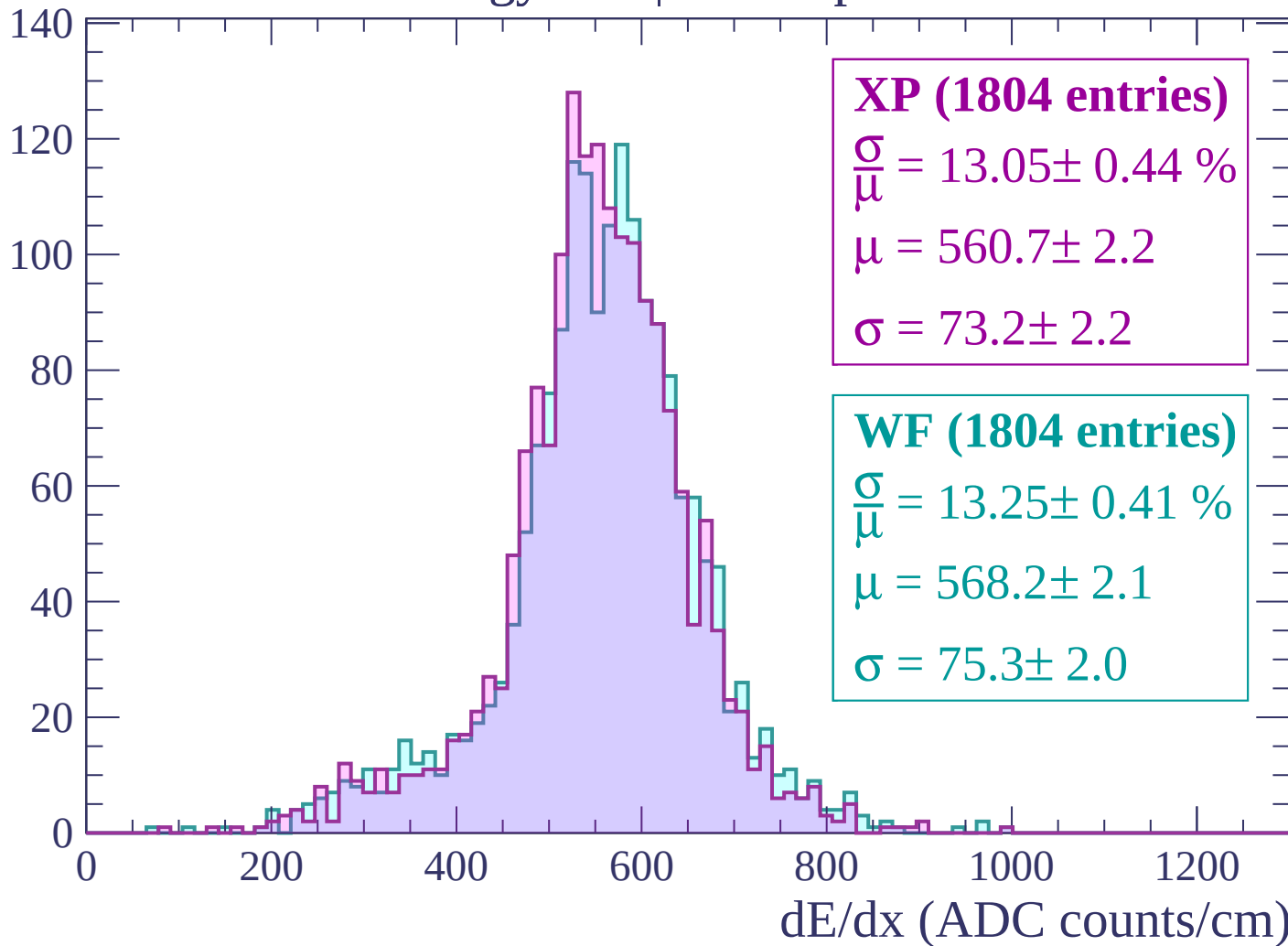
Energy loss |  $-180 < p < -120$

Count



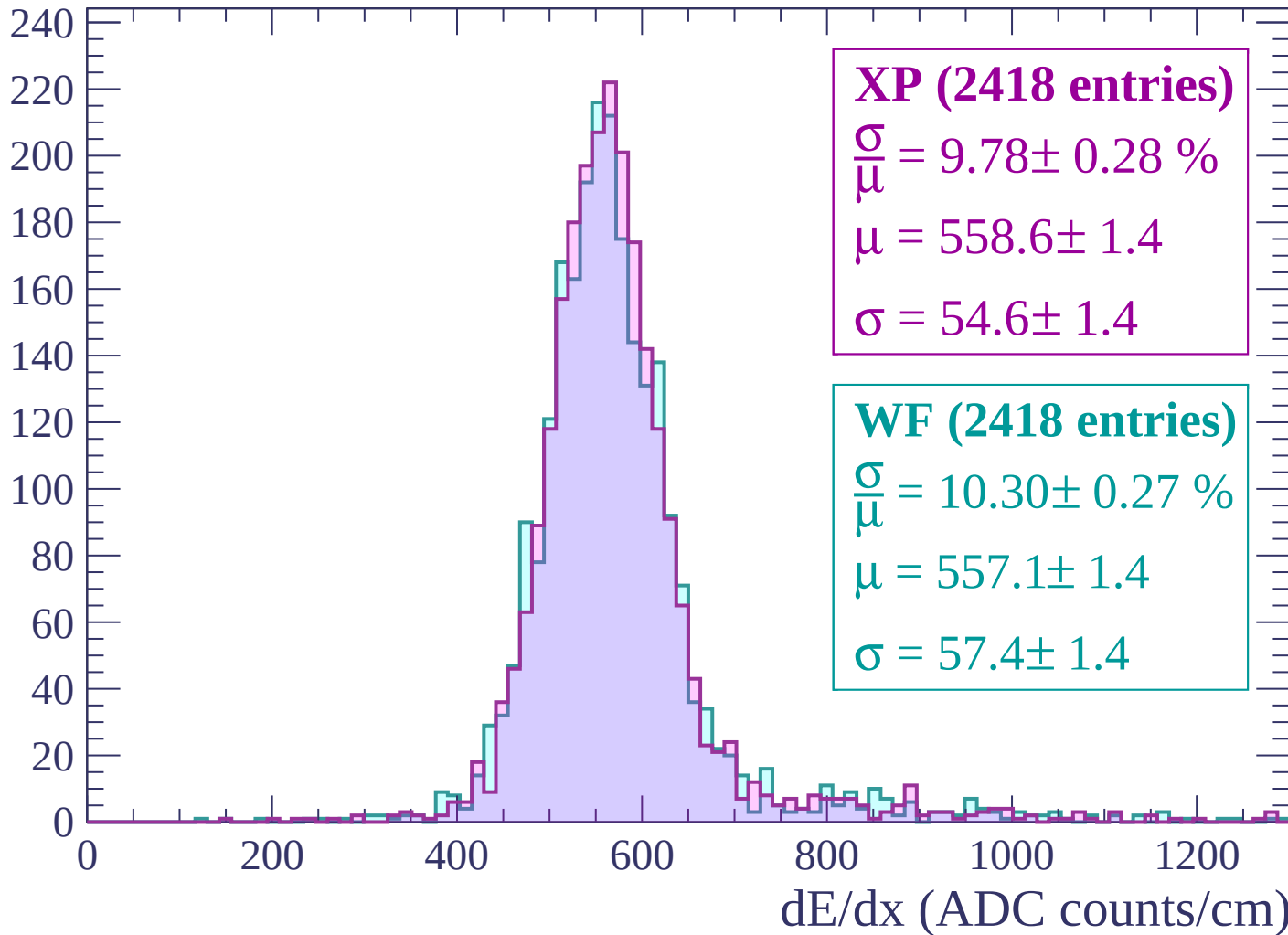
Energy loss |  $-120 < p < -60$

Count



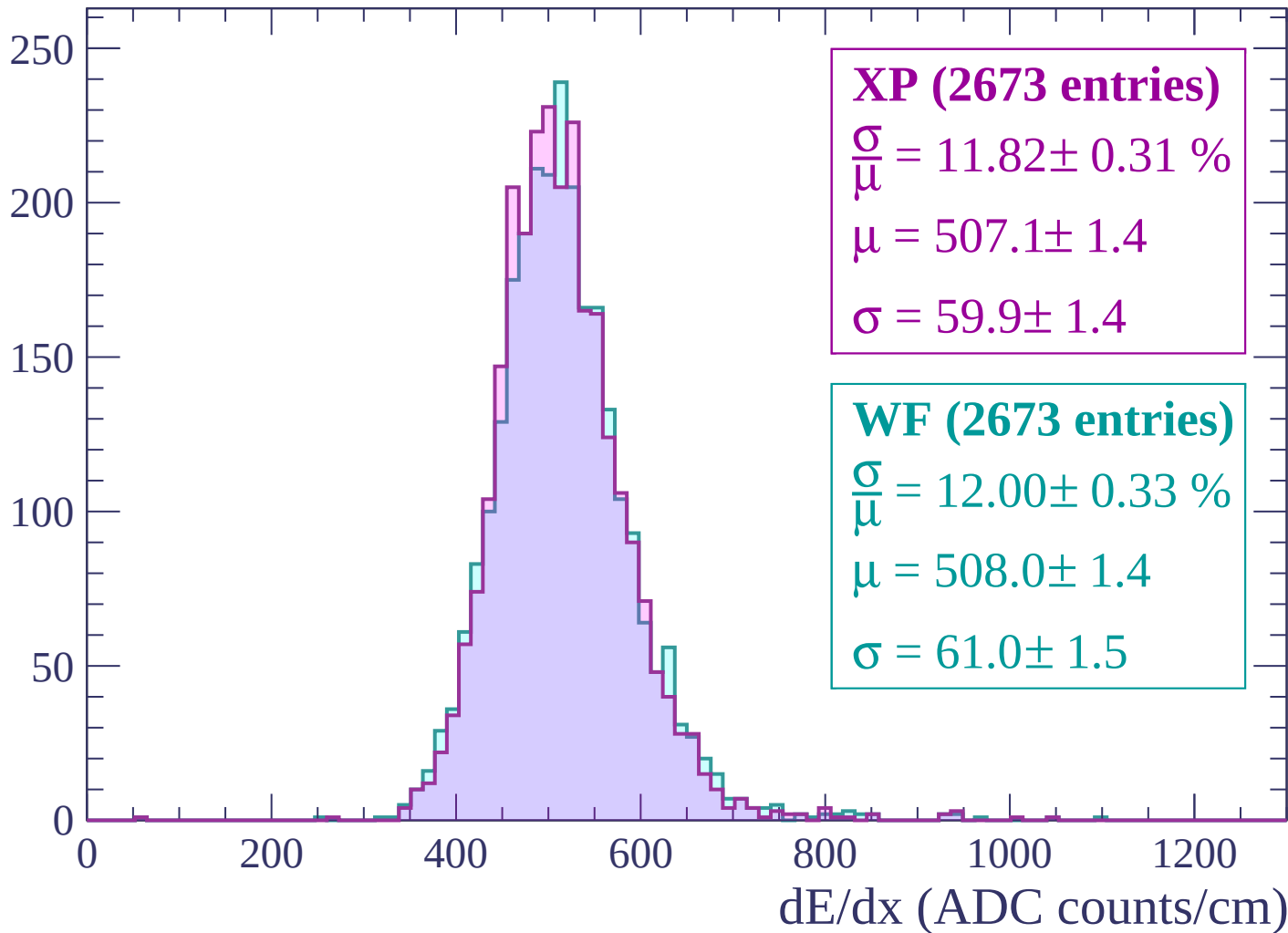
Energy loss |  $-60 < p < 0$

Count



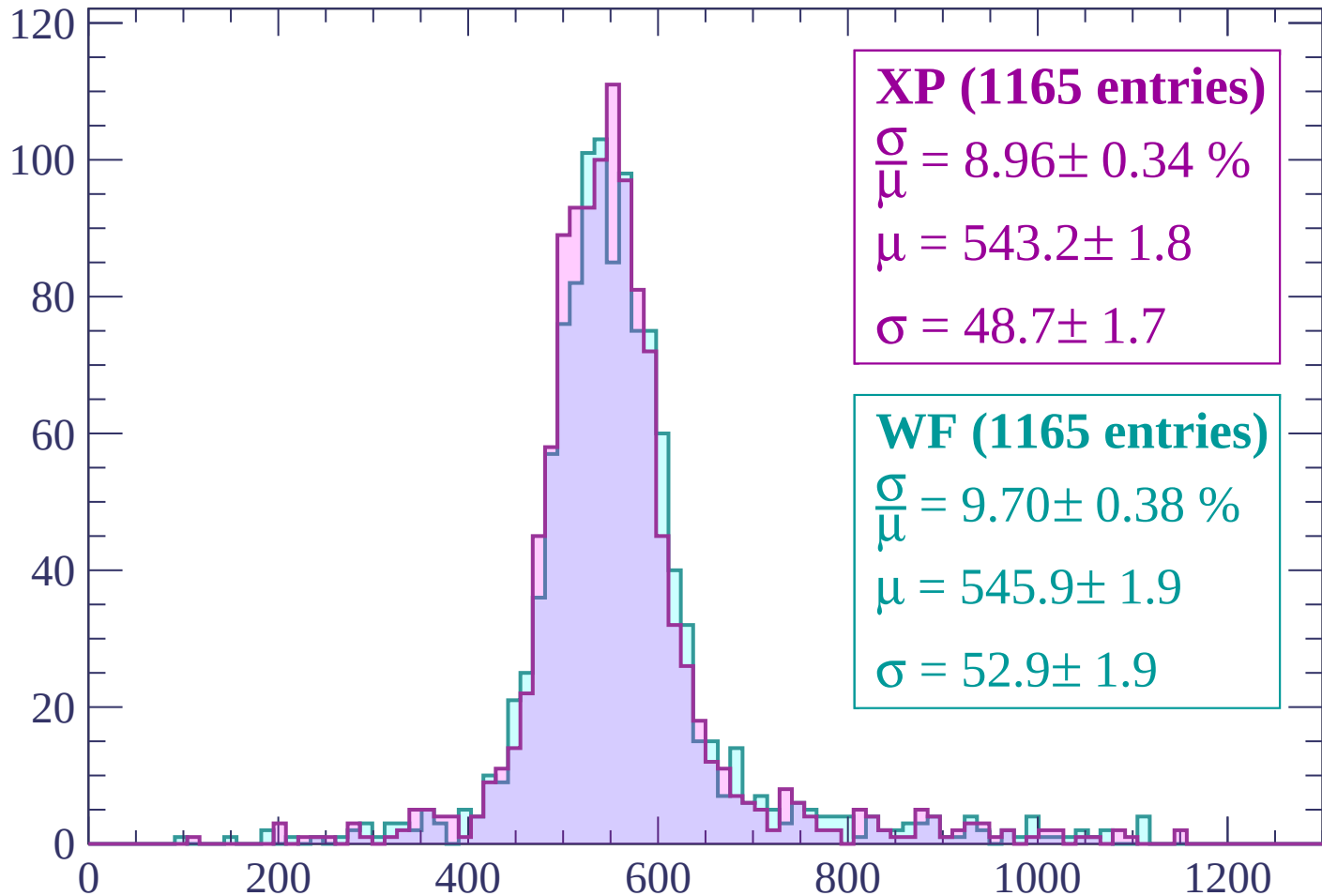
Energy loss |  $0 < p < 60$

Count



Energy loss |  $60 < p < 120$

Count



**XP (1165 entries)**

$$\frac{\sigma}{\mu} = 8.96 \pm 0.34 \%$$

$$\mu = 543.2 \pm 1.8$$

$$\sigma = 48.7 \pm 1.7$$

**WF (1165 entries)**

$$\frac{\sigma}{\mu} = 9.70 \pm 0.38 \%$$

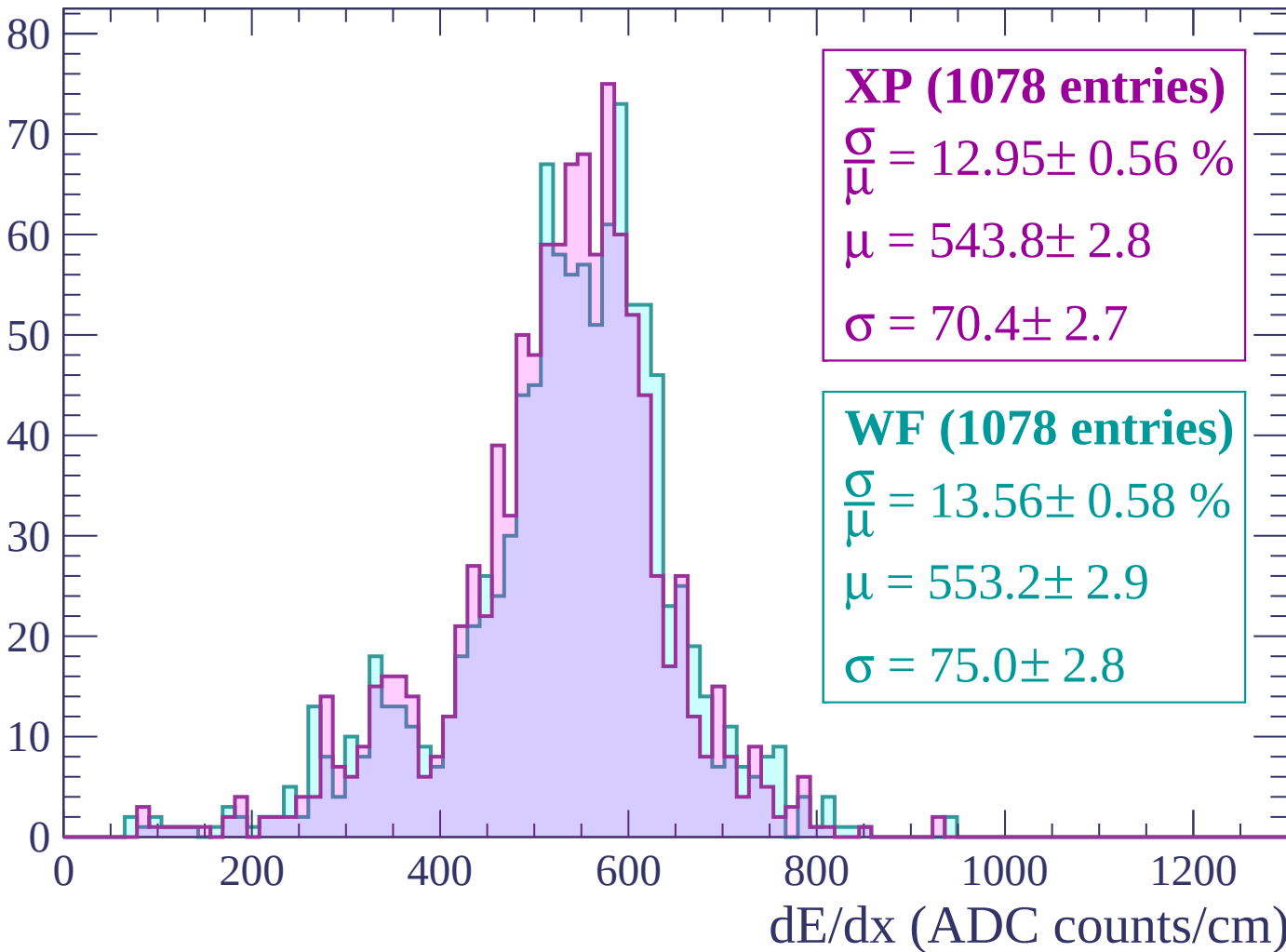
$$\mu = 545.9 \pm 1.9$$

$$\sigma = 52.9 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss |  $120 < p < 180$

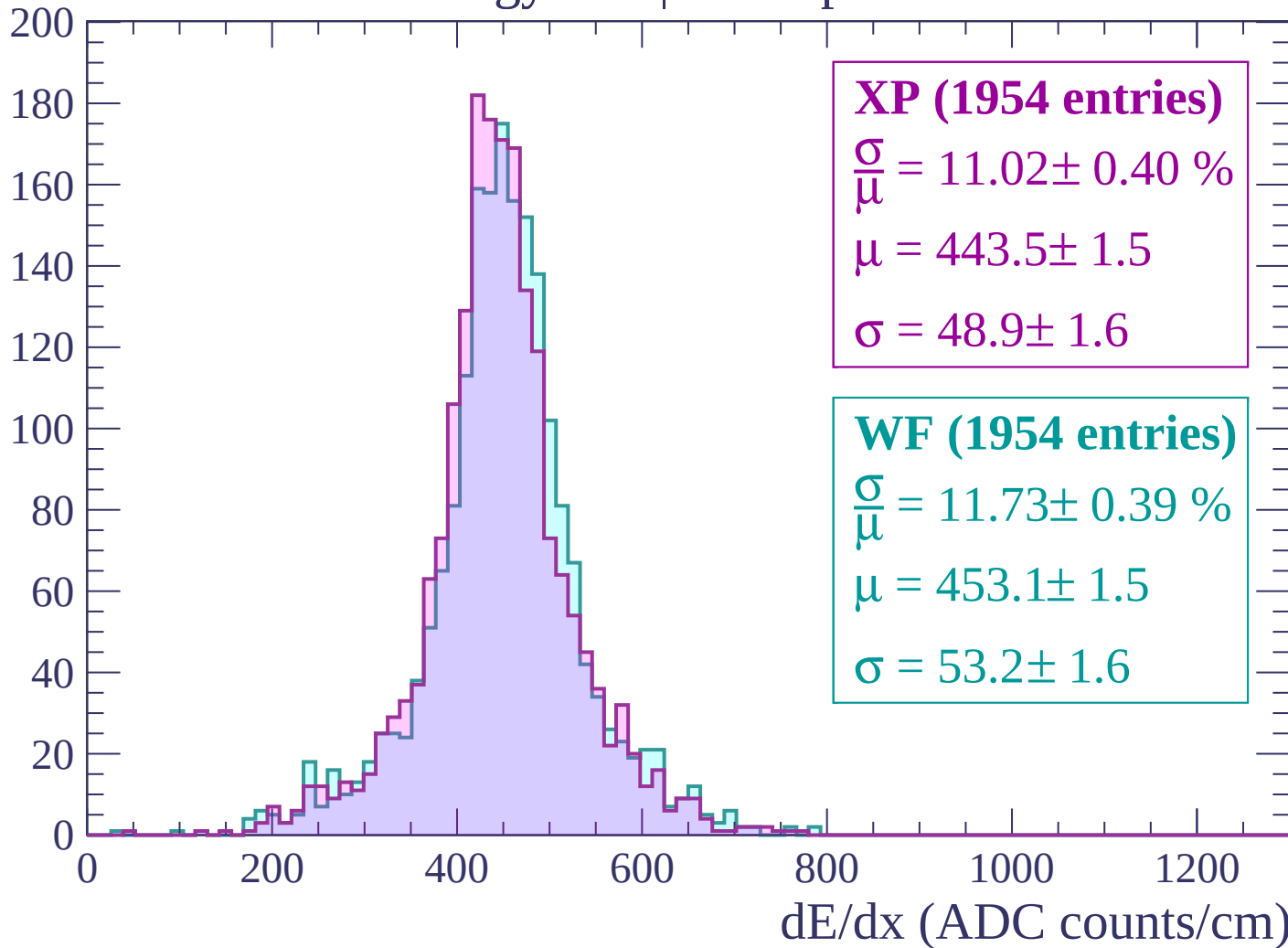
Count





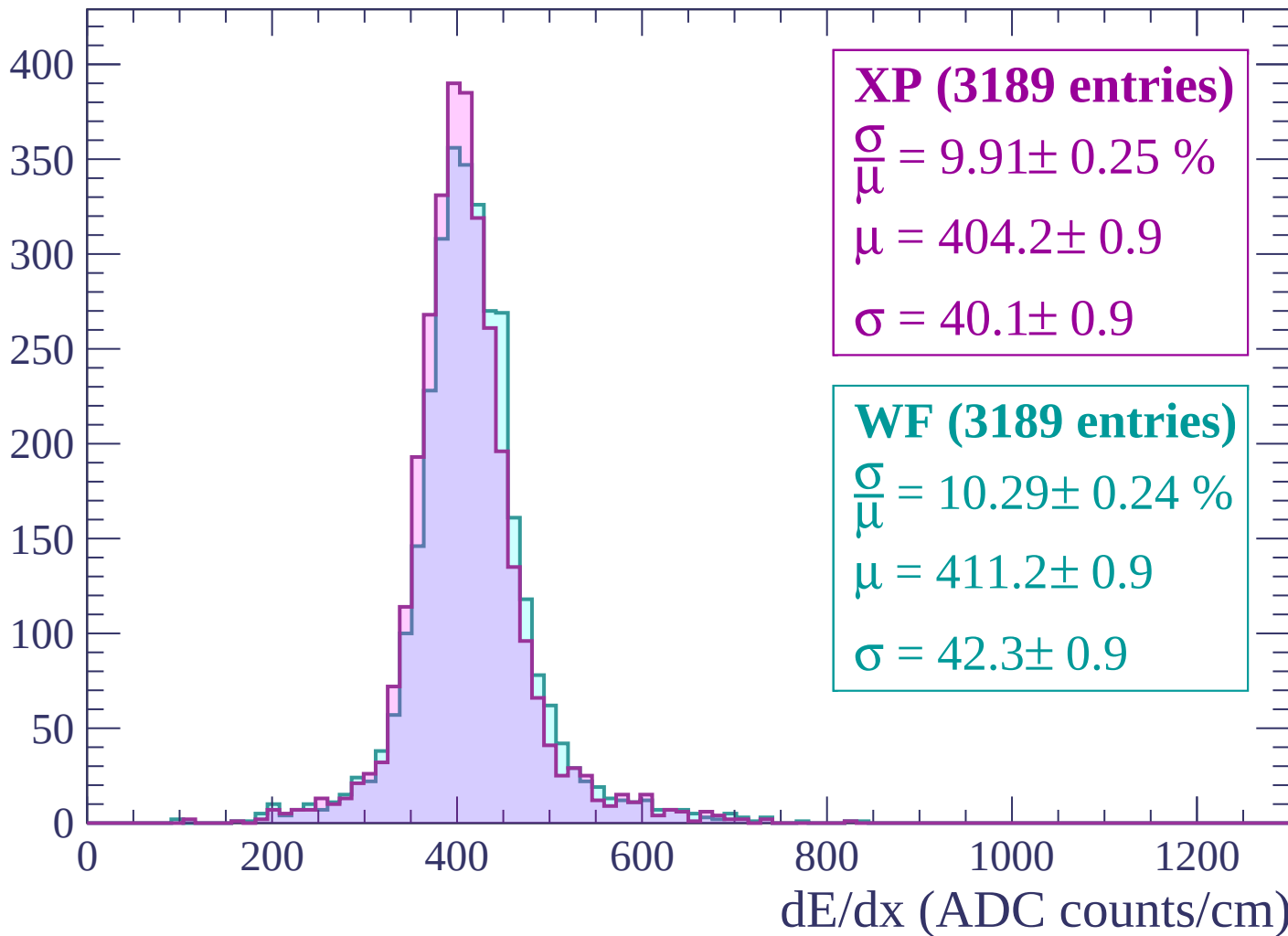
Energy loss |  $180 < p < 240$

Count



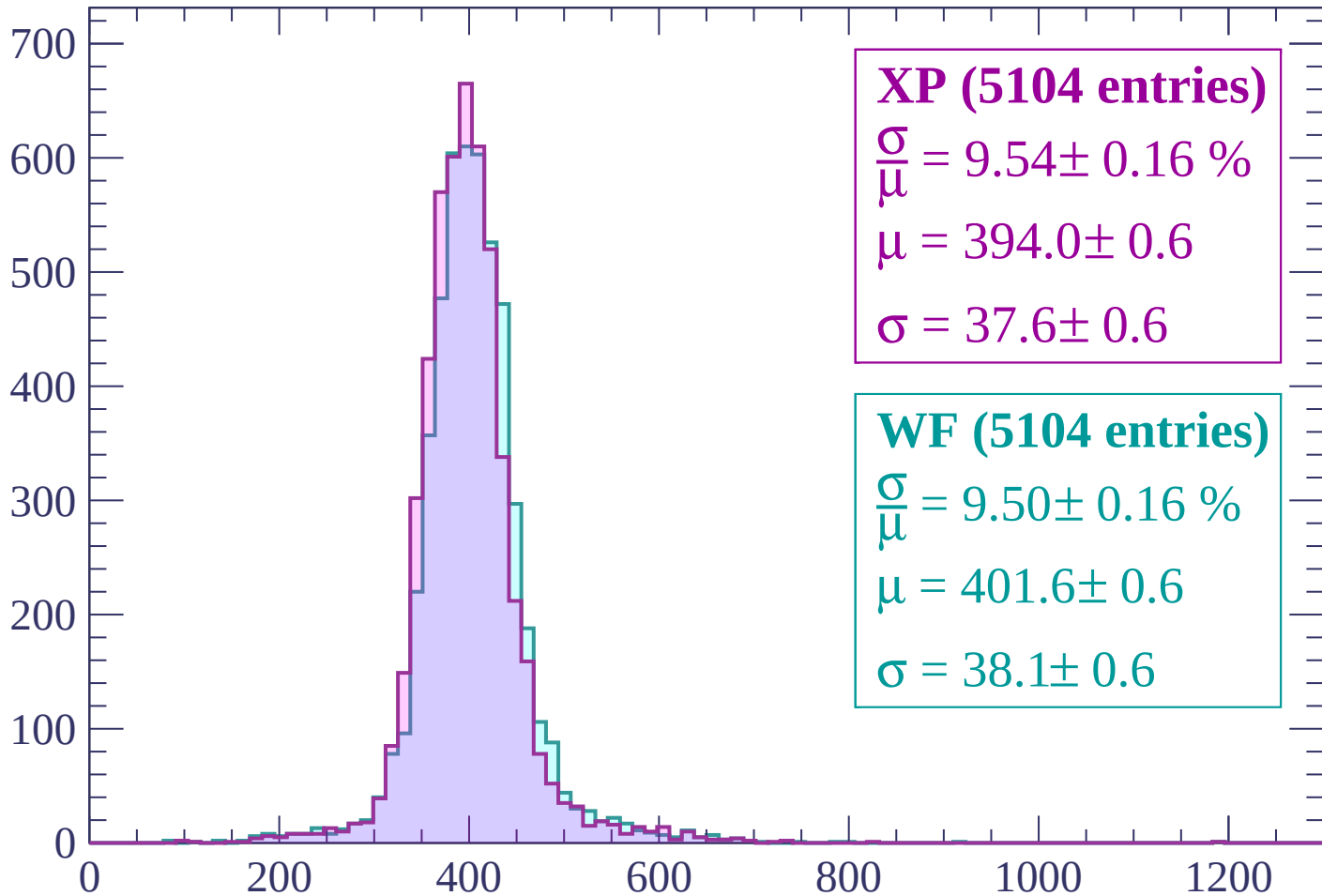
Energy loss |  $240 < p < 300$

Count



Energy loss |  $300 < p < 360$

Count



**XP (5104 entries)**

$\frac{\sigma}{\mu} = 9.54 \pm 0.16 \%$

$\mu = 394.0 \pm 0.6$

$\sigma = 37.6 \pm 0.6$

**WF (5104 entries)**

$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$

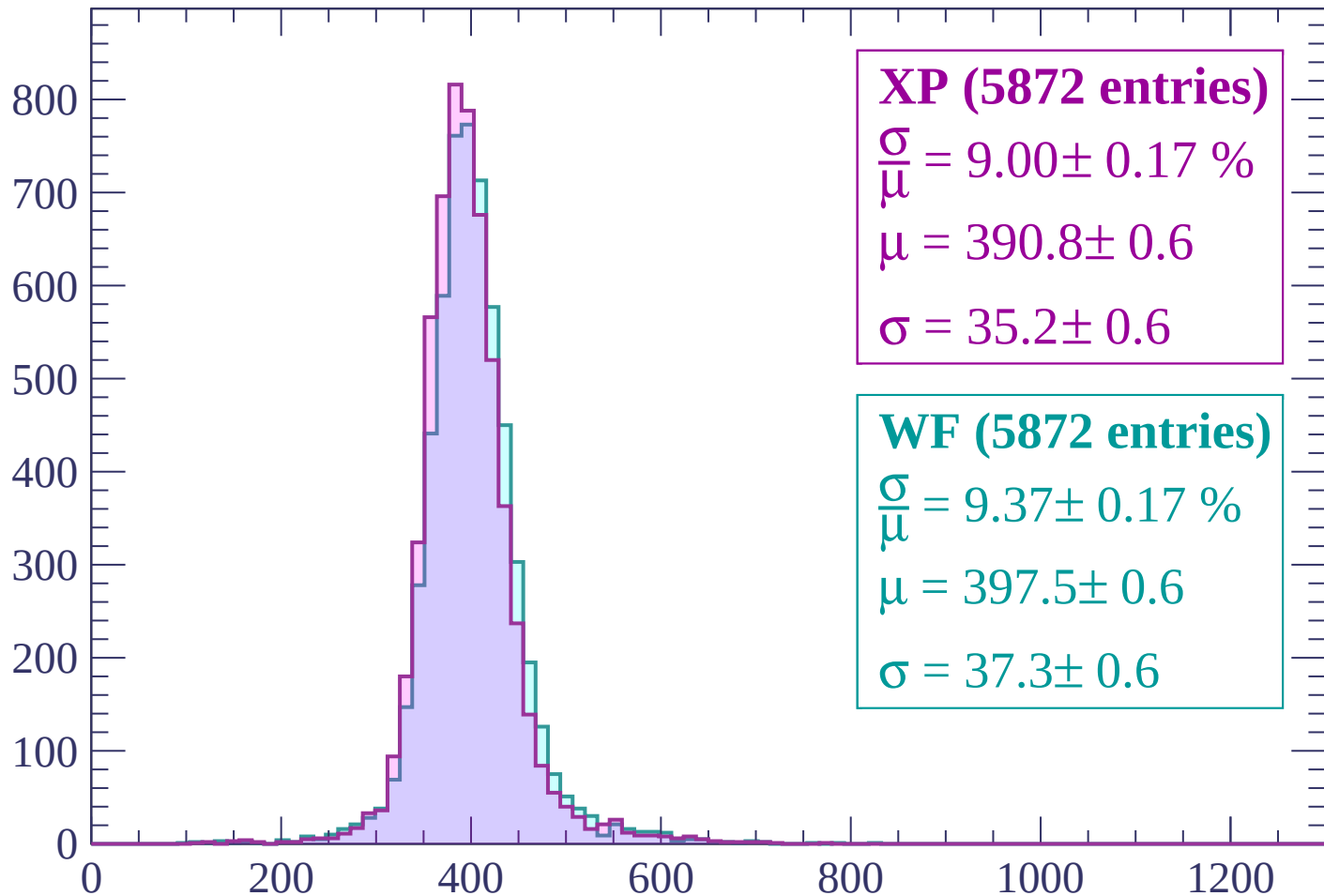
$\mu = 401.6 \pm 0.6$

$\sigma = 38.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $360 < p < 420$

Count



**XP (5872 entries)**

$\frac{\sigma}{\mu} = 9.00 \pm 0.17 \%$

$\mu = 390.8 \pm 0.6$

$\sigma = 35.2 \pm 0.6$

**WF (5872 entries)**

$\frac{\sigma}{\mu} = 9.37 \pm 0.17 \%$

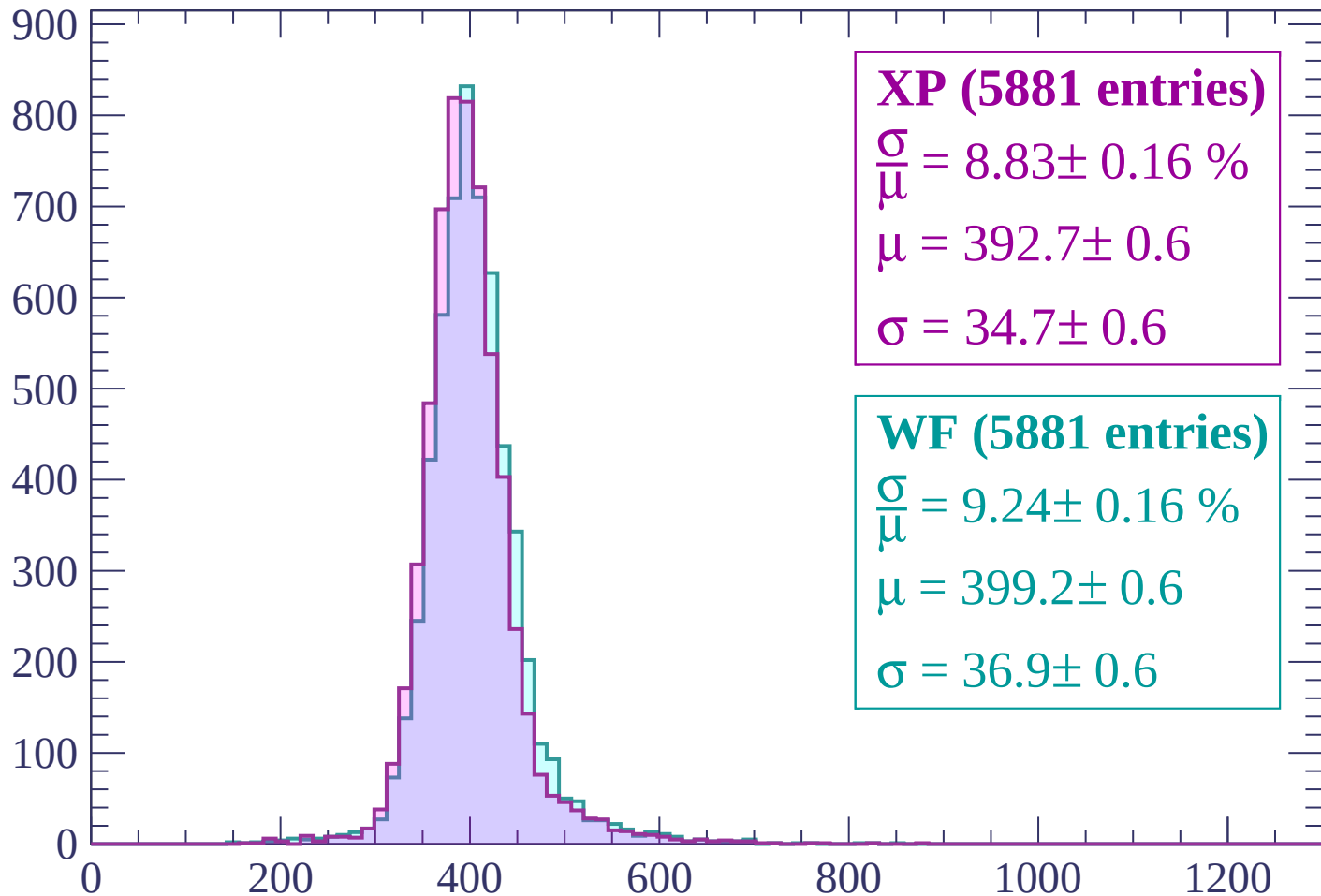
$\mu = 397.5 \pm 0.6$

$\sigma = 37.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $420 < p < 480$

Count



**XP (5881 entries)**

$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$

$\mu = 392.7 \pm 0.6$

$\sigma = 34.7 \pm 0.6$

**WF (5881 entries)**

$\frac{\sigma}{\mu} = 9.24 \pm 0.16 \%$

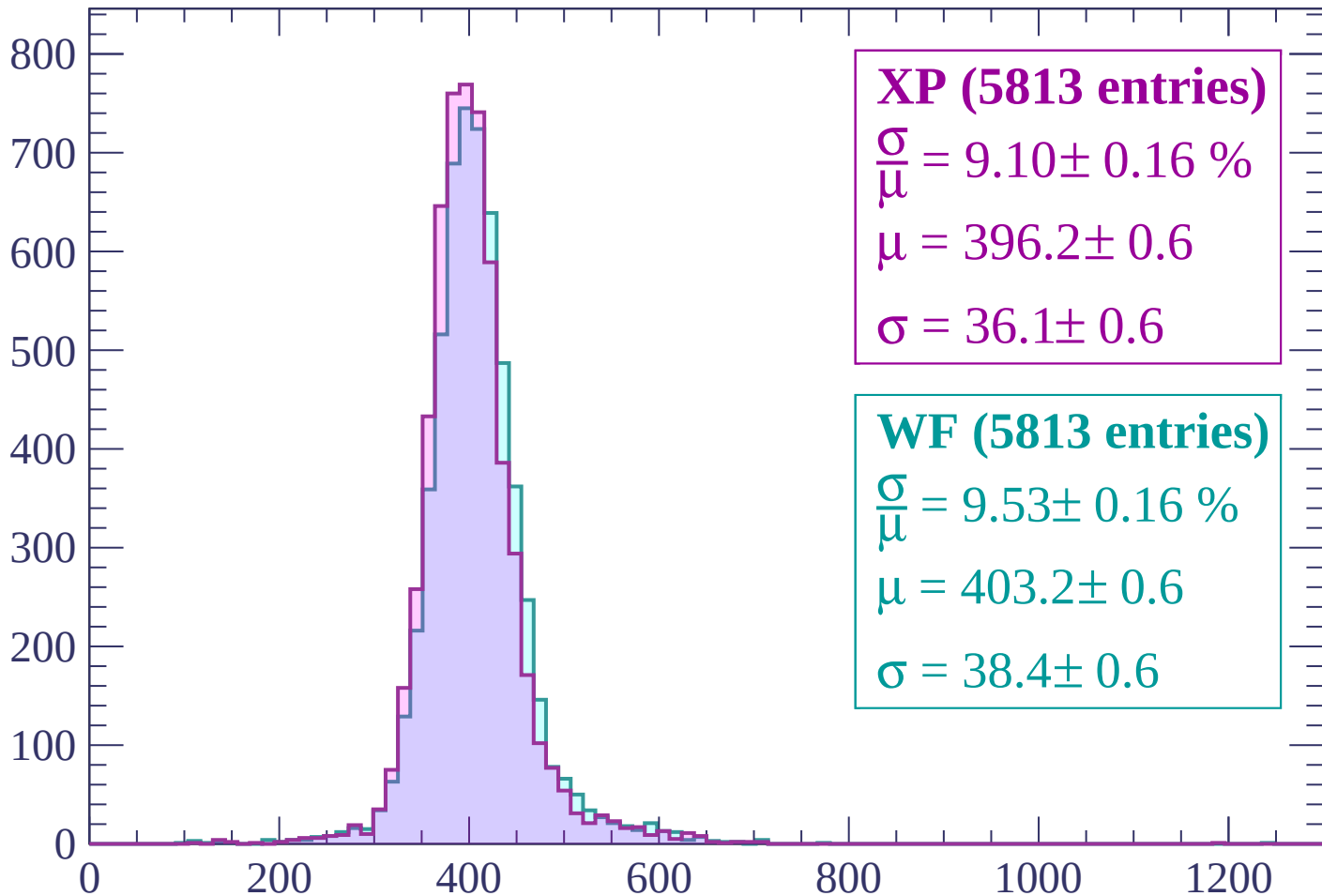
$\mu = 399.2 \pm 0.6$

$\sigma = 36.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $480 < p < 540$

Count



**XP (5813 entries)**

$$\frac{\sigma}{\mu} = 9.10 \pm 0.16 \%$$

$$\mu = 396.2 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

**WF (5813 entries)**

$$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$$

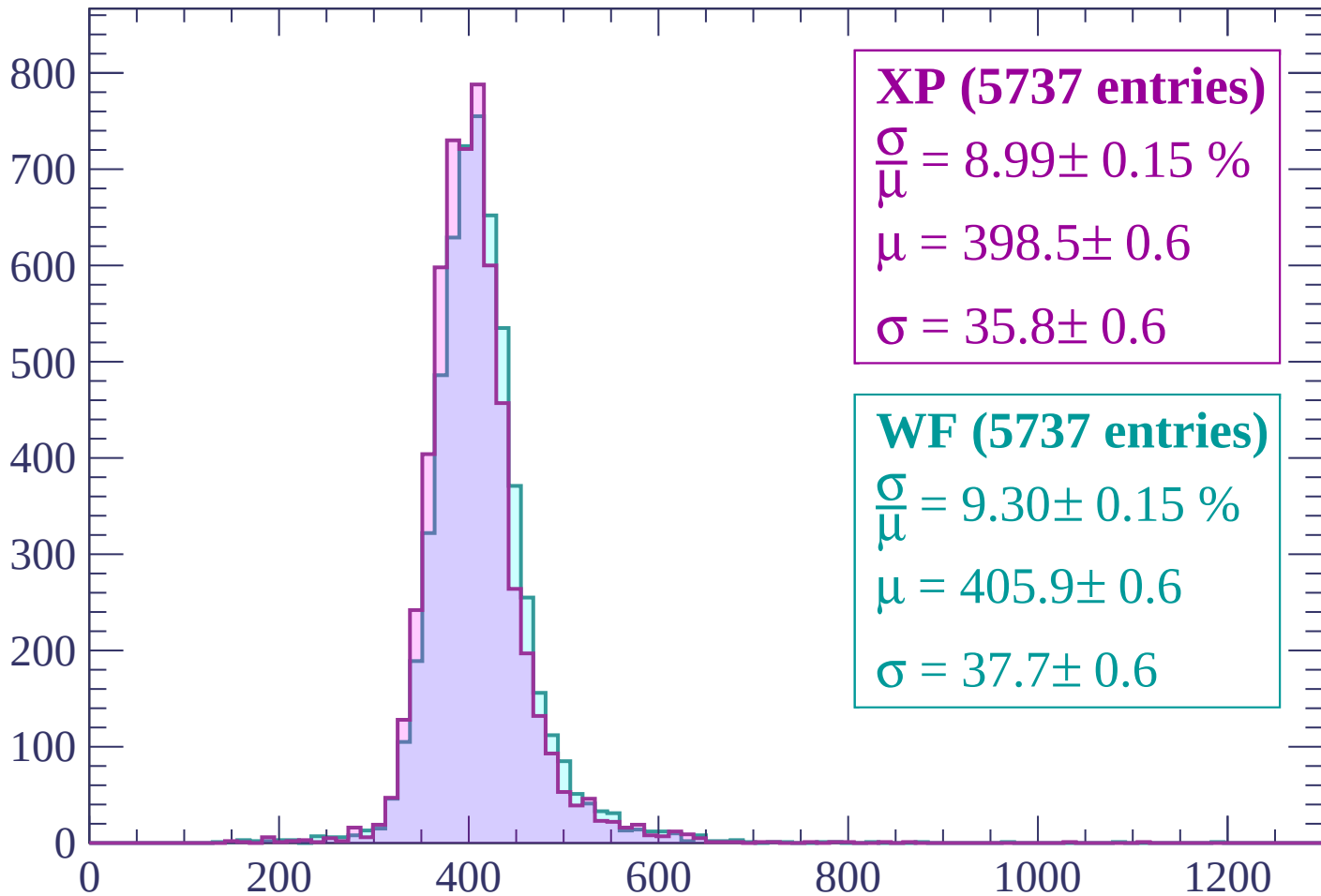
$$\mu = 403.2 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $540 < p < 600$

Count



**XP (5737 entries)**

$$\frac{\sigma}{\mu} = 8.99 \pm 0.15 \%$$

$$\mu = 398.5 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

**WF (5737 entries)**

$$\frac{\sigma}{\mu} = 9.30 \pm 0.15 \%$$

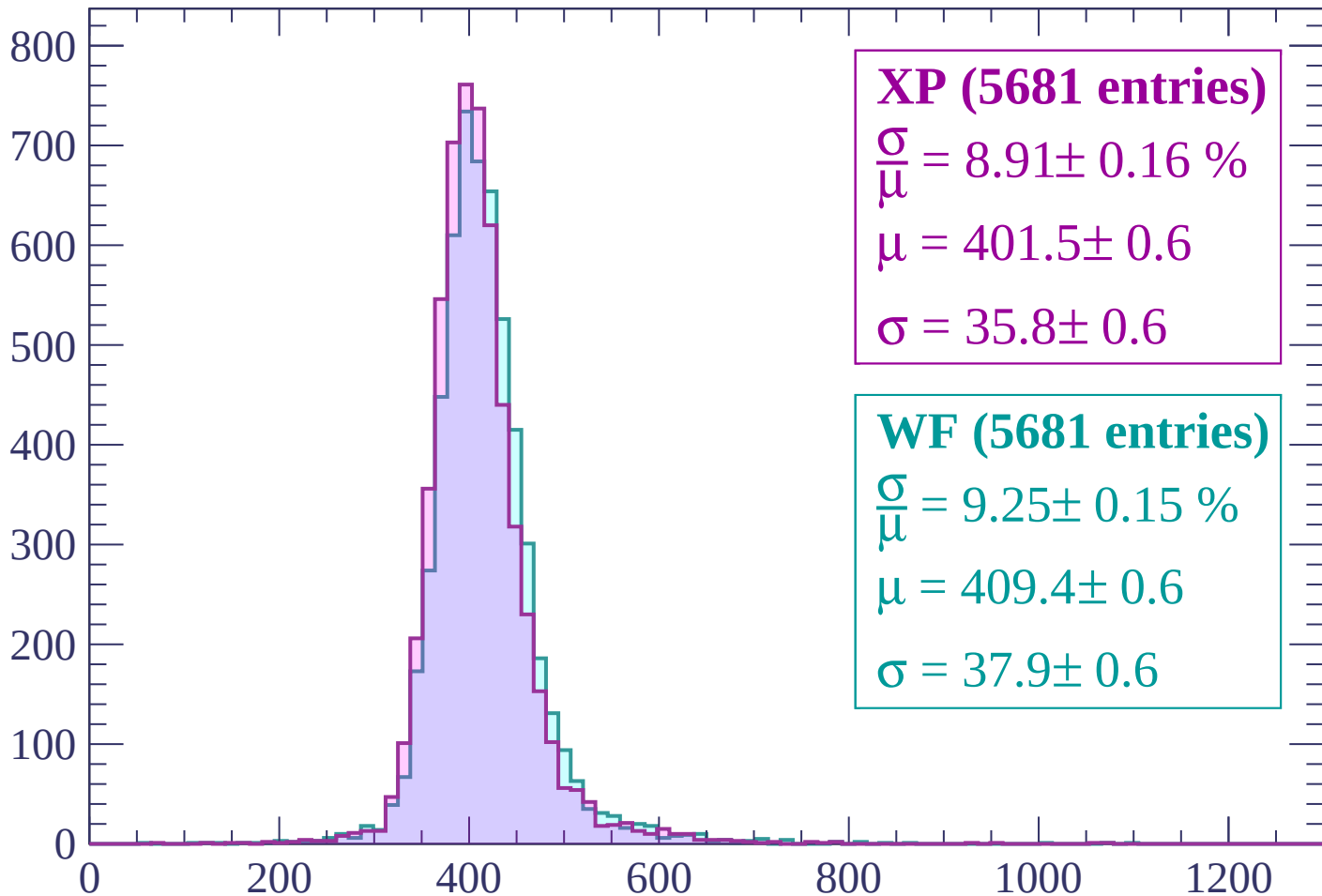
$$\mu = 405.9 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $600 < p < 660$

Count



**XP (5681 entries)**

$$\frac{\sigma}{\mu} = 8.91 \pm 0.16 \%$$

$$\mu = 401.5 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

**WF (5681 entries)**

$$\frac{\sigma}{\mu} = 9.25 \pm 0.15 \%$$

$$\mu = 409.4 \pm 0.6$$

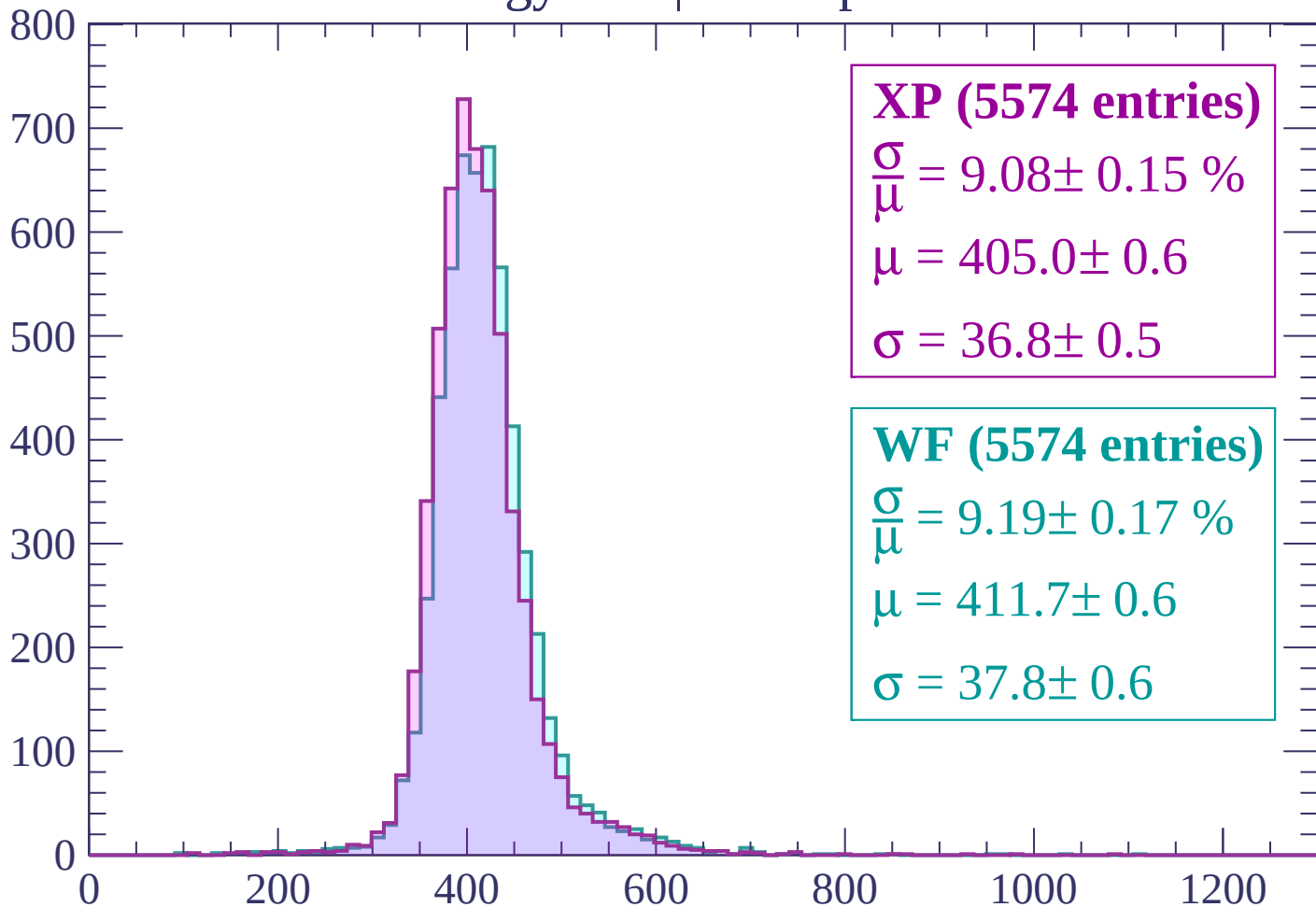
$$\sigma = 37.9 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $660 < p < 720$

Count



**XP (5574 entries)**

$$\frac{\sigma}{\mu} = 9.08 \pm 0.15 \%$$

$$\mu = 405.0 \pm 0.6$$

$$\sigma = 36.8 \pm 0.5$$

**WF (5574 entries)**

$$\frac{\sigma}{\mu} = 9.19 \pm 0.17 \%$$

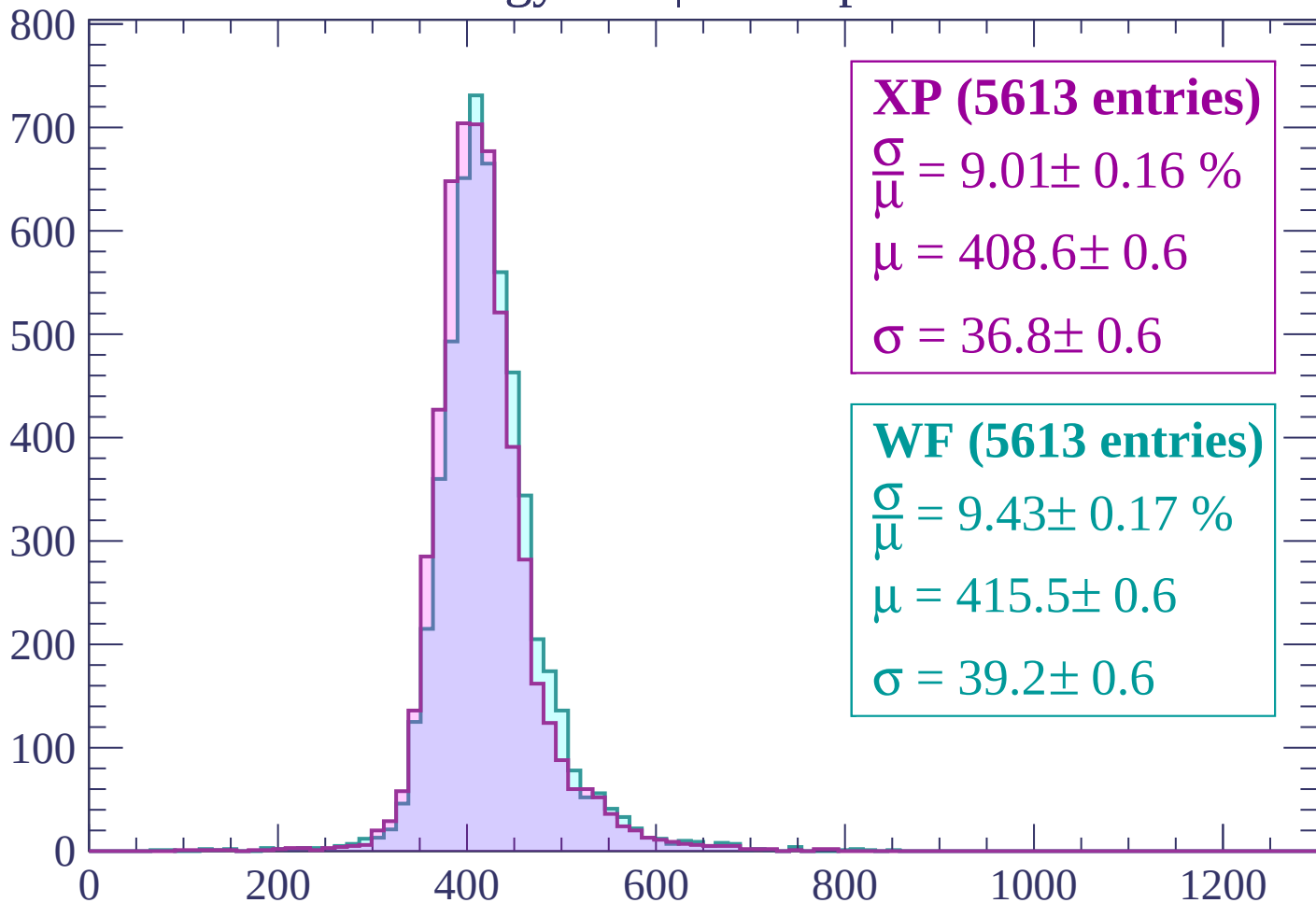
$$\mu = 411.7 \pm 0.6$$

$$\sigma = 37.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $720 < p < 780$

Count



**XP (5613 entries)**

$$\frac{\sigma}{\mu} = 9.01 \pm 0.16 \%$$

$$\mu = 408.6 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

**WF (5613 entries)**

$$\frac{\sigma}{\mu} = 9.43 \pm 0.17 \%$$

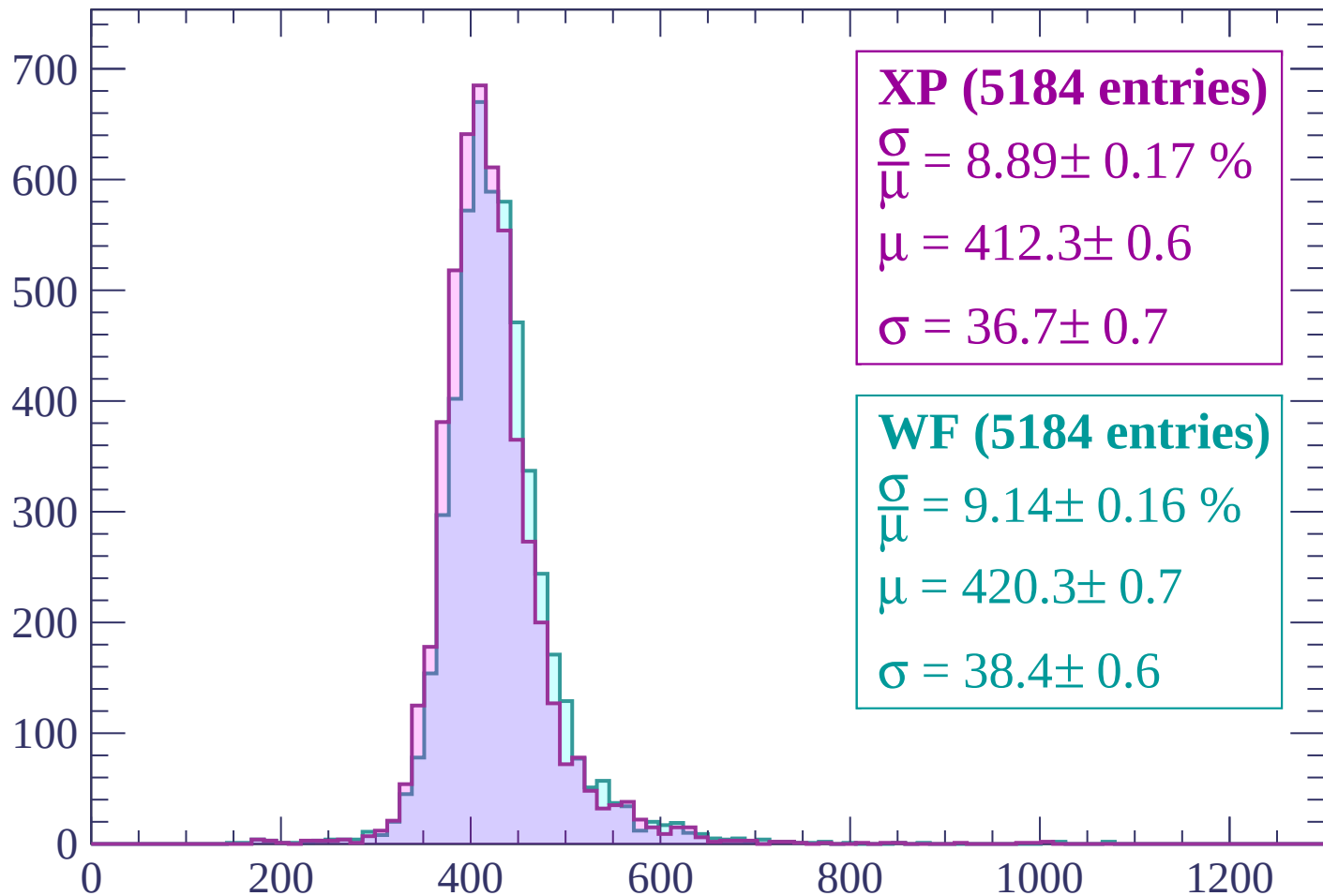
$$\mu = 415.5 \pm 0.6$$

$$\sigma = 39.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $780 < p < 840$

Count



**XP (5184 entries)**

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 412.3 \pm 0.6$$

$$\sigma = 36.7 \pm 0.7$$

**WF (5184 entries)**

$$\frac{\sigma}{\mu} = 9.14 \pm 0.16 \%$$

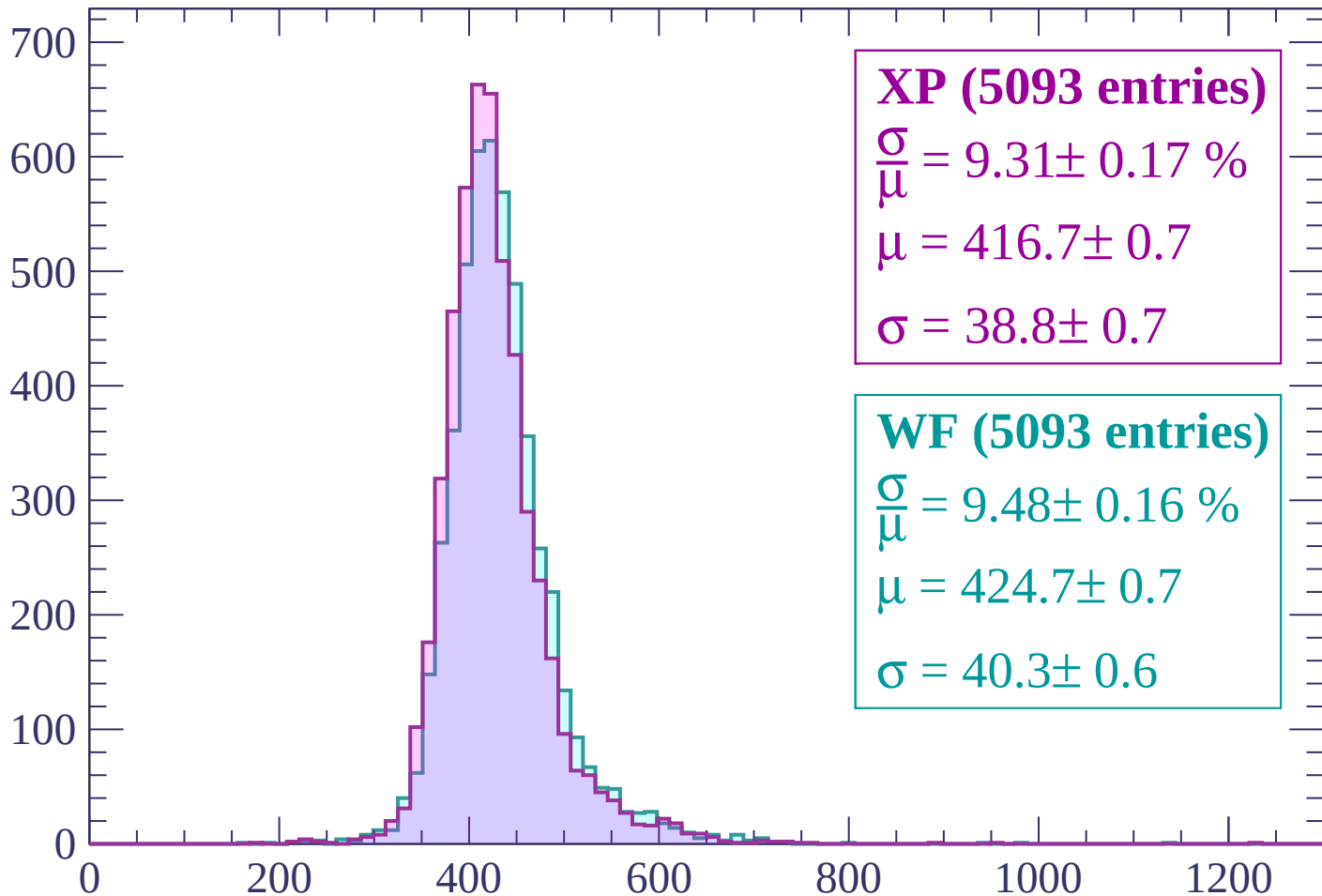
$$\mu = 420.3 \pm 0.7$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $840 < p < 900$

Count



**XP (5093 entries)**

$$\frac{\sigma}{\mu} = 9.31 \pm 0.17 \%$$

$$\mu = 416.7 \pm 0.7$$

$$\sigma = 38.8 \pm 0.7$$

**WF (5093 entries)**

$$\frac{\sigma}{\mu} = 9.48 \pm 0.16 \%$$

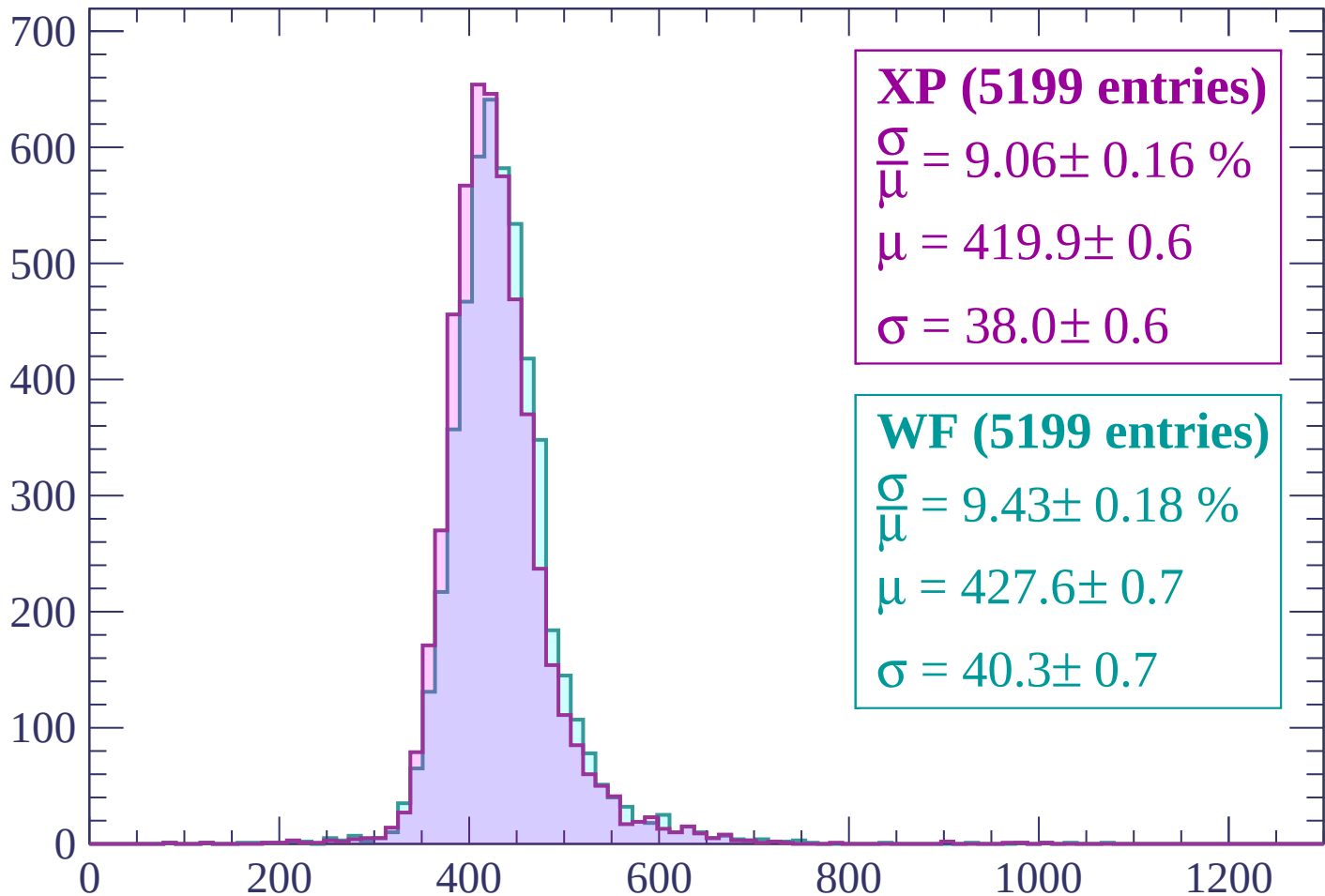
$$\mu = 424.7 \pm 0.7$$

$$\sigma = 40.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $900 < p < 960$

Count



**XP (5199 entries)**

$$\frac{\sigma}{\mu} = 9.06 \pm 0.16 \%$$

$$\mu = 419.9 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

**WF (5199 entries)**

$$\frac{\sigma}{\mu} = 9.43 \pm 0.18 \%$$

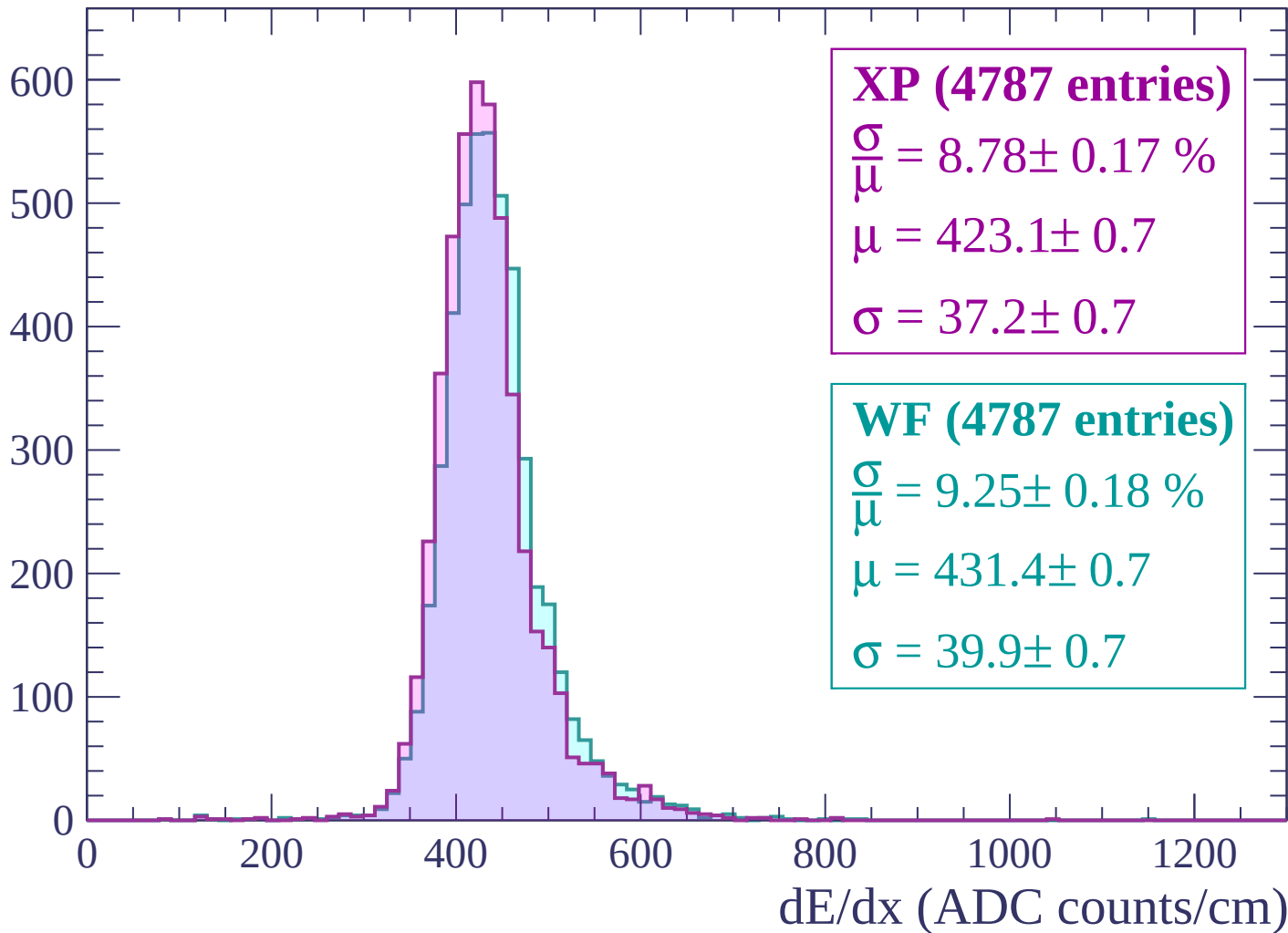
$$\mu = 427.6 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

dE/dx (ADC counts/cm)

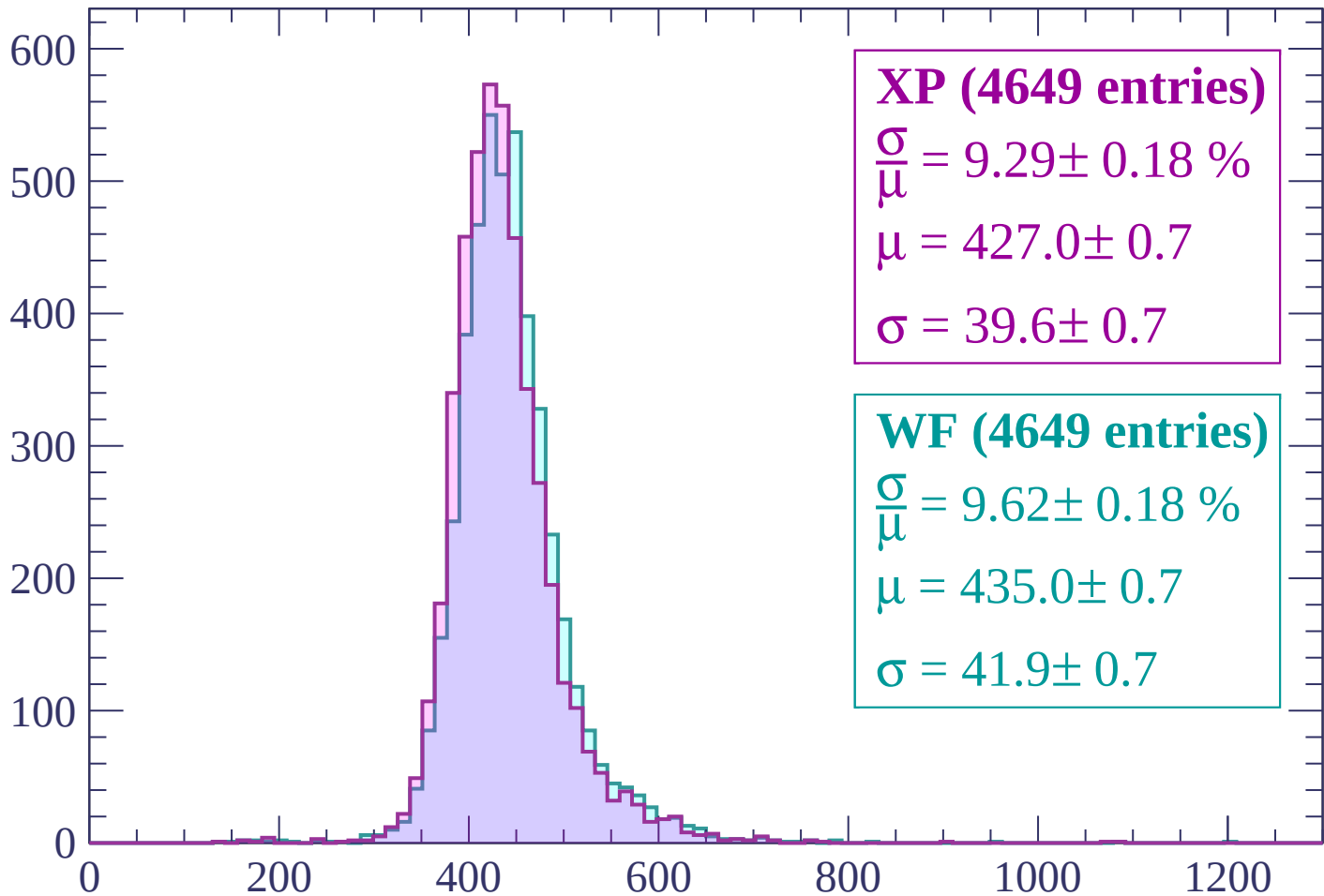
# Energy loss | $960 < p < 1020$

Count



Energy loss |  $1020 < p < 1080$

Count



**XP (4649 entries)**

$$\frac{\sigma}{\mu} = 9.29 \pm 0.18 \%$$

$$\mu = 427.0 \pm 0.7$$

$$\sigma = 39.6 \pm 0.7$$

**WF (4649 entries)**

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

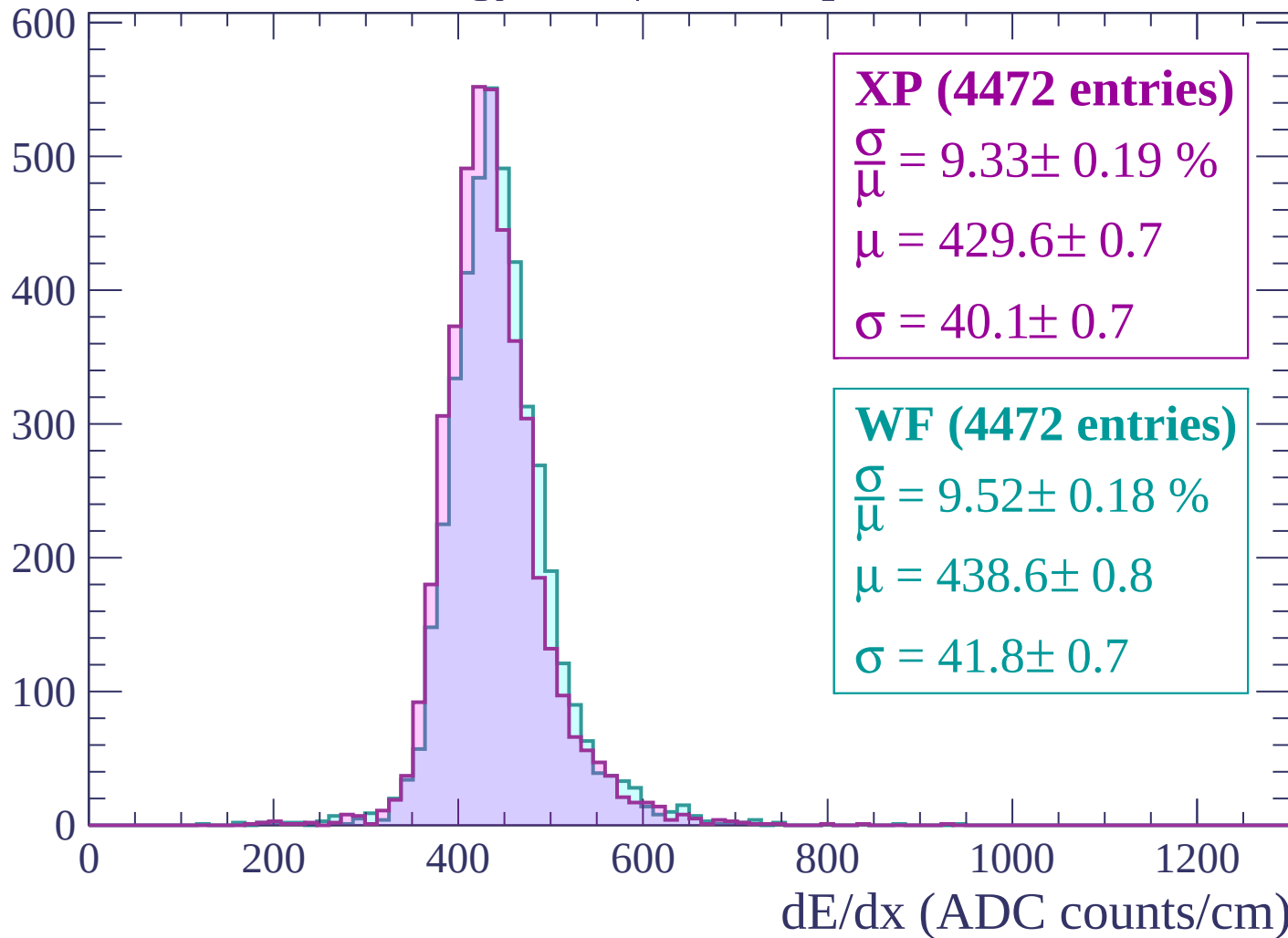
$$\mu = 435.0 \pm 0.7$$

$$\sigma = 41.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $1080 < p < 1140$

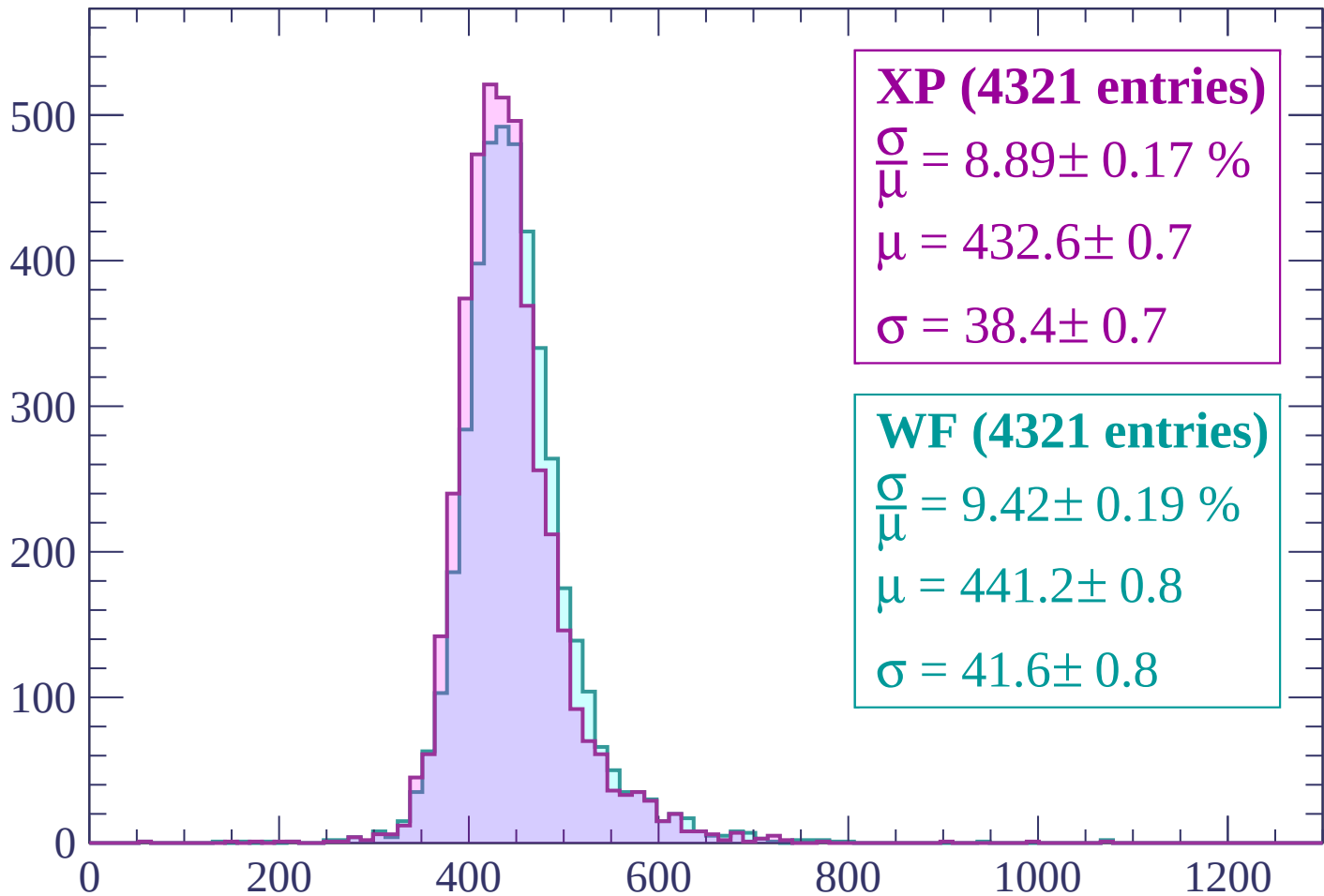
Count





Energy loss |  $1140 < p < 1200$

Count



**XP (4321 entries)**

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 432.6 \pm 0.7$$

$$\sigma = 38.4 \pm 0.7$$

**WF (4321 entries)**

$$\frac{\sigma}{\mu} = 9.42 \pm 0.19 \%$$

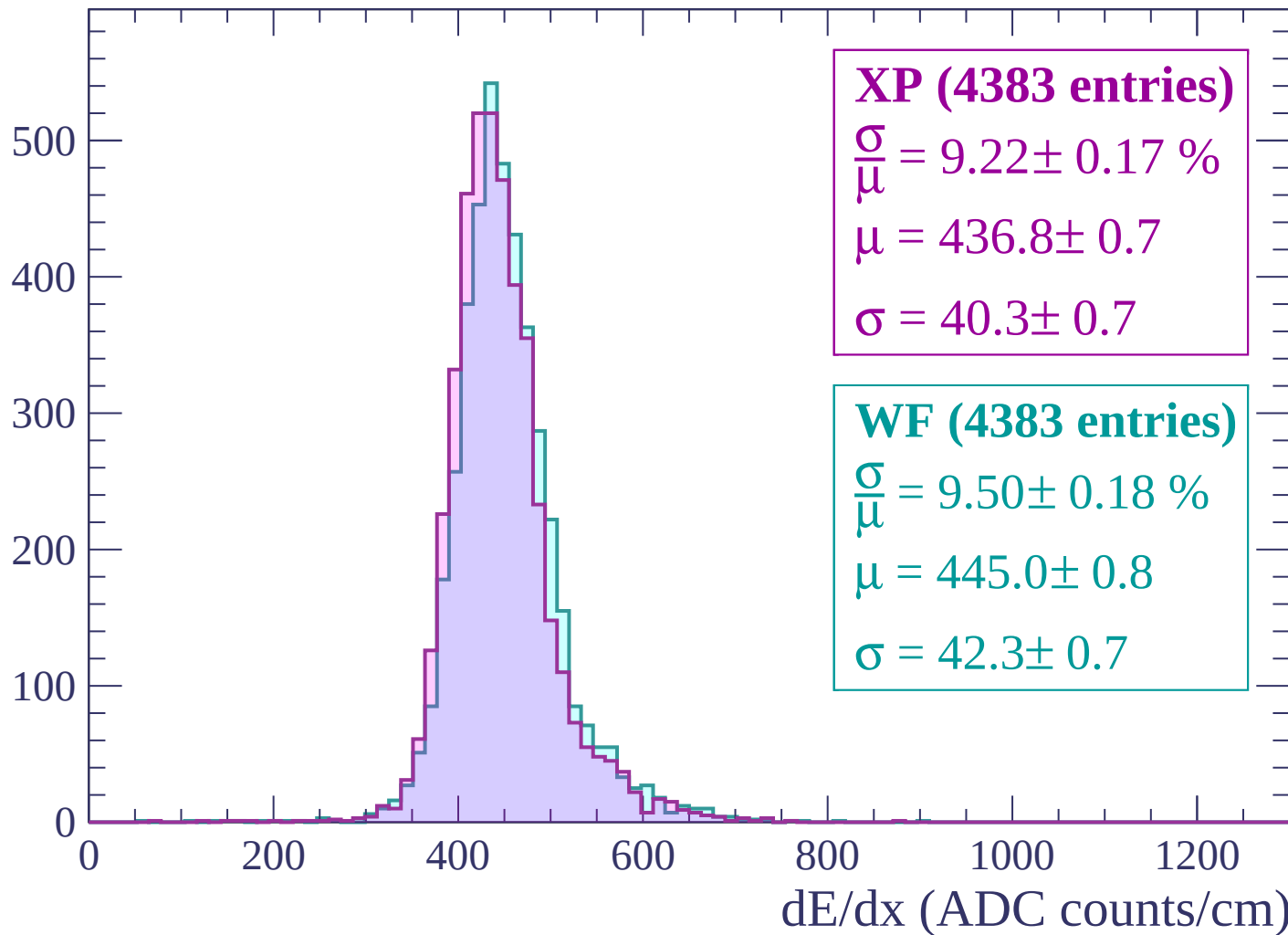
$$\mu = 441.2 \pm 0.8$$

$$\sigma = 41.6 \pm 0.8$$

dE/dx (ADC counts/cm)

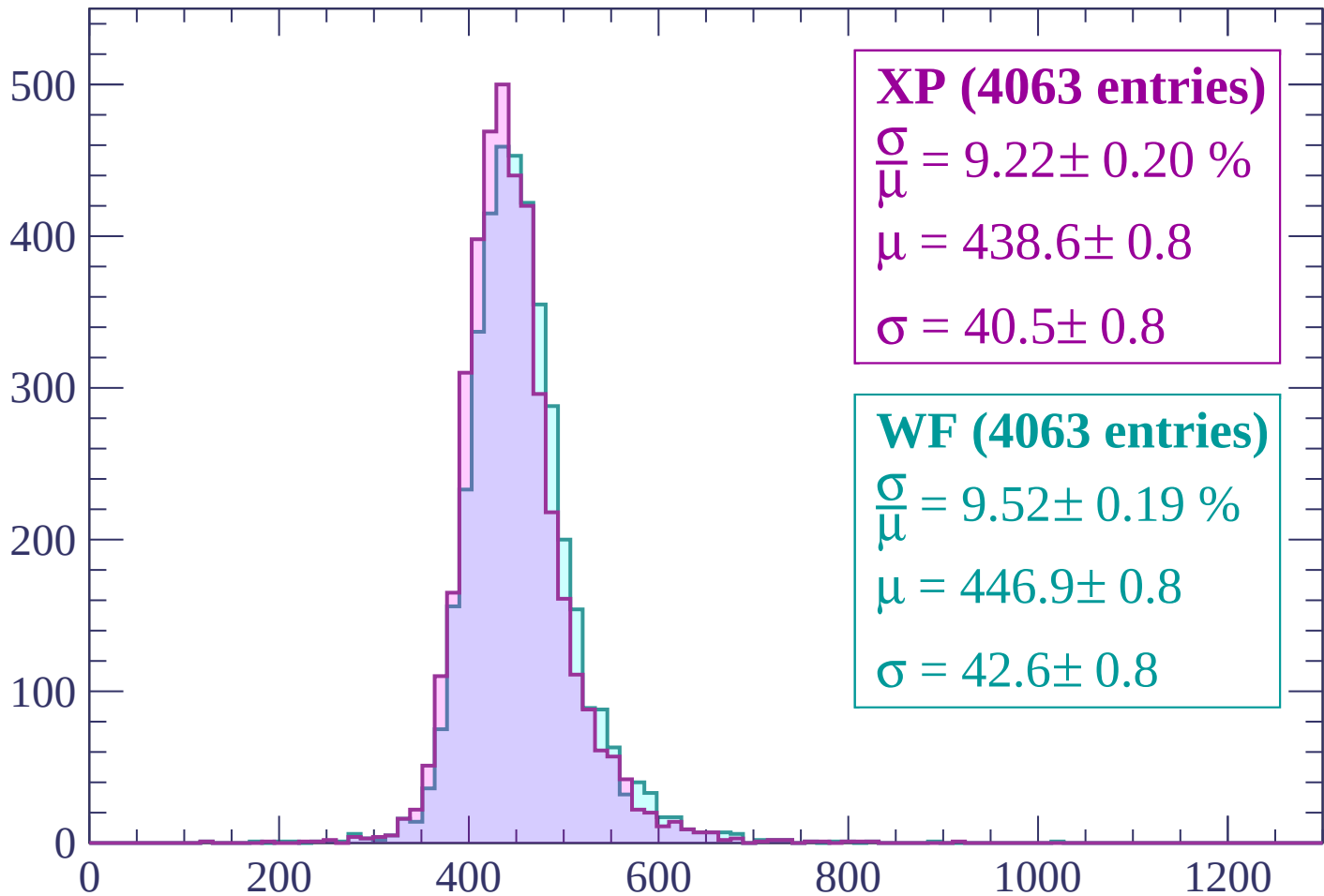
Energy loss |  $1200 < p < 1260$

Count



Energy loss |  $1260 < p < 1320$

Count



**XP (4063 entries)**

$$\frac{\sigma}{\mu} = 9.22 \pm 0.20 \%$$

$$\mu = 438.6 \pm 0.8$$

$$\sigma = 40.5 \pm 0.8$$

**WF (4063 entries)**

$$\frac{\sigma}{\mu} = 9.52 \pm 0.19 \%$$

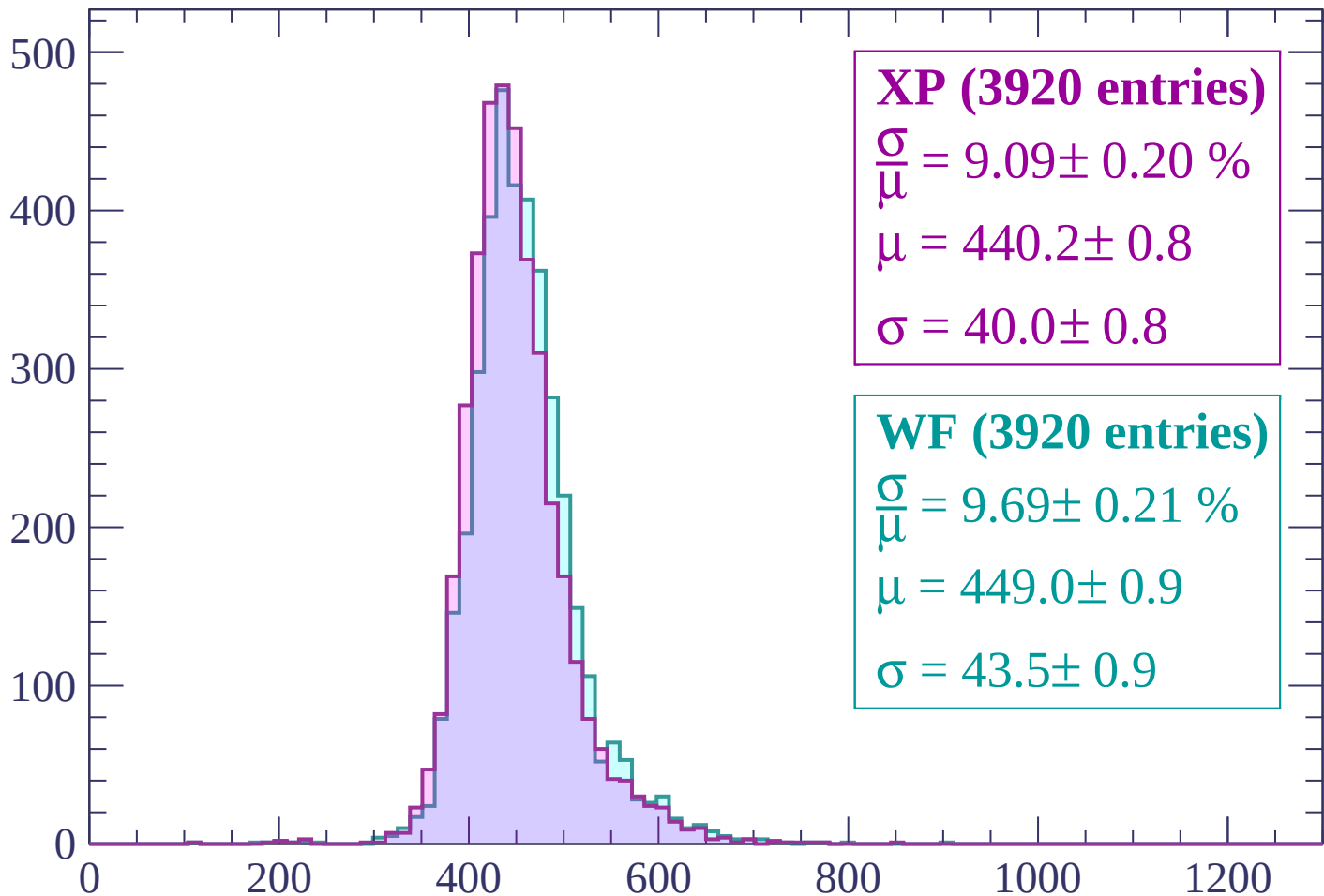
$$\mu = 446.9 \pm 0.8$$

$$\sigma = 42.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $1320 < p < 1380$

Count



**XP (3920 entries)**

$$\frac{\sigma}{\mu} = 9.09 \pm 0.20 \%$$

$$\mu = 440.2 \pm 0.8$$

$$\sigma = 40.0 \pm 0.8$$

**WF (3920 entries)**

$$\frac{\sigma}{\mu} = 9.69 \pm 0.21 \%$$

$$\mu = 449.0 \pm 0.9$$

$$\sigma = 43.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $1380 < p < 1440$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

**XP (3798 entries)**

$\frac{\sigma}{\mu} = 9.10 \pm 0.20 \%$

$\mu = 442.6 \pm 0.8$

$\sigma = 40.3 \pm 0.8$

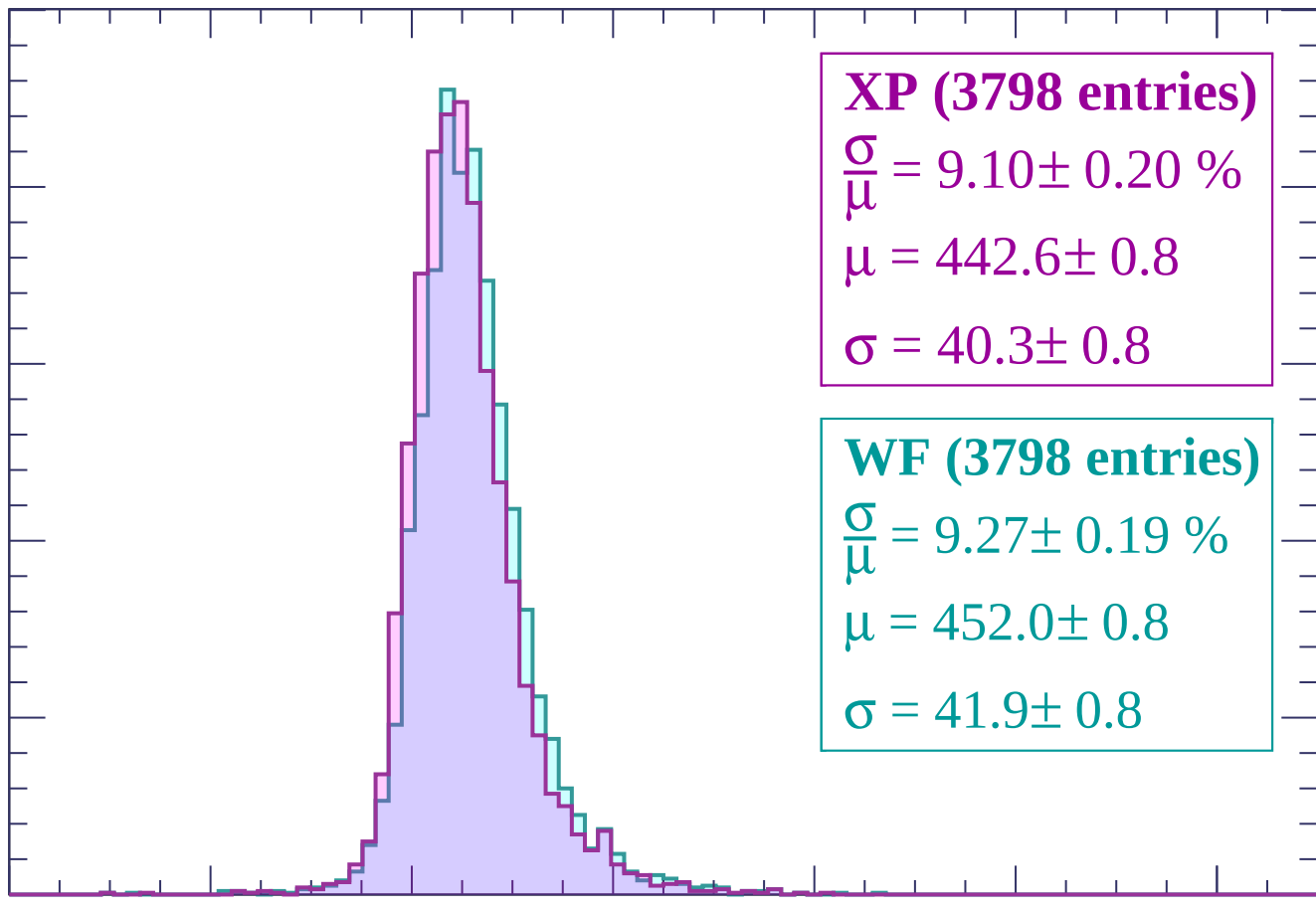
**WF (3798 entries)**

$\frac{\sigma}{\mu} = 9.27 \pm 0.19 \%$

$\mu = 452.0 \pm 0.8$

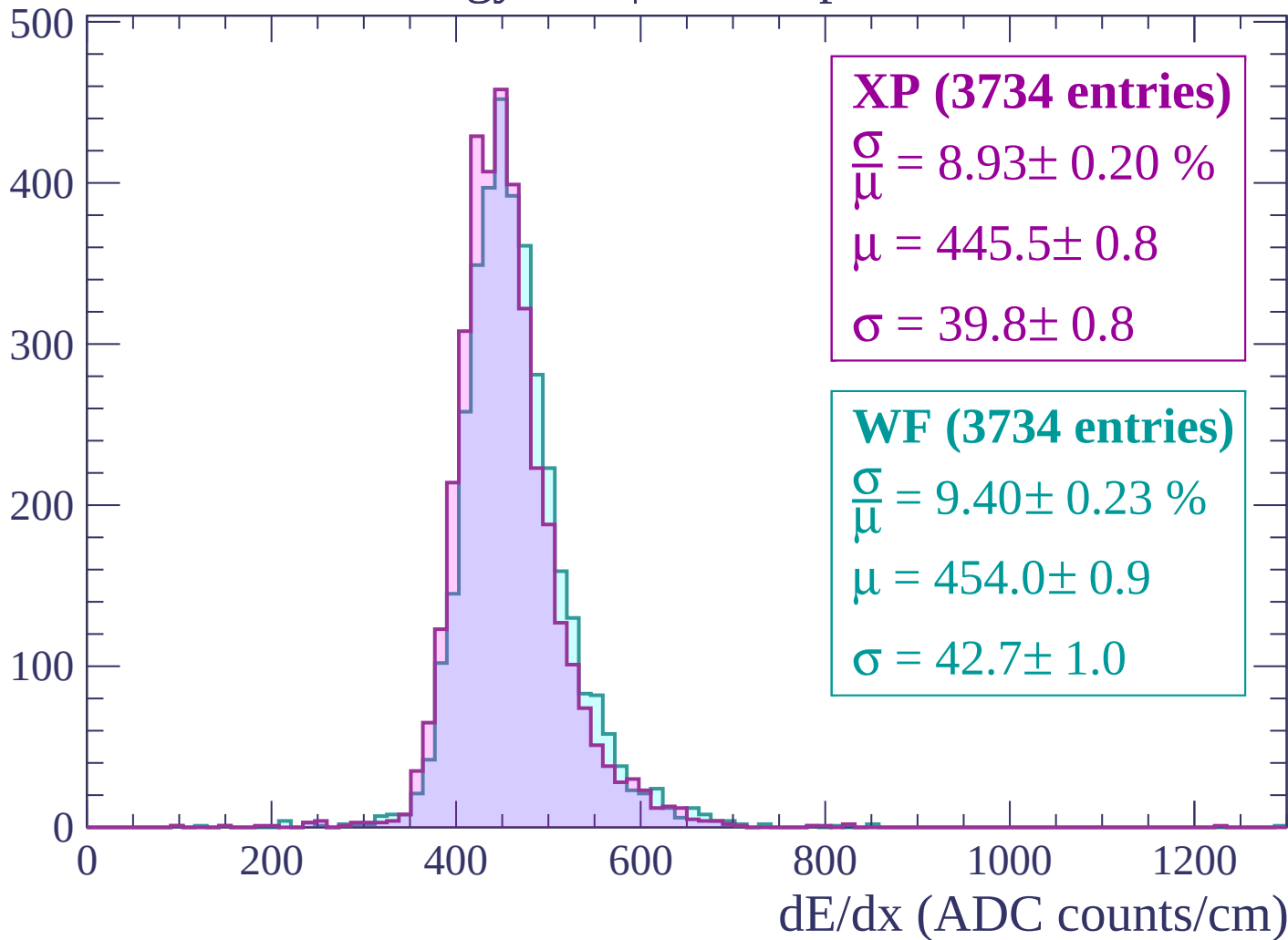
$\sigma = 41.9 \pm 0.8$

dE/dx (ADC counts/cm)



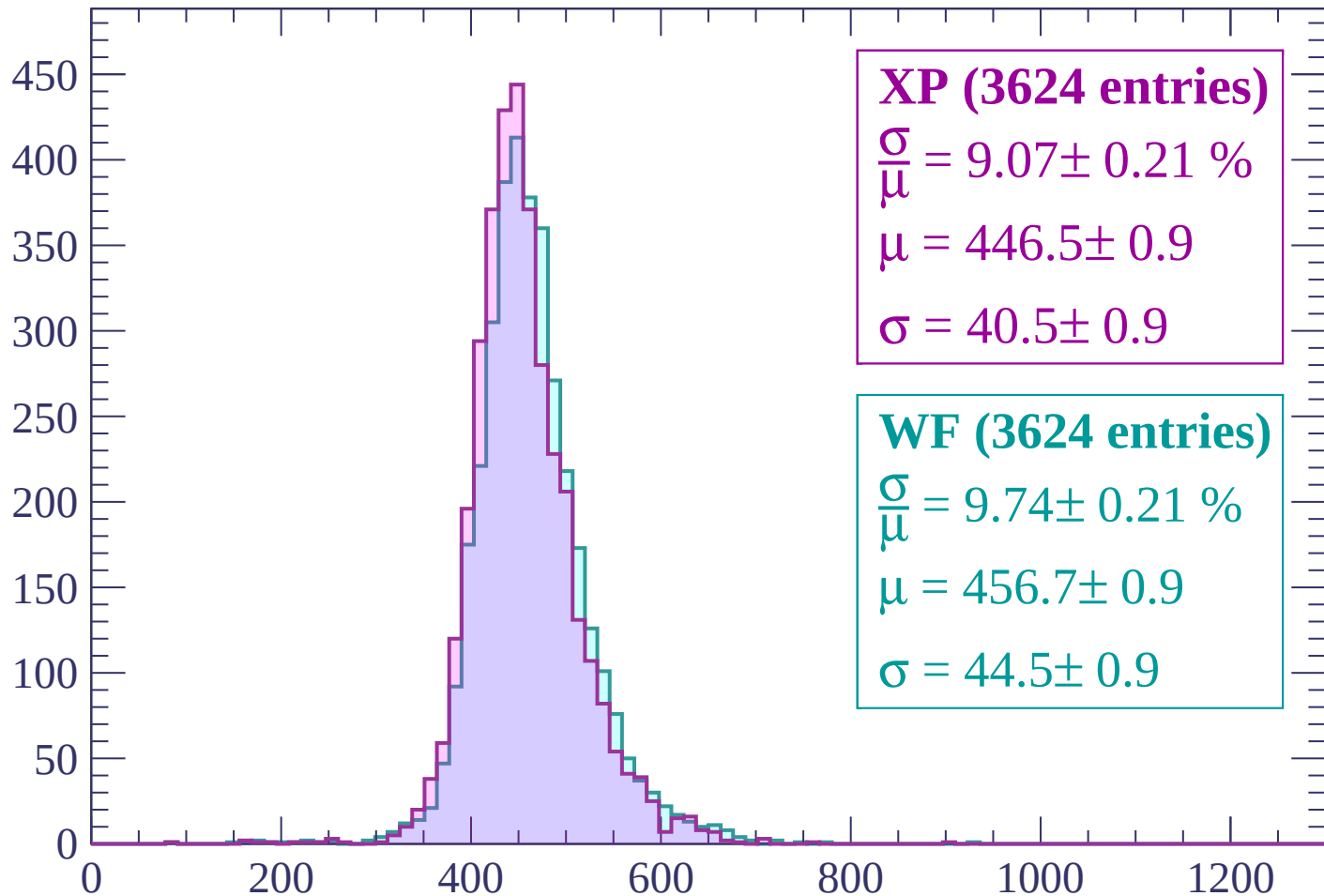
Energy loss |  $1440 < p < 1500$

Count



Energy loss |  $1500 < p < 1560$

Count



**XP (3624 entries)**

$$\frac{\sigma}{\mu} = 9.07 \pm 0.21 \%$$

$$\mu = 446.5 \pm 0.9$$

$$\sigma = 40.5 \pm 0.9$$

**WF (3624 entries)**

$$\frac{\sigma}{\mu} = 9.74 \pm 0.21 \%$$

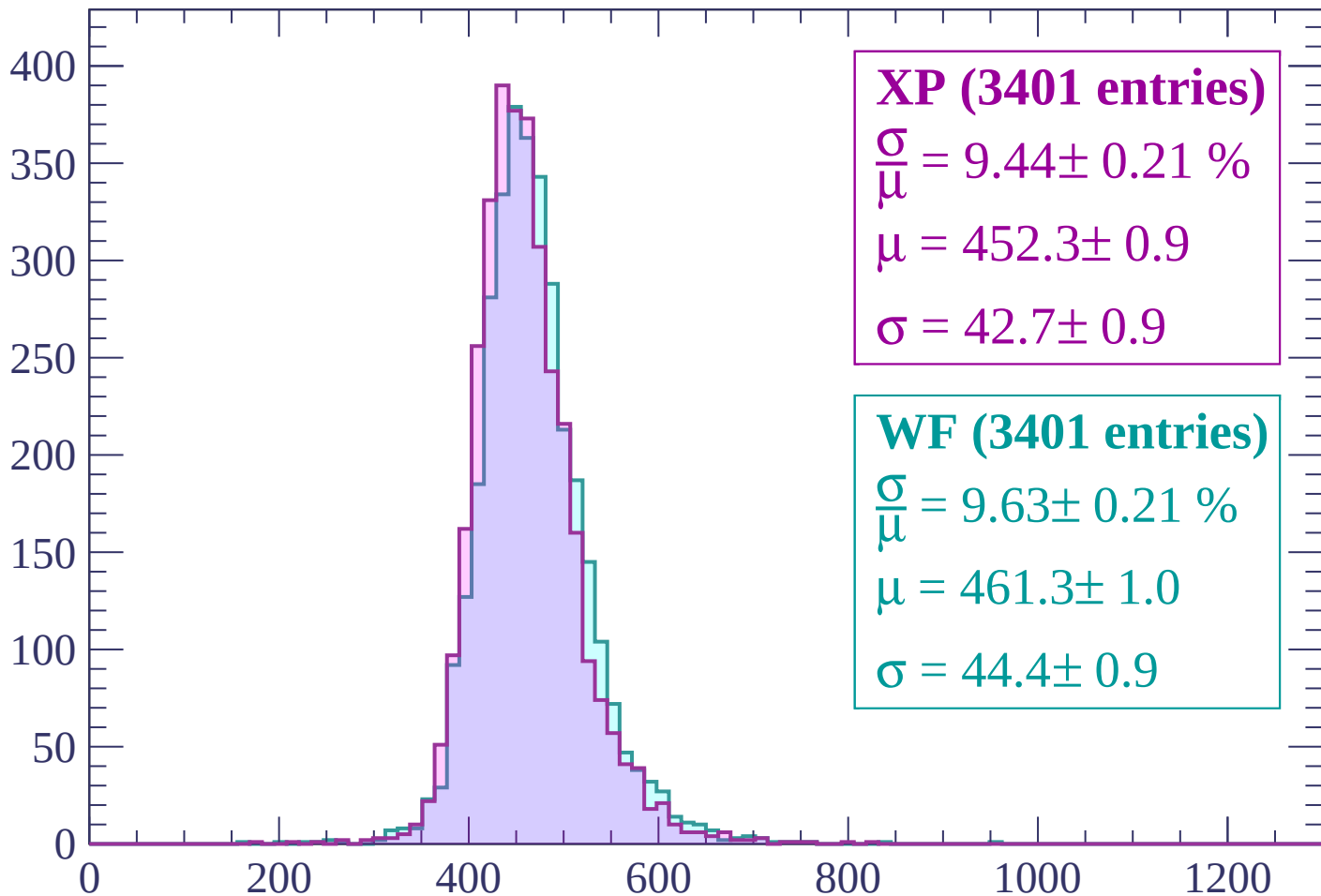
$$\mu = 456.7 \pm 0.9$$

$$\sigma = 44.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $1560 < p < 1620$

Count



**XP (3401 entries)**

$$\frac{\sigma}{\mu} = 9.44 \pm 0.21 \%$$

$$\mu = 452.3 \pm 0.9$$

$$\sigma = 42.7 \pm 0.9$$

**WF (3401 entries)**

$$\frac{\sigma}{\mu} = 9.63 \pm 0.21 \%$$

$$\mu = 461.3 \pm 1.0$$

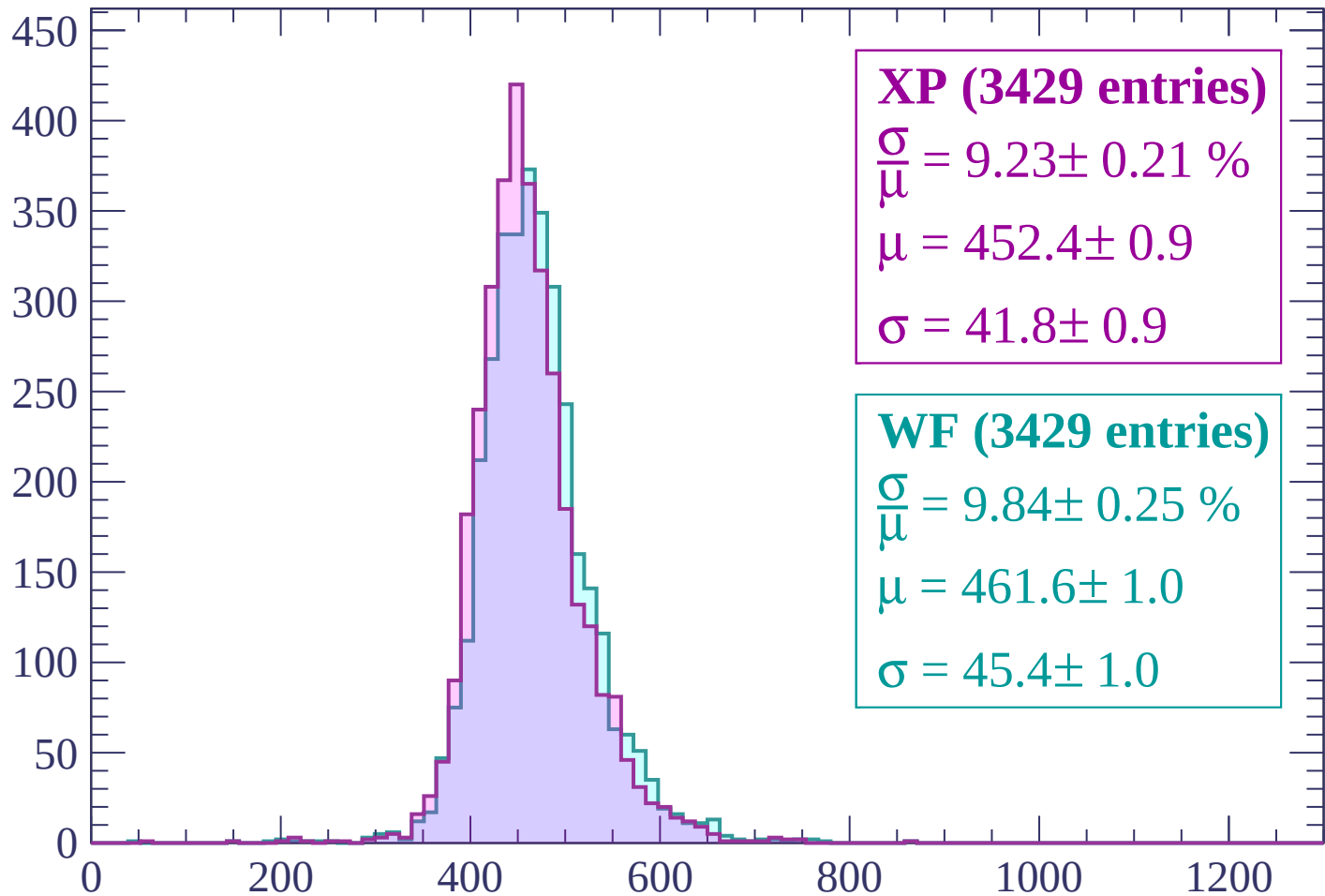
$$\sigma = 44.4 \pm 0.9$$

dE/dx (ADC counts/cm)



Energy loss |  $1620 < p < 1680$

Count



**XP (3429 entries)**

$\frac{\sigma}{\mu} = 9.23 \pm 0.21 \%$

$\mu = 452.4 \pm 0.9$

$\sigma = 41.8 \pm 0.9$

**WF (3429 entries)**

$\frac{\sigma}{\mu} = 9.84 \pm 0.25 \%$

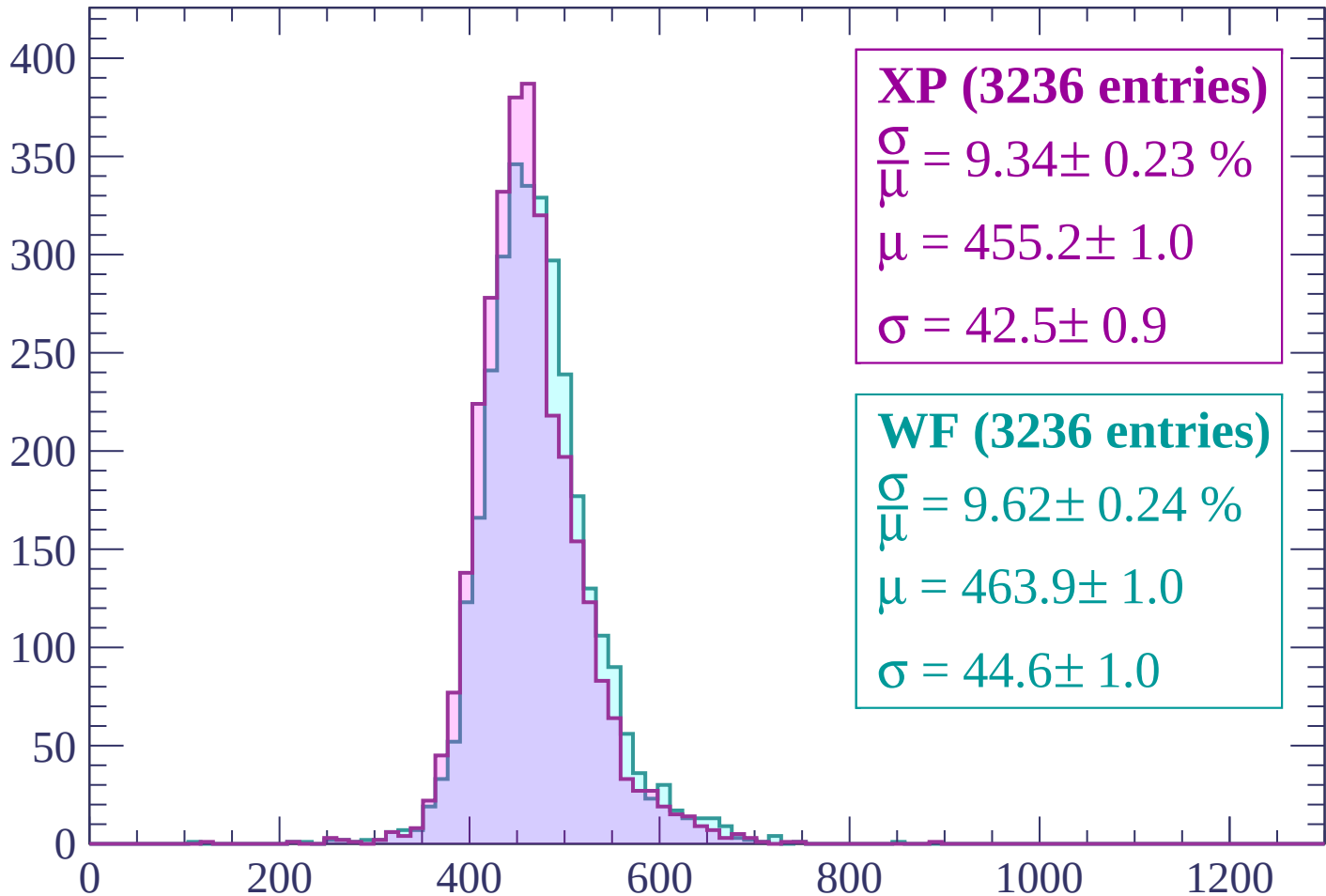
$\mu = 461.6 \pm 1.0$

$\sigma = 45.4 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss |  $1680 < p < 1740$

Count



**XP (3236 entries)**

$$\frac{\sigma}{\mu} = 9.34 \pm 0.23 \%$$

$$\mu = 455.2 \pm 1.0$$

$$\sigma = 42.5 \pm 0.9$$

**WF (3236 entries)**

$$\frac{\sigma}{\mu} = 9.62 \pm 0.24 \%$$

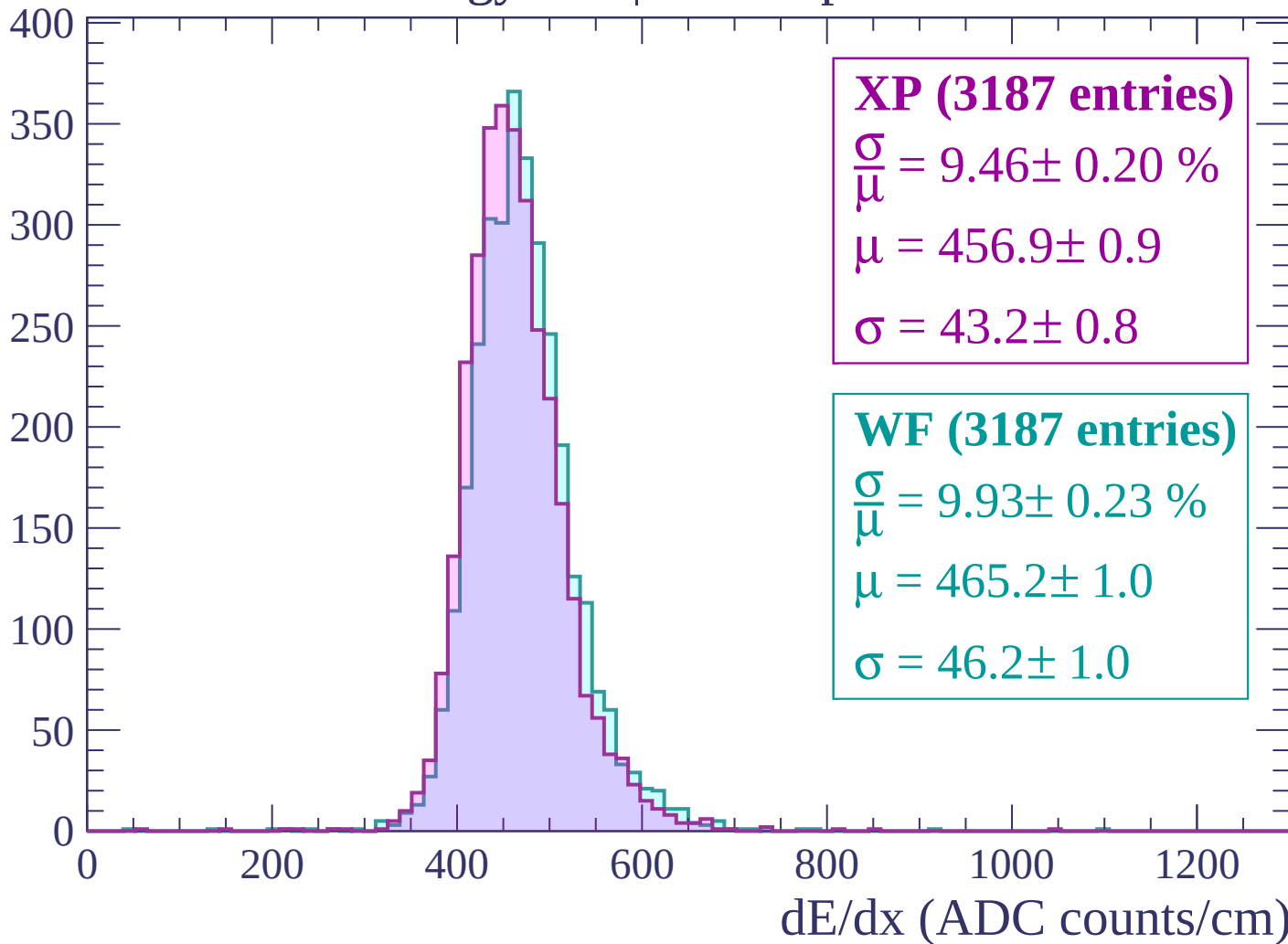
$$\mu = 463.9 \pm 1.0$$

$$\sigma = 44.6 \pm 1.0$$

dE/dx (ADC counts/cm)

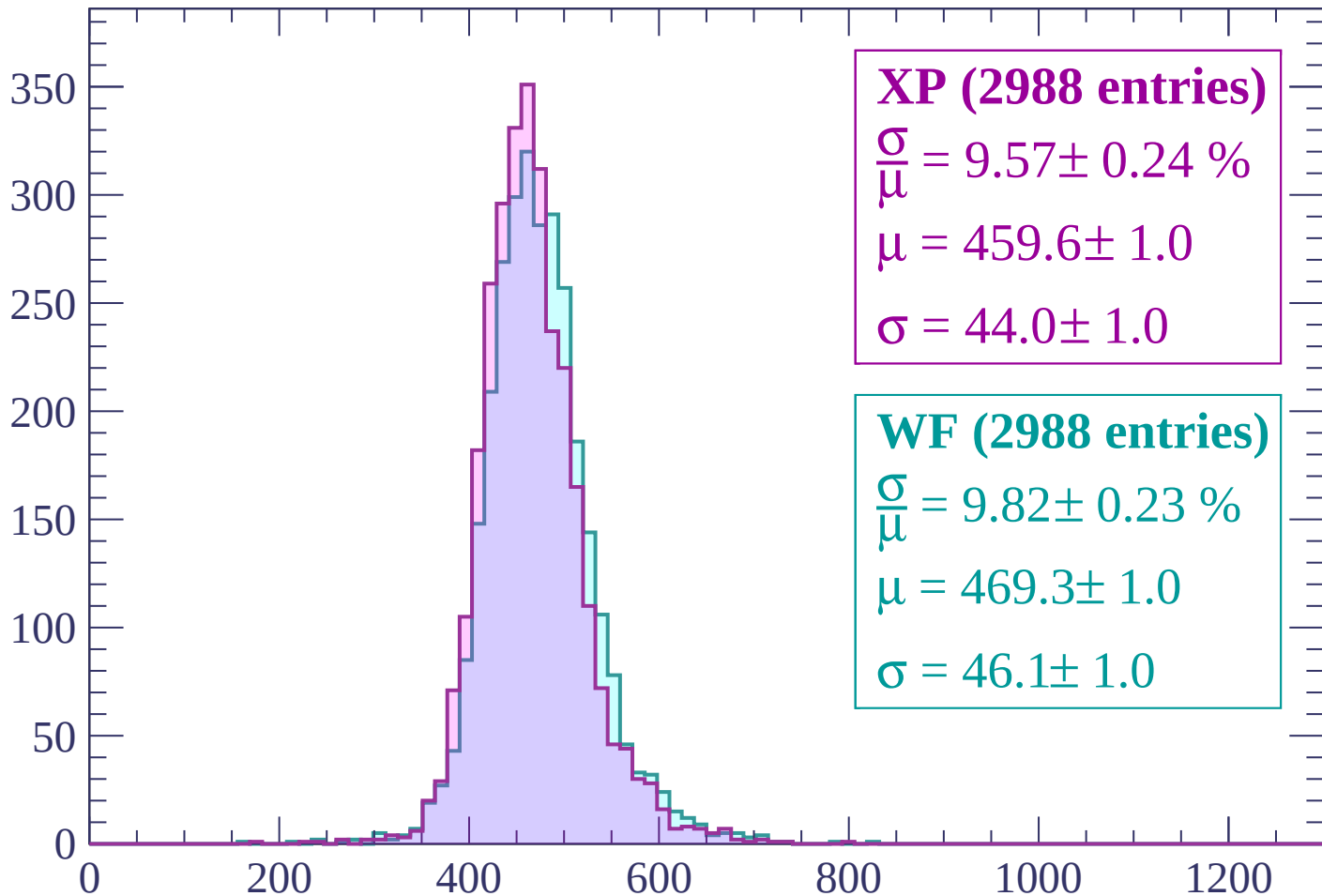
Energy loss |  $1740 < p < 1800$

Count



Energy loss |  $1800 < p < 1860$

Count



**XP (2988 entries)**

$$\frac{\sigma}{\mu} = 9.57 \pm 0.24 \%$$

$$\mu = 459.6 \pm 1.0$$

$$\sigma = 44.0 \pm 1.0$$

**WF (2988 entries)**

$$\frac{\sigma}{\mu} = 9.82 \pm 0.23 \%$$

$$\mu = 469.3 \pm 1.0$$

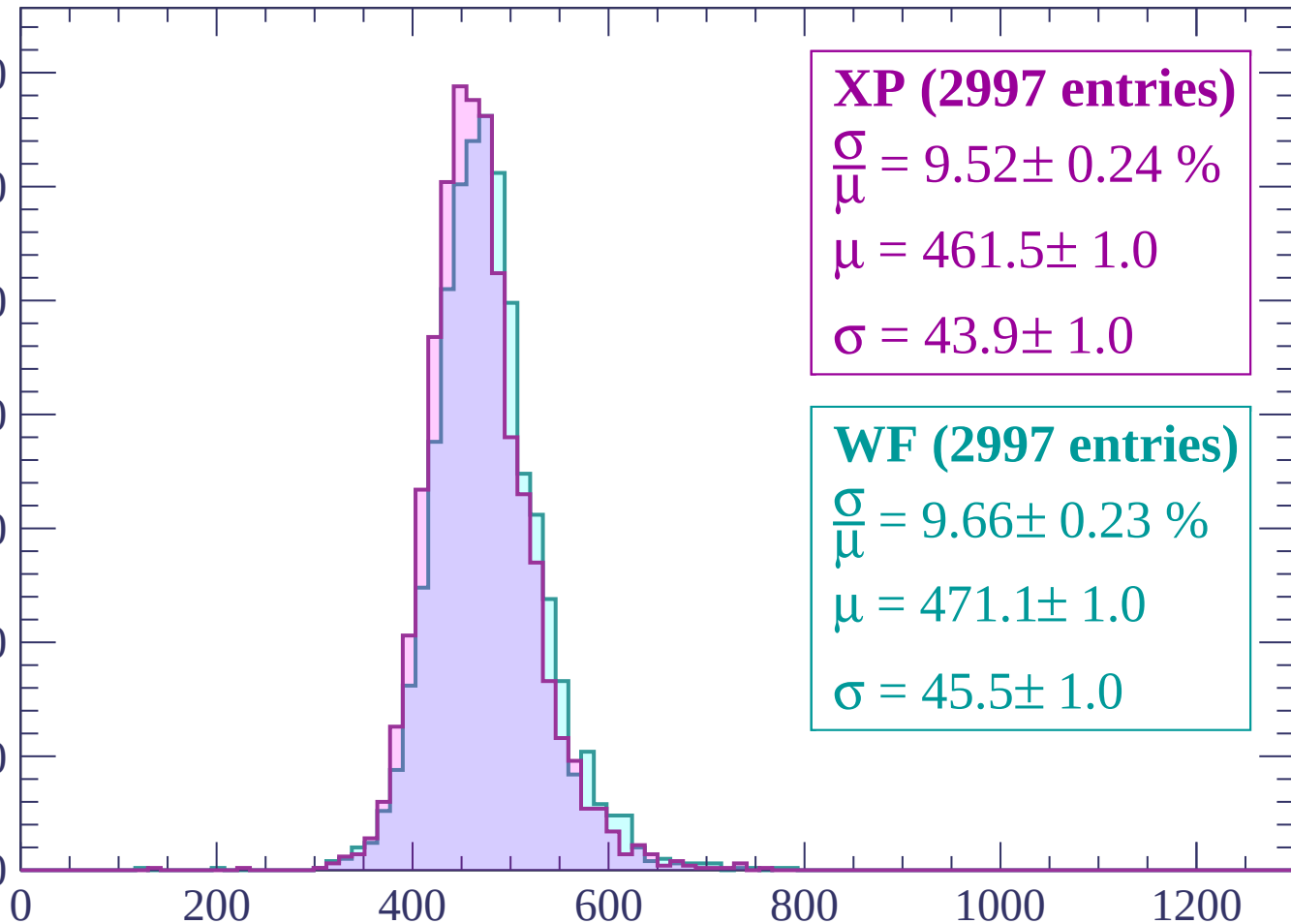
$$\sigma = 46.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $1860 < p < 1920$

Count

350  
300  
250  
200  
150  
100  
50  
0



**XP (2997 entries)**

$$\frac{\sigma}{\mu} = 9.52 \pm 0.24 \%$$

$$\mu = 461.5 \pm 1.0$$

$$\sigma = 43.9 \pm 1.0$$

**WF (2997 entries)**

$$\frac{\sigma}{\mu} = 9.66 \pm 0.23 \%$$

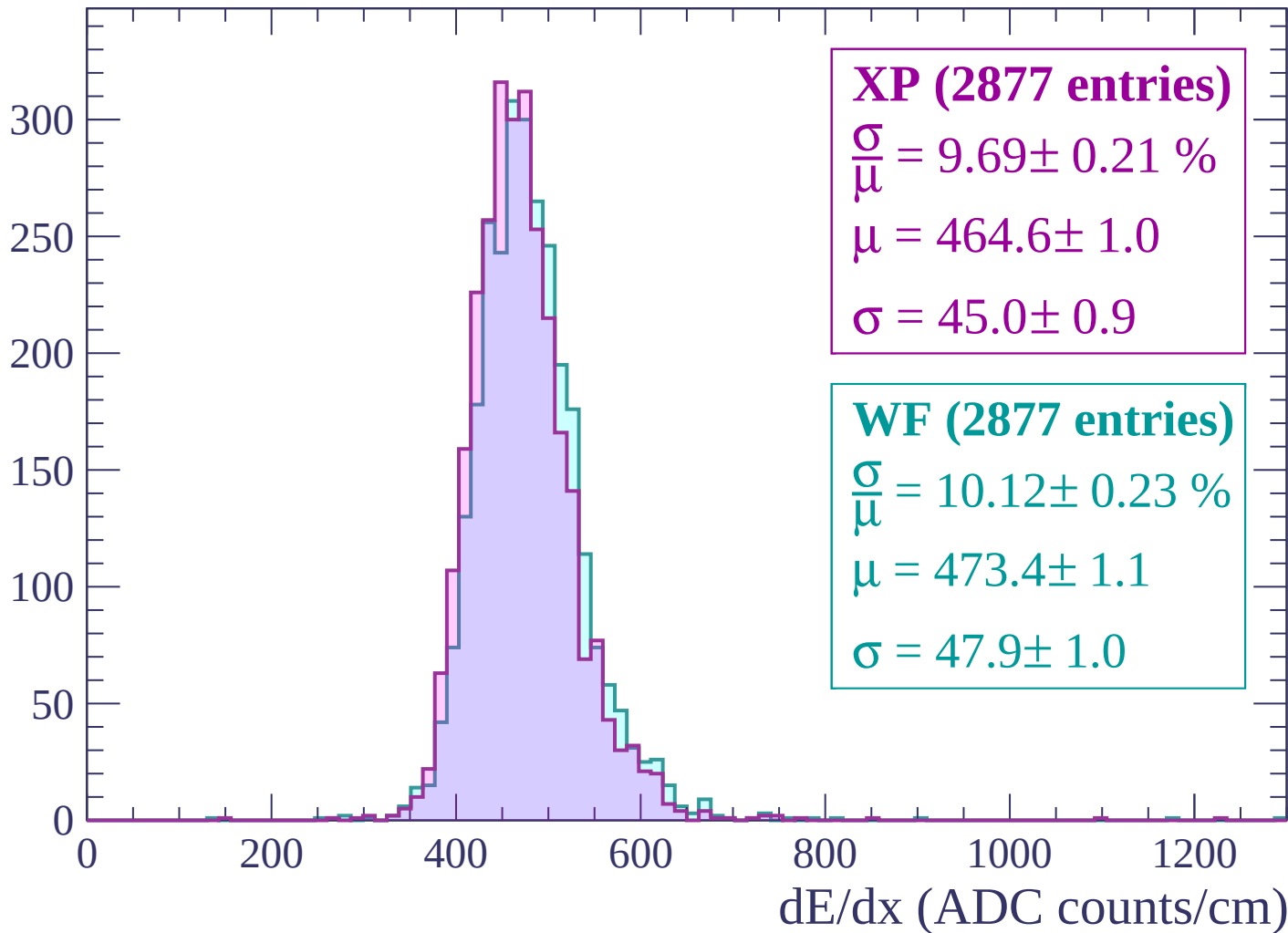
$$\mu = 471.1 \pm 1.0$$

$$\sigma = 45.5 \pm 1.0$$

dE/dx (ADC counts/cm)

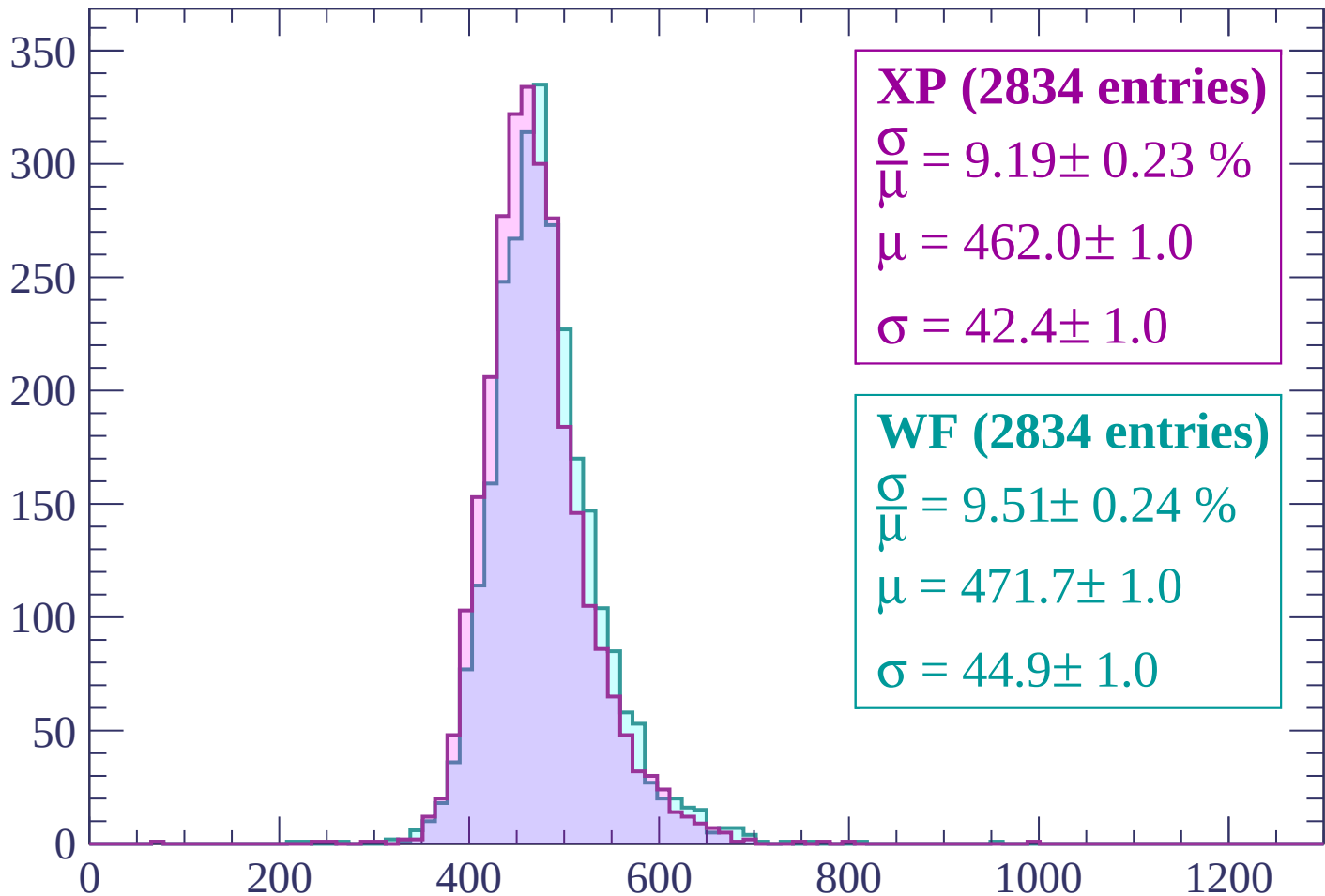
Energy loss |  $1920 < p < 1980$

Count



Energy loss |  $1980 < p < 2040$

Count



**XP (2834 entries)**

$$\frac{\sigma}{\mu} = 9.19 \pm 0.23 \%$$

$$\mu = 462.0 \pm 1.0$$

$$\sigma = 42.4 \pm 1.0$$

**WF (2834 entries)**

$$\frac{\sigma}{\mu} = 9.51 \pm 0.24 \%$$

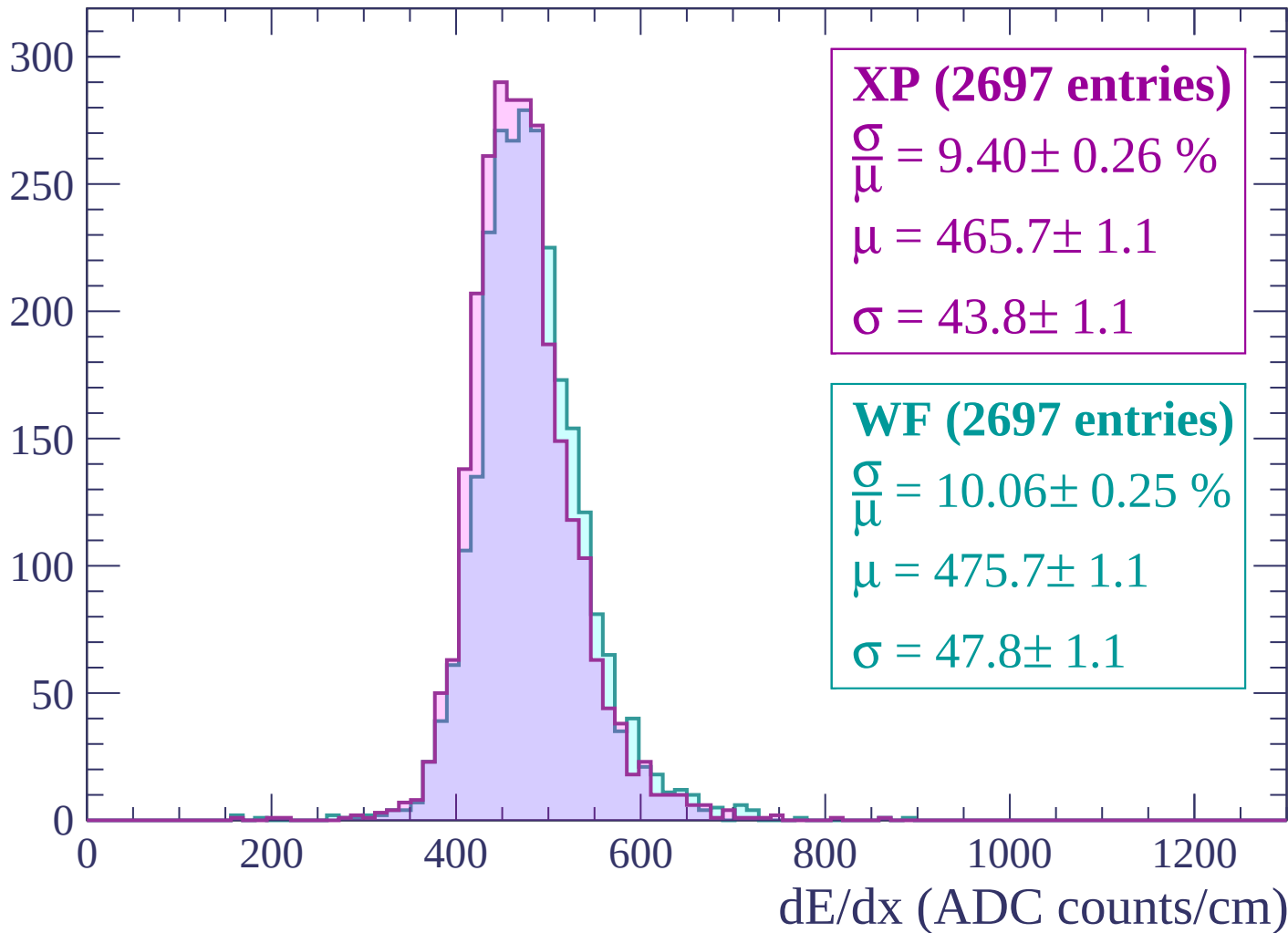
$$\mu = 471.7 \pm 1.0$$

$$\sigma = 44.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2040 < p < 2100$

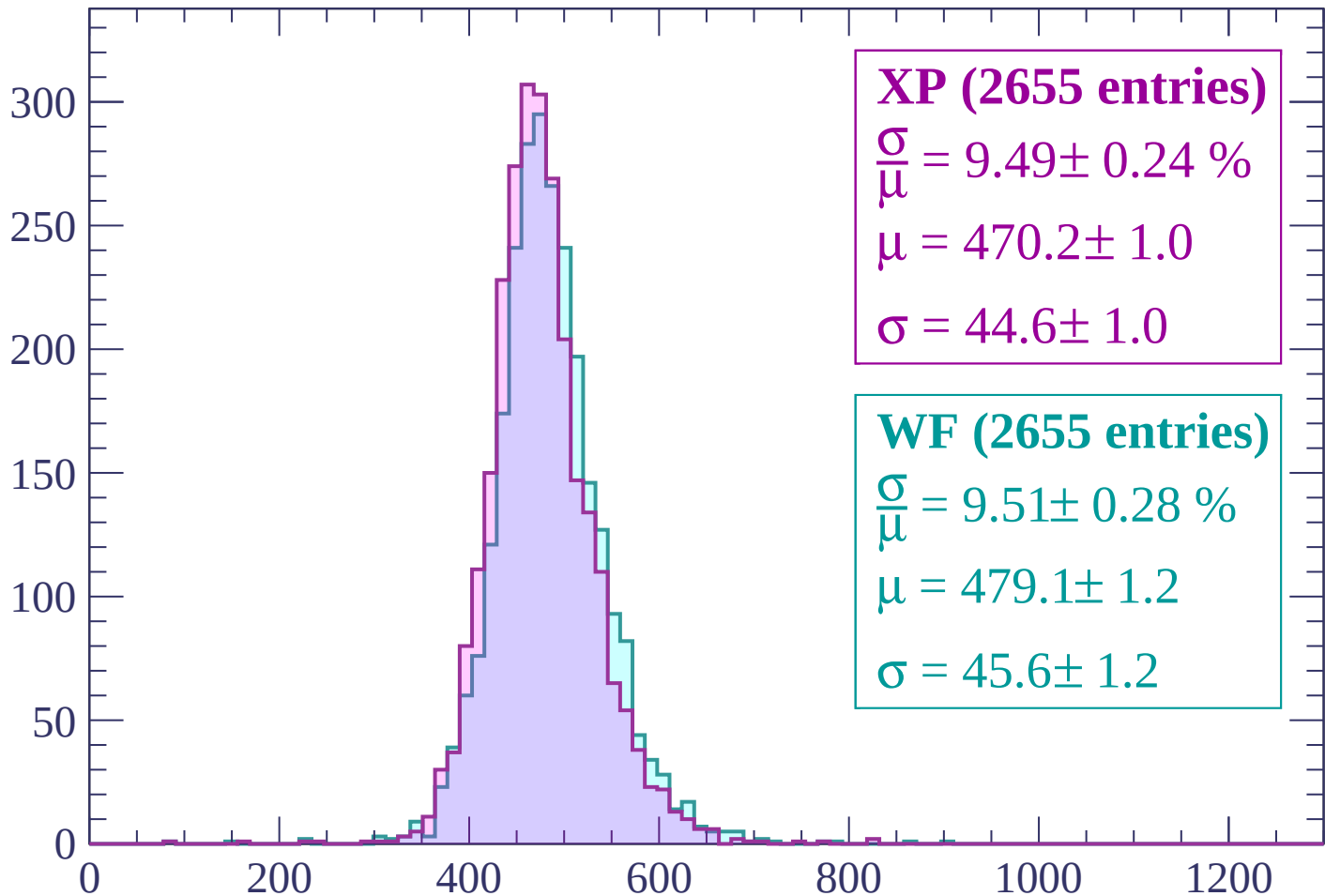
Count





Energy loss |  $2100 < p < 2160$

Count



**XP (2655 entries)**

$$\frac{\sigma}{\mu} = 9.49 \pm 0.24 \%$$

$$\mu = 470.2 \pm 1.0$$

$$\sigma = 44.6 \pm 1.0$$

**WF (2655 entries)**

$$\frac{\sigma}{\mu} = 9.51 \pm 0.28 \%$$

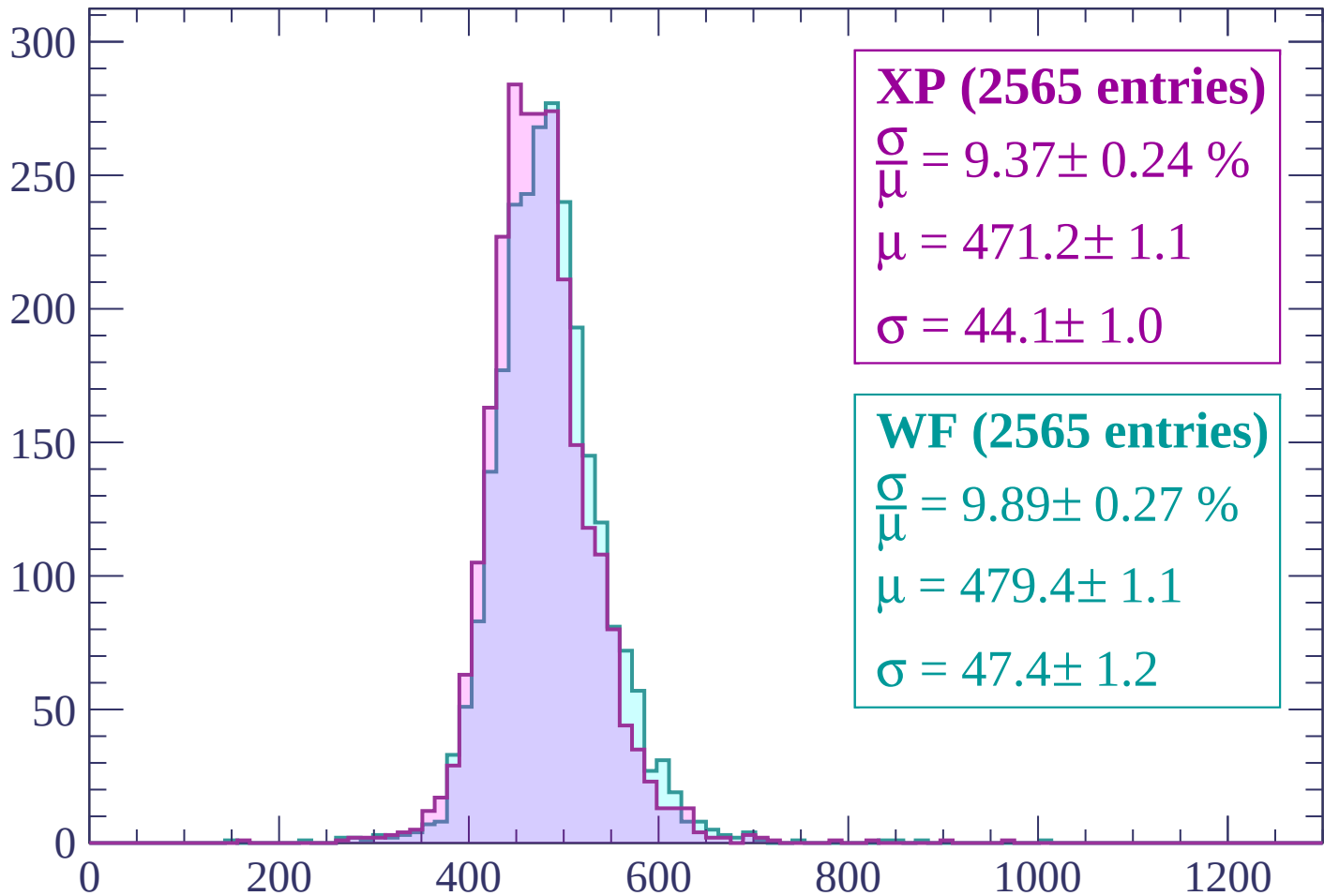
$$\mu = 479.1 \pm 1.2$$

$$\sigma = 45.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $2160 < p < 2220$

Count



**XP (2565 entries)**

$$\frac{\sigma}{\mu} = 9.37 \pm 0.24 \%$$

$$\mu = 471.2 \pm 1.1$$

$$\sigma = 44.1 \pm 1.0$$

**WF (2565 entries)**

$$\frac{\sigma}{\mu} = 9.89 \pm 0.27 \%$$

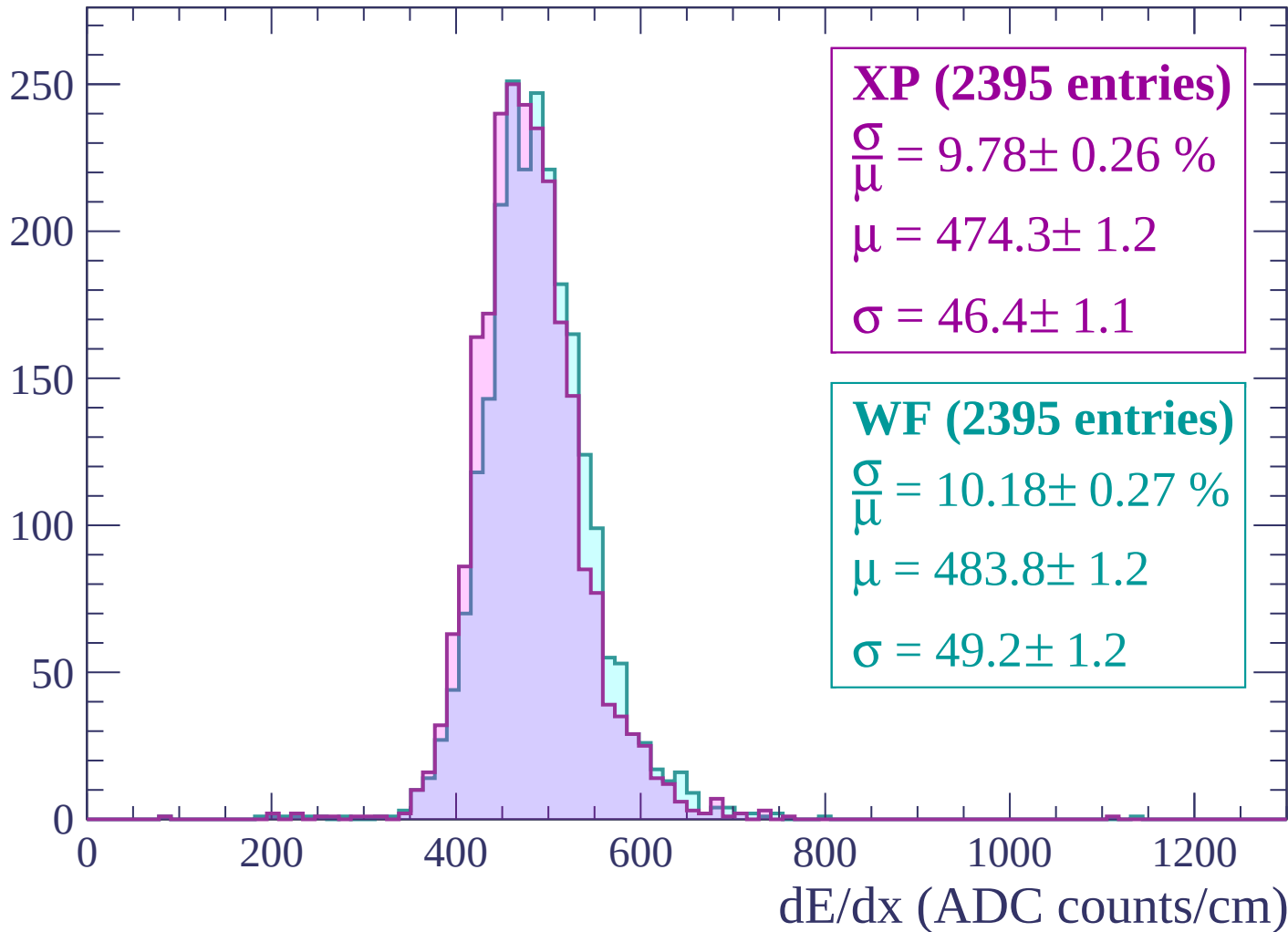
$$\mu = 479.4 \pm 1.1$$

$$\sigma = 47.4 \pm 1.2$$

dE/dx (ADC counts/cm)

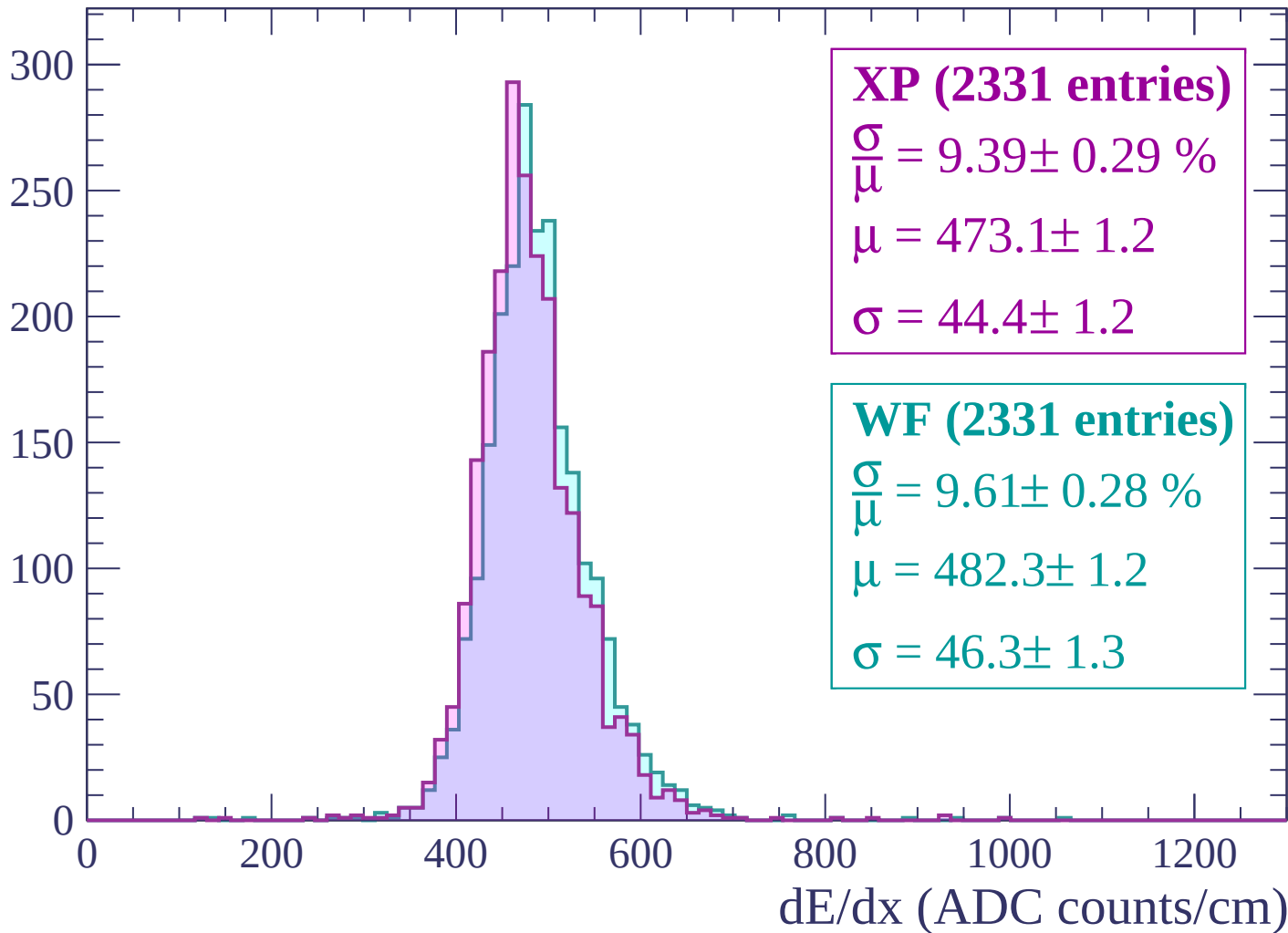
Energy loss |  $2220 < p < 2280$

Count



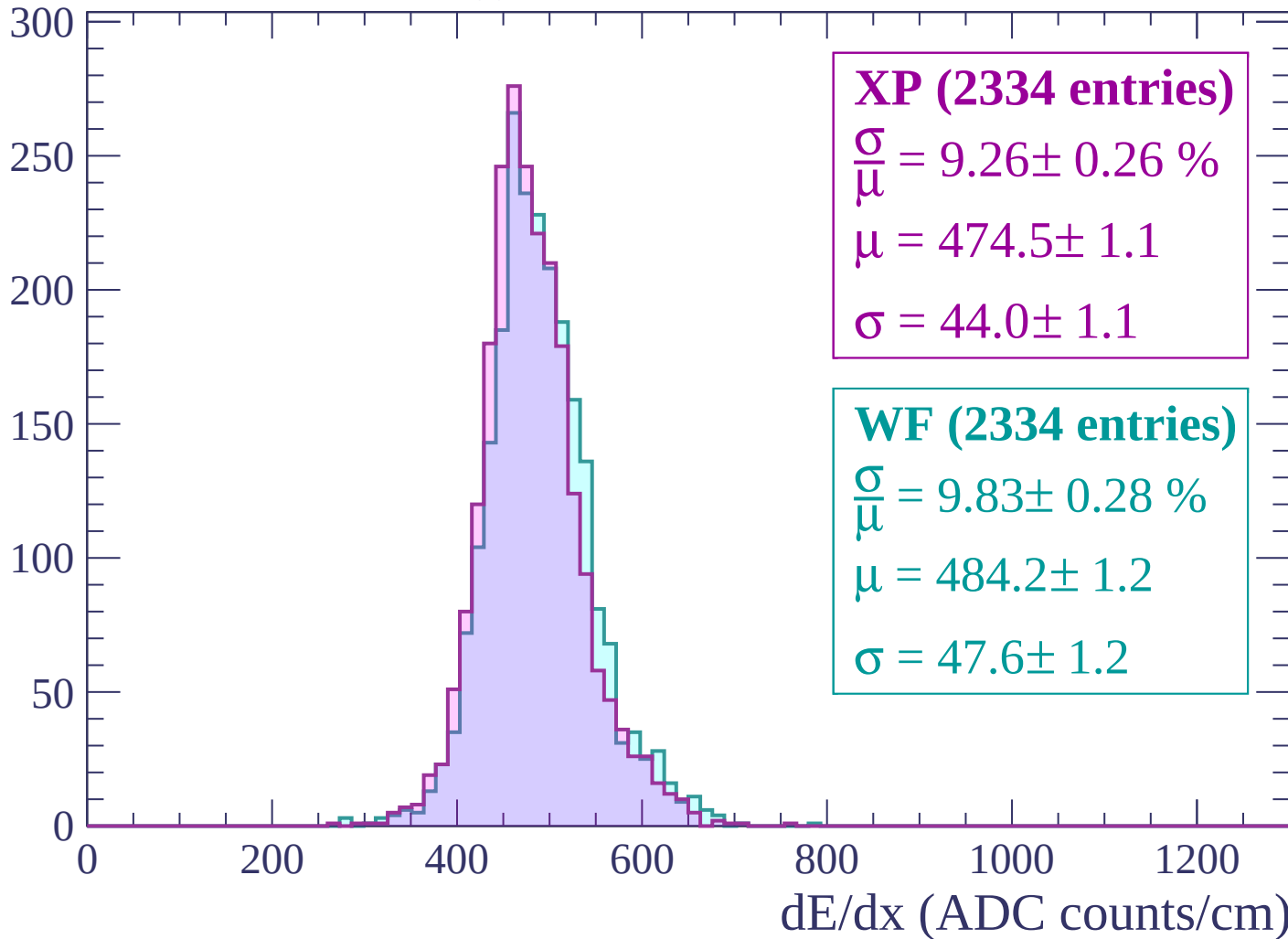
Energy loss |  $2280 < p < 2340$

Count



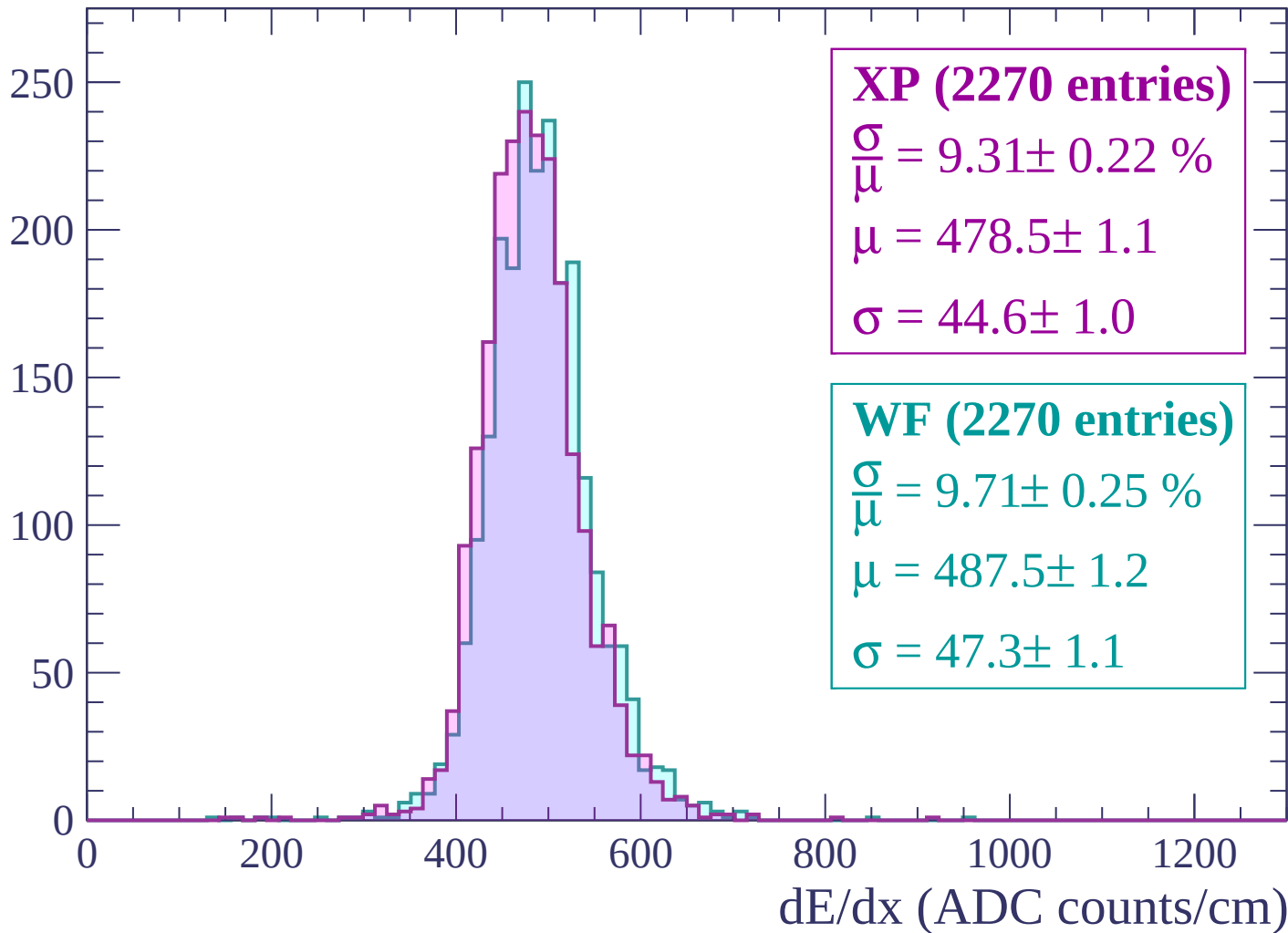
Energy loss |  $2340 < p < 2400$

Count



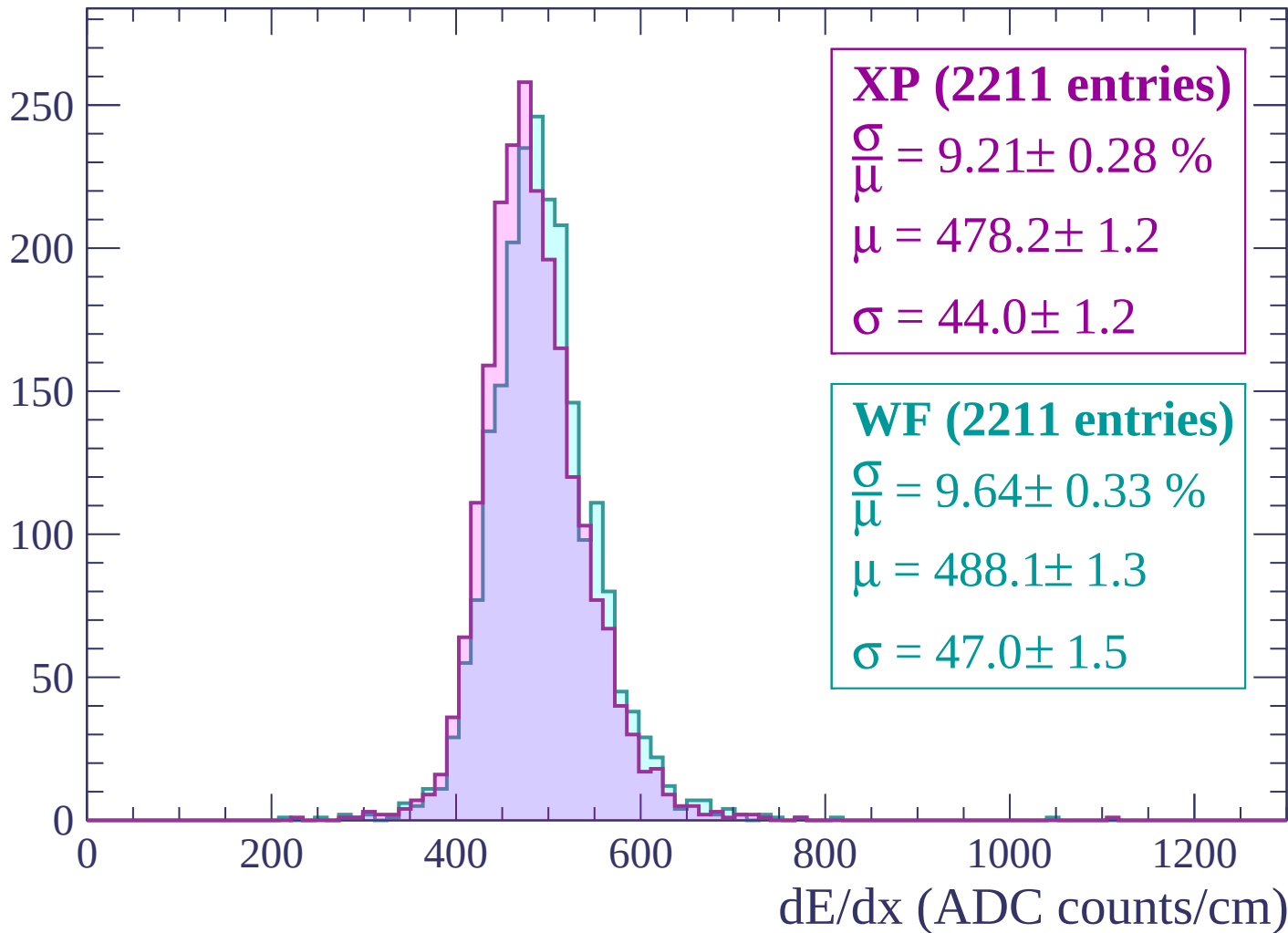
Energy loss |  $2400 < p < 2460$

Count



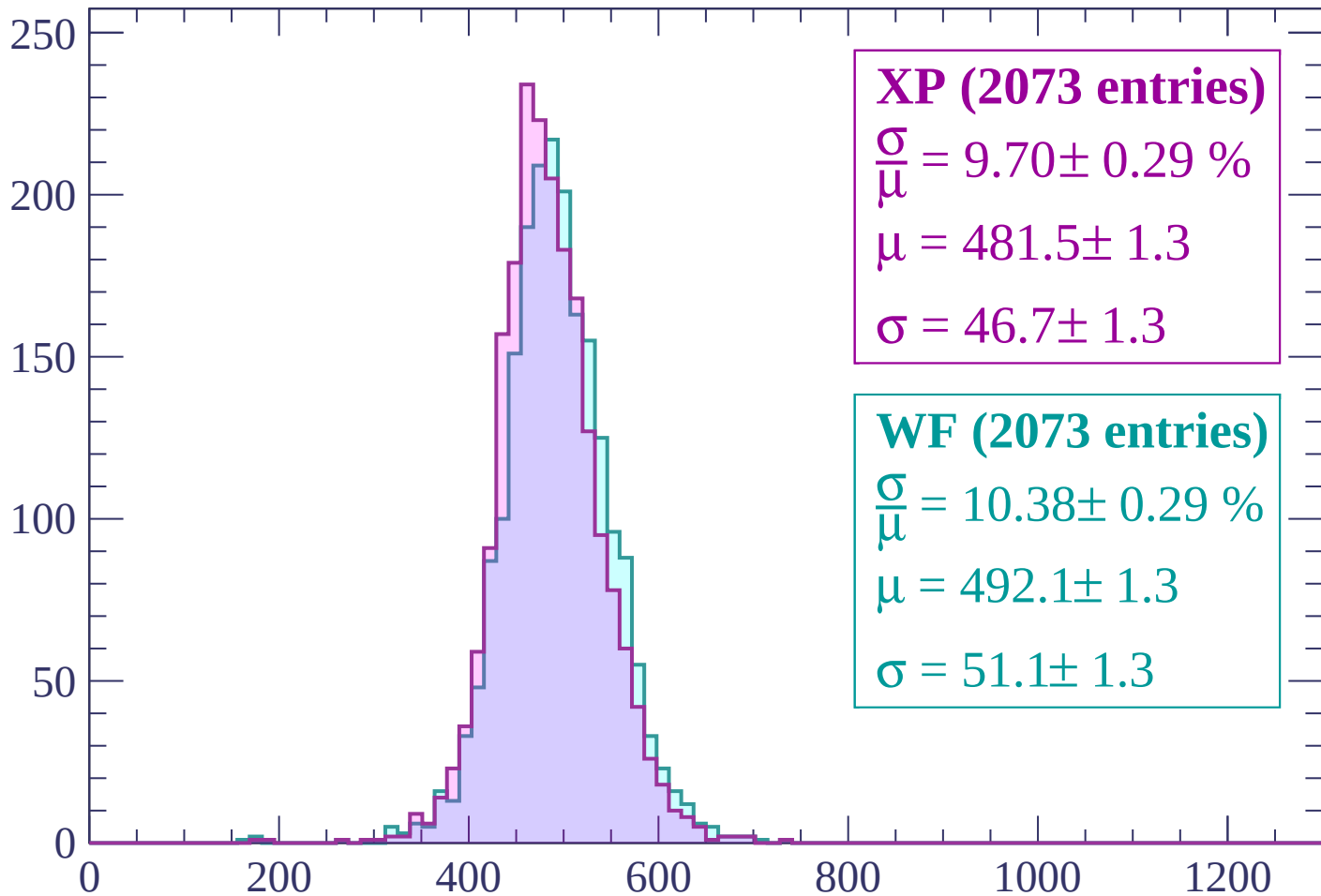
Energy loss |  $2460 < p < 2520$

Count



Energy loss |  $2520 < p < 2580$

Count



**XP (2073 entries)**

$$\frac{\sigma}{\mu} = 9.70 \pm 0.29 \%$$

$$\mu = 481.5 \pm 1.3$$

$$\sigma = 46.7 \pm 1.3$$

**WF (2073 entries)**

$$\frac{\sigma}{\mu} = 10.38 \pm 0.29 \%$$

$$\mu = 492.1 \pm 1.3$$

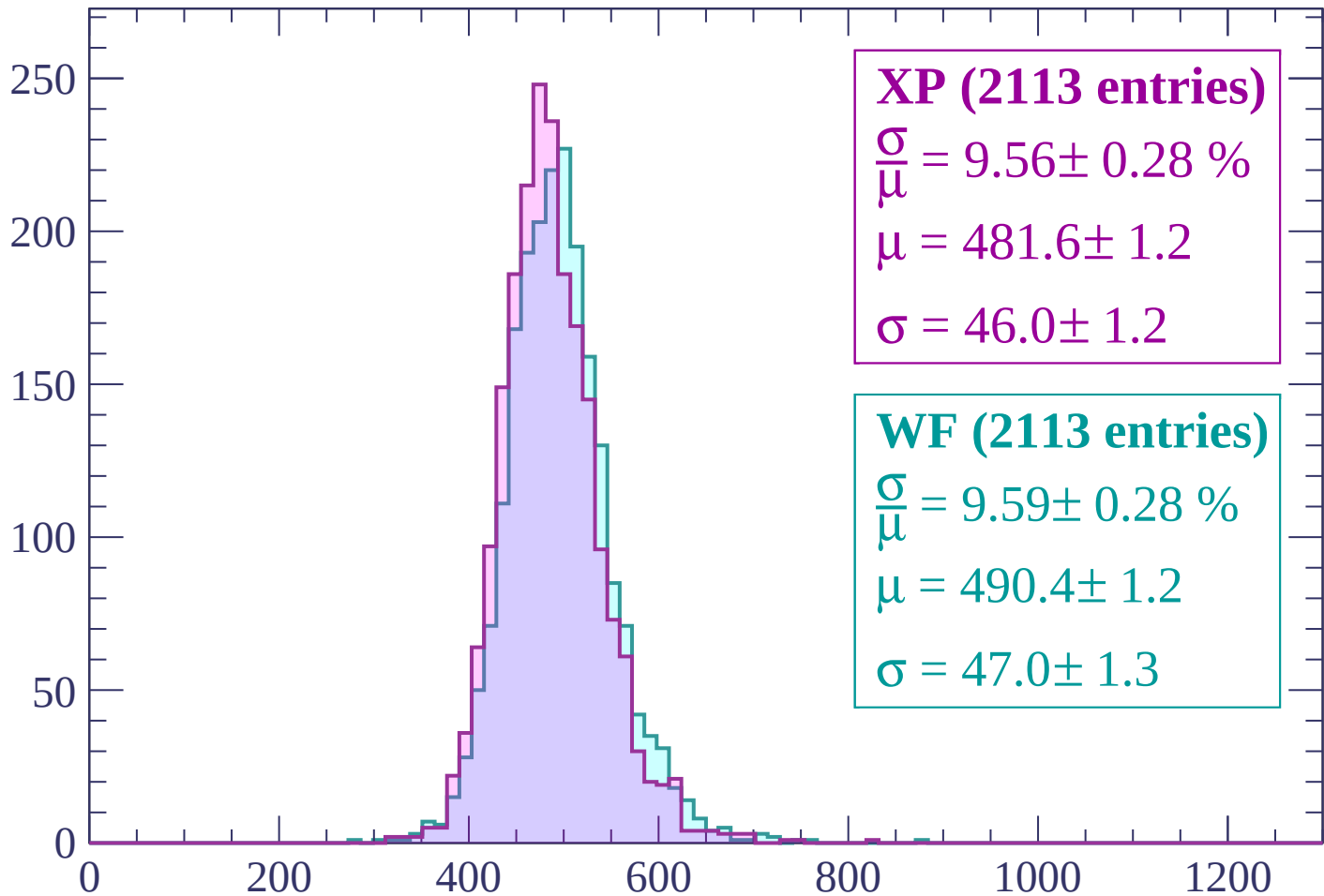
$$\sigma = 51.1 \pm 1.3$$

dE/dx (ADC counts/cm)



Energy loss |  $2580 < p < 2640$

Count



**XP (2113 entries)**

$$\frac{\sigma}{\mu} = 9.56 \pm 0.28 \%$$

$$\mu = 481.6 \pm 1.2$$

$$\sigma = 46.0 \pm 1.2$$

**WF (2113 entries)**

$$\frac{\sigma}{\mu} = 9.59 \pm 0.28 \%$$

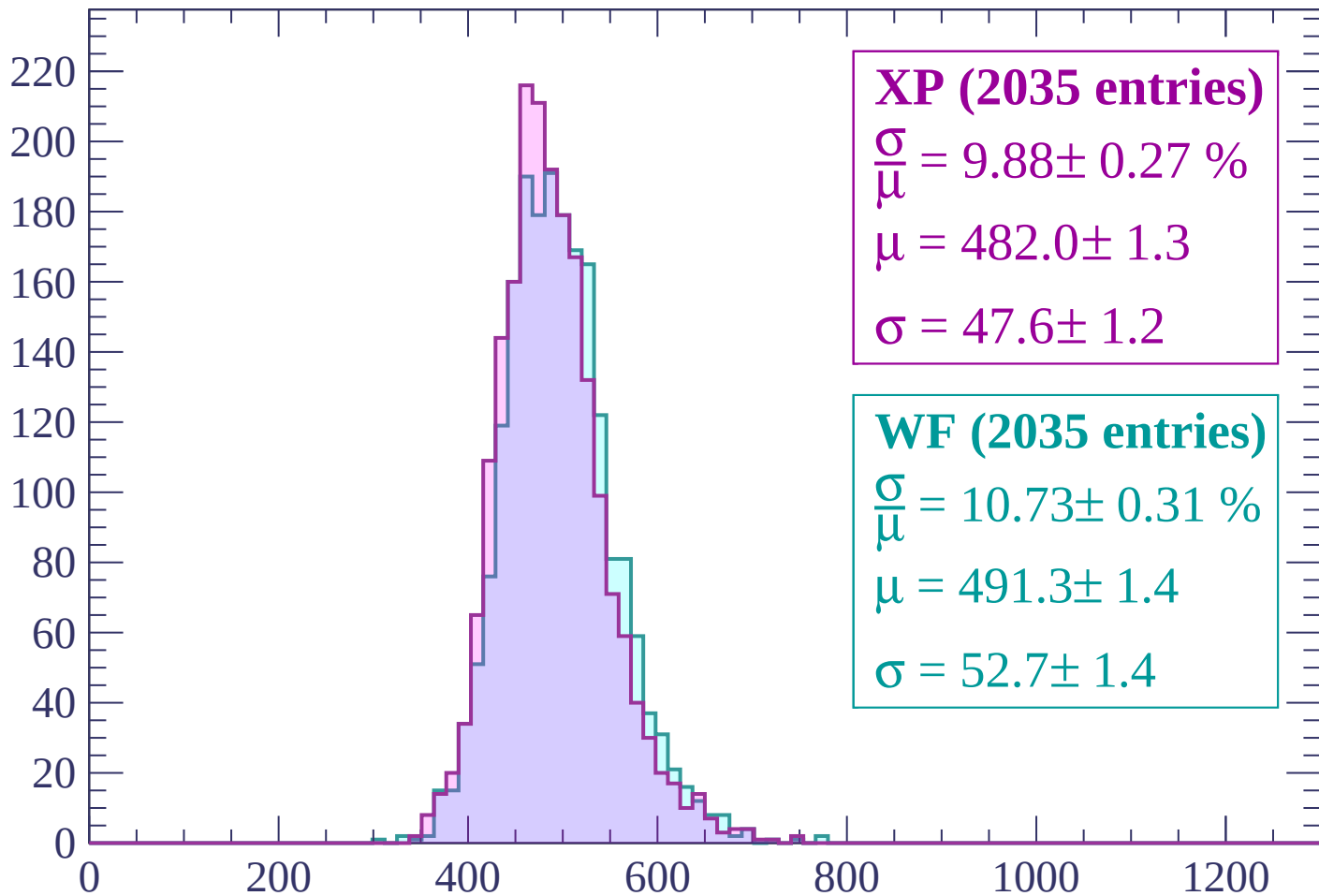
$$\mu = 490.4 \pm 1.2$$

$$\sigma = 47.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss |  $2640 < p < 2700$

Count



**XP (2035 entries)**

$$\frac{\sigma}{\mu} = 9.88 \pm 0.27 \%$$

$$\mu = 482.0 \pm 1.3$$

$$\sigma = 47.6 \pm 1.2$$

**WF (2035 entries)**

$$\frac{\sigma}{\mu} = 10.73 \pm 0.31 \%$$

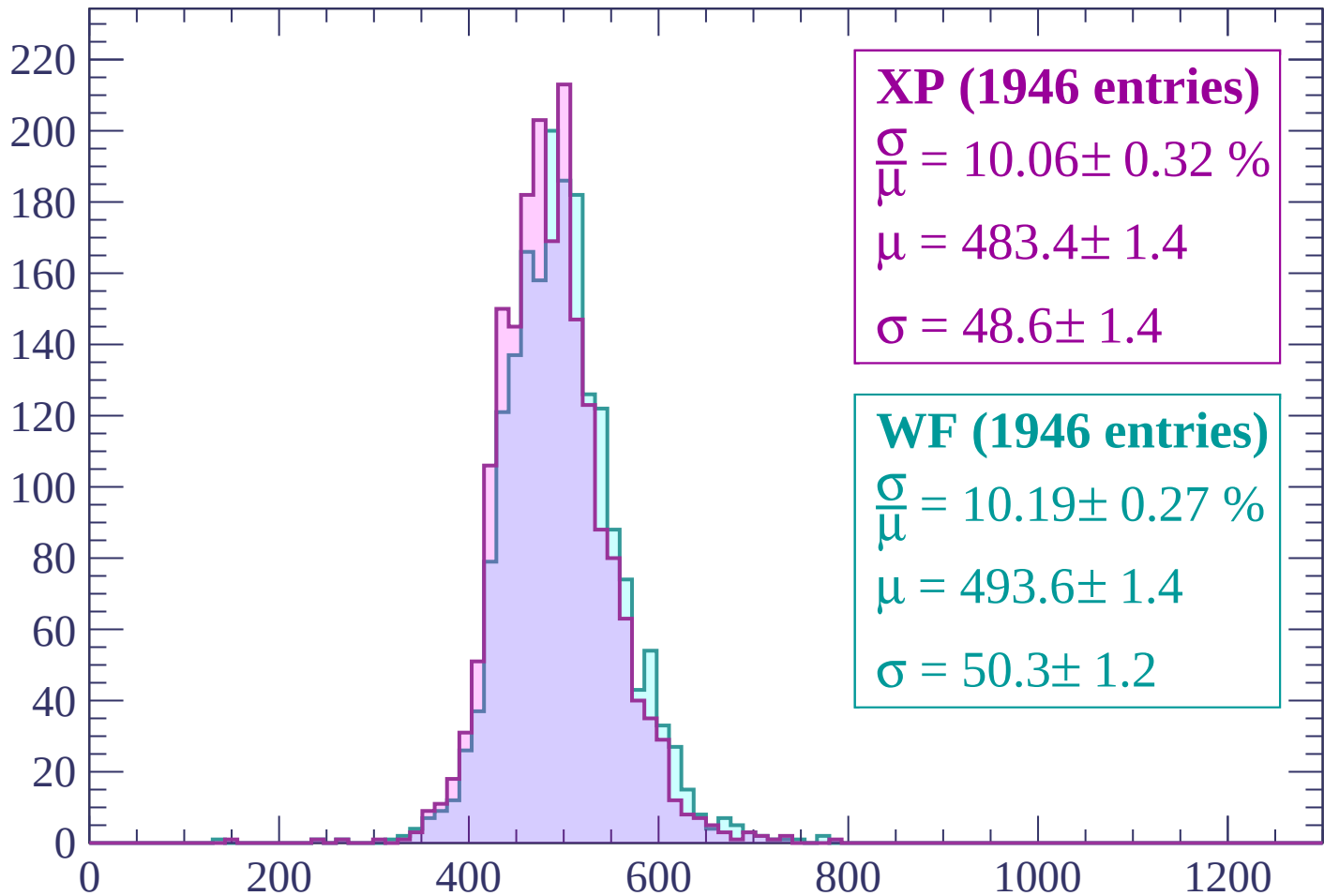
$$\mu = 491.3 \pm 1.4$$

$$\sigma = 52.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss |  $2700 < p < 2760$

Count



**XP (1946 entries)**

$$\frac{\sigma}{\mu} = 10.06 \pm 0.32 \%$$

$$\mu = 483.4 \pm 1.4$$

$$\sigma = 48.6 \pm 1.4$$

**WF (1946 entries)**

$$\frac{\sigma}{\mu} = 10.19 \pm 0.27 \%$$

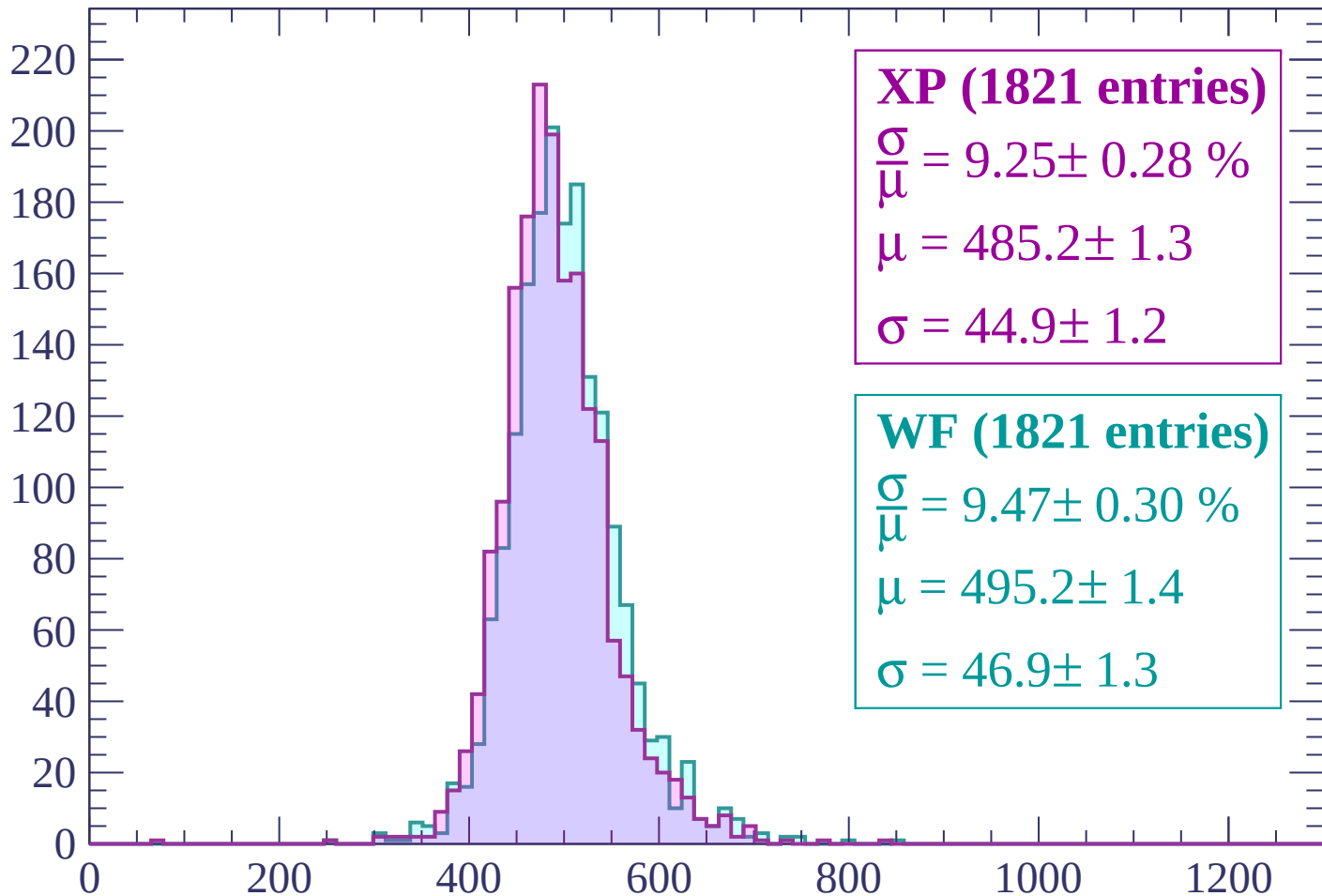
$$\mu = 493.6 \pm 1.4$$

$$\sigma = 50.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $2760 < p < 2820$

Count



**XP (1821 entries)**

$$\frac{\sigma}{\mu} = 9.25 \pm 0.28 \%$$

$$\mu = 485.2 \pm 1.3$$

$$\sigma = 44.9 \pm 1.2$$

**WF (1821 entries)**

$$\frac{\sigma}{\mu} = 9.47 \pm 0.30 \%$$

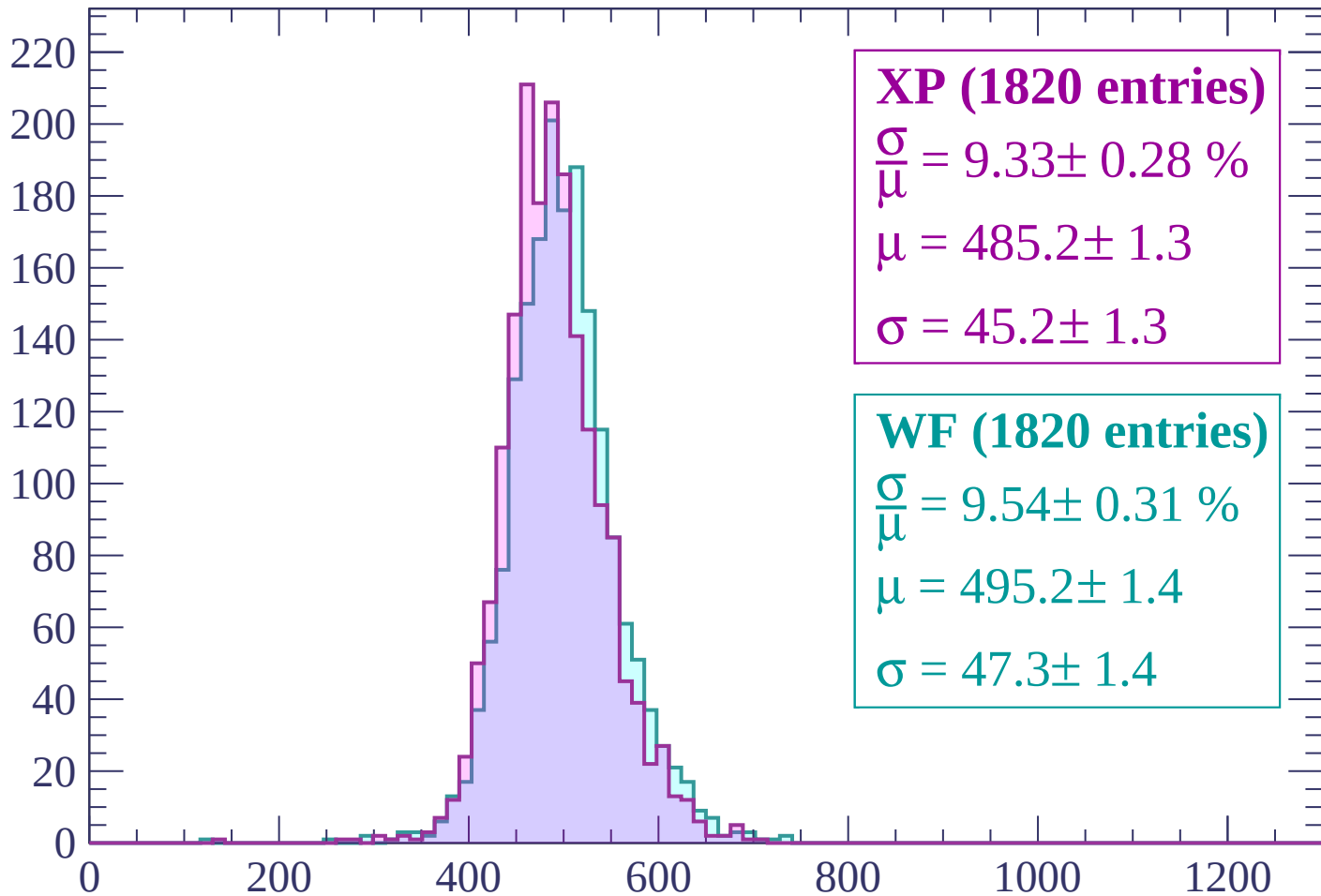
$$\mu = 495.2 \pm 1.4$$

$$\sigma = 46.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss |  $2820 < p < 2880$

Count



**XP (1820 entries)**

$$\frac{\sigma}{\mu} = 9.33 \pm 0.28 \%$$

$$\mu = 485.2 \pm 1.3$$

$$\sigma = 45.2 \pm 1.3$$

**WF (1820 entries)**

$$\frac{\sigma}{\mu} = 9.54 \pm 0.31 \%$$

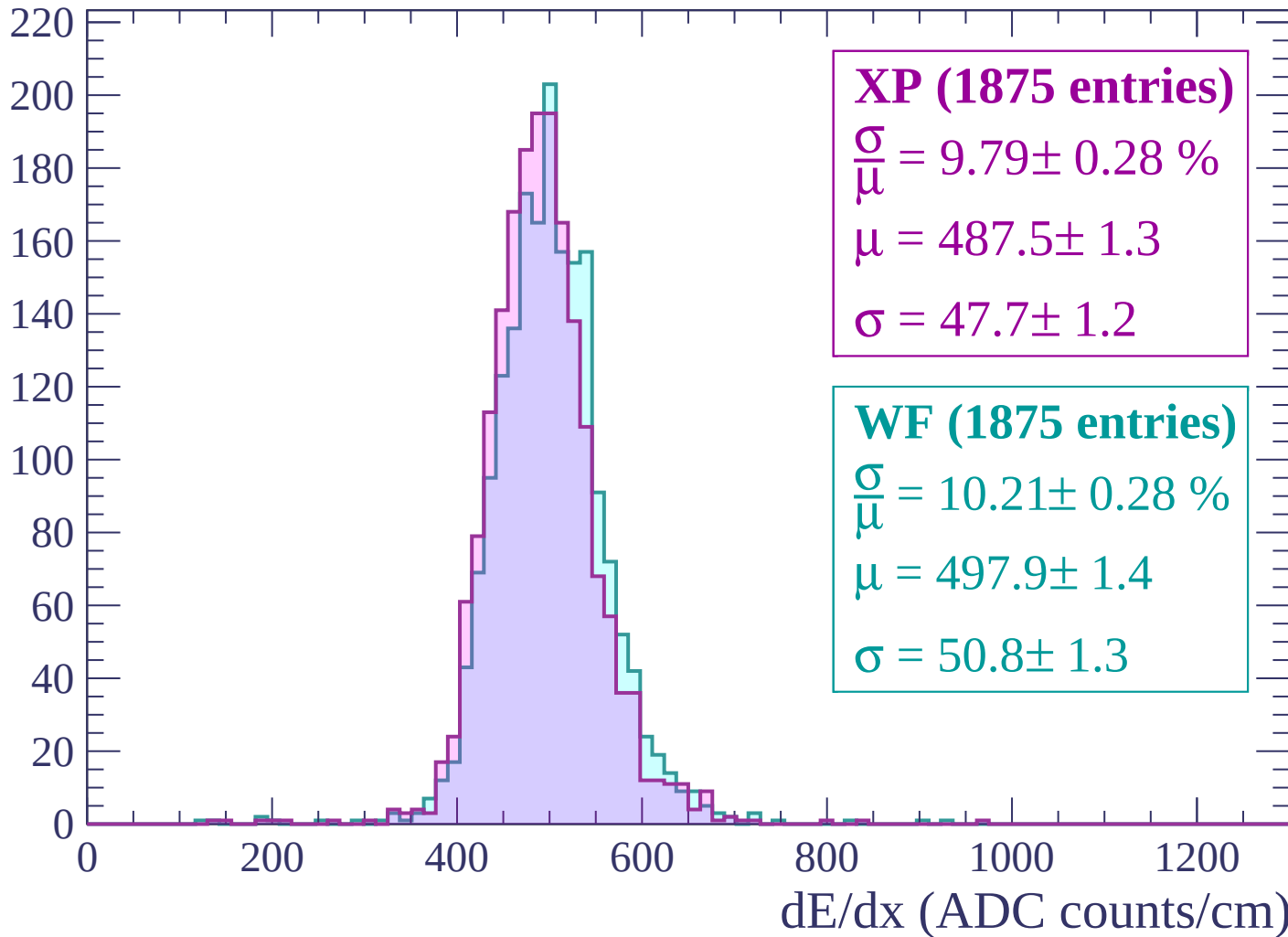
$$\mu = 495.2 \pm 1.4$$

$$\sigma = 47.3 \pm 1.4$$

dE/dx (ADC counts/cm)

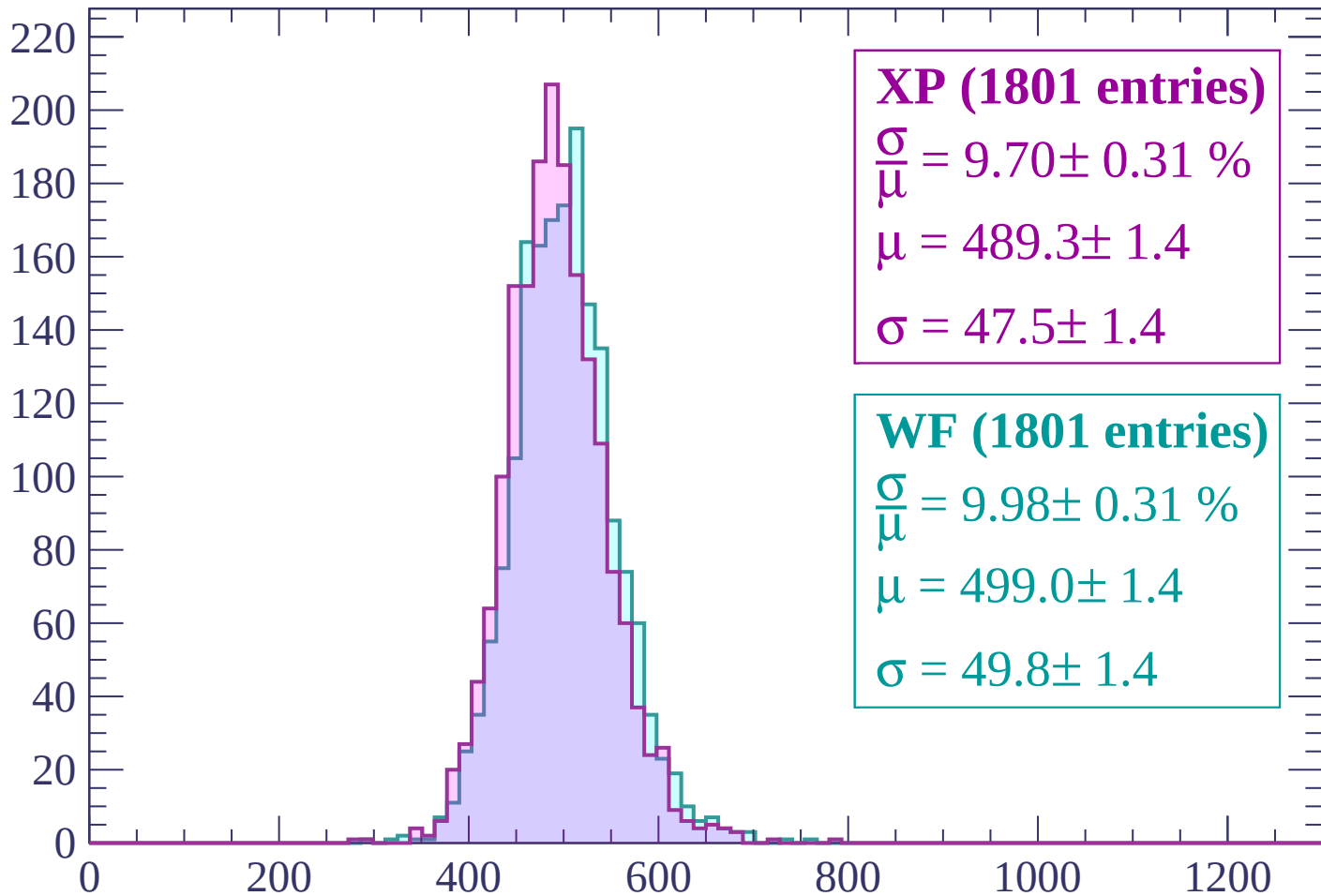
Energy loss |  $2880 < p < 2940$

Count



Energy loss |  $2940 < p < 3000$

Count



**XP (1801 entries)**

$$\frac{\sigma}{\mu} = 9.70 \pm 0.31 \%$$

$$\mu = 489.3 \pm 1.4$$

$$\sigma = 47.5 \pm 1.4$$

**WF (1801 entries)**

$$\frac{\sigma}{\mu} = 9.98 \pm 0.31 \%$$

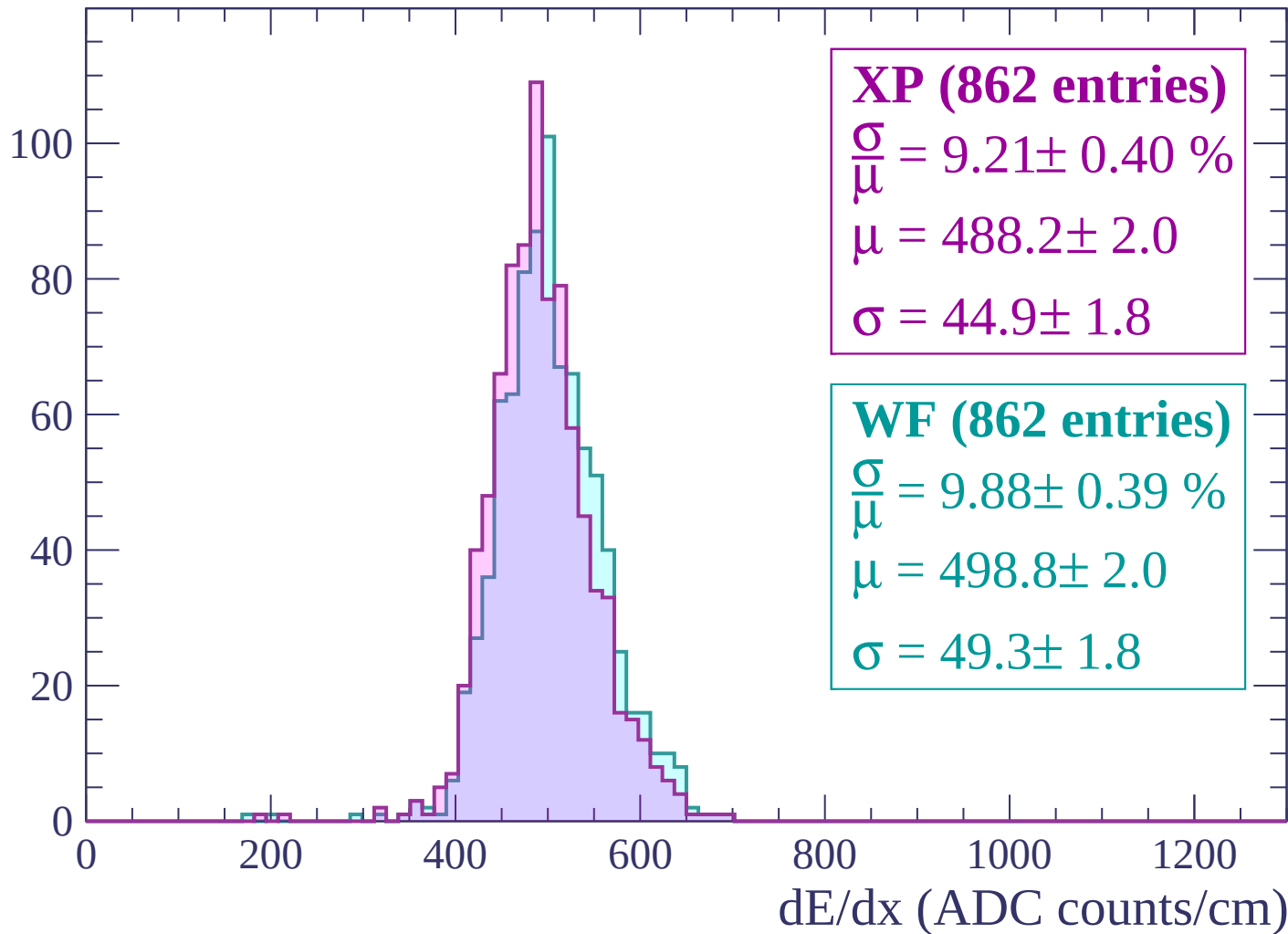
$$\mu = 499.0 \pm 1.4$$

$$\sigma = 49.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss |  $3000 < p < 3060$

Count





Energy loss |  $-90 < \theta < -88$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-88 < \theta < -86$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-86 < \theta < -84$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-84 < \theta < -82$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-82 < \theta < -80$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-80 < \theta < -78$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-78 < \theta < -76$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-76 < \theta < -74$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $-74 < \theta < -72$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-72 < \theta < -70$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-70 < \theta < -68$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-68 < \theta < -66$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-66 < \theta < -64$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-64 < \theta < -62$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-62 < \theta < -60$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-60 < \theta < -58$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $-58 < \theta < -56$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-56 < \theta < -54$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-54 < \theta < -52$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-52 < \theta < -50$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-50 < \theta < -48$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-48 < \theta < -46$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-46 < \theta < -44$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

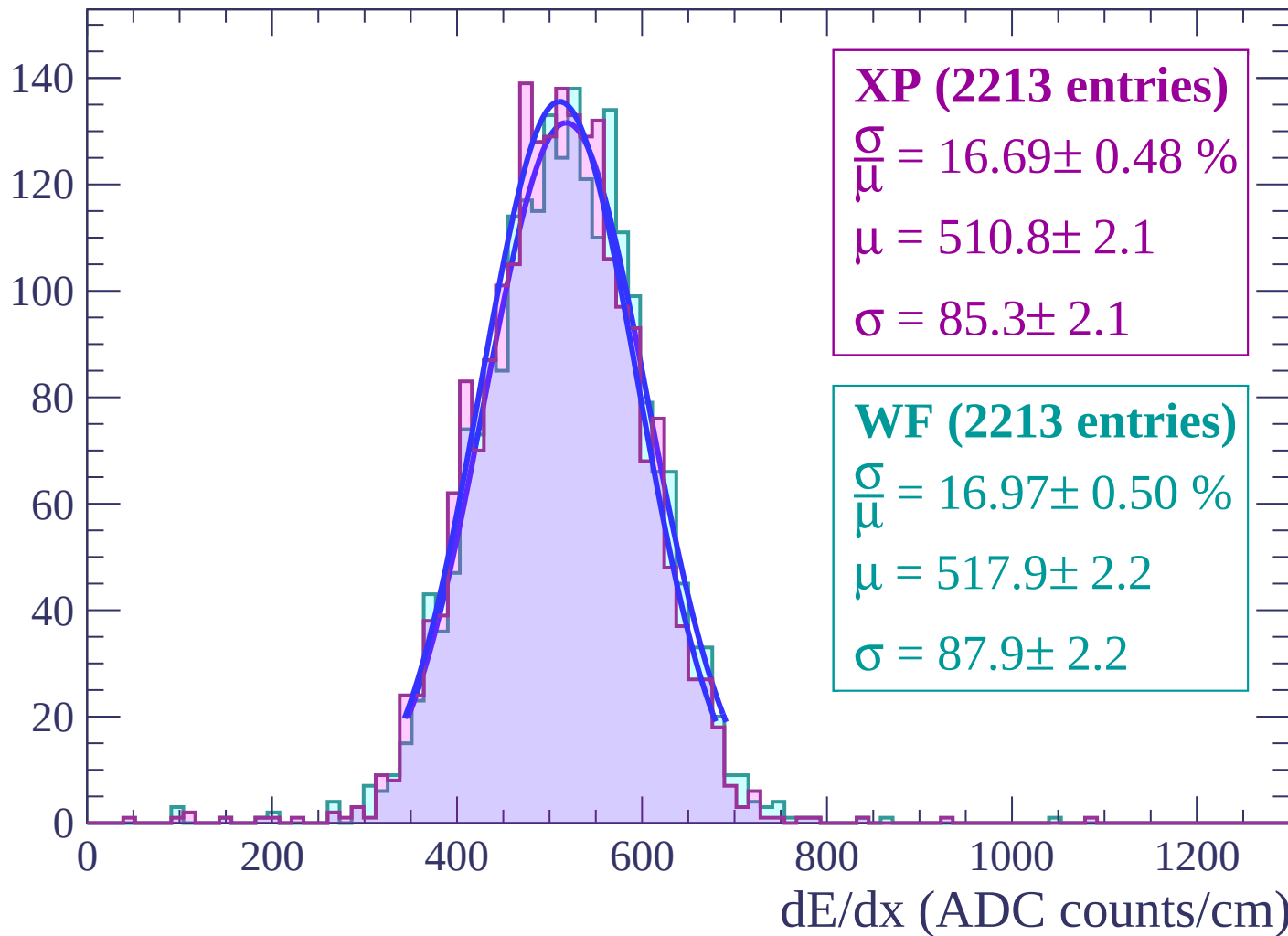
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-44 < \theta < -42$

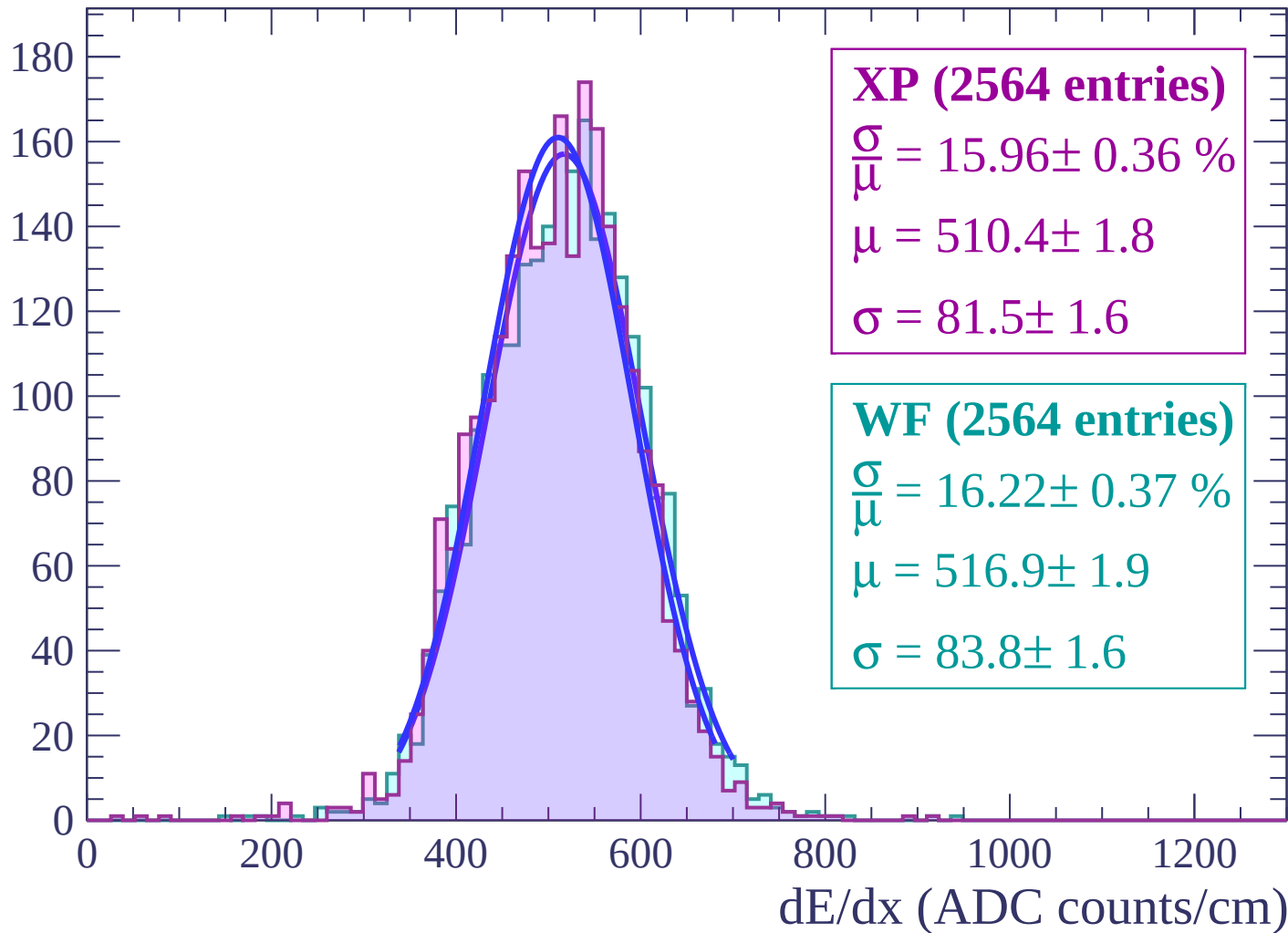
Count





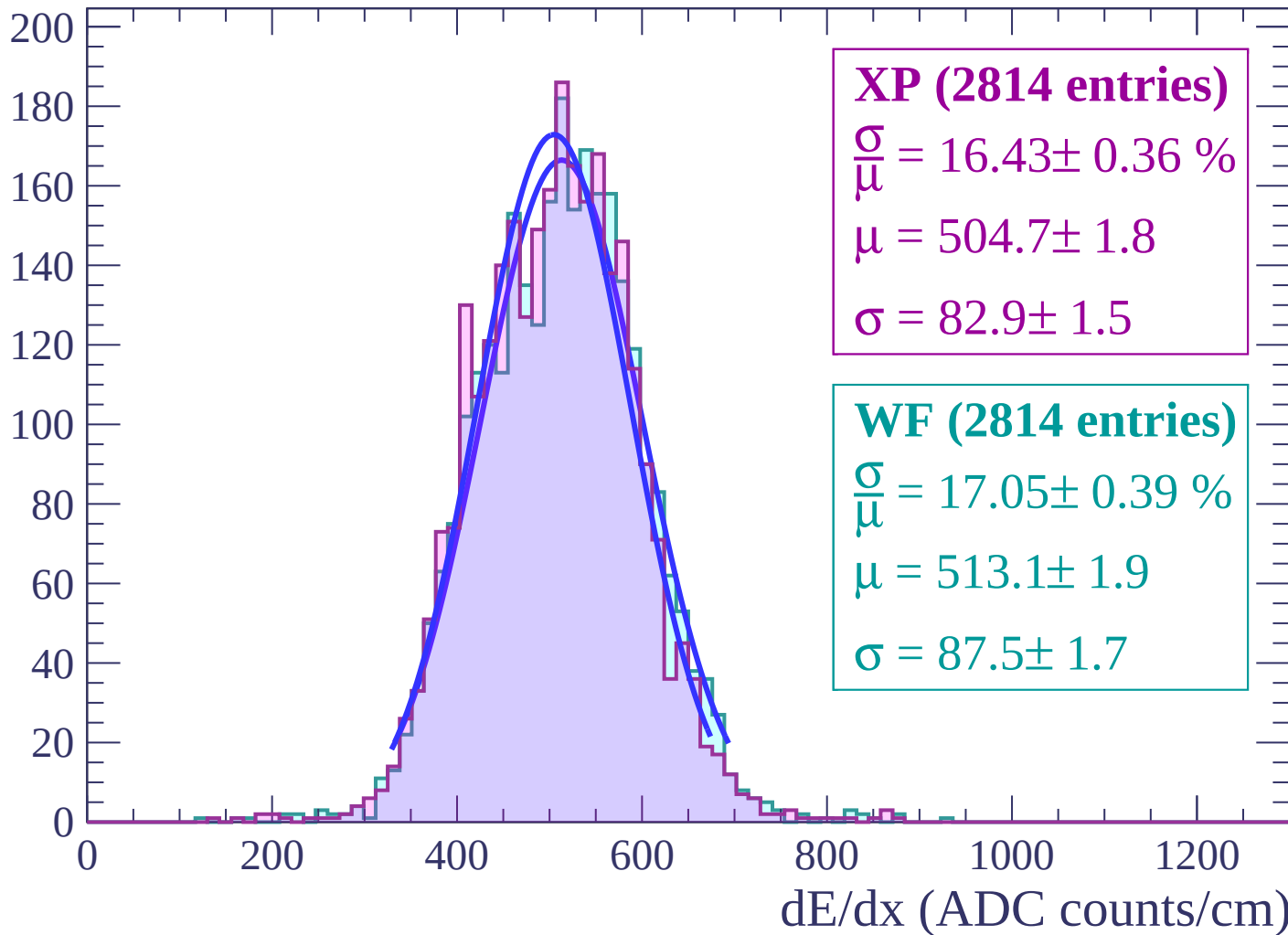
Energy loss |  $-42 < \theta < -40$

Count



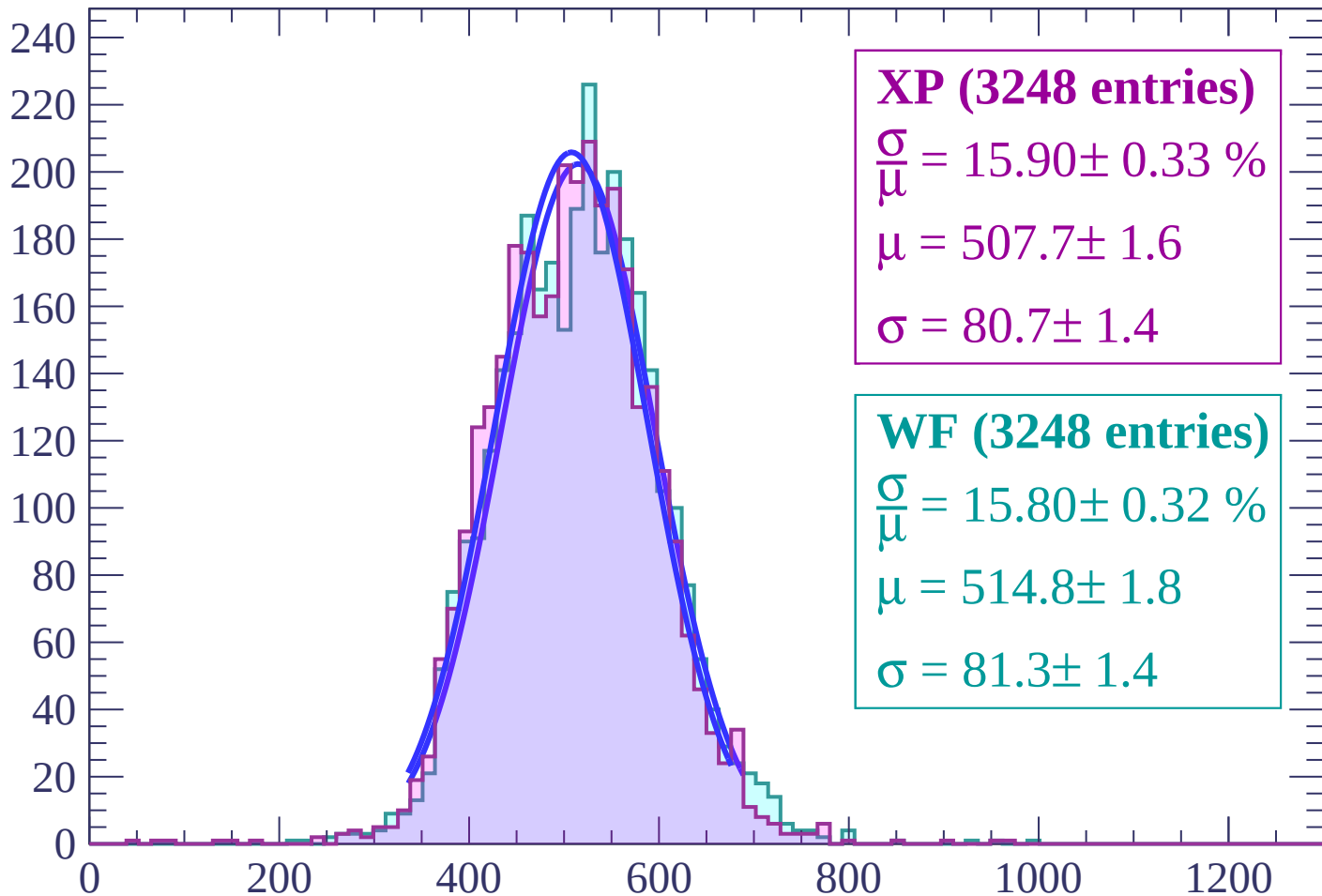
Energy loss |  $-40 < \theta < -38$

Count



Energy loss |  $-38 < \theta < -36$

Count



**XP (3248 entries)**

$$\frac{\sigma}{\mu} = 15.90 \pm 0.33 \%$$

$$\mu = 507.7 \pm 1.6$$

$$\sigma = 80.7 \pm 1.4$$

**WF (3248 entries)**

$$\frac{\sigma}{\mu} = 15.80 \pm 0.32 \%$$

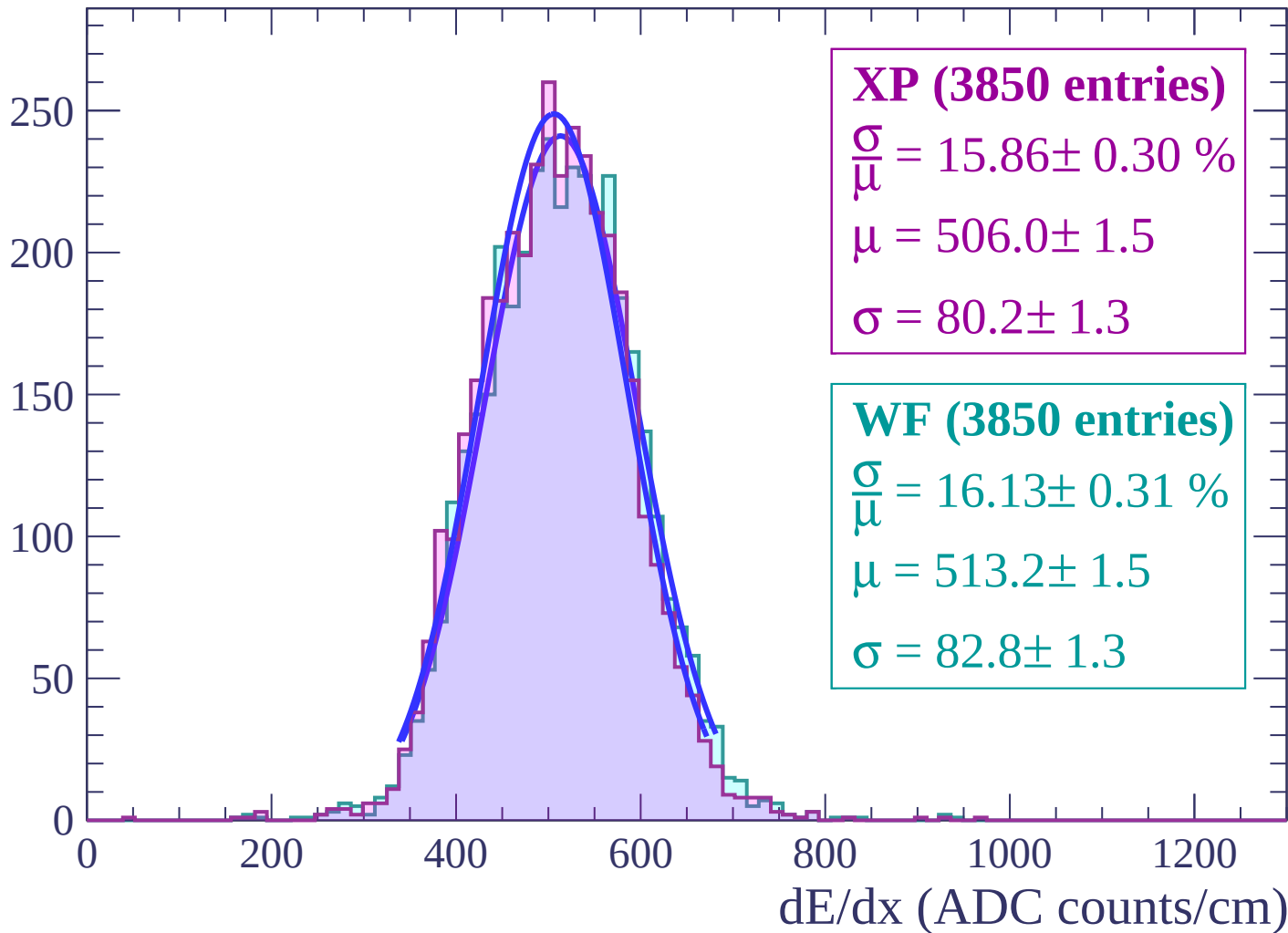
$$\mu = 514.8 \pm 1.8$$

$$\sigma = 81.3 \pm 1.4$$

dE/dx (ADC counts/cm)

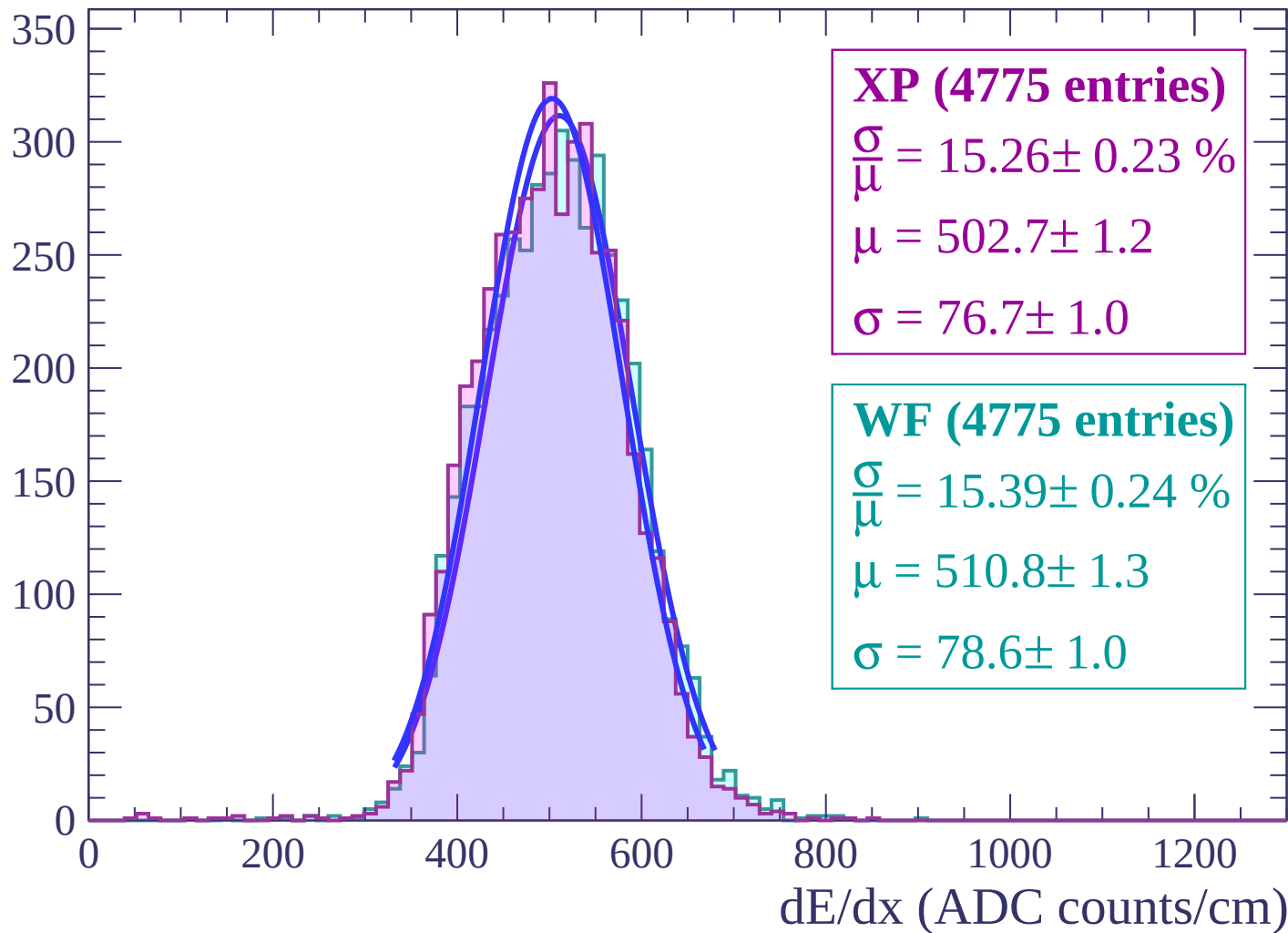
Energy loss |  $-36 < \theta < -34$

Count



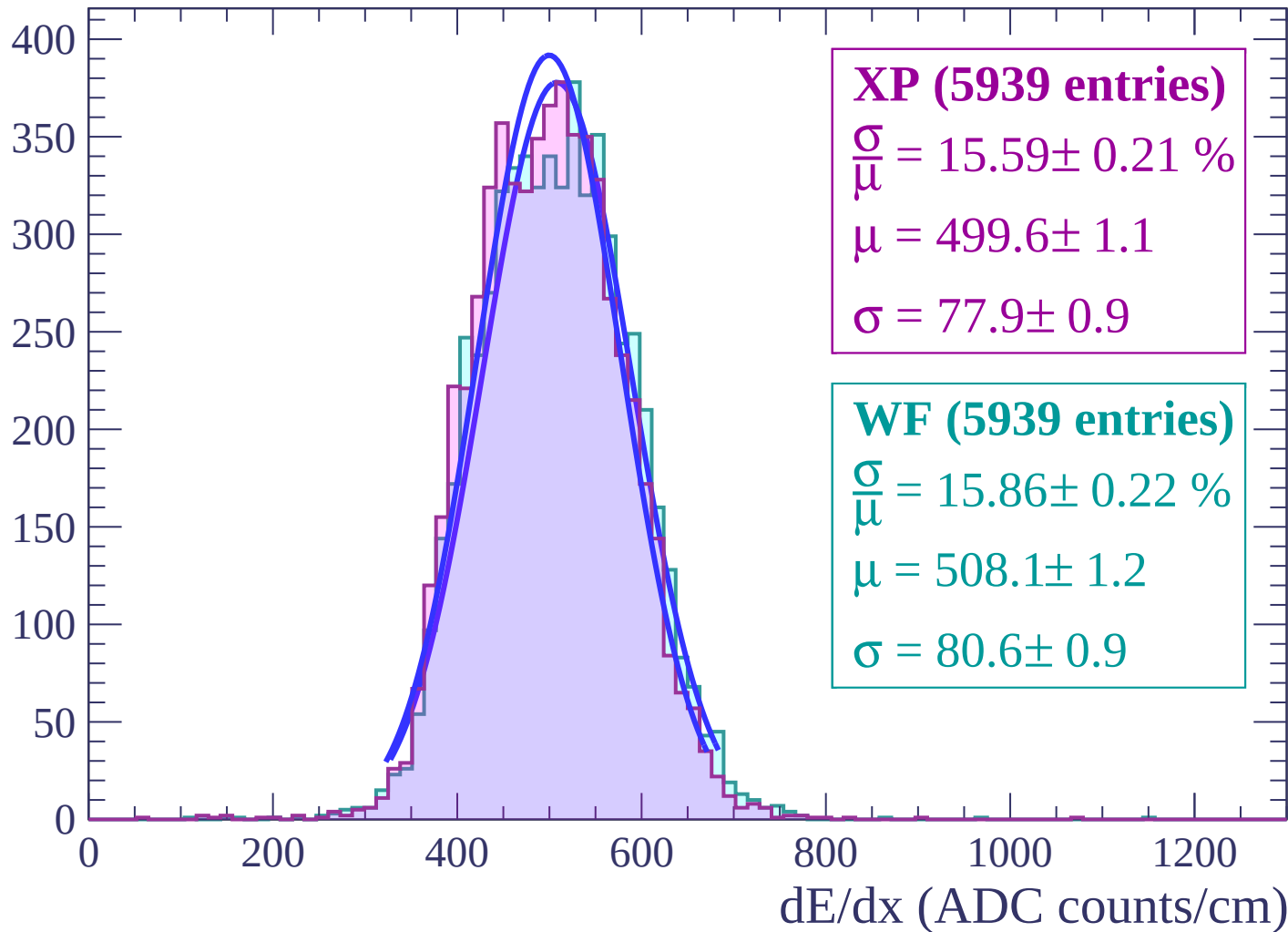
Energy loss |  $-34 < \theta < -32$

Count



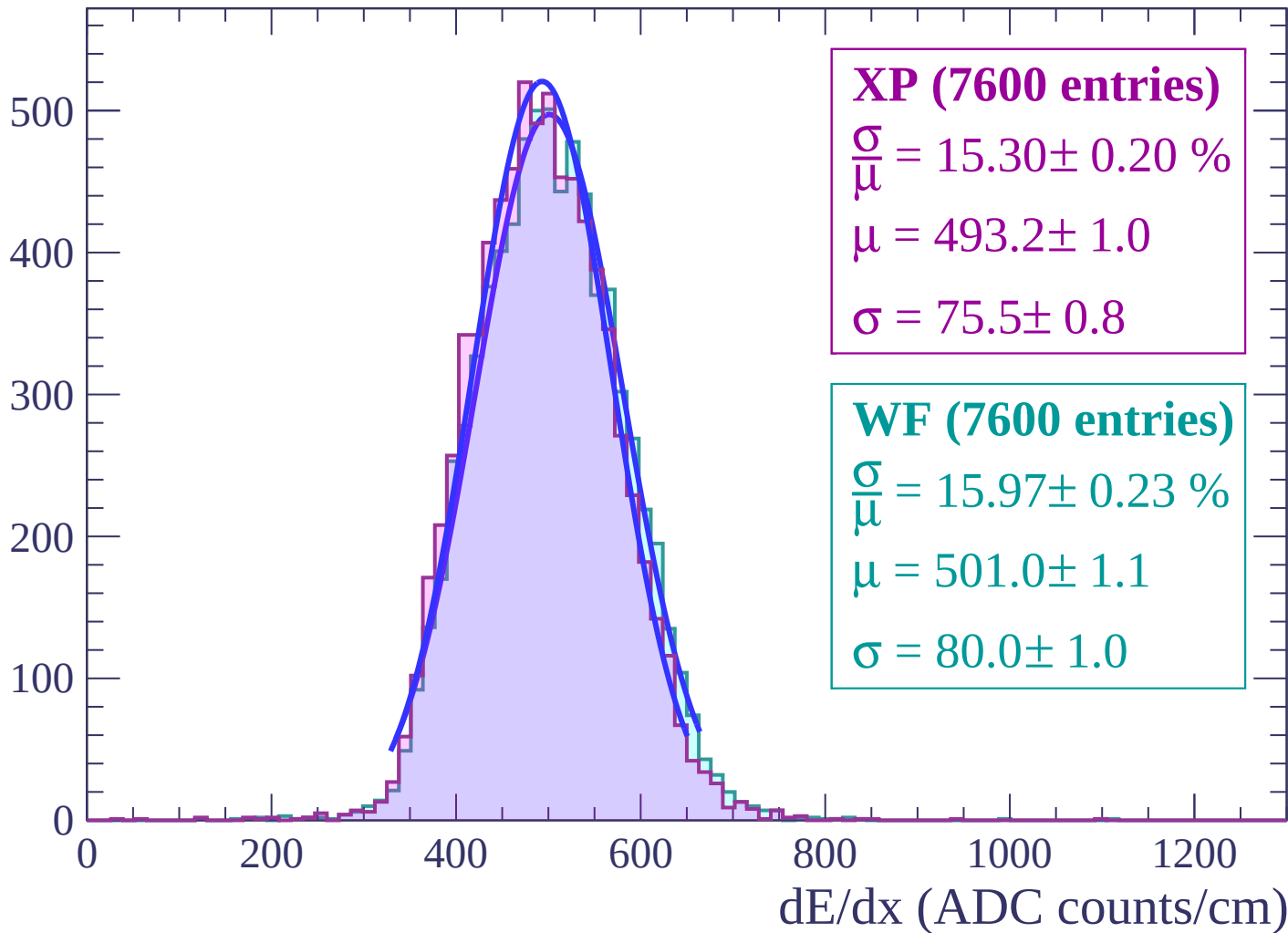
Energy loss |  $-32 < \theta < -30$

Count



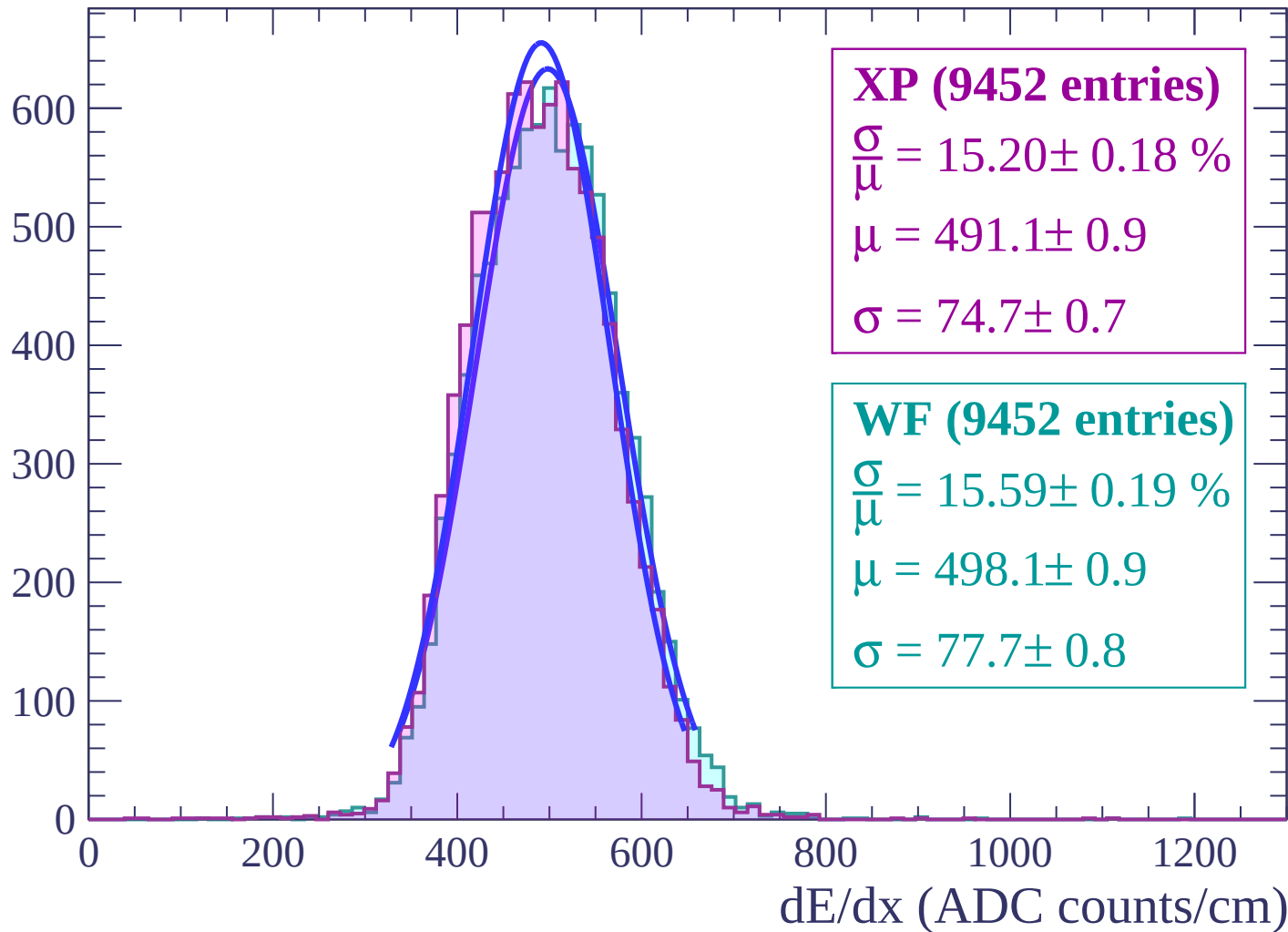
Energy loss |  $-30 < \theta < -28$

Count



Energy loss |  $-28 < \theta < -26$

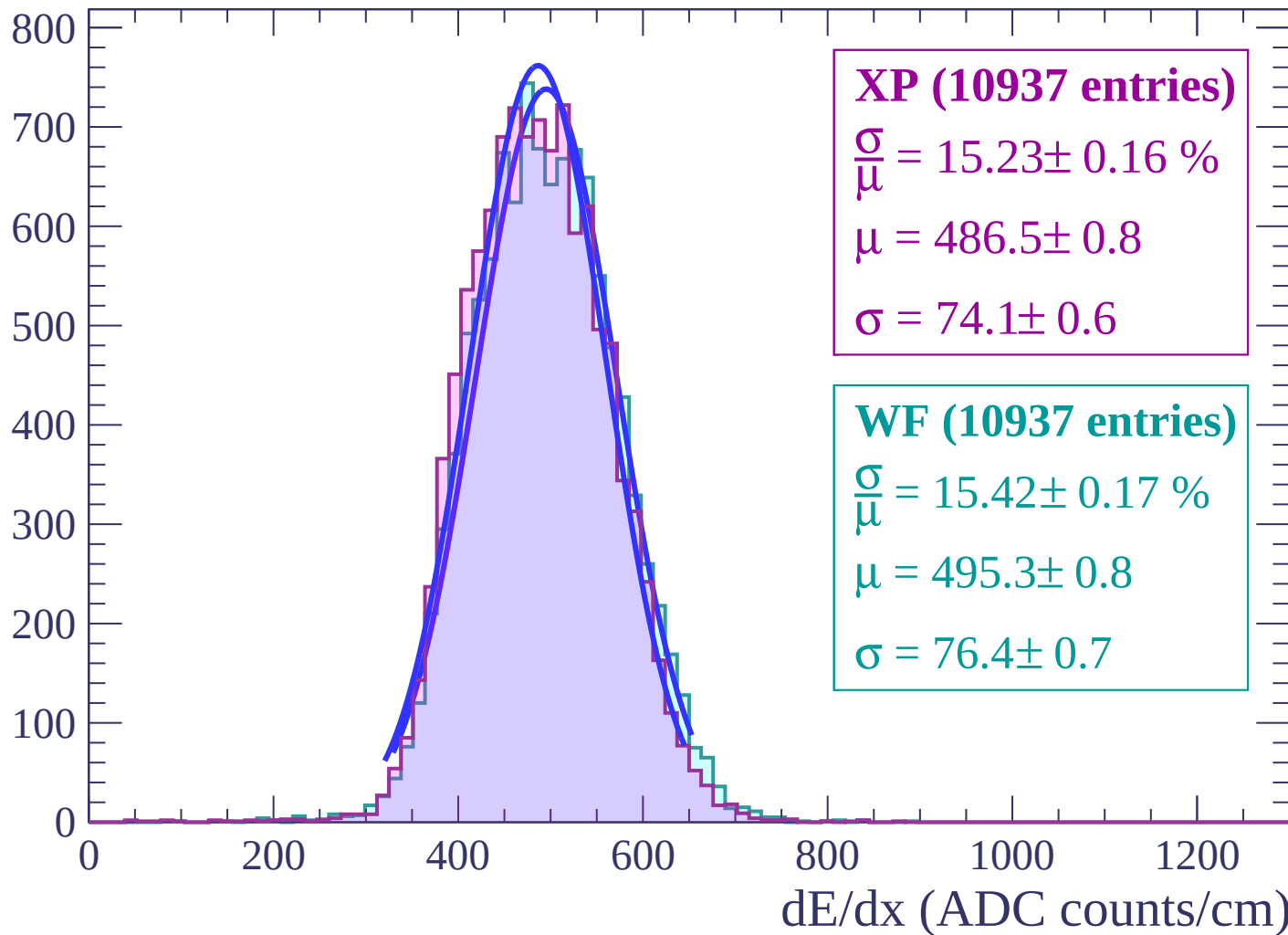
Count





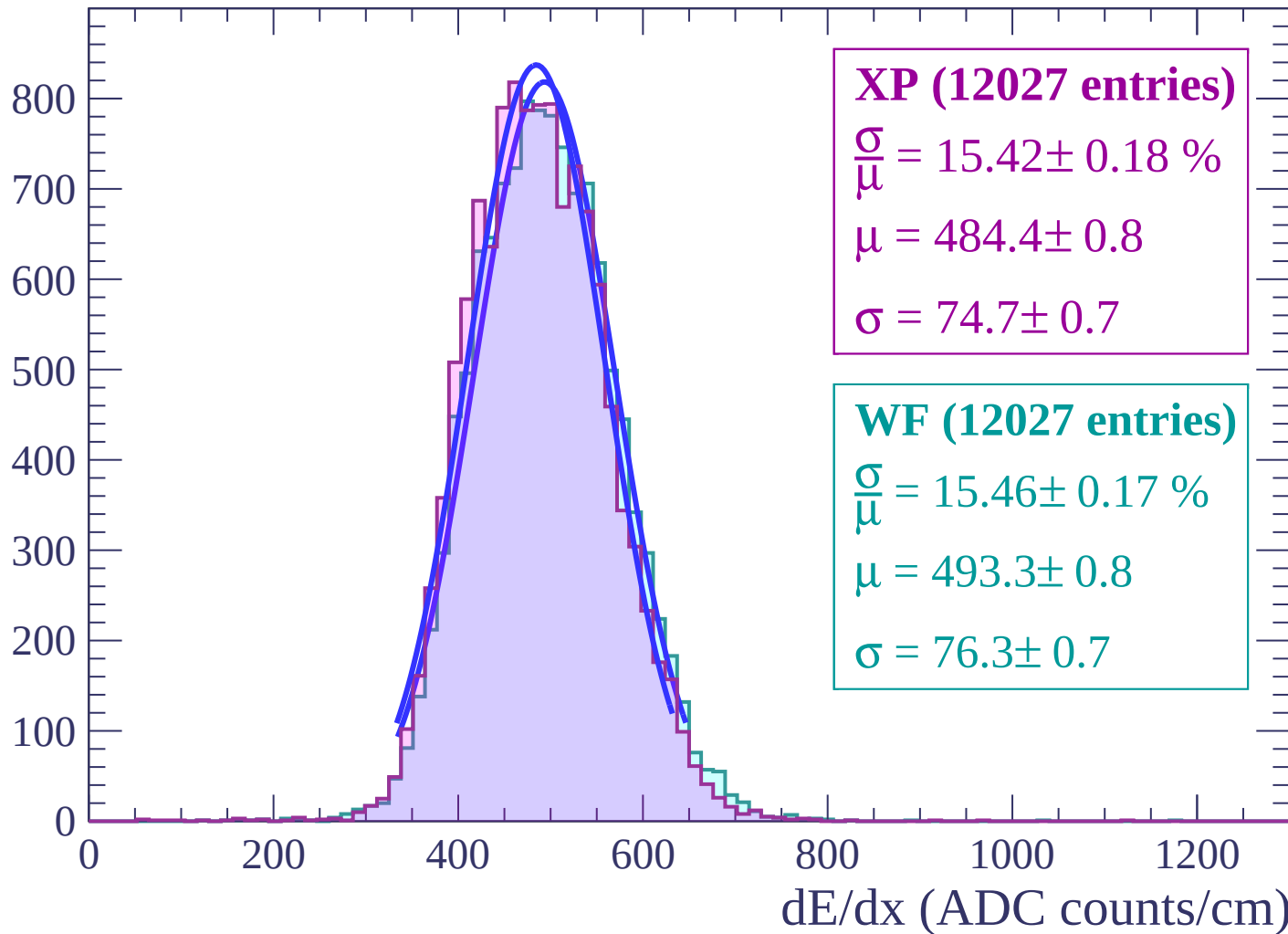
Energy loss |  $-26 < \theta < -24$

Count

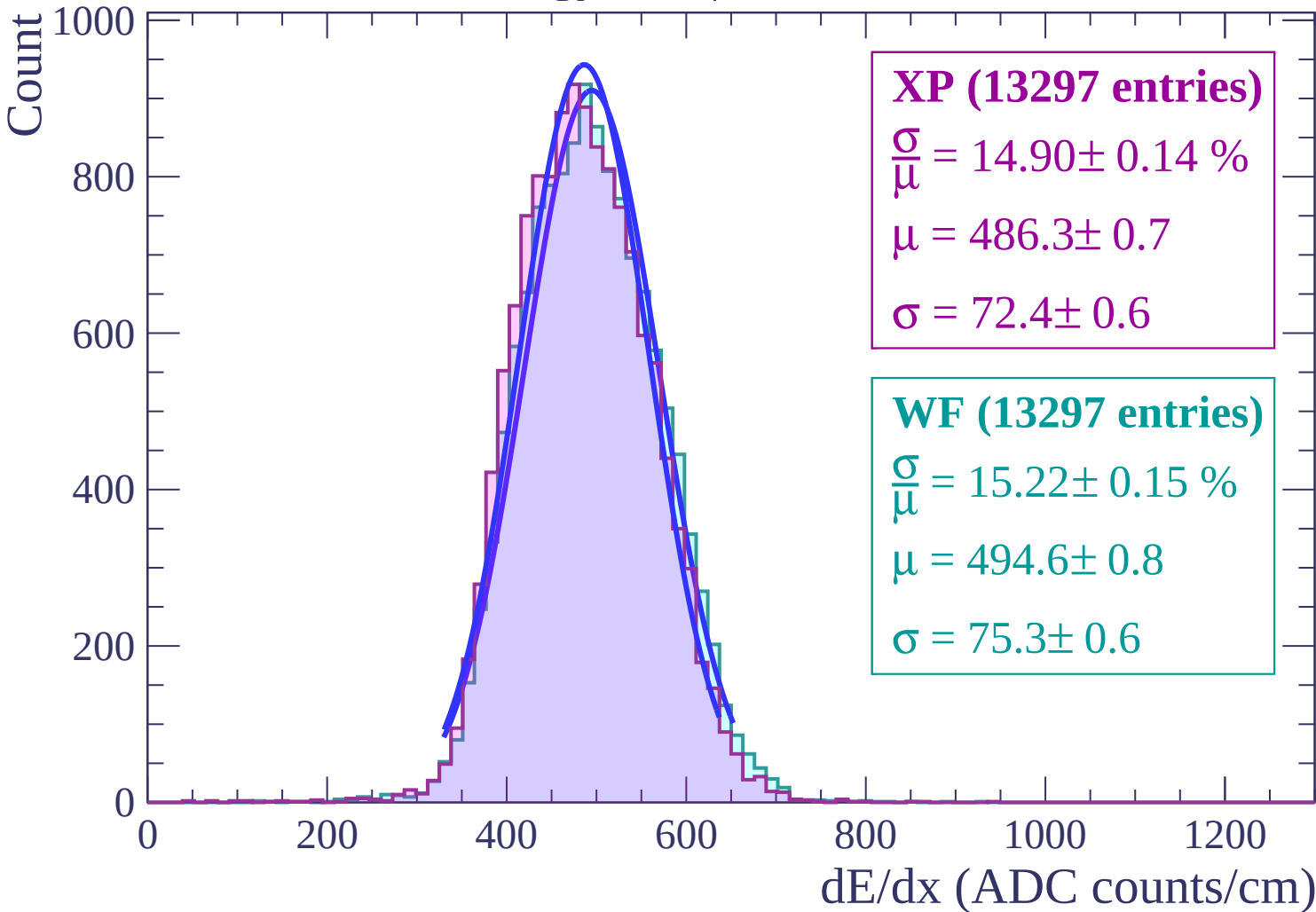


Energy loss |  $-24 < \theta < -22$

Count



Energy loss |  $-22 < \theta < -20$



Energy loss |  $-20 < \theta < -18$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

**XP (14611 entries)**

$\frac{\sigma}{\mu} = 14.94 \pm 0.14 \%$

$\mu = 485.4 \pm 0.7$

$\sigma = 72.5 \pm 0.6$

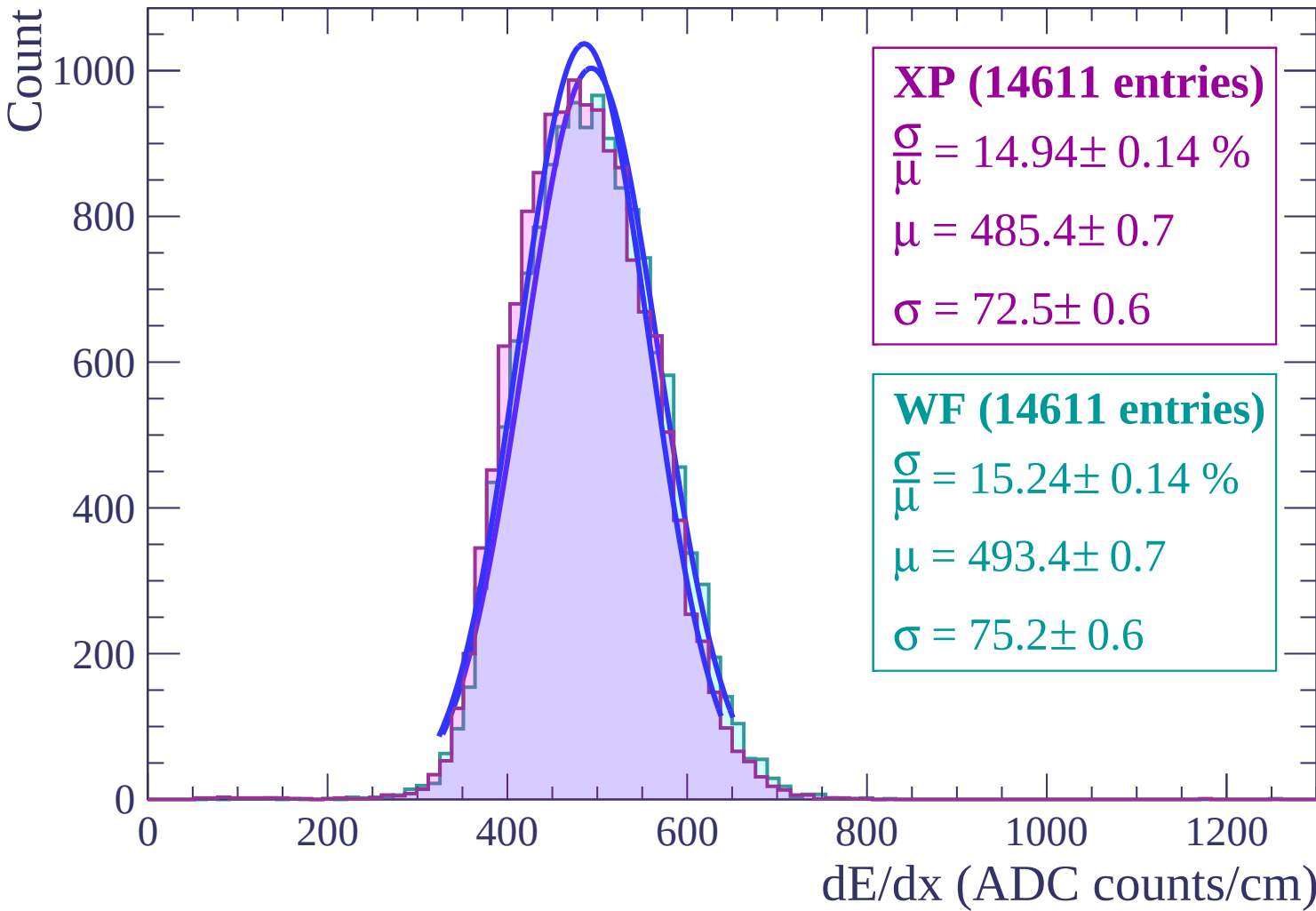
**WF (14611 entries)**

$\frac{\sigma}{\mu} = 15.24 \pm 0.14 \%$

$\mu = 493.4 \pm 0.7$

$\sigma = 75.2 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss |  $-18 < \theta < -16$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (16124 entries)**

$$\frac{\sigma}{\mu} = 14.99 \pm 0.13 \%$$

$$\mu = 483.8 \pm 0.7$$

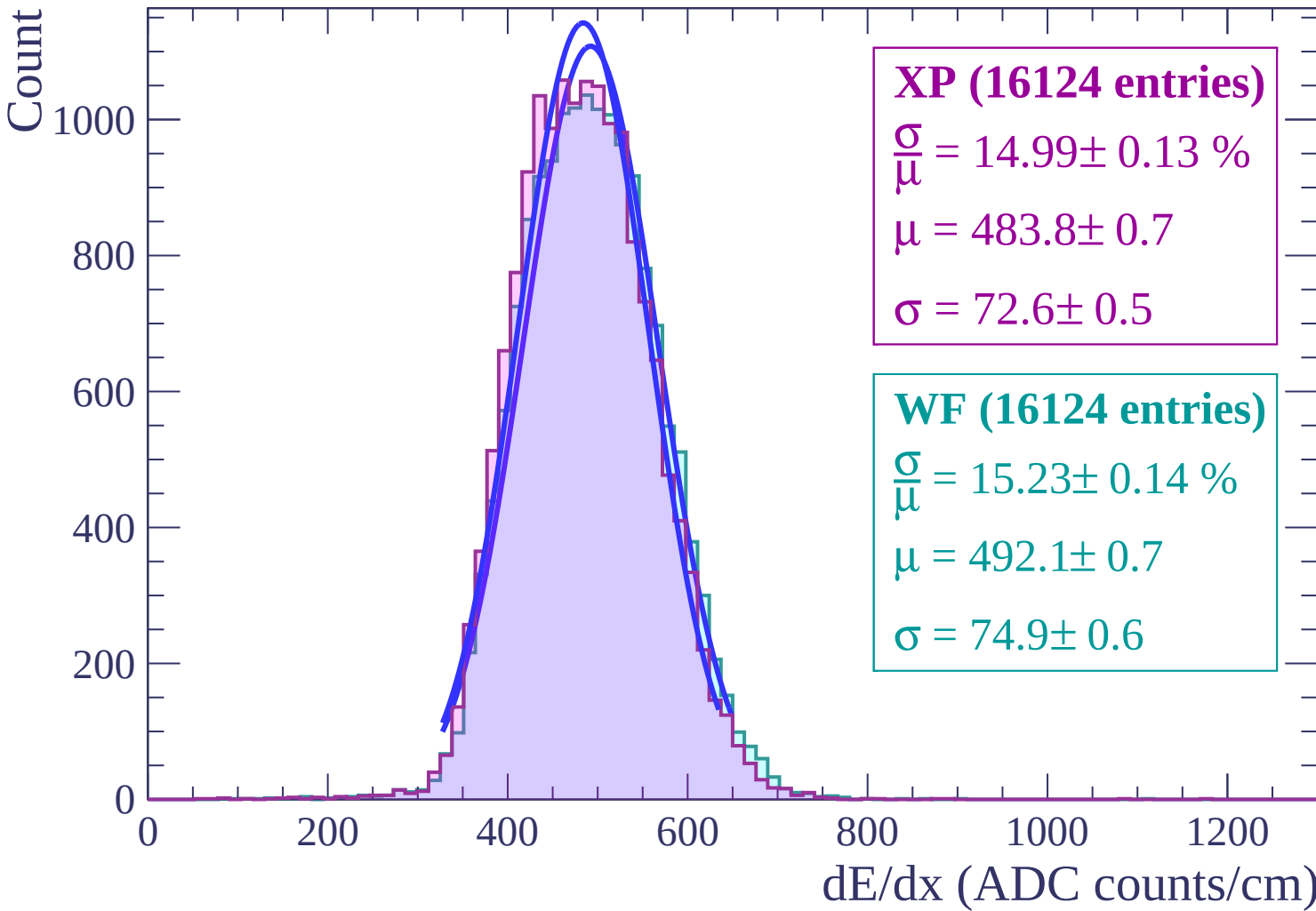
$$\sigma = 72.6 \pm 0.5$$

**WF (16124 entries)**

$$\frac{\sigma}{\mu} = 15.23 \pm 0.14 \%$$

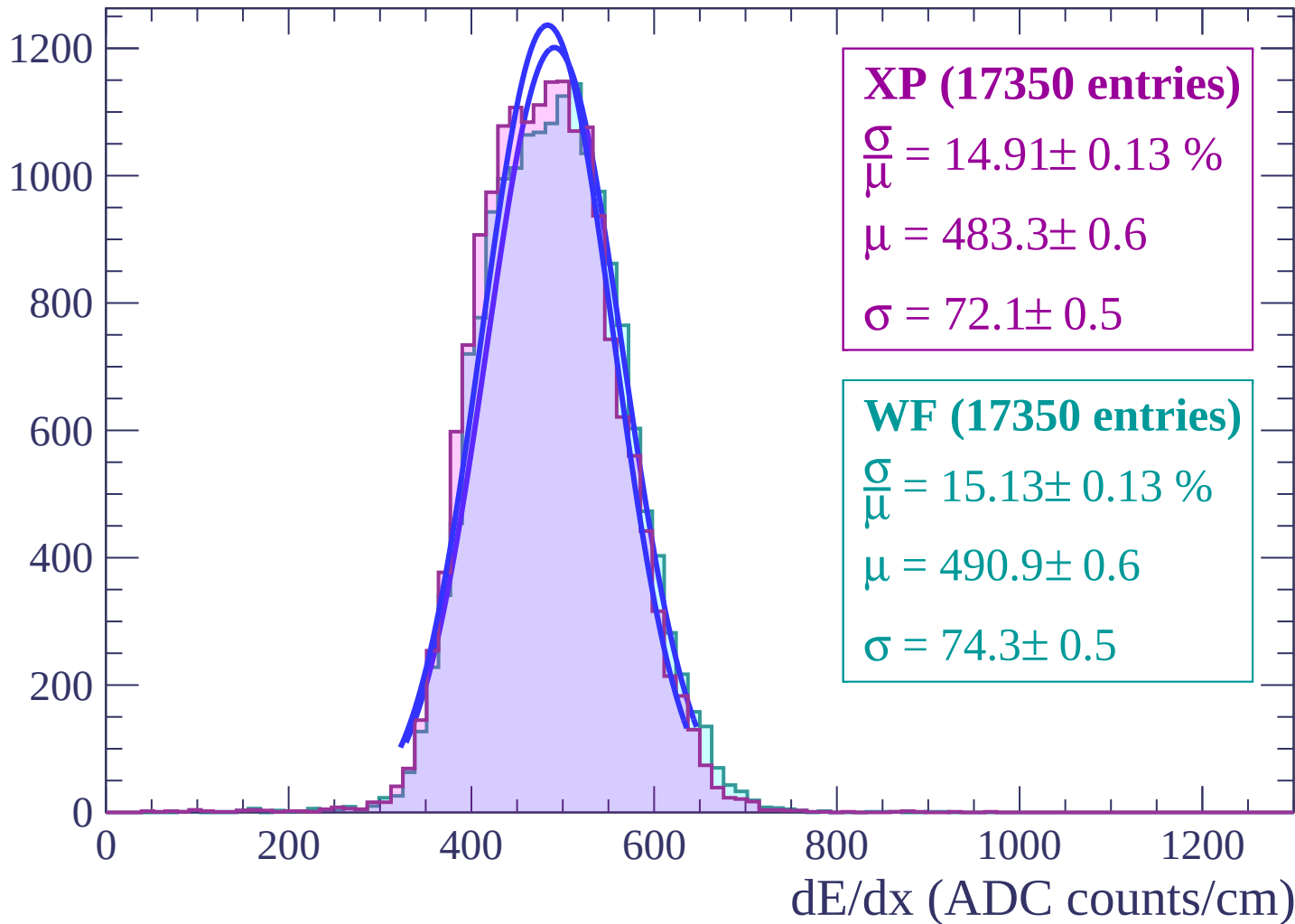
$$\mu = 492.1 \pm 0.7$$

$$\sigma = 74.9 \pm 0.6$$

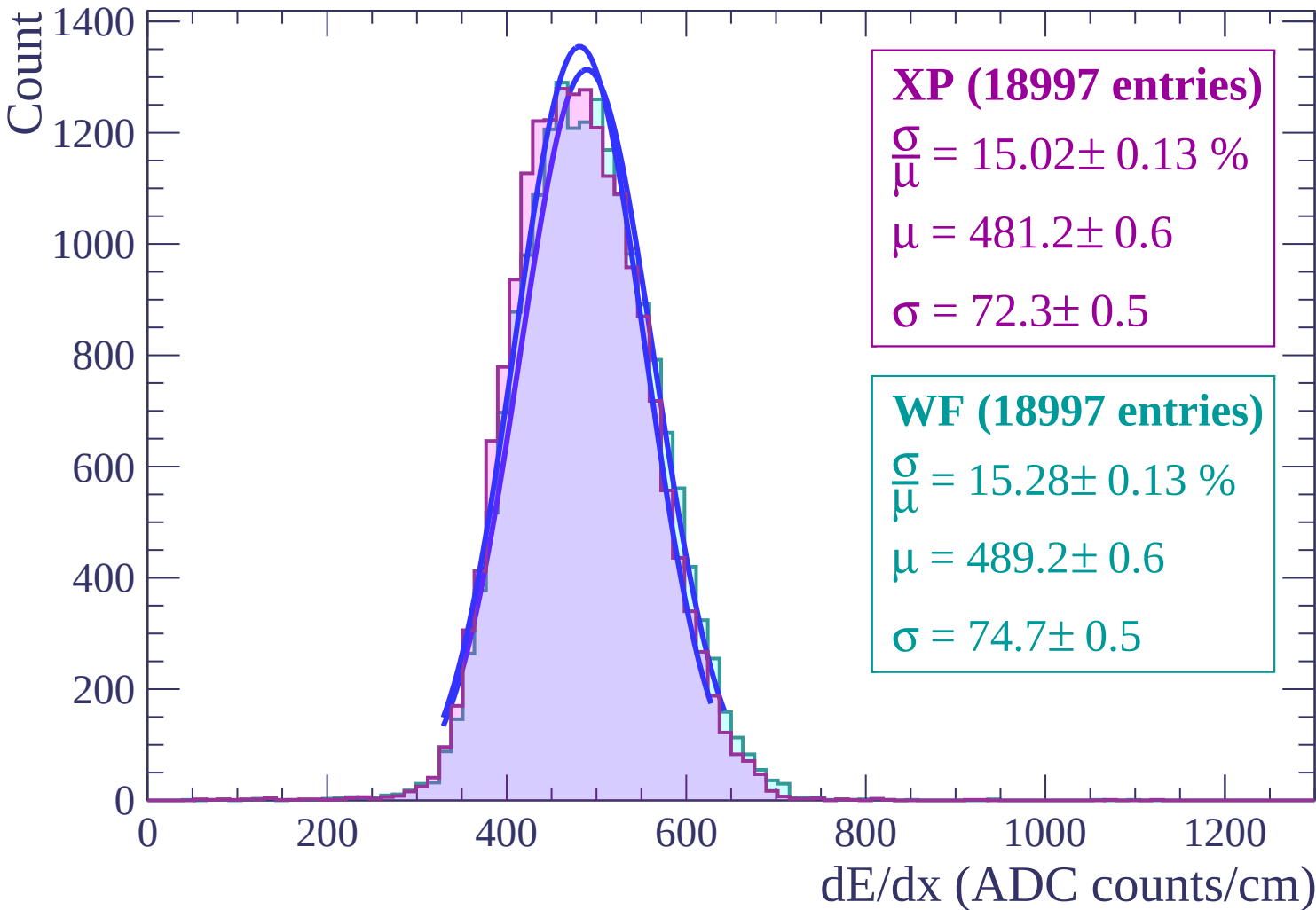


Energy loss |  $-16 < \theta < -14$

Count



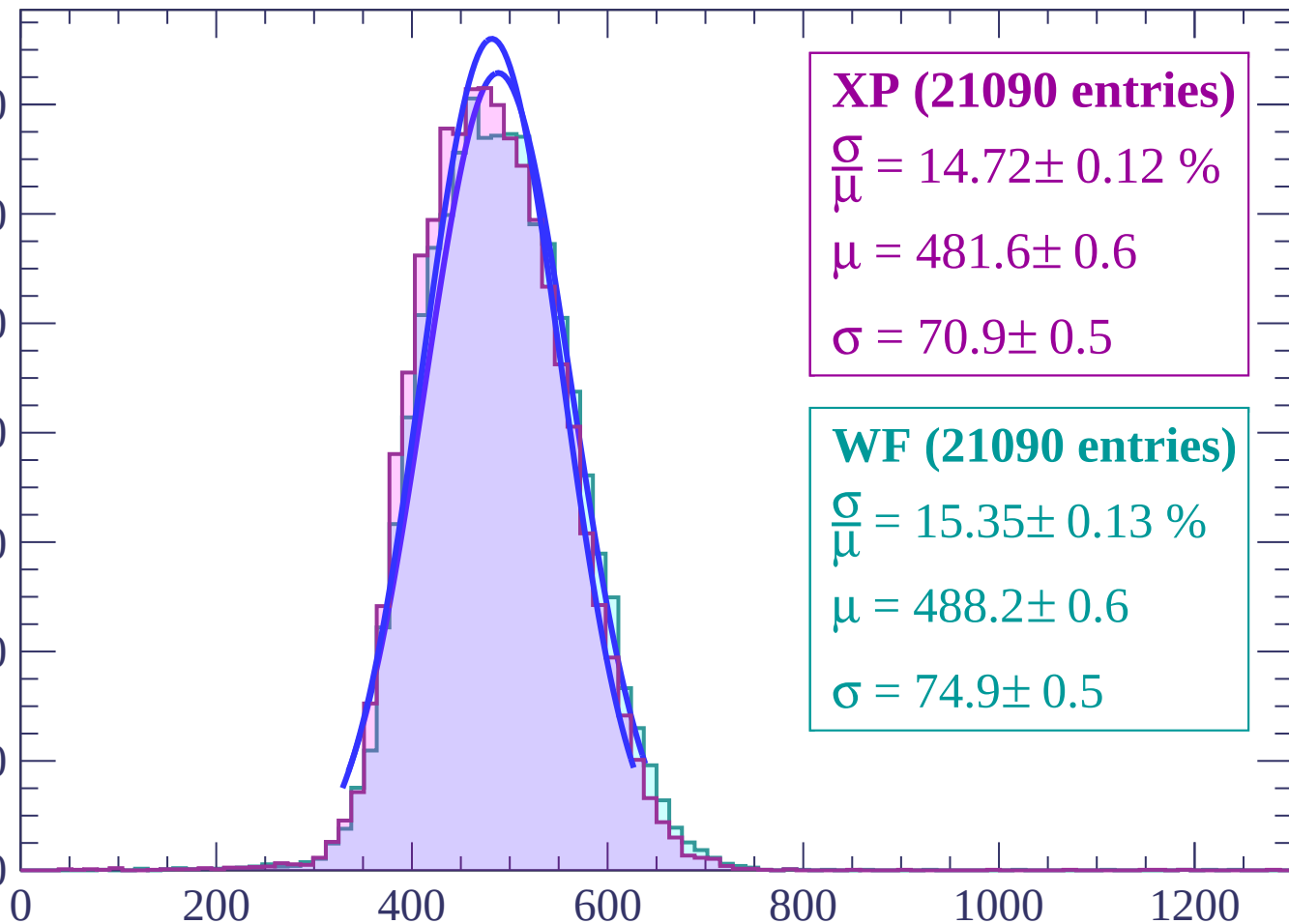
Energy loss |  $-14 < \theta < -12$



# Energy loss | $-12 < \theta < -10$

Count

1400  
1200  
1000  
800  
600  
400  
200  
0



**XP (21090 entries)**

$$\frac{\sigma}{\mu} = 14.72 \pm 0.12 \%$$

$$\mu = 481.6 \pm 0.6$$

$$\sigma = 70.9 \pm 0.5$$

**WF (21090 entries)**

$$\frac{\sigma}{\mu} = 15.35 \pm 0.13 \%$$

$$\mu = 488.2 \pm 0.6$$

$$\sigma = 74.9 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss |  $-10 < \theta < -8$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0

**XP (23000 entries)**

$$\frac{\sigma}{\mu} = 14.90 \pm 0.11 \%$$

$$\mu = 479.6 \pm 0.6$$

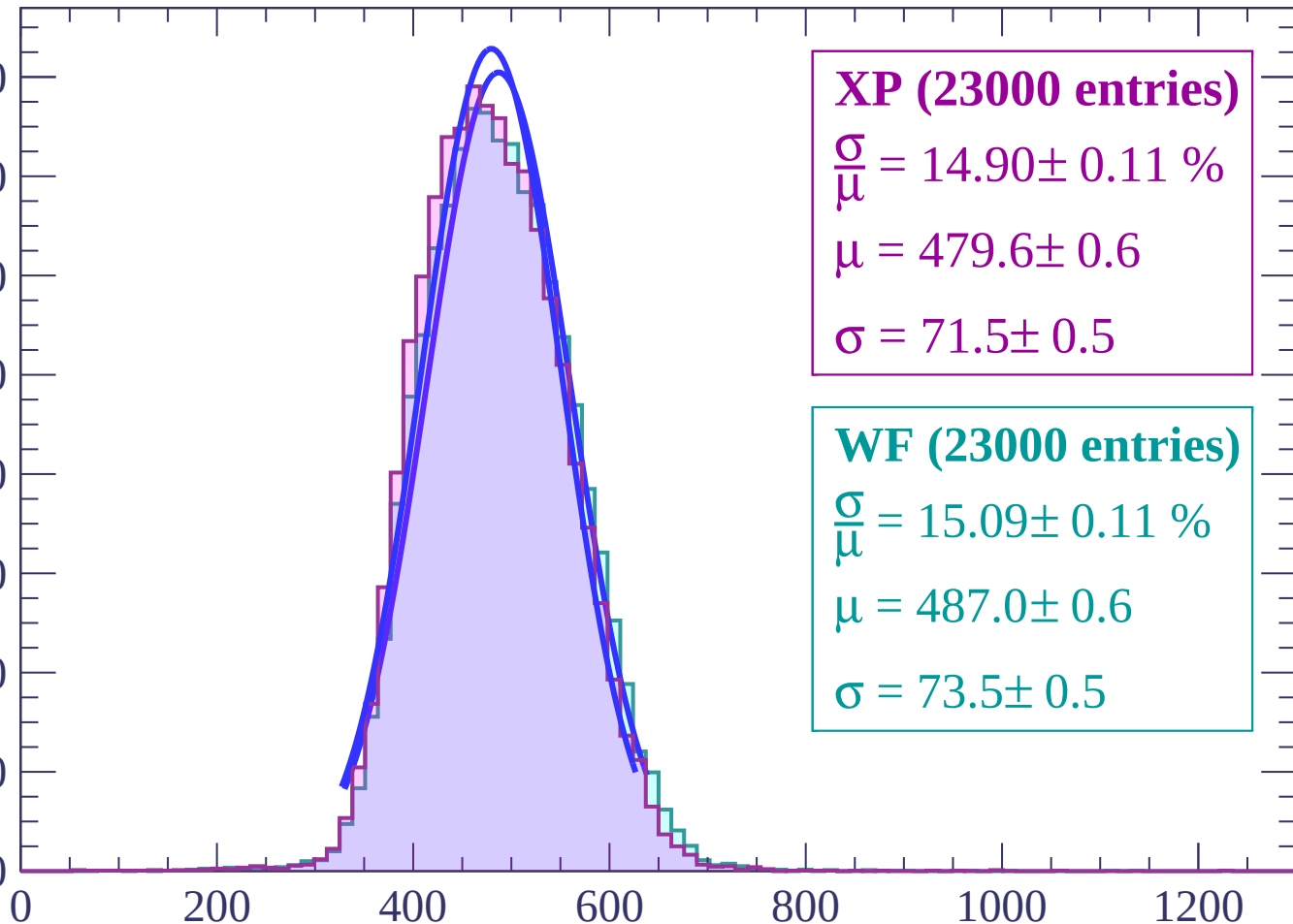
$$\sigma = 71.5 \pm 0.5$$

**WF (23000 entries)**

$$\frac{\sigma}{\mu} = 15.09 \pm 0.11 \%$$

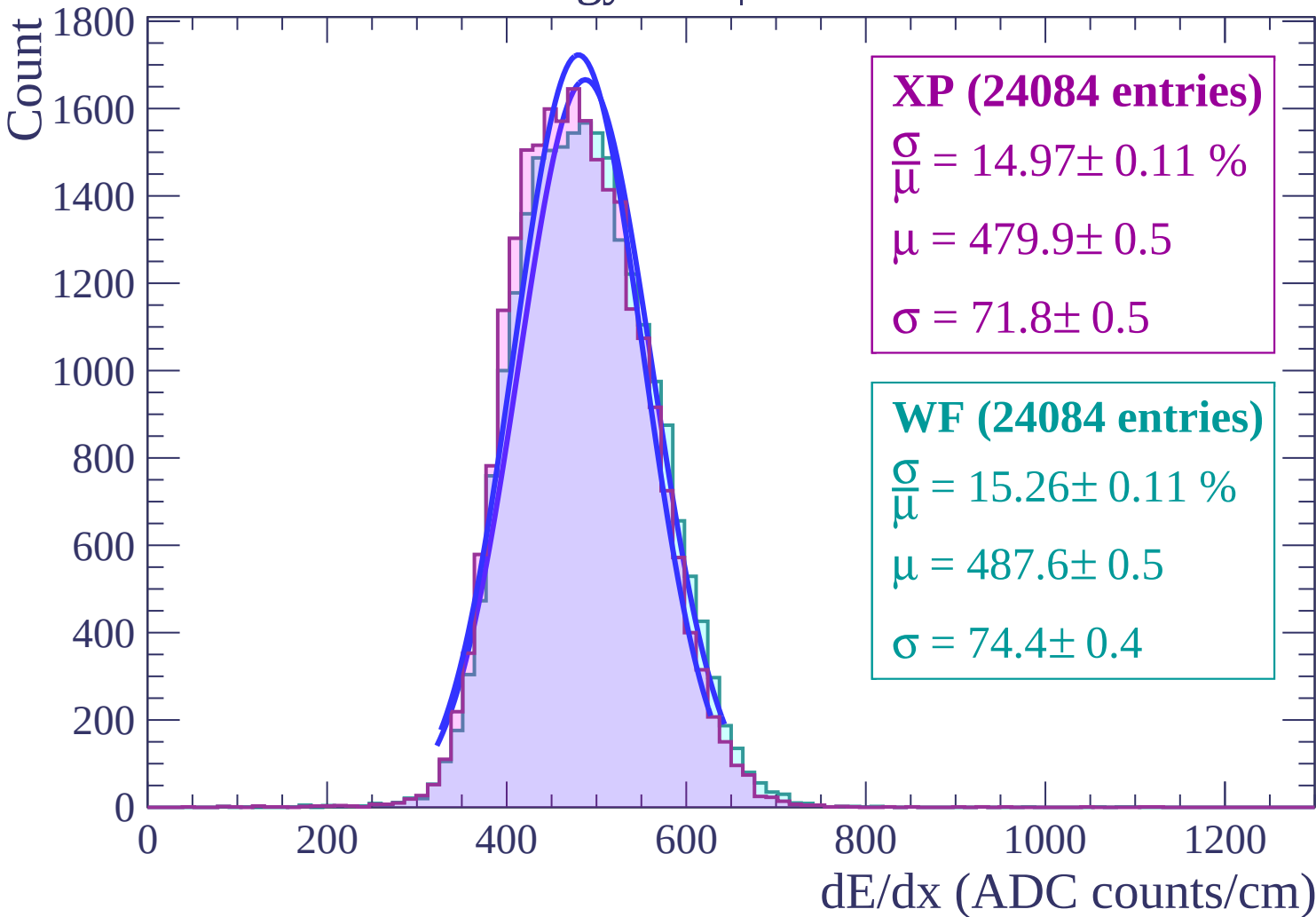
$$\mu = 487.0 \pm 0.6$$

$$\sigma = 73.5 \pm 0.5$$

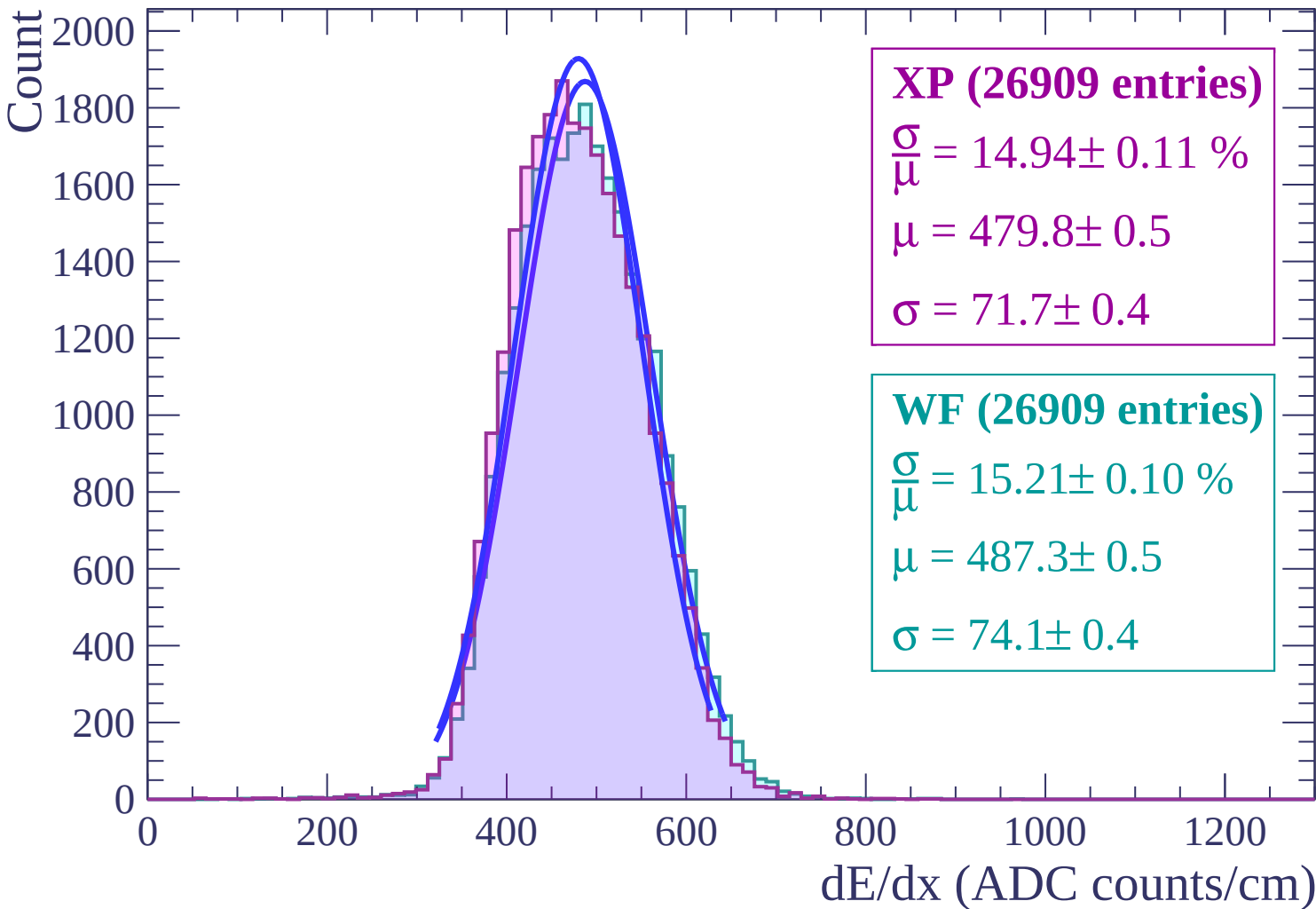


dE/dx (ADC counts/cm)

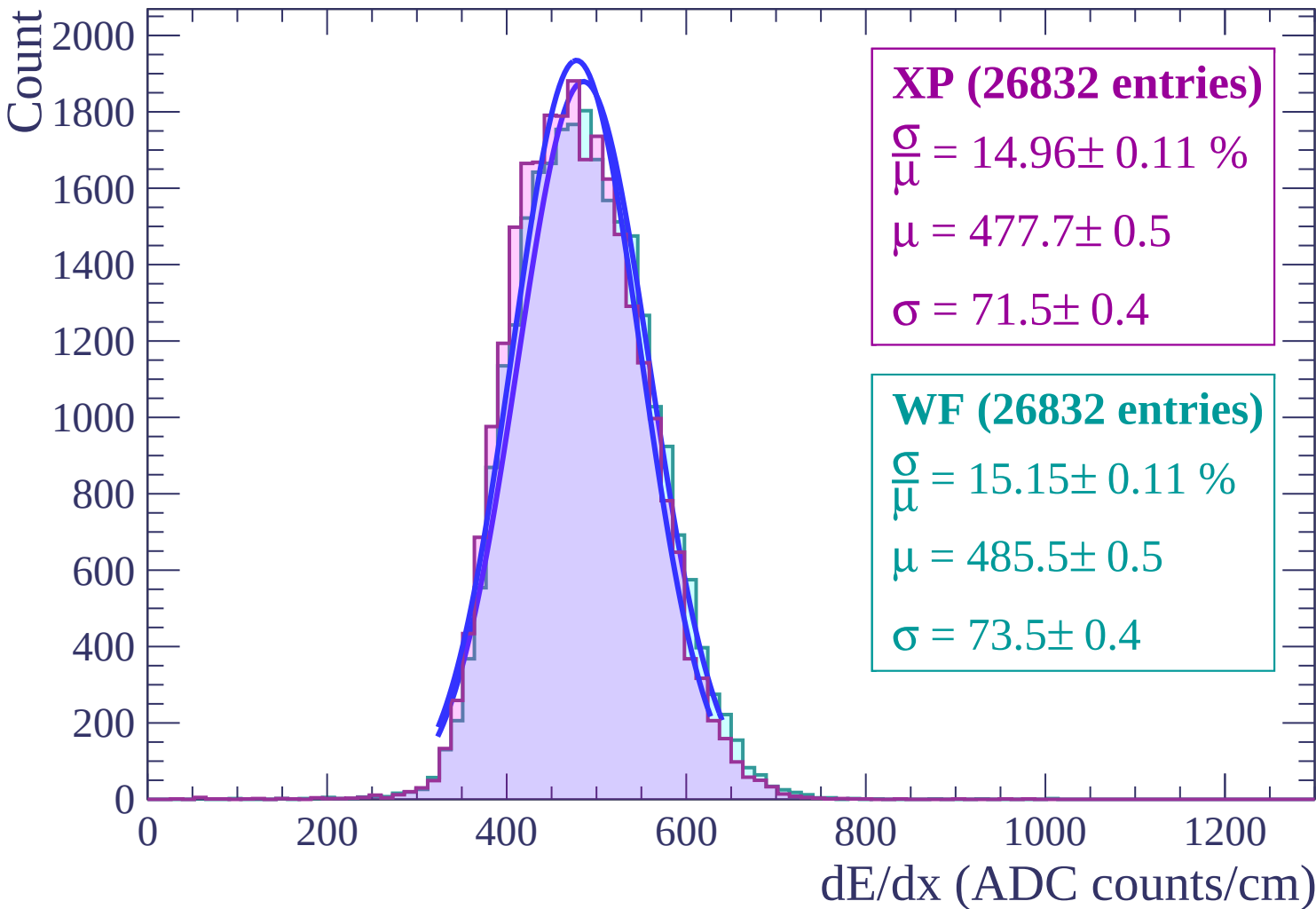
Energy loss |  $-8 < \theta < -6$



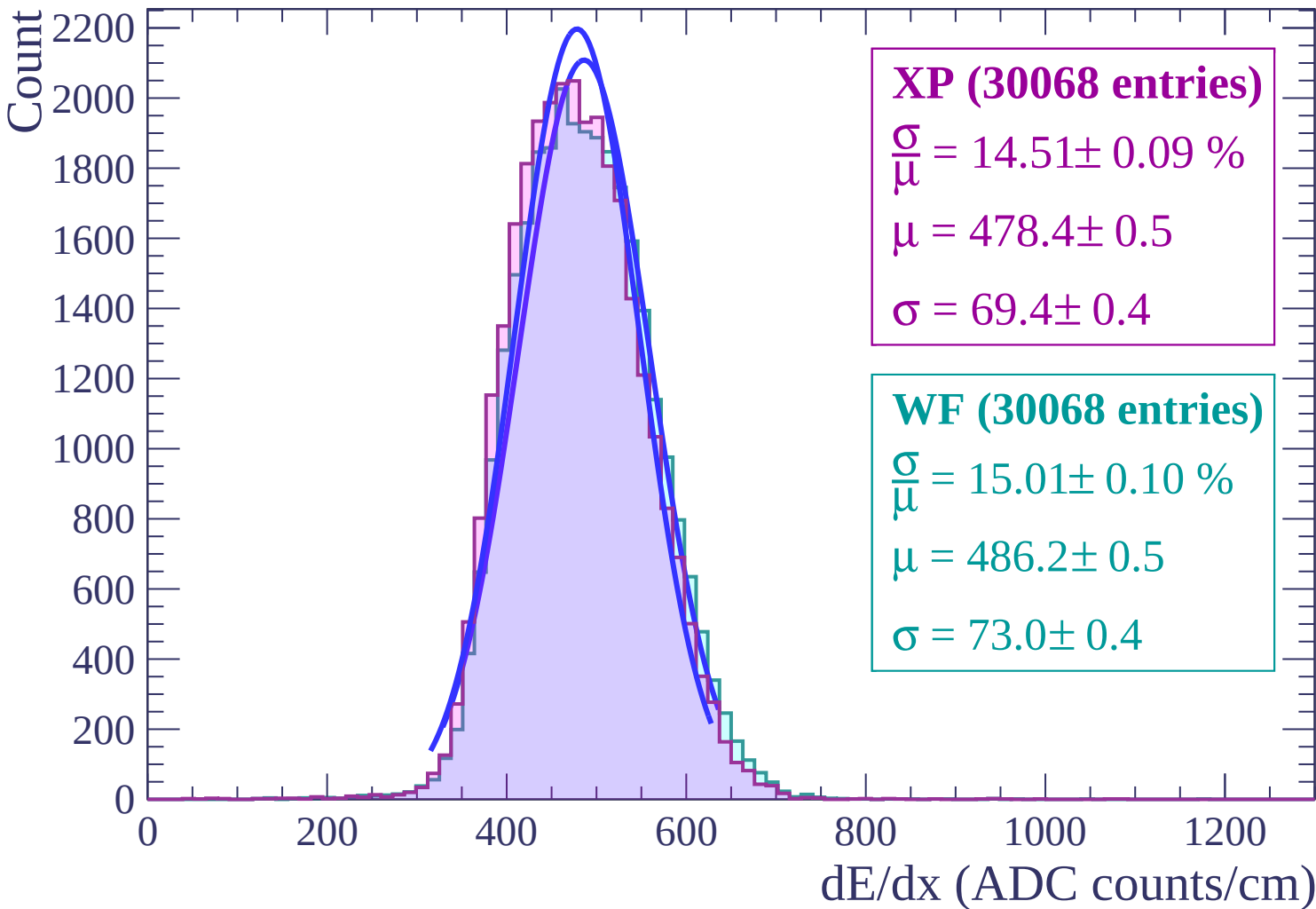
Energy loss |  $-6 < \theta < -4$



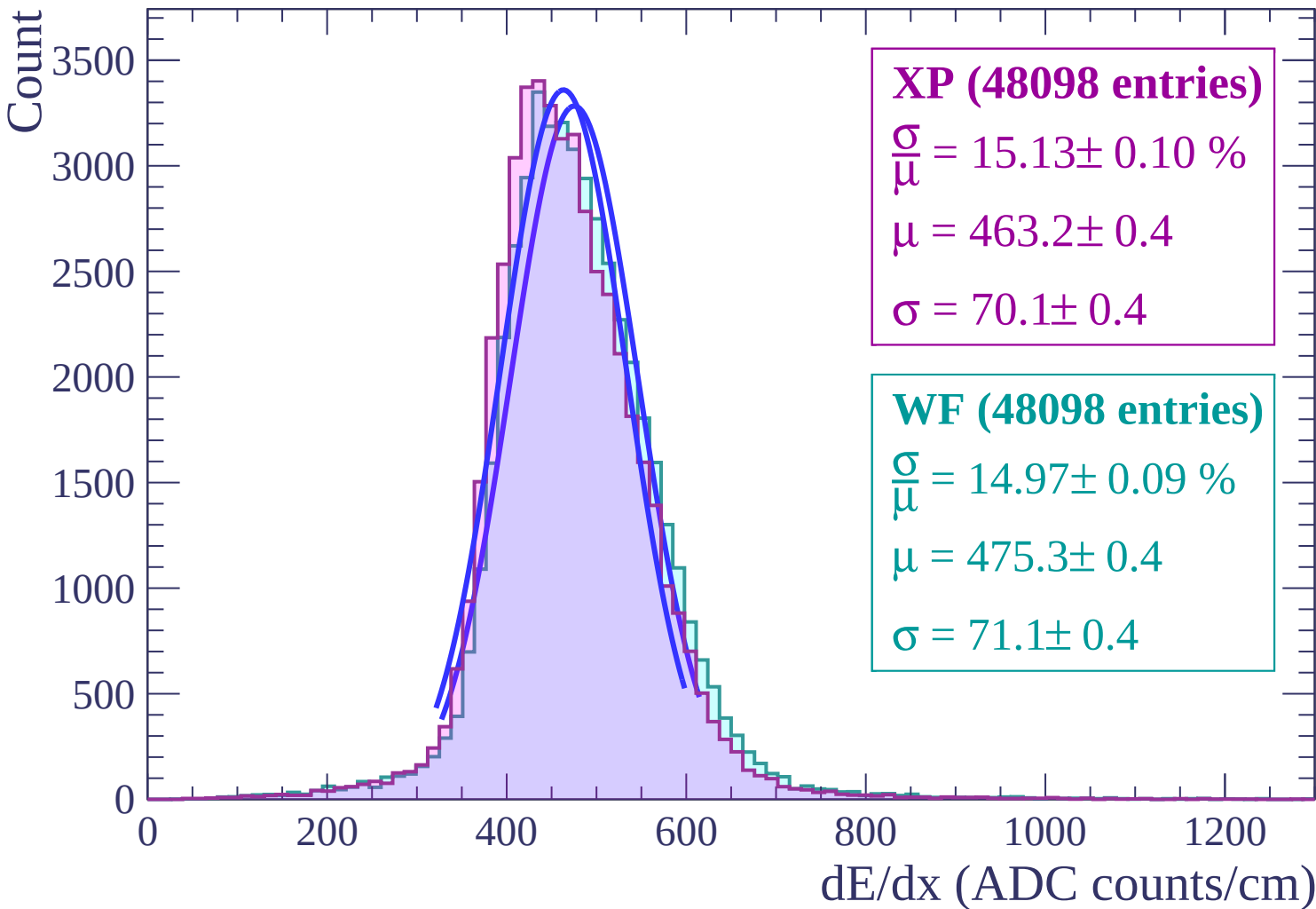
Energy loss |  $-4 < \theta < -2$



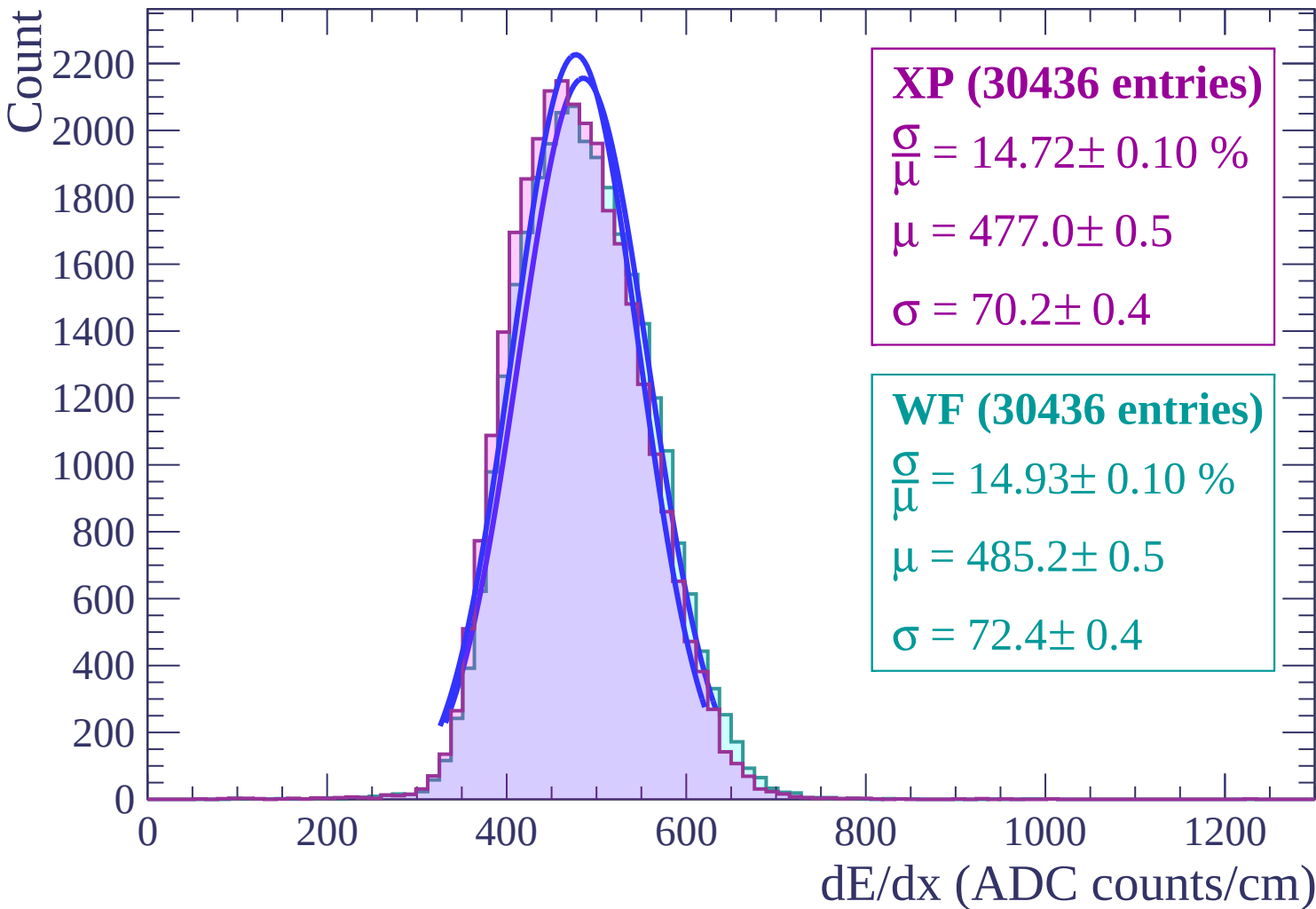
# Energy loss | $-2 < \theta < 0$



# Energy loss | $0 < \theta < 2$



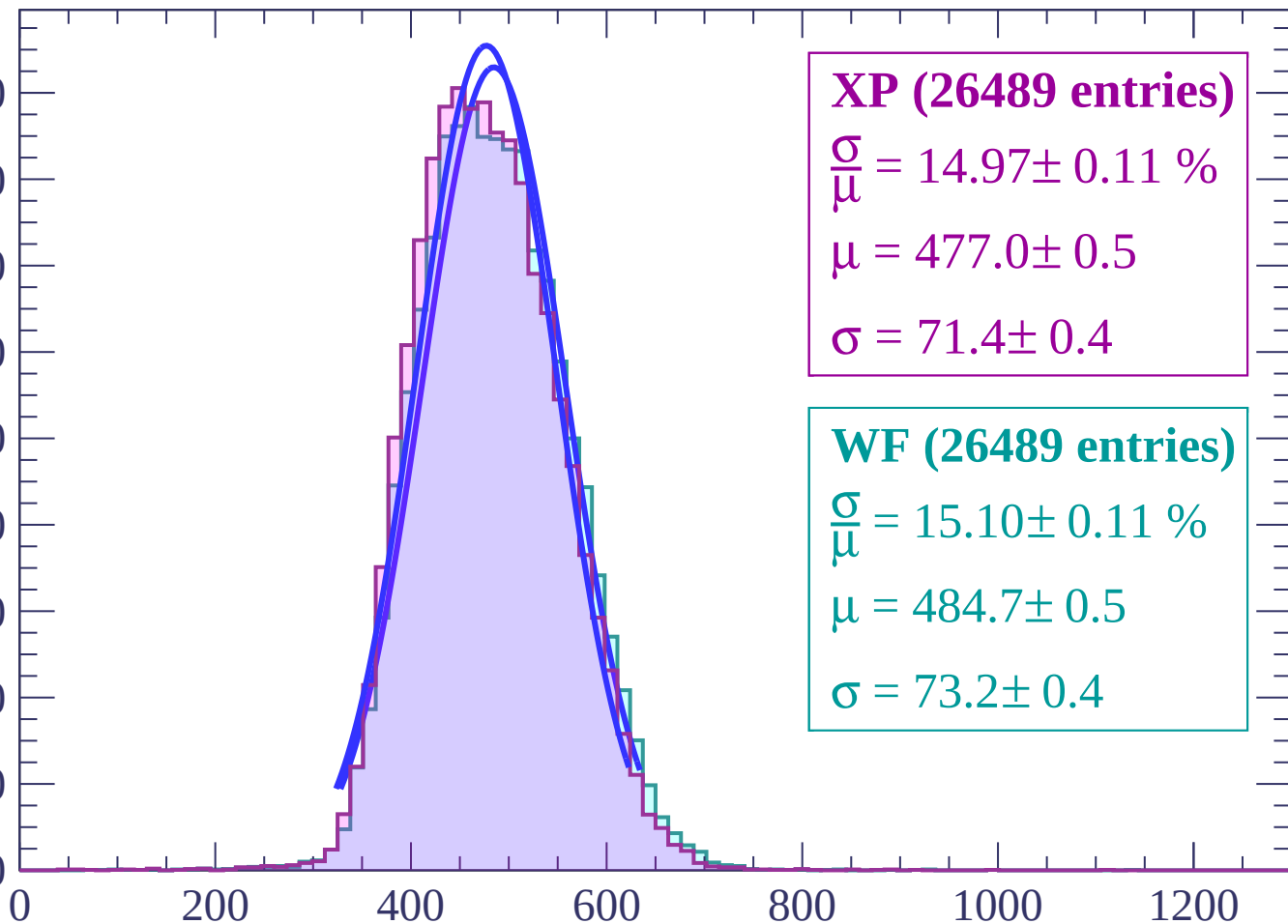
Energy loss |  $2 < \theta < 4$



Energy loss |  $4 < \theta < 6$

Count

1800  
1600  
1400  
1200  
1000  
800  
600  
400  
200  
0



**XP (26489 entries)**

$\frac{\sigma}{\mu} = 14.97 \pm 0.11 \%$

$\mu = 477.0 \pm 0.5$

$\sigma = 71.4 \pm 0.4$

**WF (26489 entries)**

$\frac{\sigma}{\mu} = 15.10 \pm 0.11 \%$

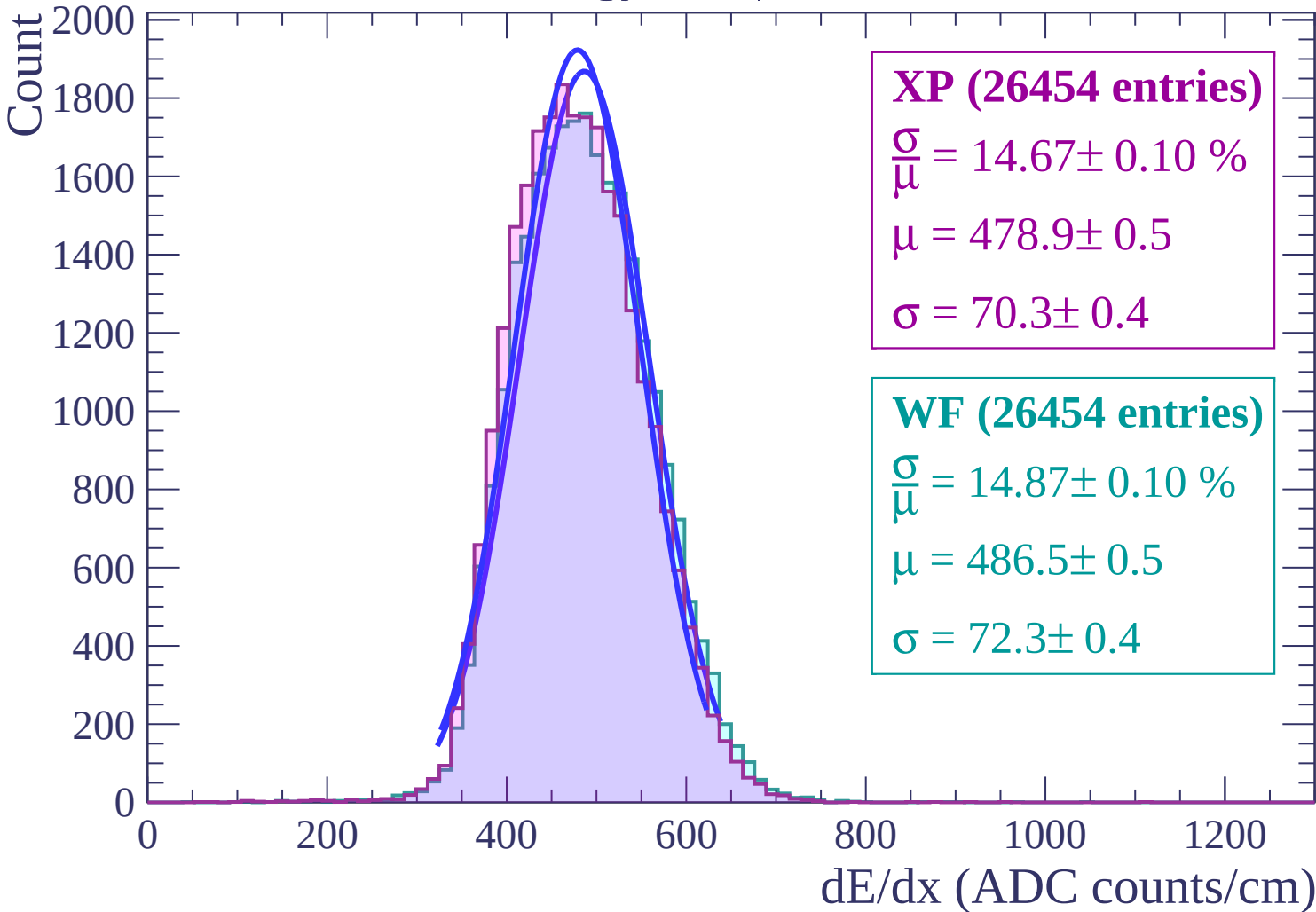
$\mu = 484.7 \pm 0.5$

$\sigma = 73.2 \pm 0.4$

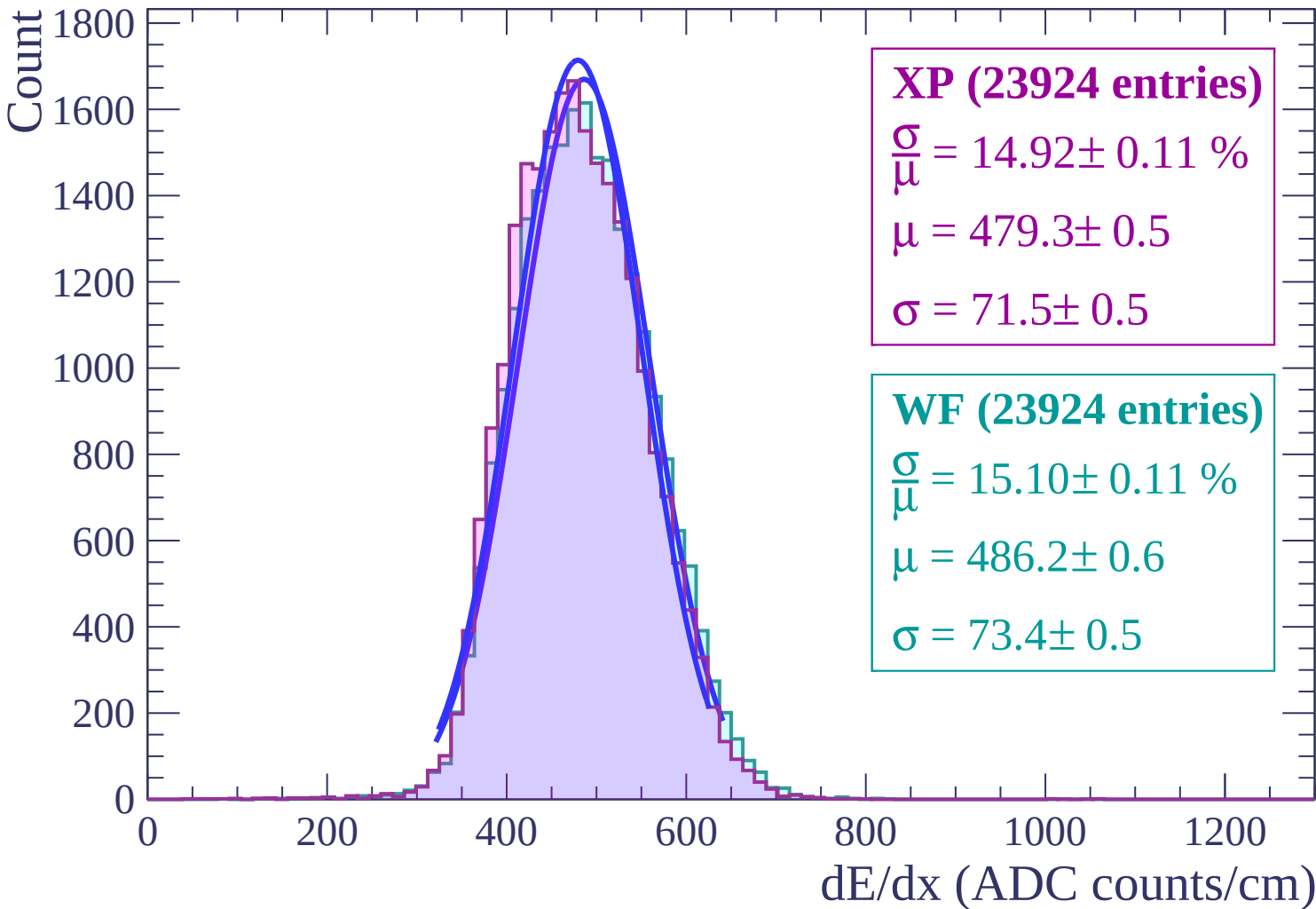
$dE/dx$  (ADC counts/cm)



Energy loss |  $6 < \theta < 8$



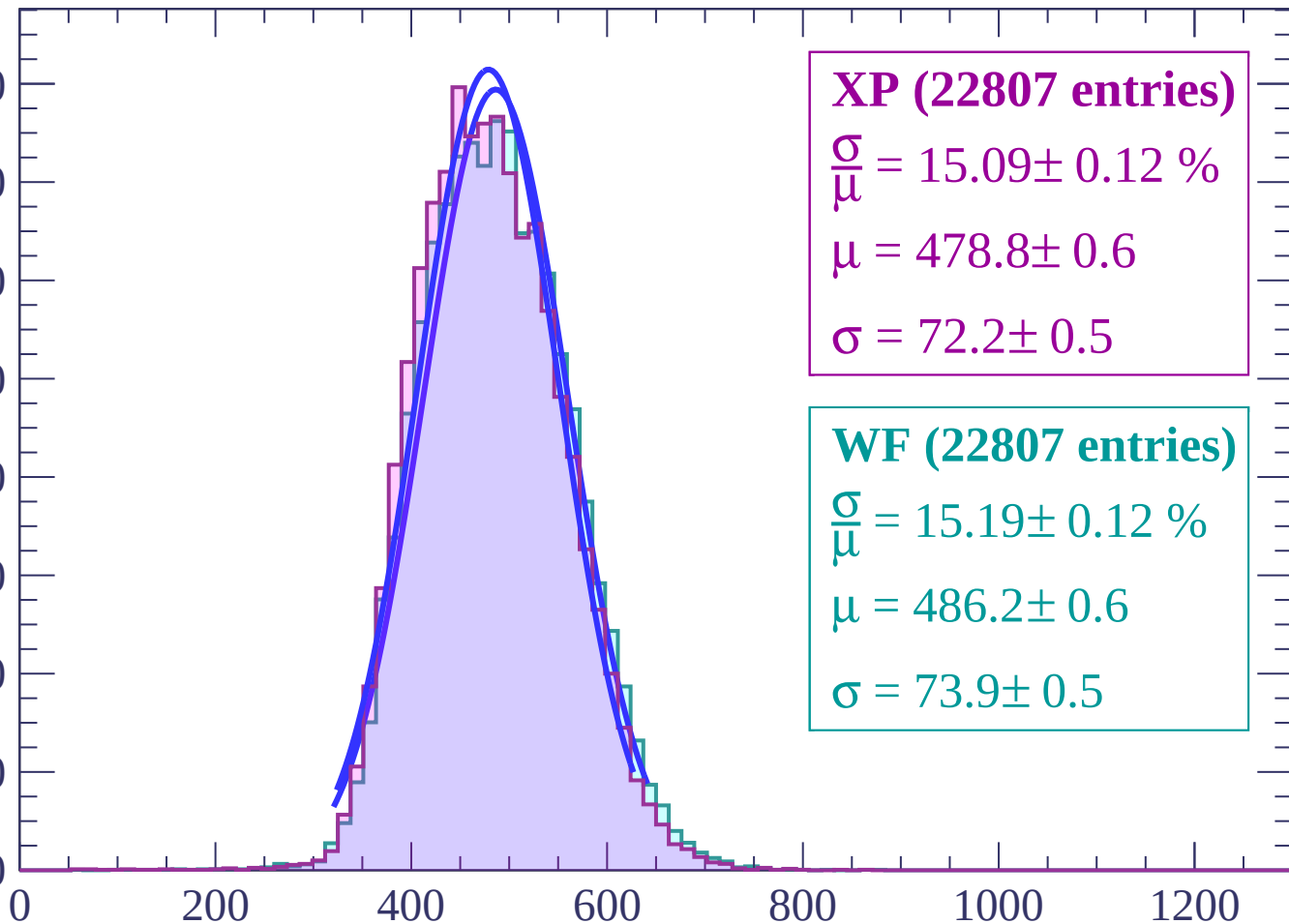
Energy loss |  $8 < \theta < 10$



Energy loss |  $10 < \theta < 12$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0



**XP (22807 entries)**

$\frac{\sigma}{\mu} = 15.09 \pm 0.12 \%$

$\mu = 478.8 \pm 0.6$

$\sigma = 72.2 \pm 0.5$

**WF (22807 entries)**

$\frac{\sigma}{\mu} = 15.19 \pm 0.12 \%$

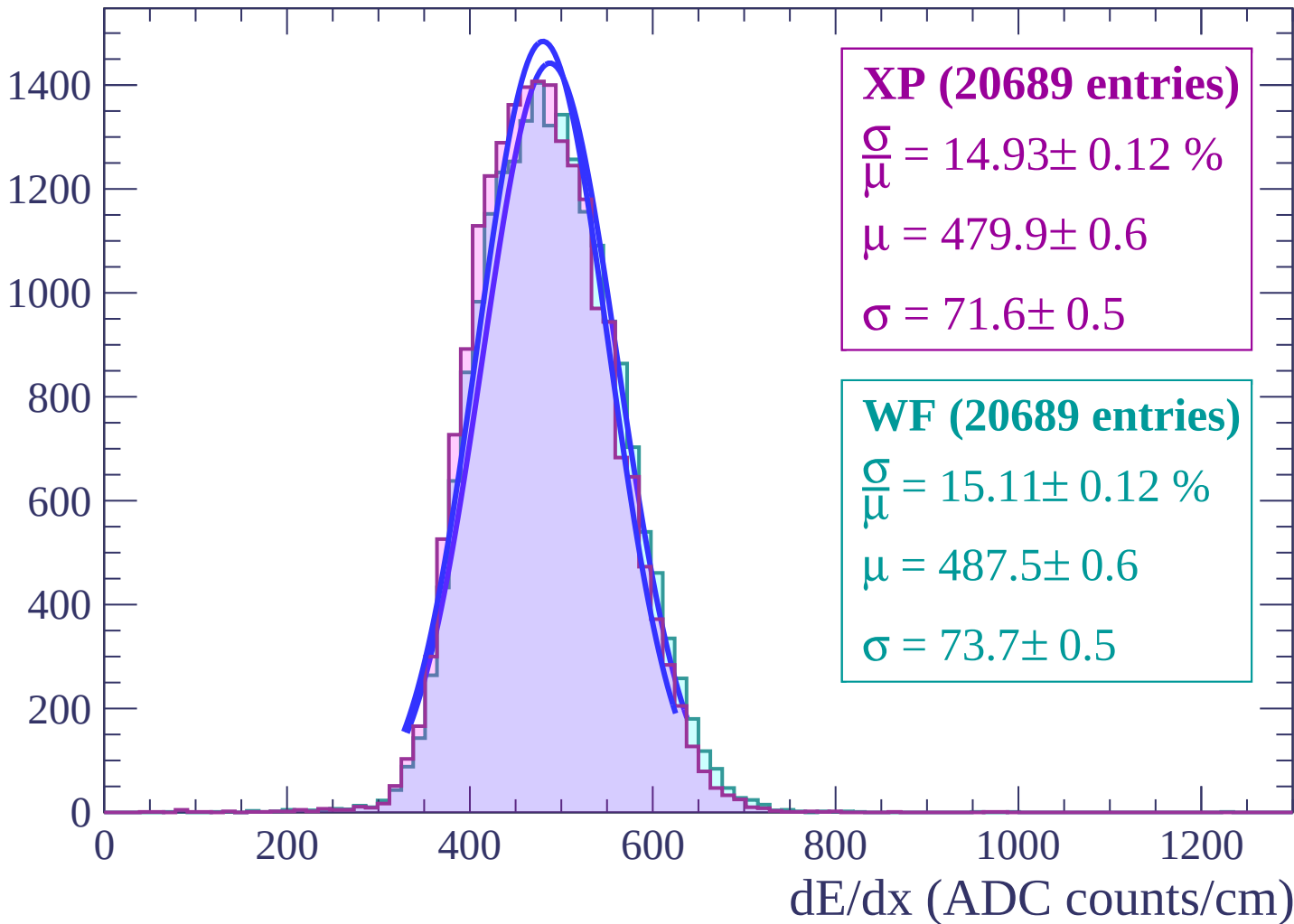
$\mu = 486.2 \pm 0.6$

$\sigma = 73.9 \pm 0.5$

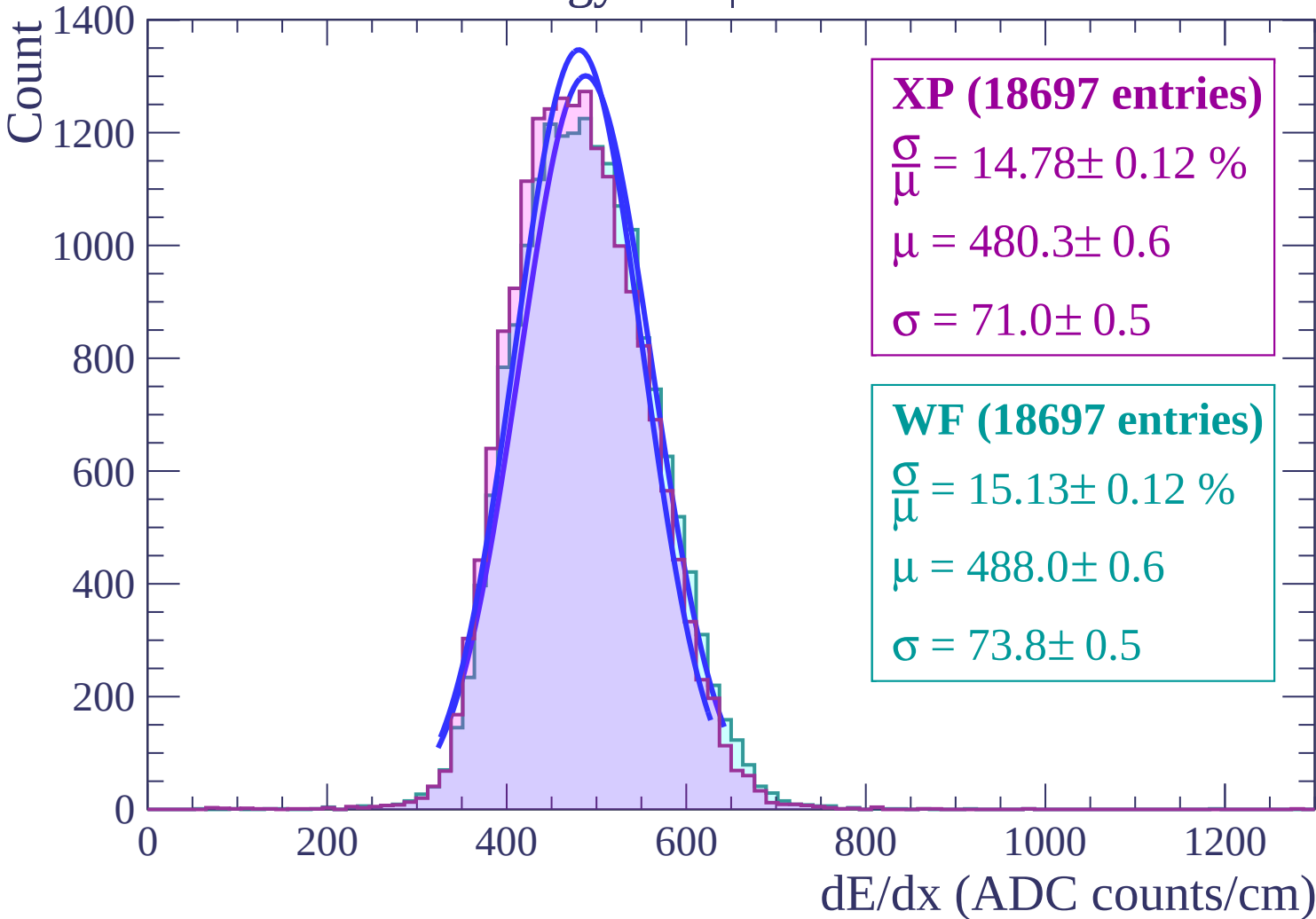
dE/dx (ADC counts/cm)

Energy loss |  $12 < \theta < 14$

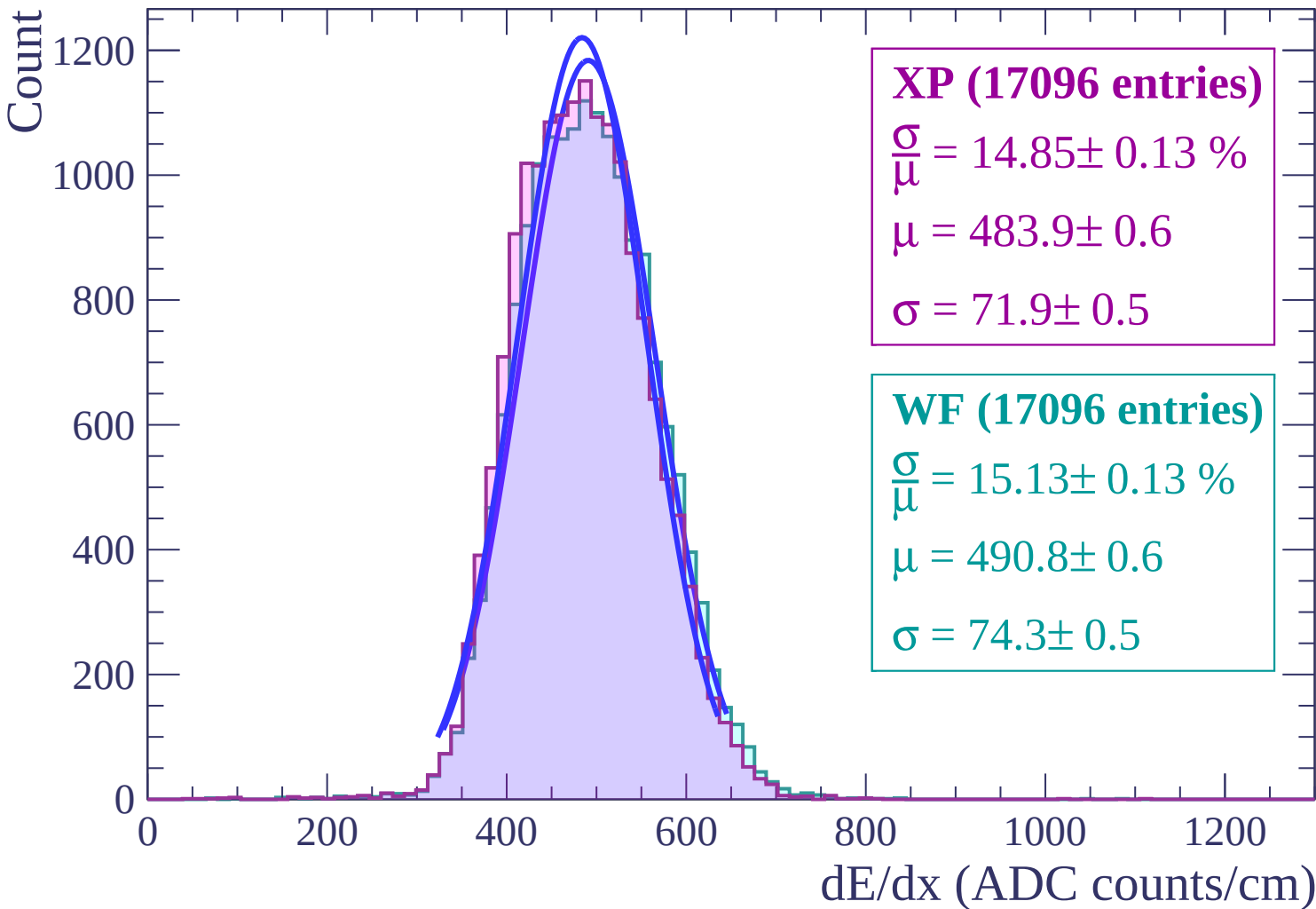
Count



# Energy loss | $14 < \theta < 16$



# Energy loss | $16 < \theta < 18$



Energy loss |  $18 < \theta < 20$

Count

1000

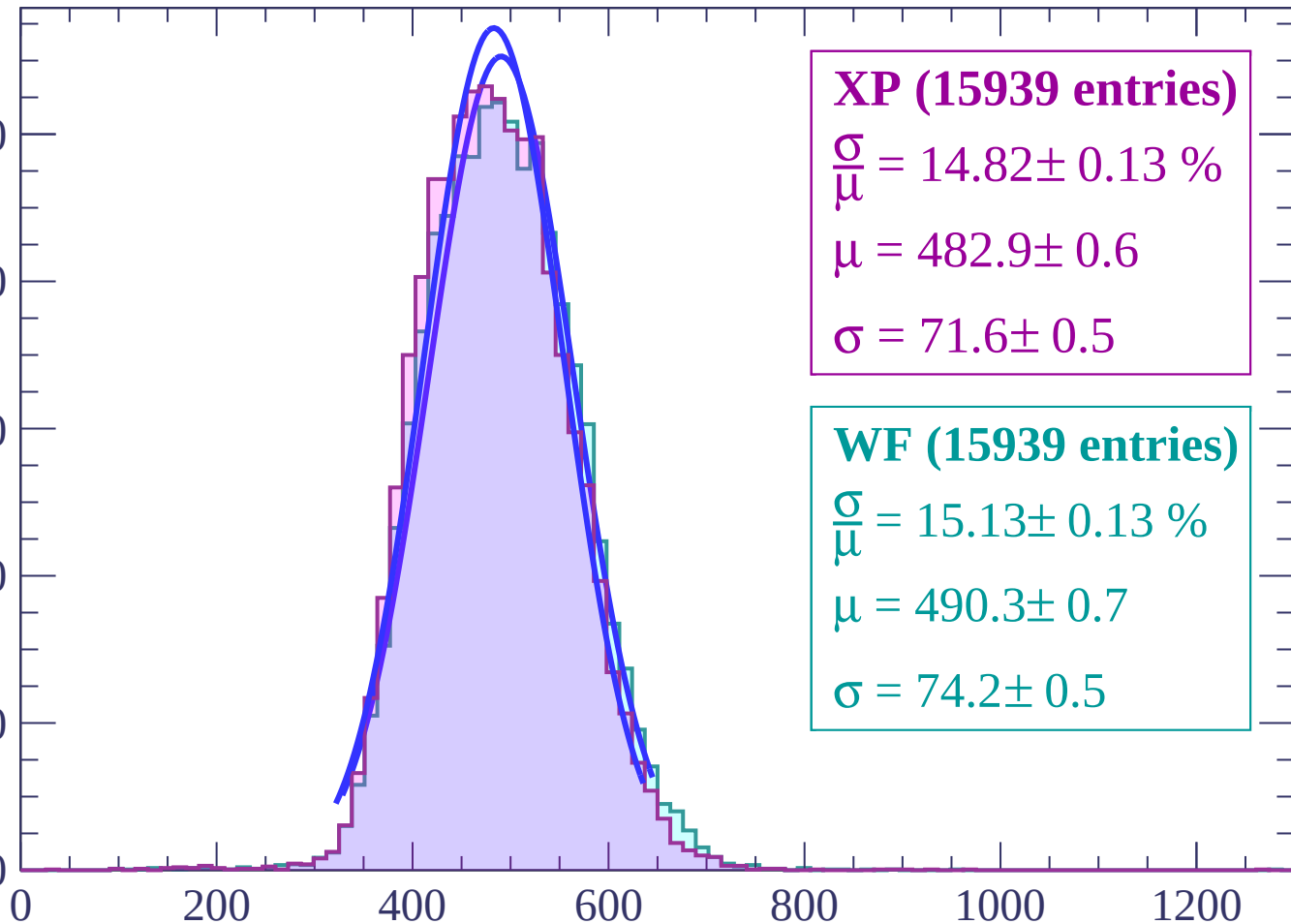
800

600

400

200

0



**XP (15939 entries)**

$\frac{\sigma}{\mu} = 14.82 \pm 0.13 \%$

$\mu = 482.9 \pm 0.6$

$\sigma = 71.6 \pm 0.5$

**WF (15939 entries)**

$\frac{\sigma}{\mu} = 15.13 \pm 0.13 \%$

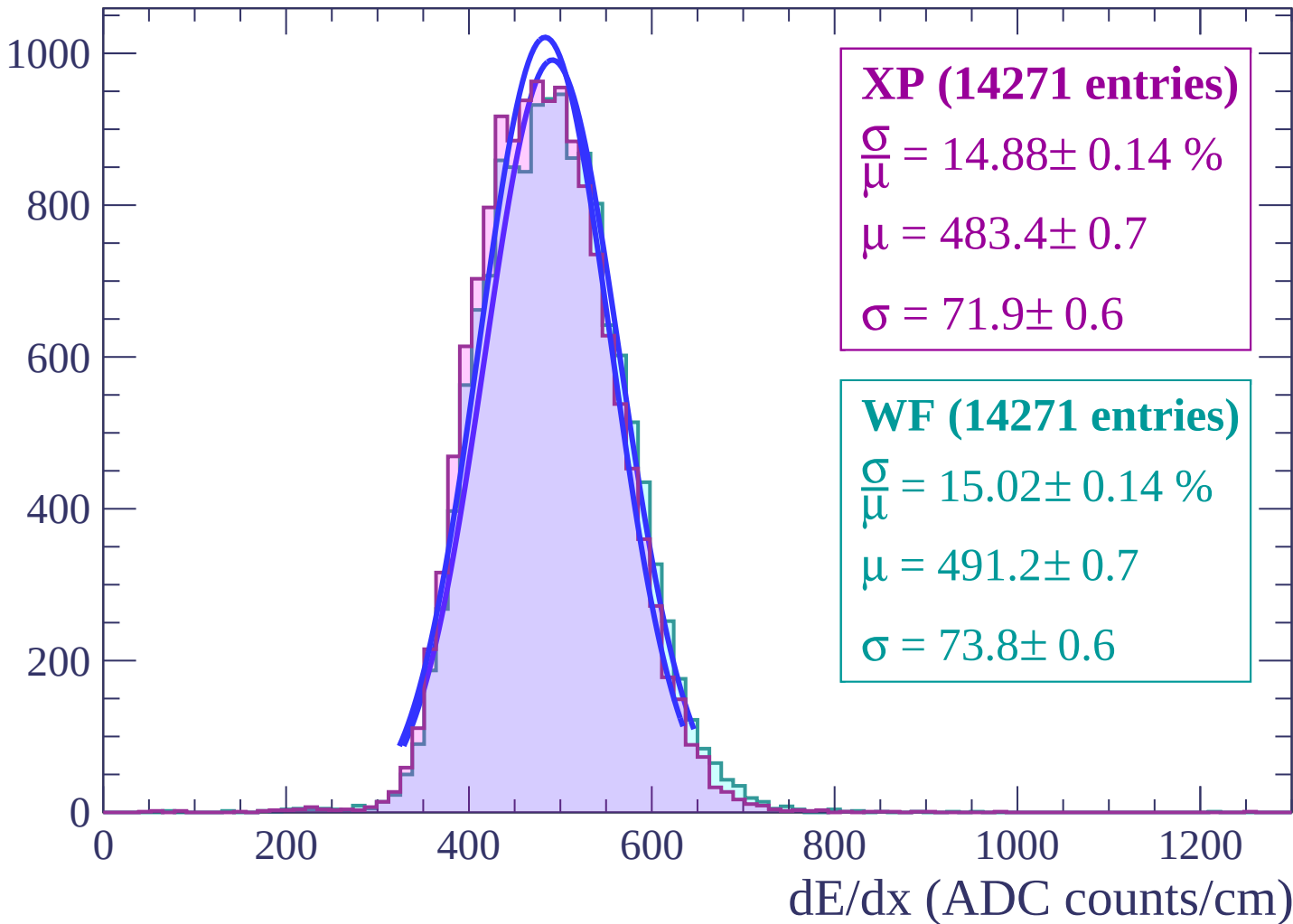
$\mu = 490.3 \pm 0.7$

$\sigma = 74.2 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $20 < \theta < 22$

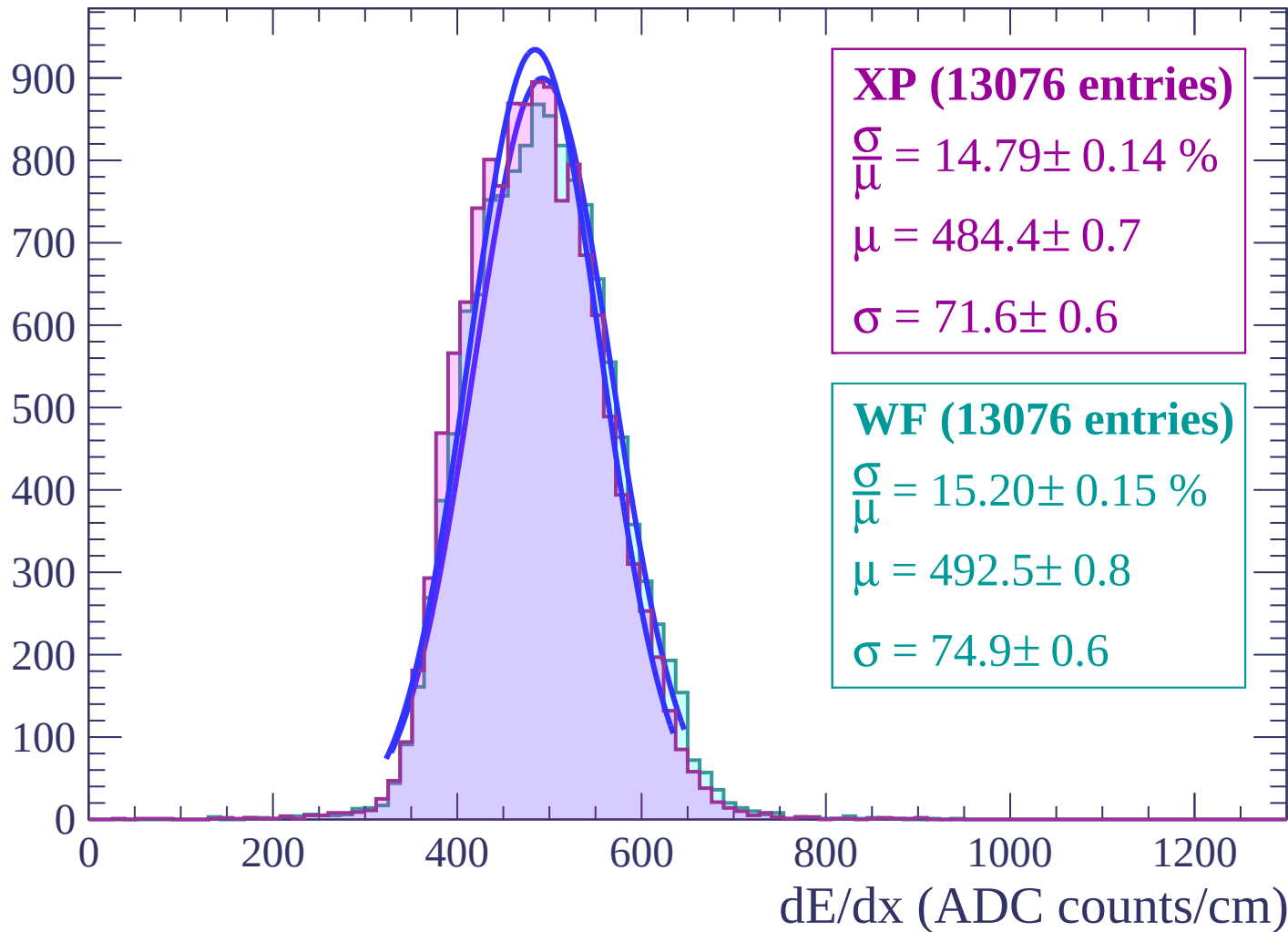
Count





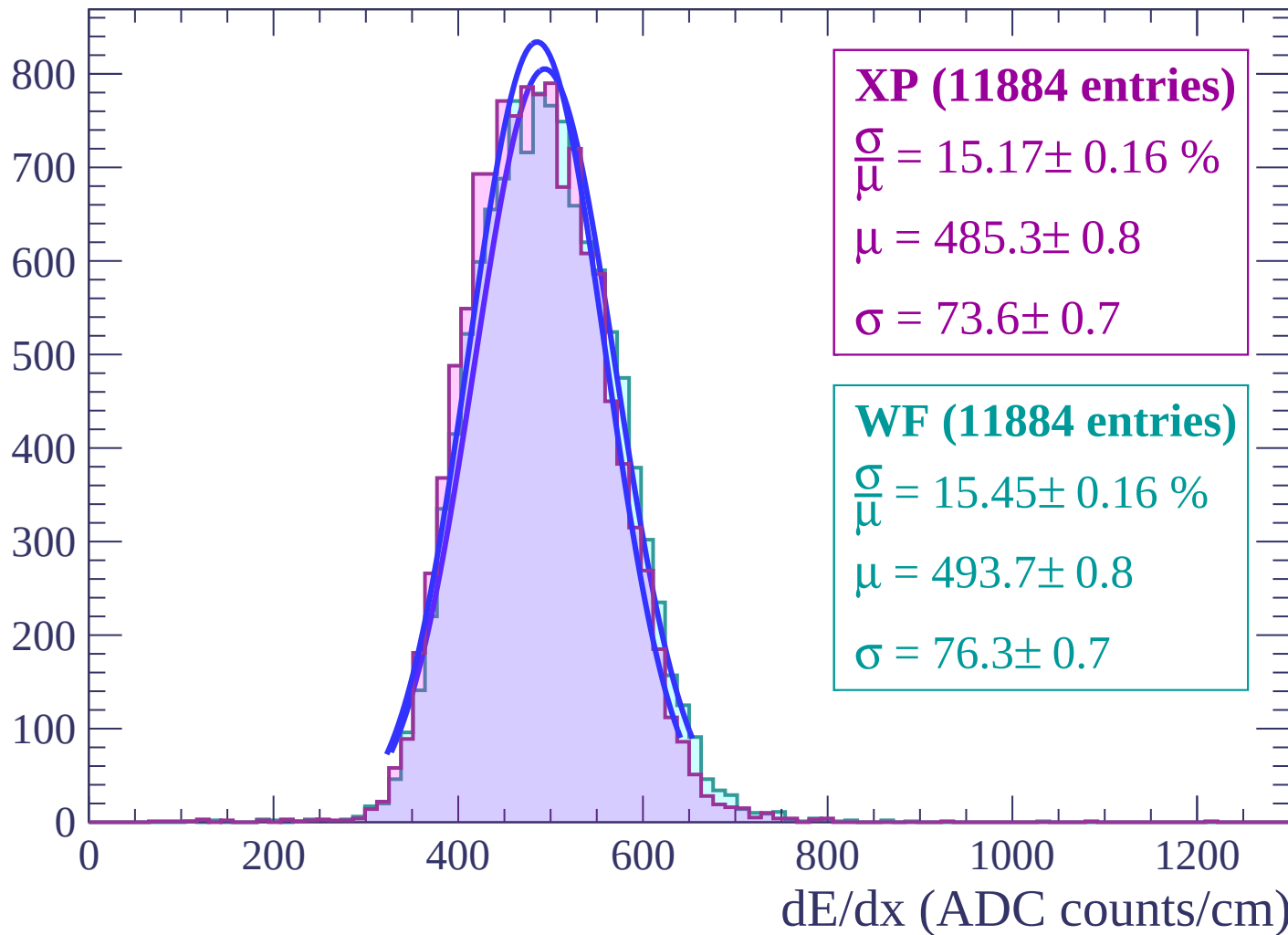
Energy loss |  $22 < \theta < 24$

Count



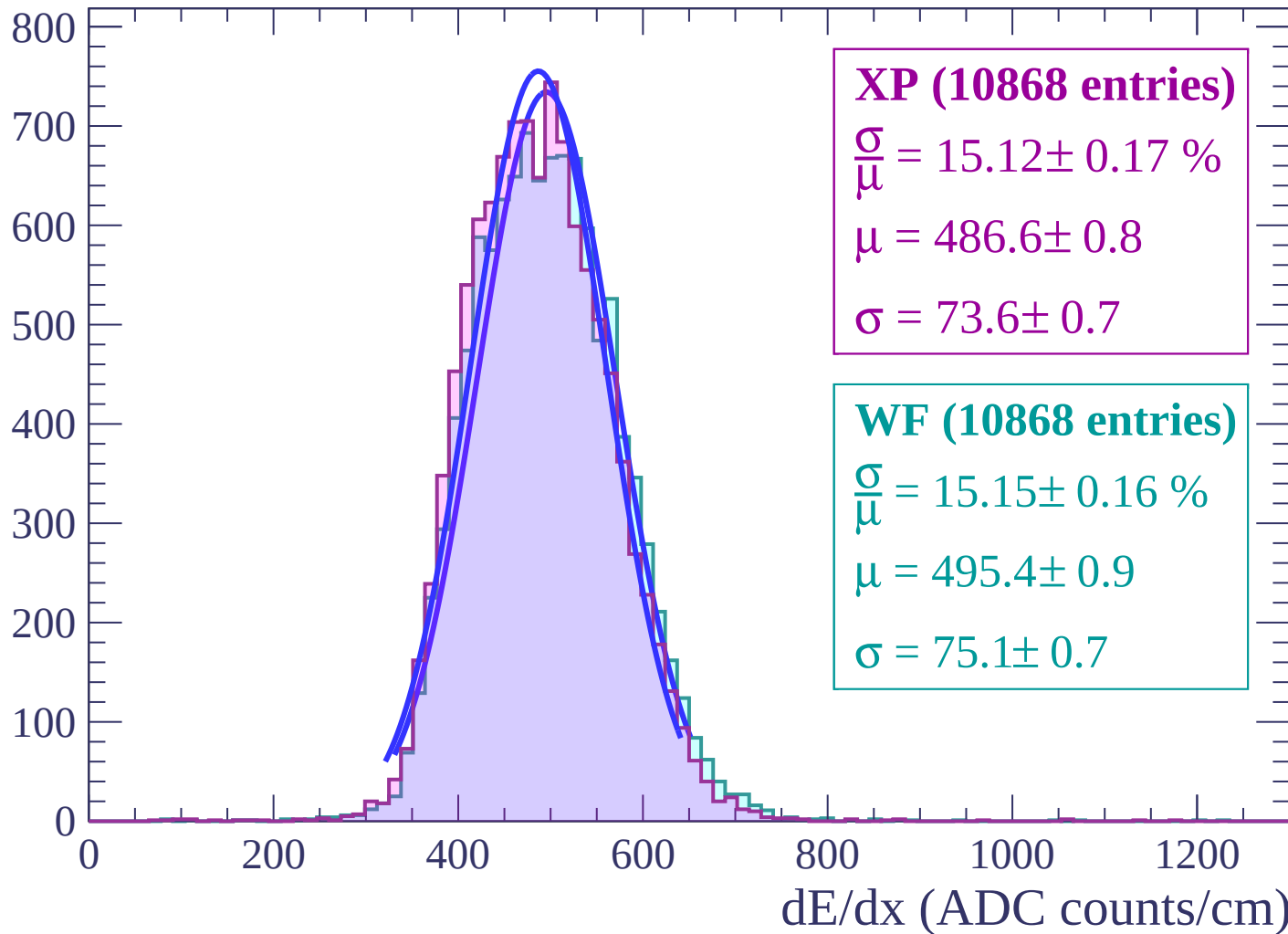
Energy loss |  $24 < \theta < 26$

Count



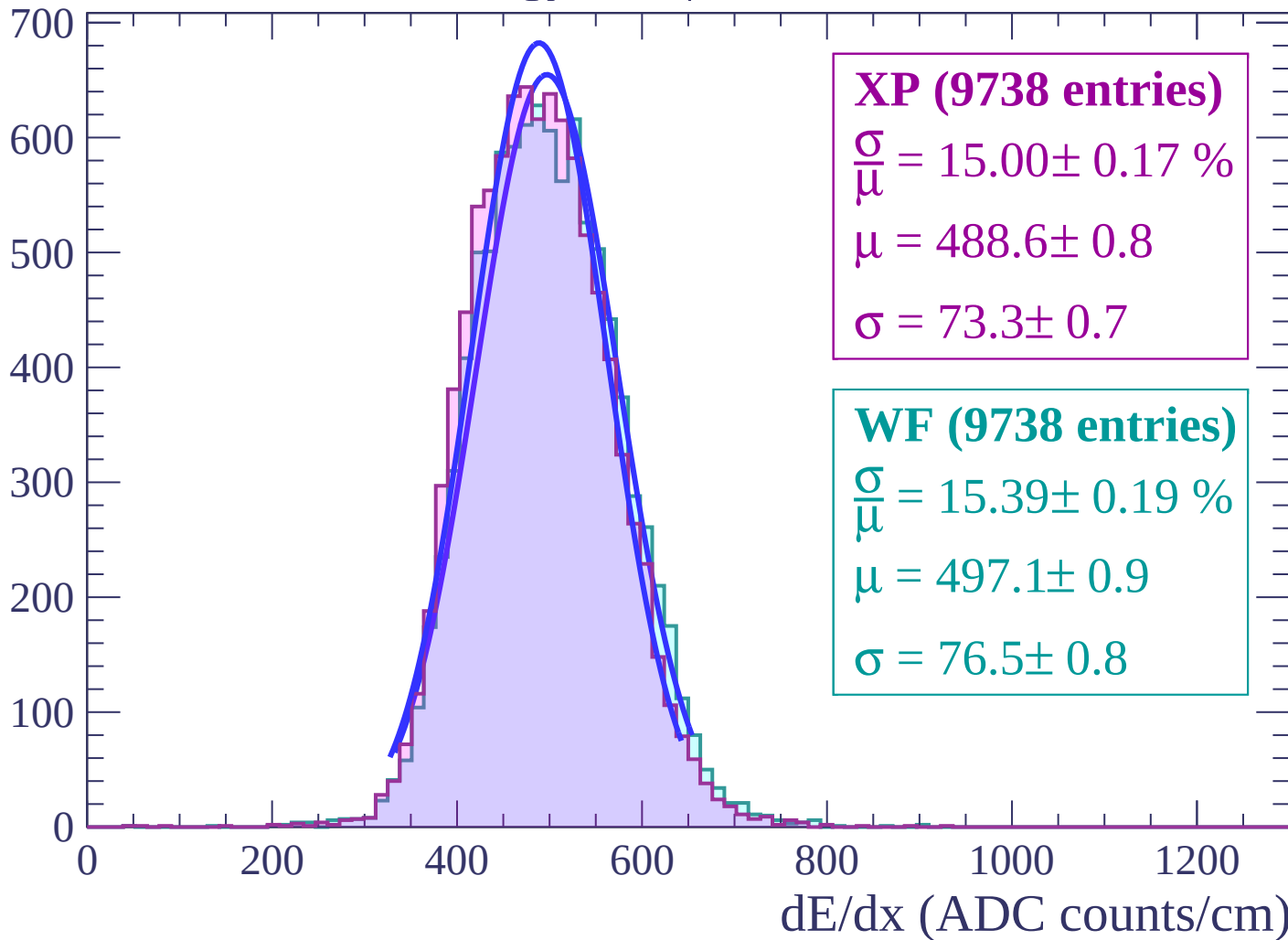
Energy loss |  $26 < \theta < 28$

Count



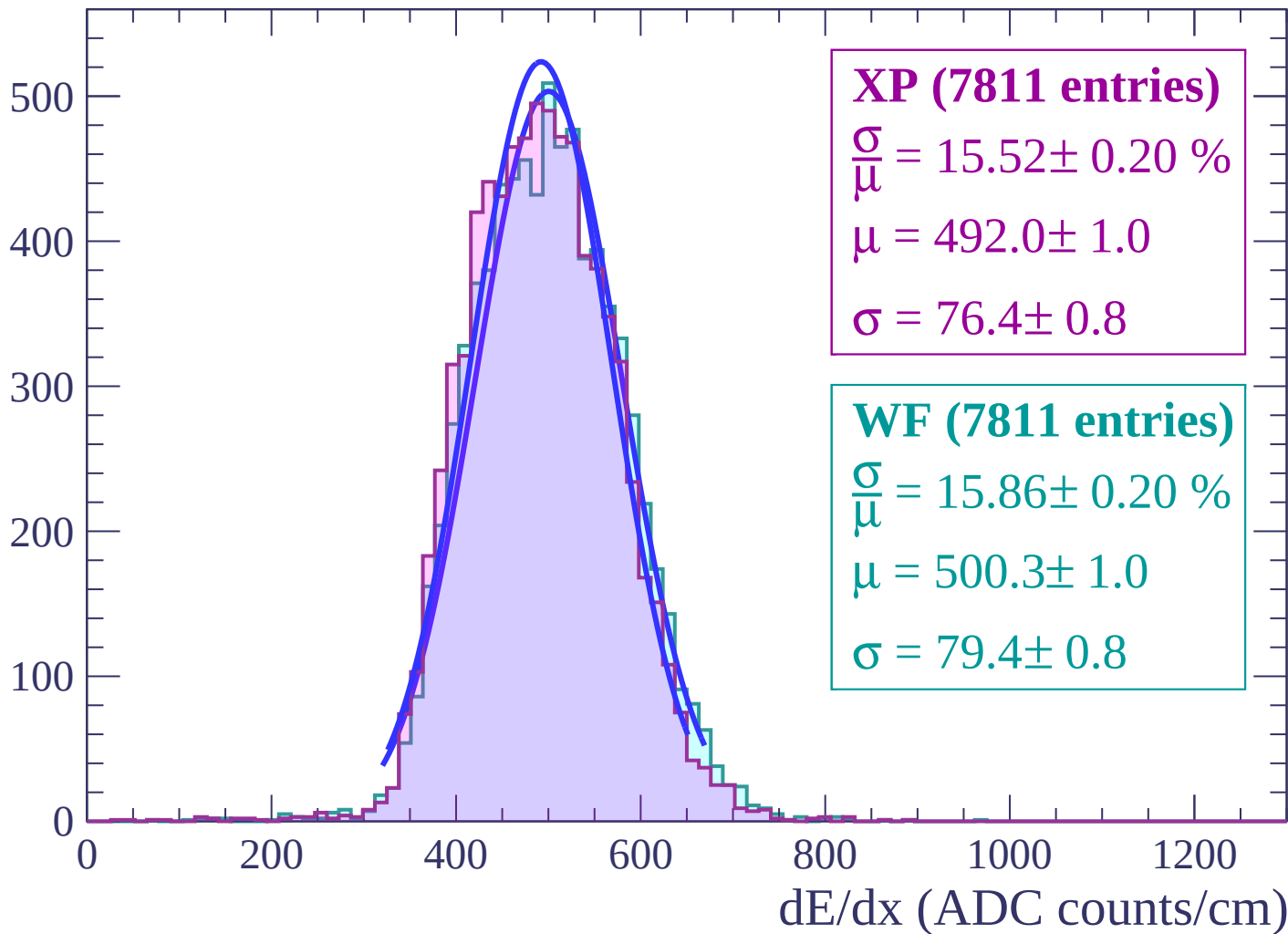
Energy loss |  $28 < \theta < 30$

Count



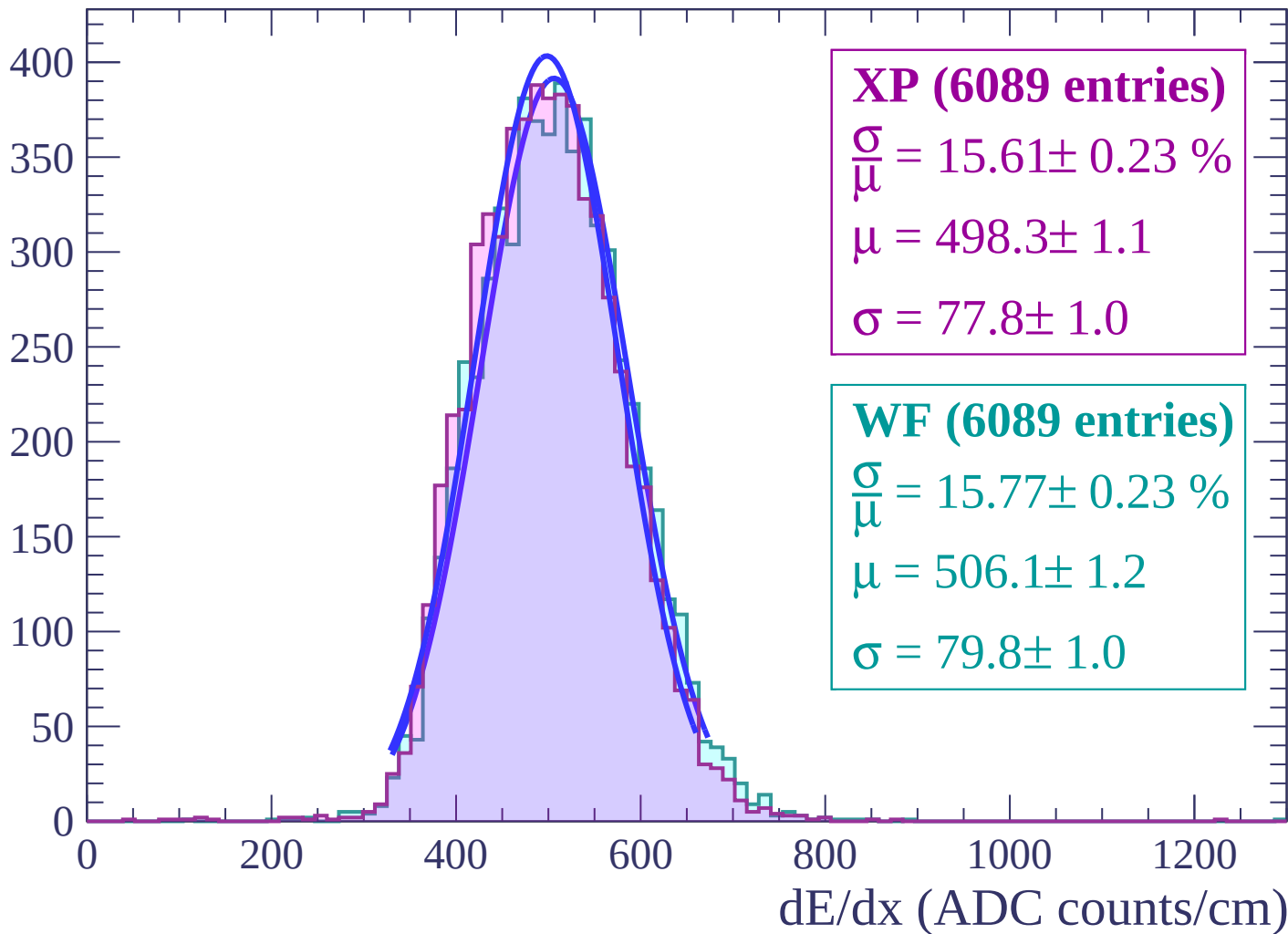
Energy loss |  $30 < \theta < 32$

Count



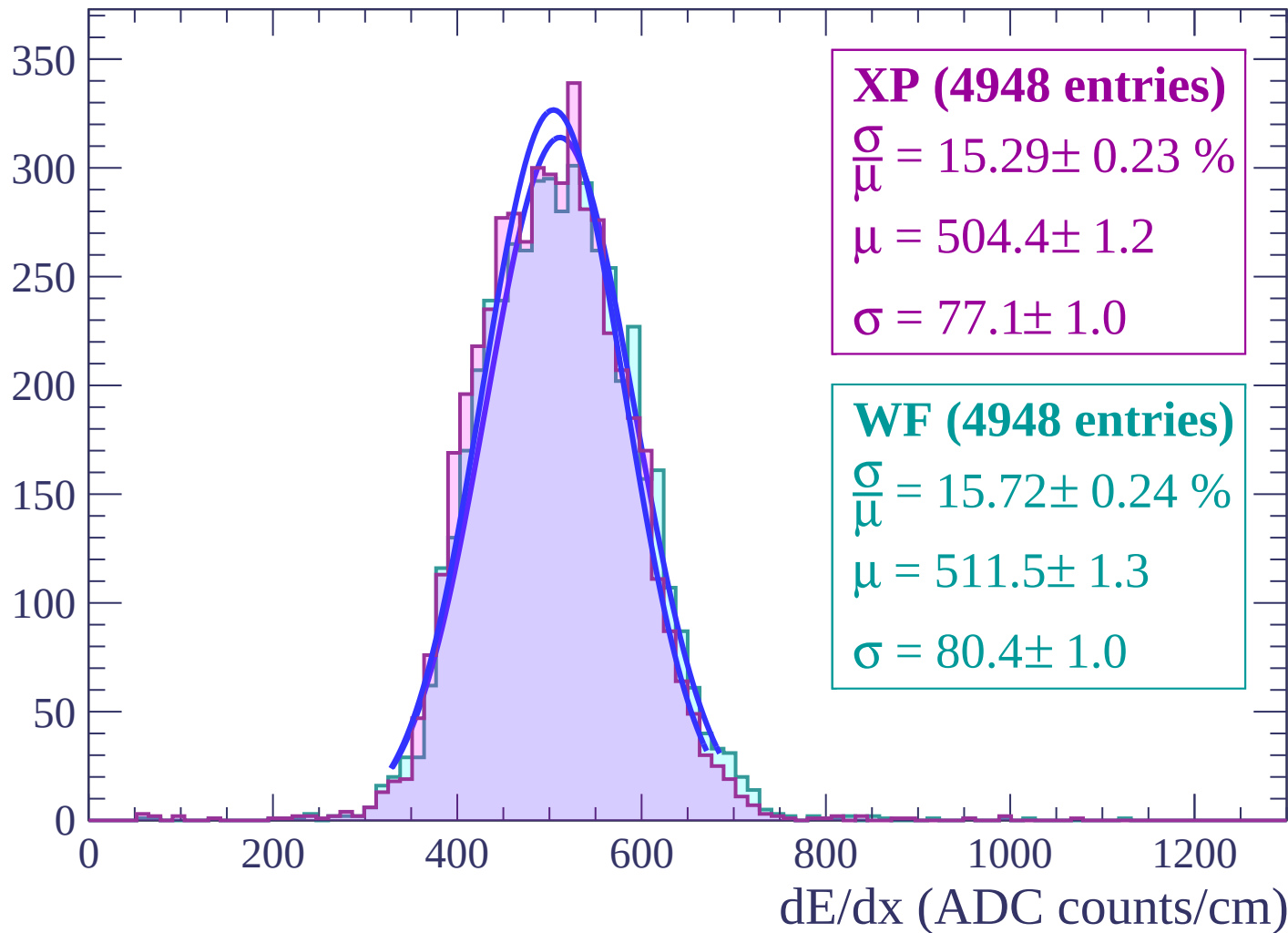
Energy loss |  $32 < \theta < 34$

Count



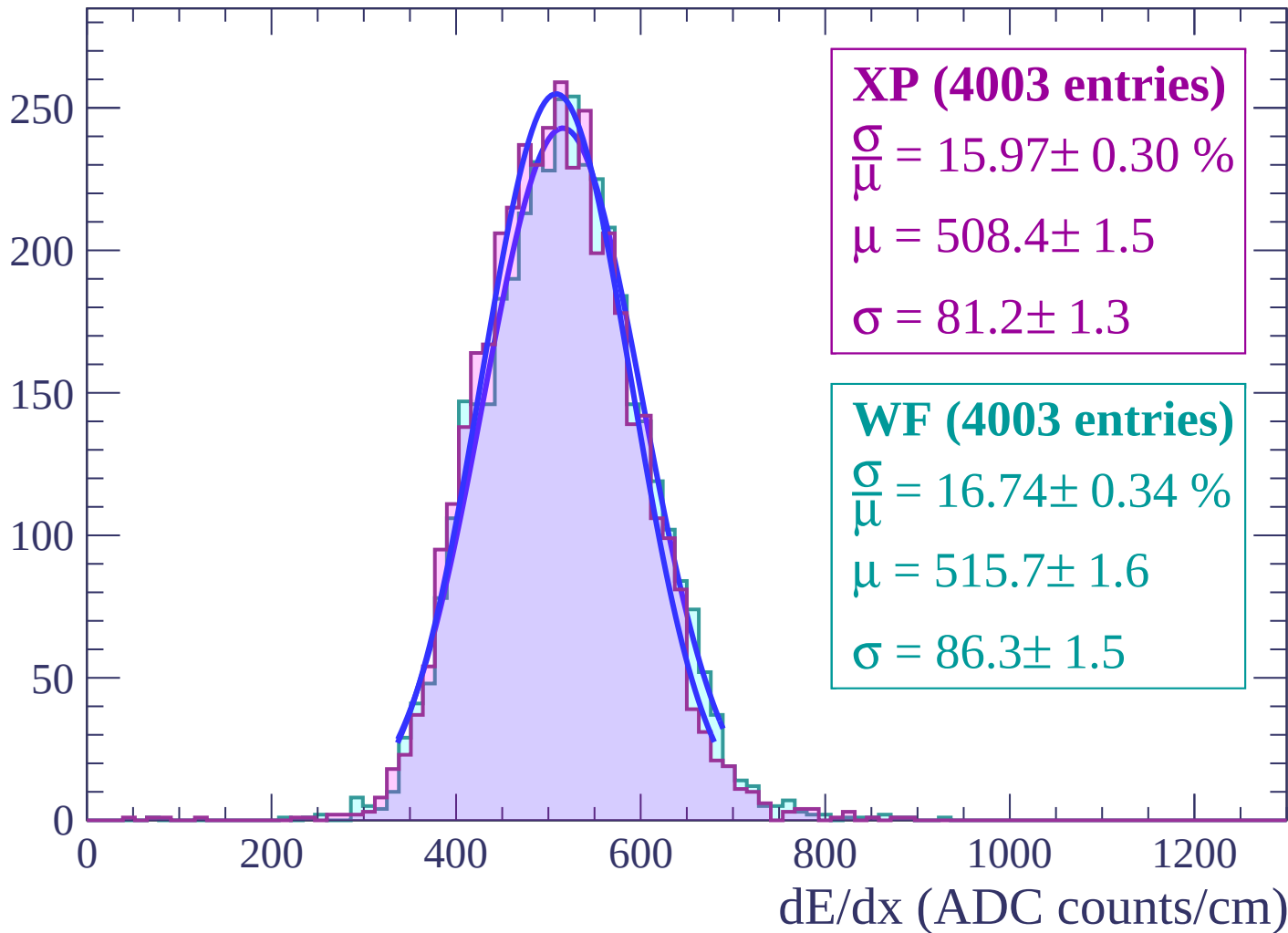
Energy loss |  $34 < \theta < 36$

Count



Energy loss |  $36 < \theta < 38$

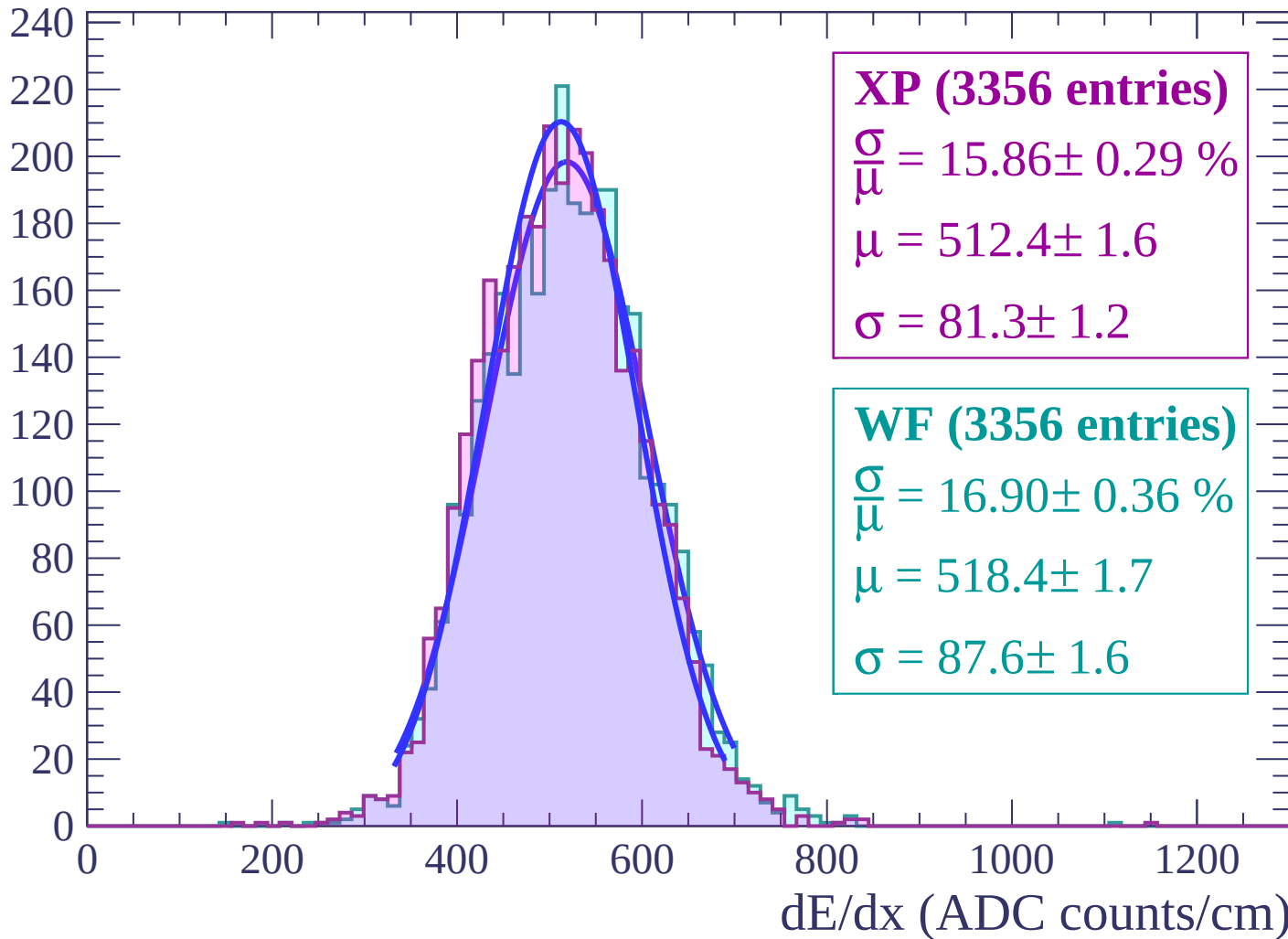
Count





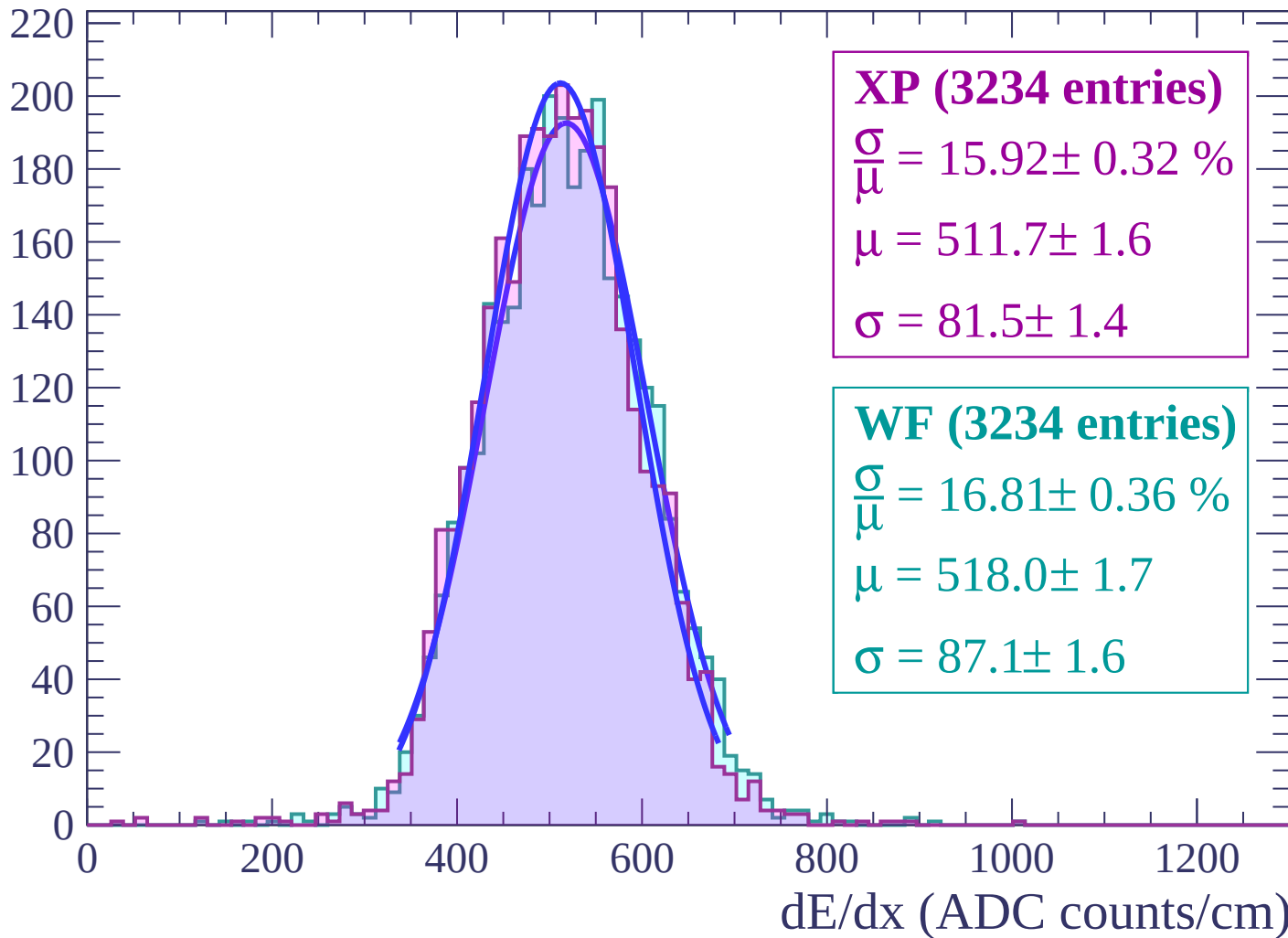
Energy loss |  $38 < \theta < 40$

Count



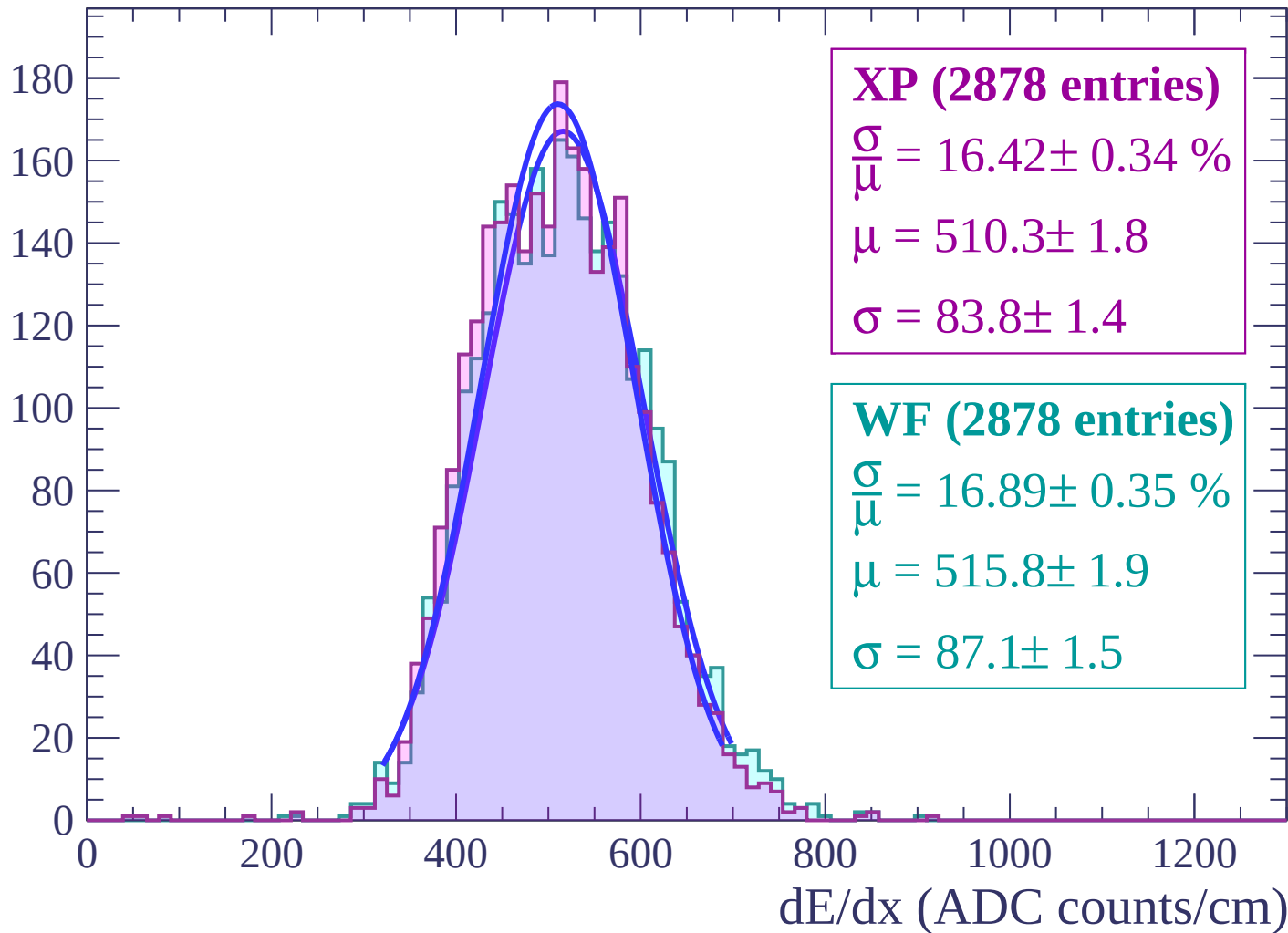
Energy loss |  $40 < \theta < 42$

Count



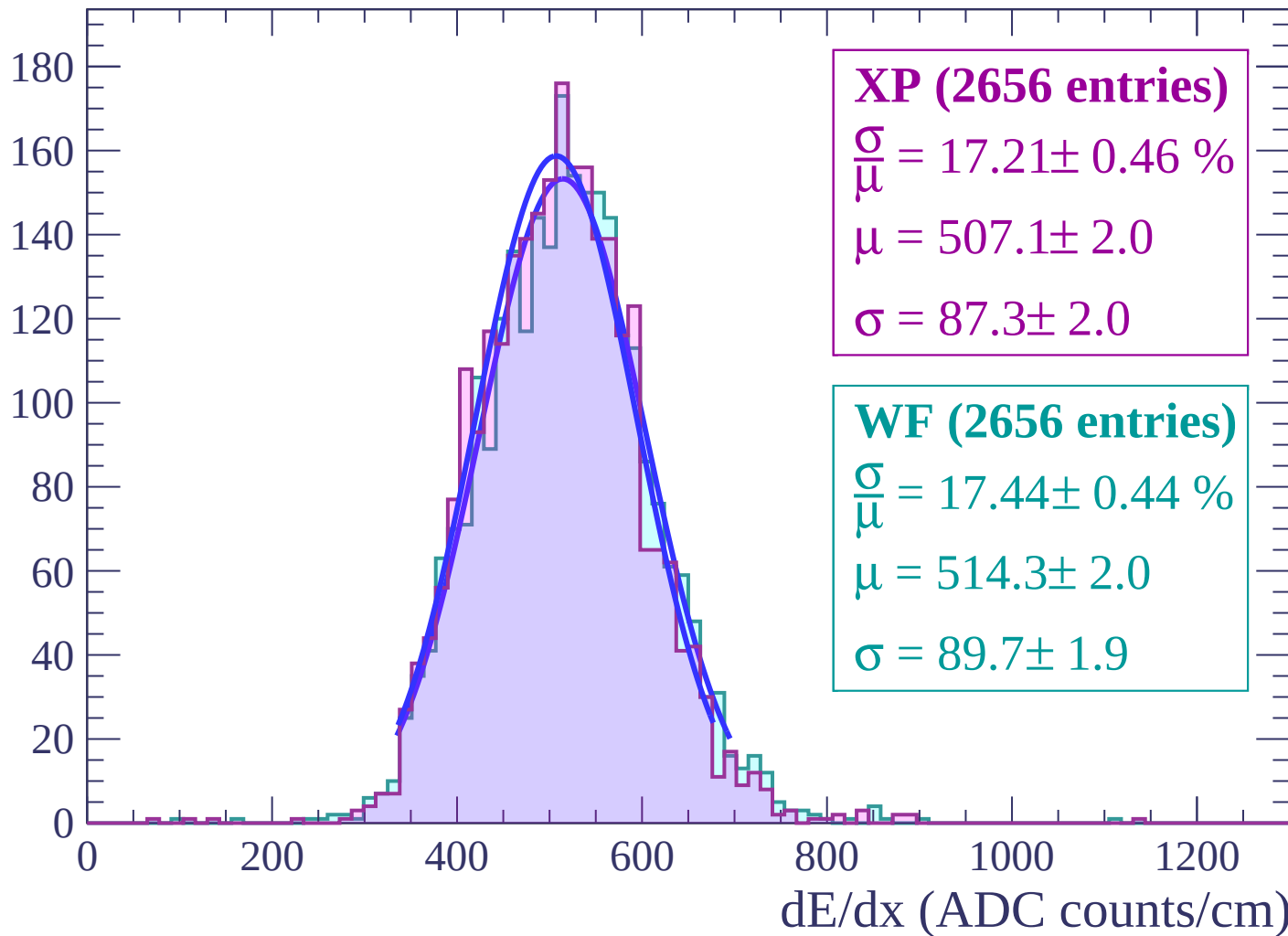
Energy loss |  $42 < \theta < 44$

Count



Energy loss |  $44 < \theta < 46$

Count



Energy loss |  $46 < \theta < 48$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $48 < \theta < 50$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $50 < \theta < 52$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $52 < \theta < 54$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $54 < \theta < 56$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $56 < \theta < 58$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $58 < \theta < 60$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $60 < \theta < 62$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $62 < \theta < 64$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $64 < \theta < 66$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $66 < \theta < 68$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $68 < \theta < 70$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $70 < \theta < 72$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $72 < \theta < 74$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $74 < \theta < 76$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $76 < \theta < 78$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $78 < \theta < 80$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $80 < \theta < 82$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $82 < \theta < 84$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $84 < \theta < 86$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $86 < \theta < 88$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $88 < \theta < 90$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $90 < \theta < 92$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF (0 entries)**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

